

FINAL TRIM AND STABILITY BOOKLET OF KB-32 (YARD NO. CHISTI 018)



AS AMENDED
SEE PAGE NO.54

SEE LETTER E-138438-190889

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GENERAL PARTICULARS

Name of the Vessel	KB-32
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE SERVICES
Yard No.	CHISTI 018
Type of Vessel	PONTOON
Official No.	MAR-2396-D
Class Notation	㊦SU, PONTOON, INDIAN COASTAL SERVICE DESCRIPTION"FOR CRANE OPERATION" "DECK STRENGTHENED FOR 10T/M ² "

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1893.06 T
Lightship	455.34 T
Deadweight	1437.72 T

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15^\circ$
- $GZC \leq 0,6 GZ_{MAX}$
- $A1 \geq 0.4 ATOT$

where:

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC , GZ_{MAX} : Defined in Fig 1

$A1$: Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle equal to the lesser of:

- heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θ_F , corresponding to flooding of unprotected openings

$ATOT$: Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = (Pd - Zz) / \Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z : Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM . The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

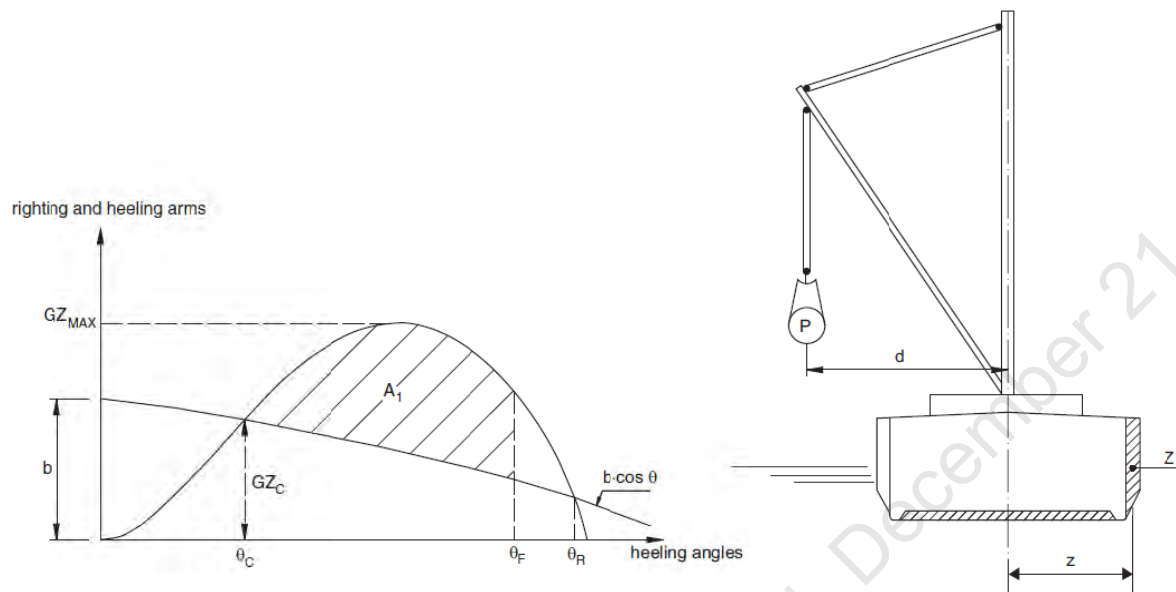


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A_2 / A_1 \geq 1$
- $\theta_2 - \theta_3 \geq 20^\circ$

where:

A_1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_1 to the heeling angle θ_C (see Fig 2)

A_2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_2 (see Fig 2)

A_3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_3 (see Fig 2)

θ_1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ_2 : Heeling angle corresponding to the lesser of θ_R and θ_F

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ_3 : Maximum heeling angle due to roll, at which $A_3 = A_1$, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θ_R : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θ_F : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$$b = Zz \cos\theta/\Delta$$

HEELING AND RIGHTING ARMS

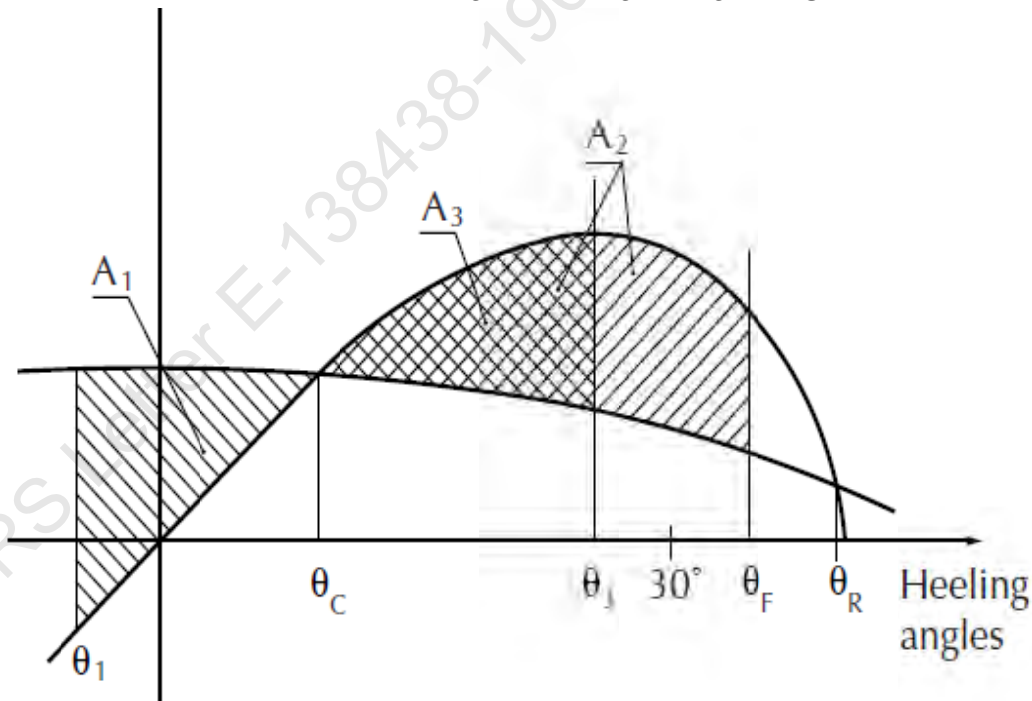


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

-1.00m, -0.50m, 0.00m, 0.50m, 1.00m

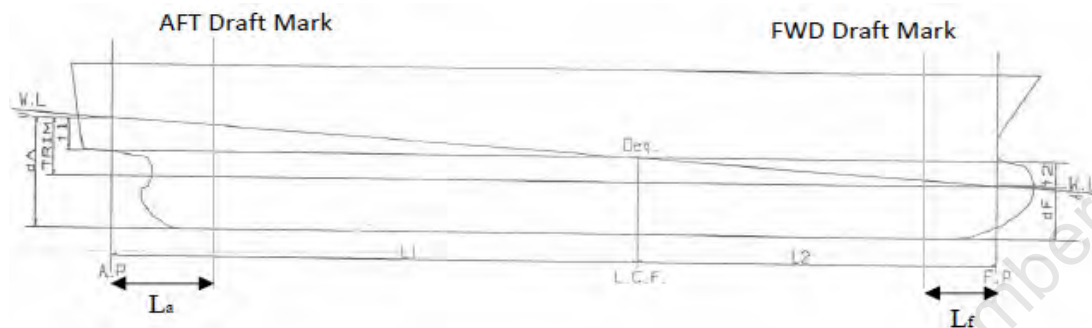
These trims are the expected operational trims.

MAX. DECK CARGO HEIGHT = 2.00M ABOVE DECK

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$\text{LCG} = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
- c) Displacement, Δ
- d) Moment to change trim, MCT1cm
- e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (\text{LCB} - \text{LCG})}{\text{MCT1cm} * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{\text{LCF} * \text{Trim}}{\text{LBP.}} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

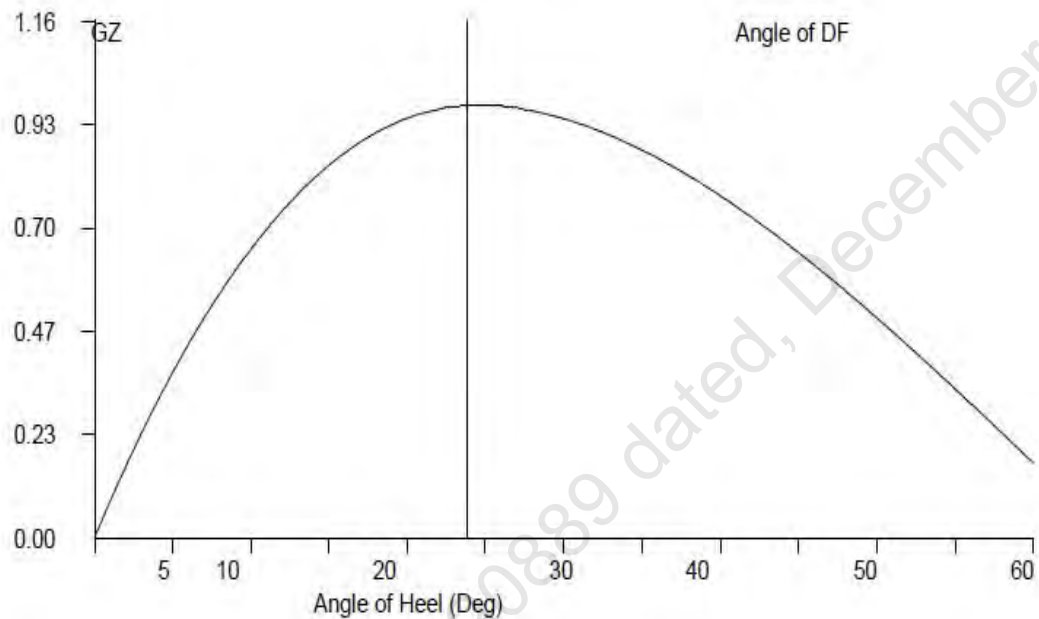
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples**1 Loading the Vessel**

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	1090.210	Tonnes
LCG	24.225	M Fwd of AP
VCG (KG)	3.931	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 1 : Lightship + Crane + Ballast

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
07_WBTK.P	48.39	1.025	49.60	47.660	2363.94	-5.971	-296.16	1.880	93.25	72.32
06_WBTK.C	132.36	1.025	135.67	42.251	5732.19	0.000	0.00	1.499	203.37	135.61
07_WBTK.S	48.39	1.025	49.60	47.660	2363.94	5.971	296.16	1.880	93.25	72.32
TOTAL TANK			234.87	44.536	10460.07	0.000	0.00	1.660	389.87	280.25

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			455.34	24.928	11,350.72	0.000	0.00	3.000	1,366.02
CRANE			400.00	11.500	4,600.00	0.000	0.00	6.325	2,530.00
TOTAL FIXED WEIGHT			855.34	18.648	15,950.72	0.000	0.00	4.555	3,896.02

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			234.870	44.536	10460.065	0.000	0.000	1.660	389.865
TOTAL FIXED WEIGHT			855.340	18.648	15950.716	0.000	0.000	4.555	3896.020
TOTAL WEIGHT			1090.210	24.225	26410.781	0.000	0.000	3.931	4285.885

Condition 1 : Lightship + Crane + Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.257	m
VCGc	4.188	m

From the MaxVCG table,

Displacement	1090.21	T
Allowable Max VCG	14.554	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1090.210	T
Draft ,Tm @ 25.0f =	1.325	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	33.186	T-m/cm
KMT =	22.431	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.254 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perpendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid	1.325	m
Draft @ 50.0f (FP),Tfwd	=Tm - Trim/2	
	1.198	m
Draft@0(AP),Taft	=Tm + Trim/2	
	1.453	m

Condition 1 : Lightship + Crane + Ballast**From HYDROSTATICS**

Hydrostatic Particulars			Drafts & Trims	
MCT	33.186	T-m	Draft (Mean)	1.325 m
LCB	25.000	m	Draft AP (USK)	1.453 m
			Draft FP (USK)	1.198 m
Metacentric Height				
KMt	22.431	m		
KG	3.931	m		
KG0	4.188	m		
GMT	18.243	m	Trim	0.254 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	32.360	T-m	Draft (Mean)	1.325 m
LCB	24.208	m	Draft AP (USK)	1.456 m
			Draft FP (USK)	1.194 m
Metacentric Height				
KMt	22.433	m		
KG	3.931	m		
KG0	4.188	m		
GMT	18.245	m	Trim	0.261 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

Condition 1 : Lightship + Crane + Ballast**Intact Stability (Primary Criterion)****Minimum Required**

Displacement	1090.21	T
KG0	4.188	M
GMt	18.245	M

0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	0.000	3.770	4.995	5.083	4.864	4.451
GZ (M)	-0.020	3.039	3.559	2.986	2.169	1.240

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.6855 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	LARGE m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	5.00 m-rad	0.00 Deg.

Plot GZ values against the angles of heel as above.

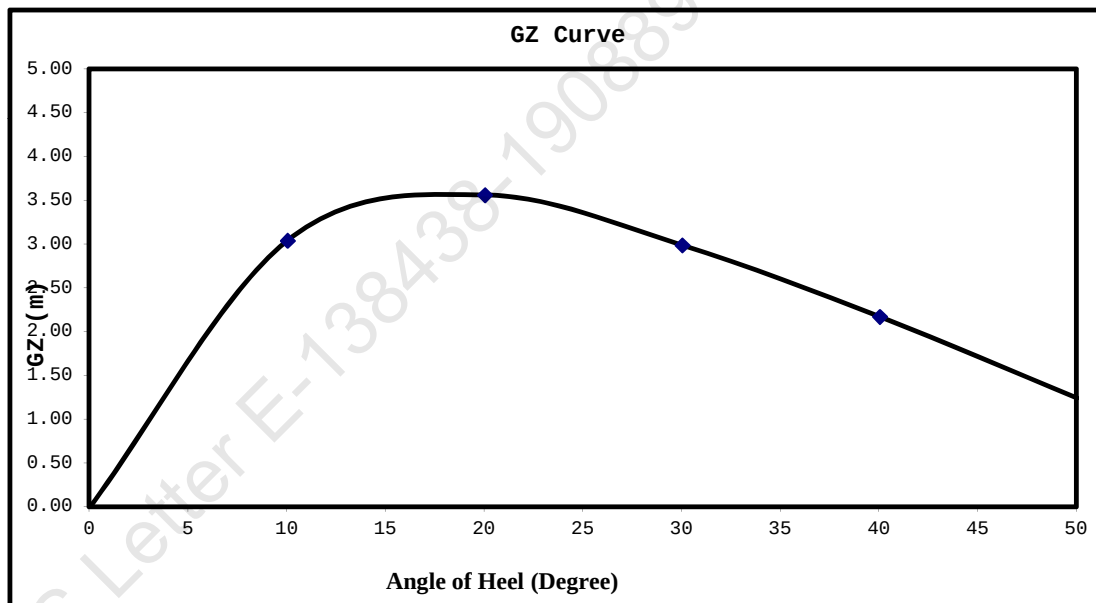
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees

then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	1.325	M
LCF from AP	24.756	M
LCB from AP	24.208	M
MCT	32.360	T-M/Cm
Trim	0.261	M
Draft (AP)	1.456	M
Draft (FP)	1.194	M
KG	3.931	M
KGo	4.188	M
GoMt	18.245	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

[illegible]

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			455.340	24.928	11,350.716	0.000	0.000	3.000	1,366.020
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

[illegible]

	m
VCGc	m

Displacement	T
Allowable Max VCG	m

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	T
Draft ,Tm @ 25.0f =	m
LCB =	m
LCF =	m
MCT =	T-m/cm
KMT =	m

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCTX} \times 100)$$

m, (+) is aft trim, (-) is fwd trim

LBP = 50m

Draft @ 25.00 (MS), T_{mid} m

Draft @ 50.00 (FP), T_{fwd} = $T_m - \frac{Trim}{2}$ m

Draft @ 0 (AP), T_{aft} = $T_m + \frac{Trim}{2}$ m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

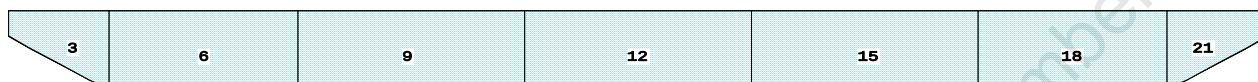
1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

01-08-20 21:16:44
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00N

CAPACITY PLAN AND TABLE

Profile View



Plan View

1	4	7	10	13	16	19
2	5	8	11	14	17	20
3	6	9	12	15	18	21

Tanks

1 01_WBTK.P	4 02_WBTK.P	8 03_WBTK.C	12 04_WBTK.S	16 06_WBTK.P	20 07_WBTK.C
2 01_WBTK.C	5 02_WBTK.C	9 03_WBTK.S	13 05_WBTK.P	17 06_WBTK.C	21 07_WBTK.S
3 01_WBTK.S	6 02_WBTK.S	10 04_WBTK.P	14 05_WBTK.C	18 06_WBTK.S	
	7 03_WBTK.P	11 04_WBTK.C	15 05_WBTK.S	19 07_WBTK.P	

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GHS 17.22 PONT00N

TANK STATUS							
Trim: zero, Heel: zero							
Part-----	Cu.M.----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
01_WBTK.P	48.4	1.025	49.60	2.340f	5.971p	1.880	72.32*
01_WBTK.C	48.9	1.025	50.11	2.336f	0.000	1.872	72.33*
01_WBTK.S	48.4	1.025	49.60	2.340f	5.971s	1.880	72.32*
02_WBTK.P	131.4	1.025	134.72	7.749f	5.980p	1.509	135.61*
02_WBTK.C	132.4	1.025	135.67	7.749f	0.000	1.499	135.61*
02_WBTK.S	131.4	1.025	134.72	7.749f	5.980s	1.509	135.61*
03_WBTK.P	157.7	1.025	161.60	16.000f	5.980p	1.509	162.73*
03_WBTK.C	158.8	1.025	162.73	16.000f	0.000	1.500	162.73*
03_WBTK.S	157.7	1.025	161.60	16.000f	5.980s	1.509	162.73*
04_WBTK.P	157.7	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	158.8	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	157.7	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.P	157.7	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.C	158.8	1.025	162.73	34.000f	0.000	1.500	162.73*
05_WBTK.S	157.7	1.025	161.60	34.000f	5.980s	1.509	162.73*
06_WBTK.P	131.4	1.025	134.72	42.251f	5.980p	1.509	135.61*
06_WBTK.C	132.4	1.025	135.67	42.251f	0.000	1.499	135.61*
06_WBTK.S	131.4	1.025	134.72	42.251f	5.980s	1.509	135.61*
07_WBTK.P	48.4	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	48.9	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	48.4	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			2,566.63	25.000f	0.000	1.549	2712.14
Distances in METERS.-----				Moments in m.-MT.-----			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

HYDROSTATIC PARTICULARS

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	---LCB---	---VCB---	---cm---	---LCF---	cm trim---	---KML---	---KMT	
0.112	133.29	37.689f	0.175	5.02	33.490f	6.52	244.36	96.881	
0.212	188.30	36.107f	0.211	5.98	31.124f	10.90	289.46	82.456	
0.312	252.92	34.523f	0.246	6.94	28.751f	16.92	334.36	71.772	
0.412	327.16	32.938f	0.281	7.91	26.375f	24.82	379.30	63.564	
0.512	407.76	31.583f	0.319	8.13	26.034f	26.65	326.70	53.012	
0.612	489.54	30.651f	0.361	8.23	25.975f	27.51	280.91	45.226	
0.712	572.35	29.971f	0.406	8.33	25.923f	28.38	247.92	39.491	
0.812	656.10	29.452f	0.452	8.42	25.886f	29.26	222.97	35.068	
0.912	740.70	29.044f	0.499	8.50	25.870f	30.12	203.29	31.535	
1.012	826.02	28.717f	0.548	8.57	25.880f	30.81	186.46	28.540	
1.112	912.01	28.450f	0.597	8.63	25.878f	31.57	173.02	26.147	
1.212	998.69	28.226f	0.646	8.70	25.875f	32.35	161.94	24.171	
1.312	1,086.07	28.037f	0.696	8.78	25.872f	33.17	152.69	22.519	
1.412	1,174.15	27.875f	0.746	8.85	25.869f	34.03	144.90	21.124	
1.512	1,262.91	27.735f	0.797	8.91	25.870f	34.71	137.41	19.834	
1.612	1,352.12	27.609f	0.848	8.94	25.783f	35.14	129.91	18.700	
1.712	1,441.67	27.493f	0.899	8.97	25.694f	35.58	123.36	17.712	
1.812	1,531.58	27.384f	0.950	9.01	25.603f	36.03	117.61	16.848	
1.912	1,621.83	27.283f	1.000	9.05	25.508f	36.51	112.52	16.088	
2.012	1,712.42	27.187f	1.051	9.08	25.436f	36.88	107.68	15.374	
2.112	1,803.32	27.097f	1.102	9.11	25.351f	37.30	103.41	14.750	
2.212	1,894.54	27.011f	1.153	9.14	25.263f	37.74	99.58	14.190	
2.312	1,986.08	26.928f	1.203	9.17	25.174f	38.19	96.11	13.689	
2.412	2,077.95	26.849f	1.254	9.21	25.081f	38.65	92.98	13.238	
2.512	2,170.15	26.772f	1.305	9.23	25.013f	39.01	89.86	12.803	
2.612	2,258.72	26.664f	1.353	8.40	22.760f	29.59	65.48	11.408	
2.712	2,338.10	26.490f	1.395	7.48	20.257f	21.09	45.09	10.043	
2.812	2,408.27	26.272f	1.431	6.56	17.755f	14.46	30.01	8.792	
2.912	2,469.20	26.032f	1.462	5.63	15.252f	9.46	19.15	7.632	
3.012	2,520.92	25.786f	1.488	4.71	12.750f	5.86	11.62	6.543	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

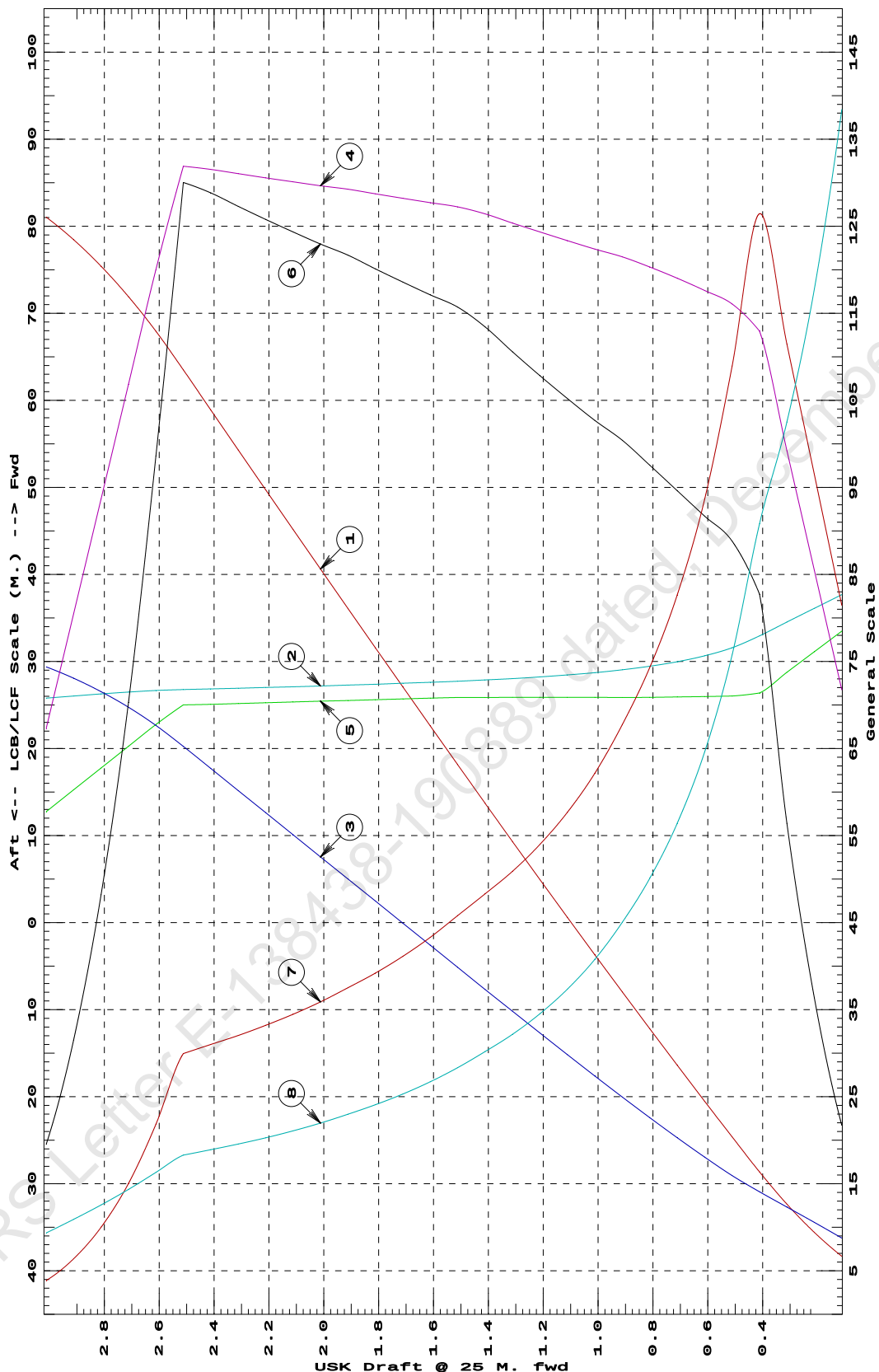
Trim is per 50.00m.

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31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 1 M. FWD TRIM



- ① Displacement 1=20 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.02 M.
- ④ Immersion 1=.07 MT/cm
- ④ WPA 1=6.83 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.3 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.7 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement ---Weight(MT)---	Buoyancy-Ctr. ----LCB-----VCB-----	Weight/ -----cm-----	Moment/ -----LCF-----cm trim-----	KML-----	KMT
0.112	94.61	35.657f 0.098	5.88 30.439f	10.72	566.32	155.843
0.212	162.65	32.417f 0.132	7.74 25.617f	24.15	742.47	120.816
0.312	240.79	30.208f 0.173	7.89 25.602f	25.06	520.34	85.097
0.412	320.44	29.059f 0.218	8.04 25.566f	25.94	404.70	66.389
0.512	401.45	28.349f 0.266	8.16 25.509f	26.81	333.87	54.610
0.612	483.61	27.862f 0.315	8.27 25.457f	27.70	286.38	46.395
0.712	566.72	27.508f 0.365	8.35 25.449f	28.52	251.60	40.187
0.812	650.57	27.242f 0.415	8.42 25.440f	29.21	224.49	35.343
0.912	735.11	27.034f 0.466	8.49 25.439f	29.96	203.76	31.632
1.012	820.35	26.868f 0.517	8.56 25.438f	30.74	187.34	28.689
1.112	906.31	26.733f 0.567	8.63 25.436f	31.56	174.08	26.309
1.212	992.99	26.620f 0.619	8.71 25.451f	32.35	162.89	24.309
1.312	1,080.33	26.524f 0.670	8.77 25.439f	33.07	153.05	22.576
1.412	1,168.34	26.443f 0.722	8.83 25.438f	33.86	144.89	21.132
1.512	1,257.02	26.372f 0.773	8.90 25.437f	34.68	137.92	19.898
1.612	1,346.39	26.310f 0.825	8.97 25.435f	35.53	131.95	18.836
1.712	1,436.46	26.255f 0.877	9.04 25.448f	36.37	126.58	17.889
1.812	1,527.11	26.206f 0.929	9.08 25.389f	36.90	120.81	16.991
1.912	1,618.11	26.157f 0.981	9.12 25.304f	37.34	115.37	16.198
2.012	1,709.44	26.109f 1.033	9.15 25.217f	37.79	110.53	15.496
2.112	1,801.11	26.062f 1.085	9.19 25.127f	38.26	106.21	14.872
2.212	1,893.13	26.014f 1.137	9.22 25.034f	38.75	102.33	14.317
2.312	1,985.42	25.968f 1.189	9.23 25.012f	38.91	97.97	13.759
2.412	2,077.73	25.925f 1.240	9.23 25.009f	38.96	93.75	13.255
2.512	2,170.06	25.886f 1.292	9.23 25.006f	39.02	89.90	12.799
2.612	2,262.39	25.850f 1.343	9.23 25.004f	39.08	86.36	12.384
2.712	2,354.73	25.817f 1.394	9.24 25.001f	39.14	83.10	12.006
2.812	2,445.61	25.772f 1.445	8.50 23.000f	30.67	62.71	10.846
2.912	2,521.33	25.617f 1.485	6.65 18.000f	15.11	29.97	8.622
3.012	2,578.59	25.395f 1.516	4.80 13.000f	6.19	12.01	6.555

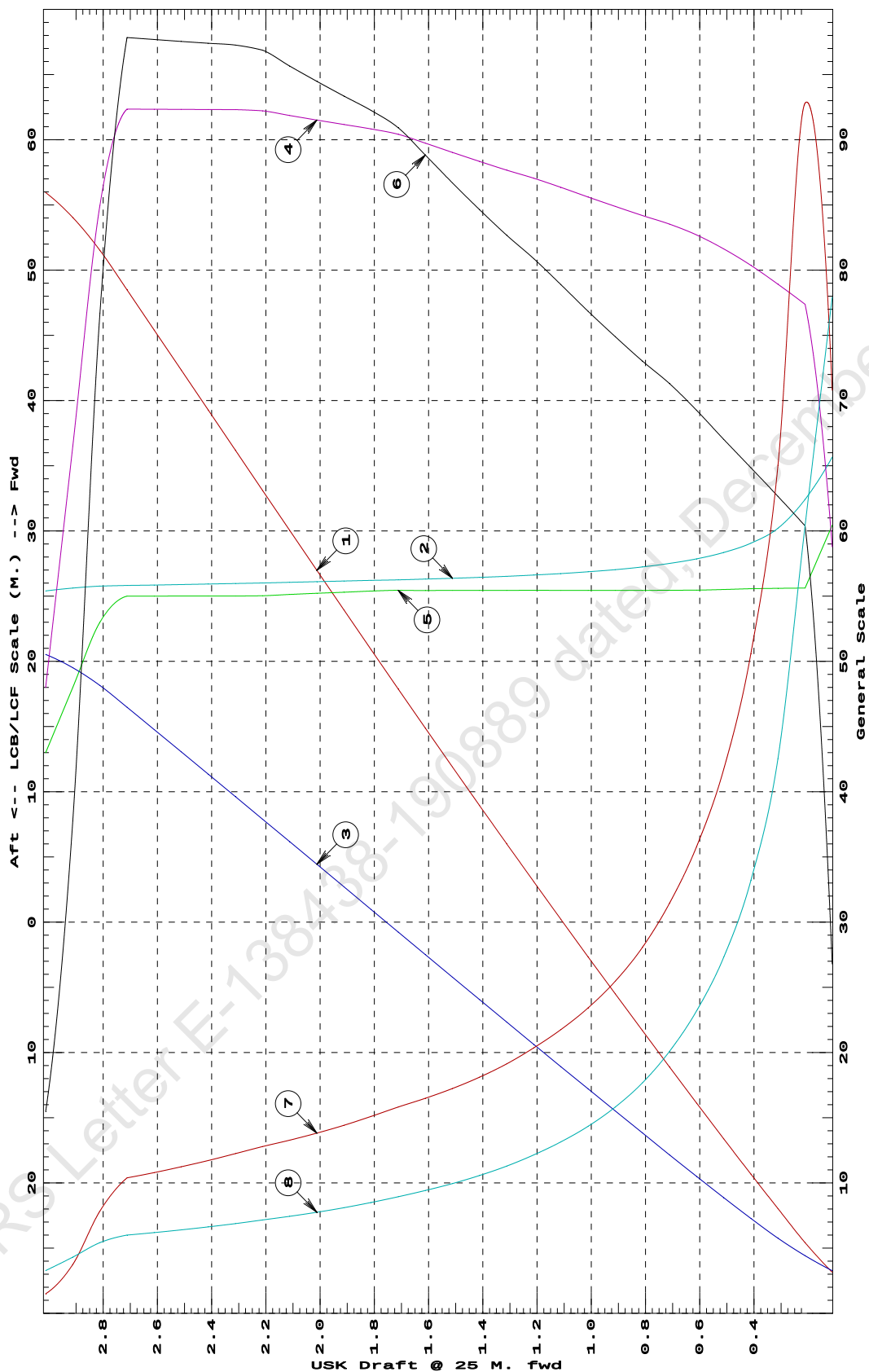
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM



- ① Displacement 1=30 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.03 M.
 ④ Immersion 1=.1 MT/cm
 ⑤ WPA 1=9.76 Sq.M.
 ⑥ LCF (use top scale)
 ⑦ Moment/Trim 1=.4 M.-MT/cm
 ⑧ KML 1=8 M.
 ⑨ KMT 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	83.79	25.000f	0.044	7.57	25.000f	23.17	1382.58	223.643	
0.212	160.29	25.000f	0.095	7.73	25.000f	24.11	752.15	122.246	
0.312	238.38	25.000f	0.145	7.89	25.000f	25.08	526.13	85.934	
0.412	318.09	25.000f	0.196	8.06	25.000f	26.09	410.04	67.317	
0.512	399.40	25.000f	0.248	8.20	25.000f	26.98	337.80	55.685	
0.612	481.78	25.000f	0.300	8.27	25.000f	27.70	287.43	46.641	
0.712	564.87	25.000f	0.351	8.35	25.000f	28.44	251.70	40.223	
0.812	648.69	25.000f	0.403	8.42	25.000f	29.21	225.14	35.449	
0.912	733.24	25.000f	0.454	8.50	25.000f	30.02	204.72	31.774	
1.012	818.51	25.000f	0.506	8.56	25.000f	30.72	187.65	28.736	
1.112	904.44	25.000f	0.558	8.63	25.000f	31.47	173.99	26.304	
1.212	991.06	25.000f	0.610	8.70	25.000f	32.26	162.74	24.297	
1.312	1,078.37	25.000f	0.661	8.77	25.000f	33.07	153.35	22.621	
1.412	1,166.40	25.000f	0.713	8.84	25.000f	33.93	145.46	21.208	
1.512	1,255.11	25.000f	0.765	8.90	25.000f	34.66	138.07	19.915	
1.612	1,344.46	25.000f	0.818	8.97	25.000f	35.45	131.83	18.822	
1.712	1,434.46	25.000f	0.870	9.03	25.000f	36.27	126.41	17.872	
1.812	1,525.13	25.000f	0.922	9.10	25.000f	37.12	121.69	17.042	
1.912	1,616.48	25.000f	0.974	9.17	25.000f	38.01	117.57	16.315	
2.012	1,708.51	25.000f	1.027	9.23	25.000f	38.71	113.30	15.610	
2.112	1,800.77	25.000f	1.079	9.23	25.000f	38.78	107.66	14.924	
2.212	1,893.06	25.000f	1.132	9.23	25.000f	38.84	102.58	14.309	
2.312	1,985.36	25.000f	1.184	9.23	25.000f	38.90	97.98	13.756	
2.412	2,077.68	25.000f	1.236	9.23	25.000f	38.97	93.78	13.257	
2.512	2,170.03	25.000f	1.287	9.24	25.000f	39.04	89.95	12.804	
2.612	2,262.38	25.000f	1.339	9.24	25.000f	39.09	86.38	12.385	
2.712	2,354.73	25.000f	1.390	9.24	25.000f	39.14	83.10	12.003	
2.812	2,447.09	25.000f	1.442	9.24	25.000f	39.19	80.07	11.654	
2.912	2,539.44	25.000f	1.493	9.24	25.000f	39.24	77.26	11.334	
3.012	2,631.79	25.000f	1.544	9.24	25.000f	39.29	74.65	11.040	

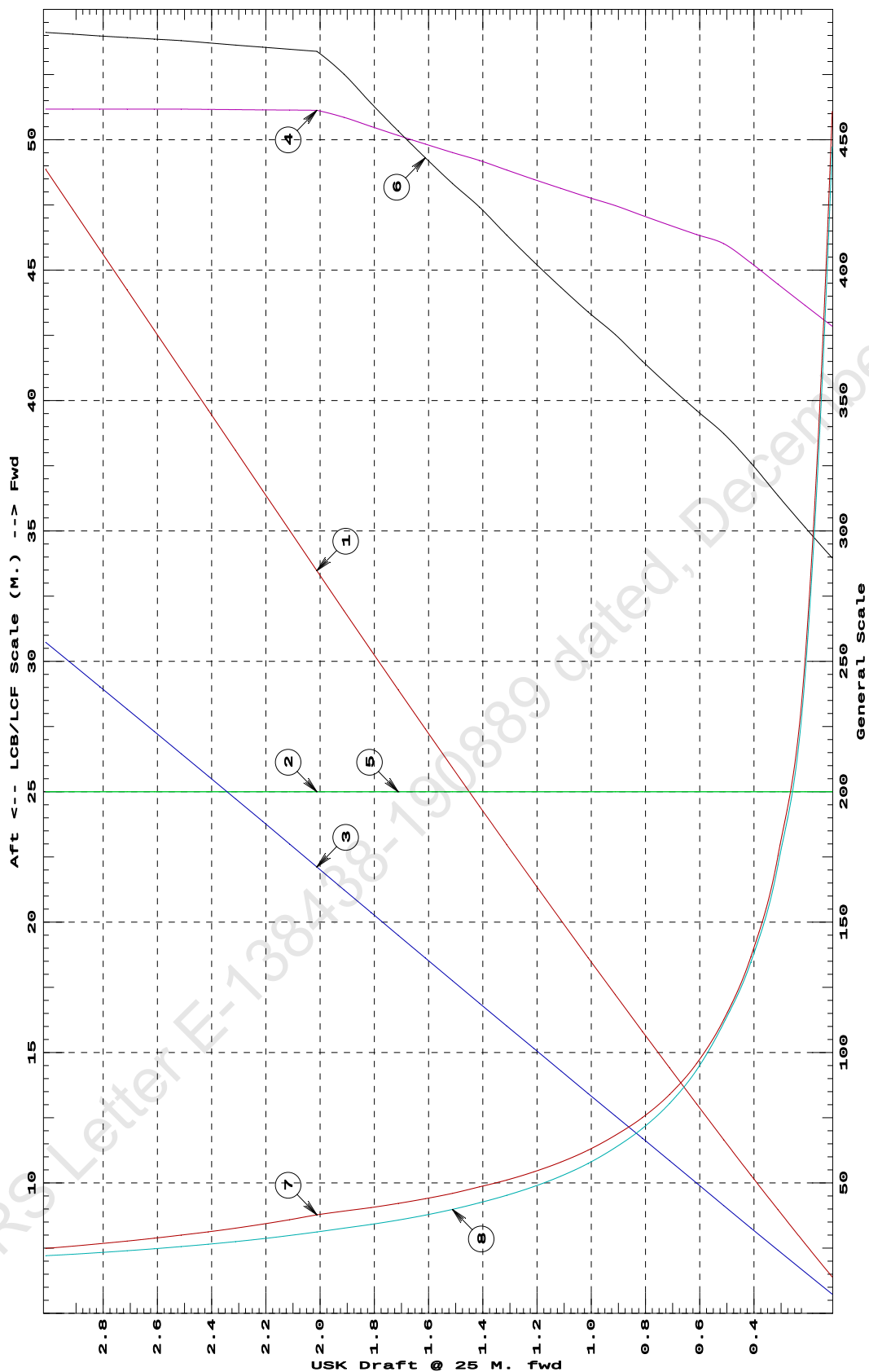
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at LEVEL TRIM



- ① Displacement 1=6 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.006 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.08 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.5 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	94.61	14.343f	0.098	5.88	19.561f	10.72	566.32	155.836	
0.212	162.65	17.583f	0.132	7.74	24.383f	24.15	742.47	120.811	
0.312	240.79	19.792f	0.173	7.89	24.398f	25.06	520.34	85.098	
0.412	320.44	20.941f	0.218	8.04	24.434f	25.94	404.70	66.390	
0.512	401.45	21.651f	0.266	8.16	24.491f	26.81	333.87	54.611	
0.612	483.61	22.138f	0.315	8.27	24.543f	27.70	286.38	46.395	
0.712	566.72	22.492f	0.365	8.35	24.551f	28.52	251.60	40.186	
0.812	650.57	22.758f	0.415	8.42	24.560f	29.21	224.49	35.343	
0.912	735.11	22.966f	0.466	8.49	24.561f	29.96	203.76	31.632	
1.012	820.35	23.132f	0.517	8.56	24.562f	30.74	187.34	28.689	
1.112	906.31	23.267f	0.567	8.63	24.564f	31.56	174.08	26.309	
1.212	992.99	23.380f	0.619	8.71	24.549f	32.35	162.89	24.308	
1.312	1,080.33	23.476f	0.670	8.77	24.561f	33.07	153.05	22.576	
1.412	1,168.34	23.557f	0.722	8.83	24.562f	33.86	144.89	21.132	
1.512	1,257.02	23.628f	0.773	8.90	24.563f	34.68	137.92	19.898	
1.612	1,346.39	23.690f	0.825	8.97	24.565f	35.53	131.95	18.836	
1.712	1,436.46	23.745f	0.877	9.04	24.552f	36.37	126.58	17.889	
1.812	1,527.11	23.794f	0.929	9.08	24.611f	36.90	120.81	16.991	
1.912	1,618.10	23.843f	0.981	9.12	24.696f	37.34	115.37	16.198	
2.012	1,709.44	23.891f	1.033	9.15	24.783f	37.79	110.53	15.496	
2.112	1,801.11	23.938f	1.085	9.19	24.873f	38.26	106.21	14.872	
2.212	1,893.13	23.986f	1.137	9.22	24.966f	38.75	102.33	14.317	
2.312	1,985.42	24.032f	1.189	9.23	24.988f	38.91	97.97	13.758	
2.412	2,077.73	24.075f	1.240	9.23	24.991f	38.96	93.75	13.255	
2.512	2,170.06	24.114f	1.292	9.23	24.994f	39.02	89.90	12.799	
2.612	2,262.39	24.150f	1.343	9.23	24.996f	39.08	86.36	12.384	
2.712	2,354.73	24.183f	1.394	9.24	24.999f	39.14	83.10	12.006	
2.812	2,445.61	24.228f	1.445	8.50	27.000f	30.67	62.71	10.846	
2.912	2,521.33	24.383f	1.485	6.65	32.000f	15.11	29.97	8.622	
3.012	2,578.59	24.605f	1.516	4.80	37.000f	6.19	12.01	6.555	

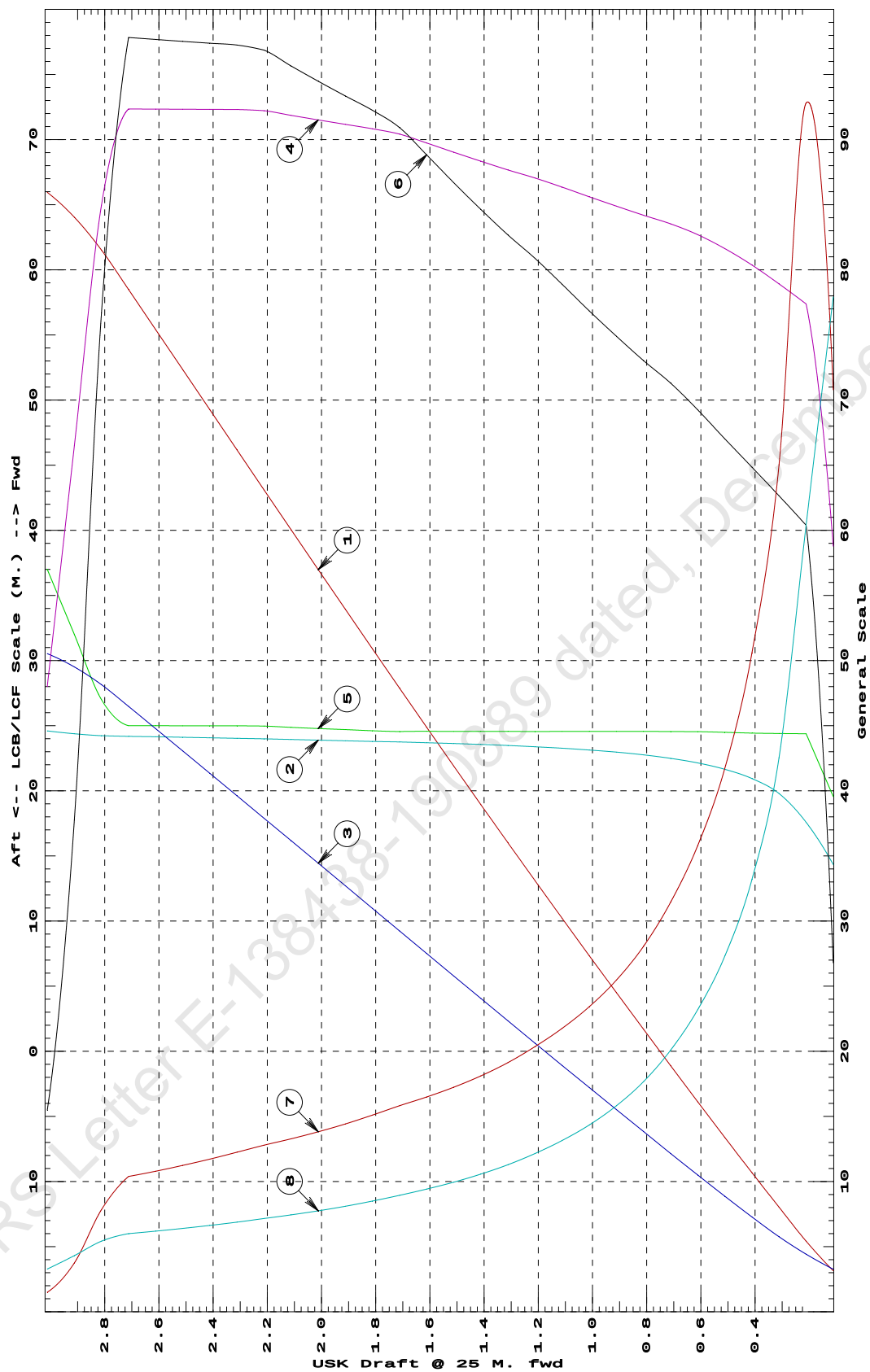
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM



- ① Displacement 1=30 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=0.03 M.
 ④ Immersion 1=1.1 MT/cm
 ④ WPA 1=9.76 Sq.M.
 ⑤ LCF (use top scale)
 ⑥ Moment/Trim 1=.4 M.-MT/cm
 ⑦ KML 1=8 M.
 ⑧ KMT 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement ---Weight(MT)	Buoyancy-Ctr. ----LCB-----VCB	Weight/ -----cm	Moment/ -----LCF-----cm trim	KML	KMT
0.112	133.29	12.311f 0.175	5.02 16.510f	6.52 244.36	96.880	
0.212	188.30	13.893f 0.211	5.98 18.876f	10.90 289.46	82.454	
0.312	252.92	15.477f 0.246	6.94 21.249f	16.92 334.36	71.771	
0.412	327.16	17.062f 0.281	7.91 23.625f	24.82 379.31	63.562	
0.512	407.76	18.417f 0.319	8.13 23.966f	26.65 326.70	53.012	
0.612	489.54	19.349f 0.361	8.23 24.026f	27.51 280.91	45.226	
0.712	572.35	20.029f 0.406	8.33 24.077f	28.38 247.92	39.491	
0.812	656.10	20.548f 0.452	8.42 24.114f	29.26 222.97	35.069	
0.912	740.70	20.956f 0.499	8.50 24.130f	30.12 203.29	31.535	
1.012	826.02	21.283f 0.548	8.57 24.120f	30.81 186.46	28.540	
1.112	912.01	21.550f 0.597	8.63 24.122f	31.57 173.02	26.147	
1.212	998.69	21.774f 0.646	8.70 24.125f	32.35 161.94	24.171	
1.312	1,086.07	21.963f 0.696	8.78 24.128f	33.17 152.69	22.519	
1.412	1,174.15	22.125f 0.746	8.85 24.131f	34.03 144.90	21.125	
1.512	1,262.91	22.265f 0.797	8.91 24.130f	34.71 137.41	19.834	
1.612	1,352.12	22.391f 0.848	8.94 24.217f	35.14 129.91	18.700	
1.712	1,441.67	22.507f 0.899	8.97 24.306f	35.58 123.36	17.712	
1.812	1,531.58	22.616f 0.950	9.01 24.397f	36.03 117.61	16.848	
1.912	1,621.83	22.717f 1.000	9.05 24.492f	36.51 112.52	16.088	
2.012	1,712.42	22.813f 1.051	9.08 24.564f	36.88 107.68	15.374	
2.112	1,803.32	22.903f 1.102	9.11 24.649f	37.30 103.41	14.749	
2.212	1,894.54	22.989f 1.153	9.14 24.737f	37.74 99.58	14.190	
2.312	1,986.08	23.072f 1.203	9.17 24.826f	38.19 96.11	13.689	
2.412	2,077.95	23.151f 1.254	9.21 24.919f	38.65 92.98	13.238	
2.512	2,170.15	23.228f 1.305	9.23 24.987f	39.01 89.86	12.803	
2.612	2,258.72	23.336f 1.353	8.40 27.240f	29.59 65.48	11.408	
2.712	2,338.10	23.510f 1.395	7.48 29.743f	21.09 45.09	10.042	
2.812	2,408.27	23.728f 1.431	6.56 32.245f	14.46 30.01	8.792	
2.912	2,469.21	23.968f 1.462	5.63 34.748f	9.46 19.15	7.632	
3.012	2,520.92	24.214f 1.488	4.71 37.250f	5.86 11.62	6.543	

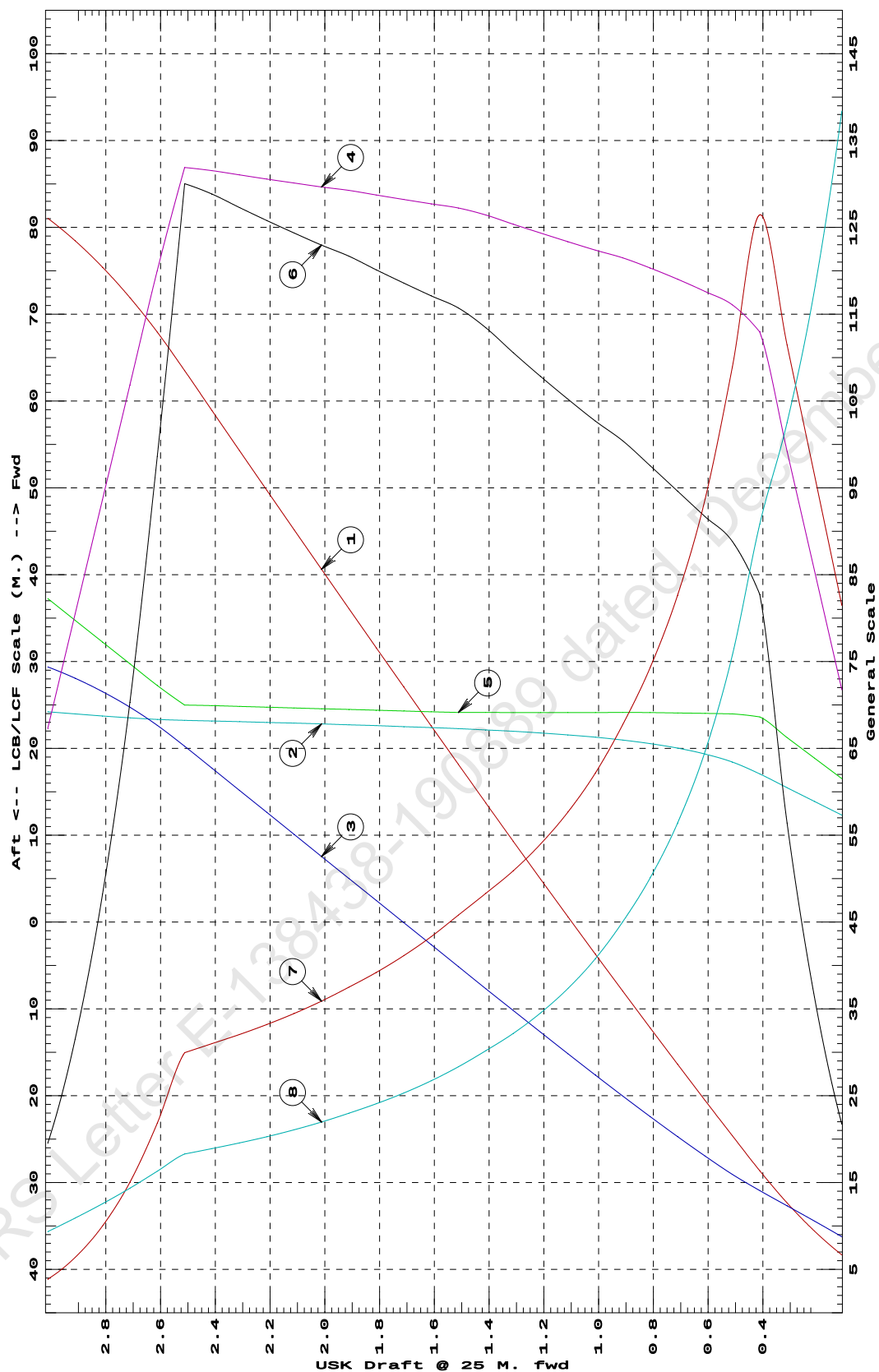
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 1 M. AFT TRIM



- ① Displacement 1=200 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.02 M.
 ④ Immersion 1=.07 MT/cm
 ④ WPA 1=6.83 Sq.M.
 ⑤ LCF (use top scale)
 ⑥ Moment/Trim 1=.3 M.-MT/cm
 ⑦ KML 1=3 M.
 ⑧ KMT 1=.7 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

$$\underline{GZ = KN - KG' \sin \theta - TCG \cos \theta}$$

01-08-20 21:16:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

VOLUME USED FOR KN COMPUTATION

Condition Graphic

Profile View



Plan View



CROSS CURVE TABLES

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

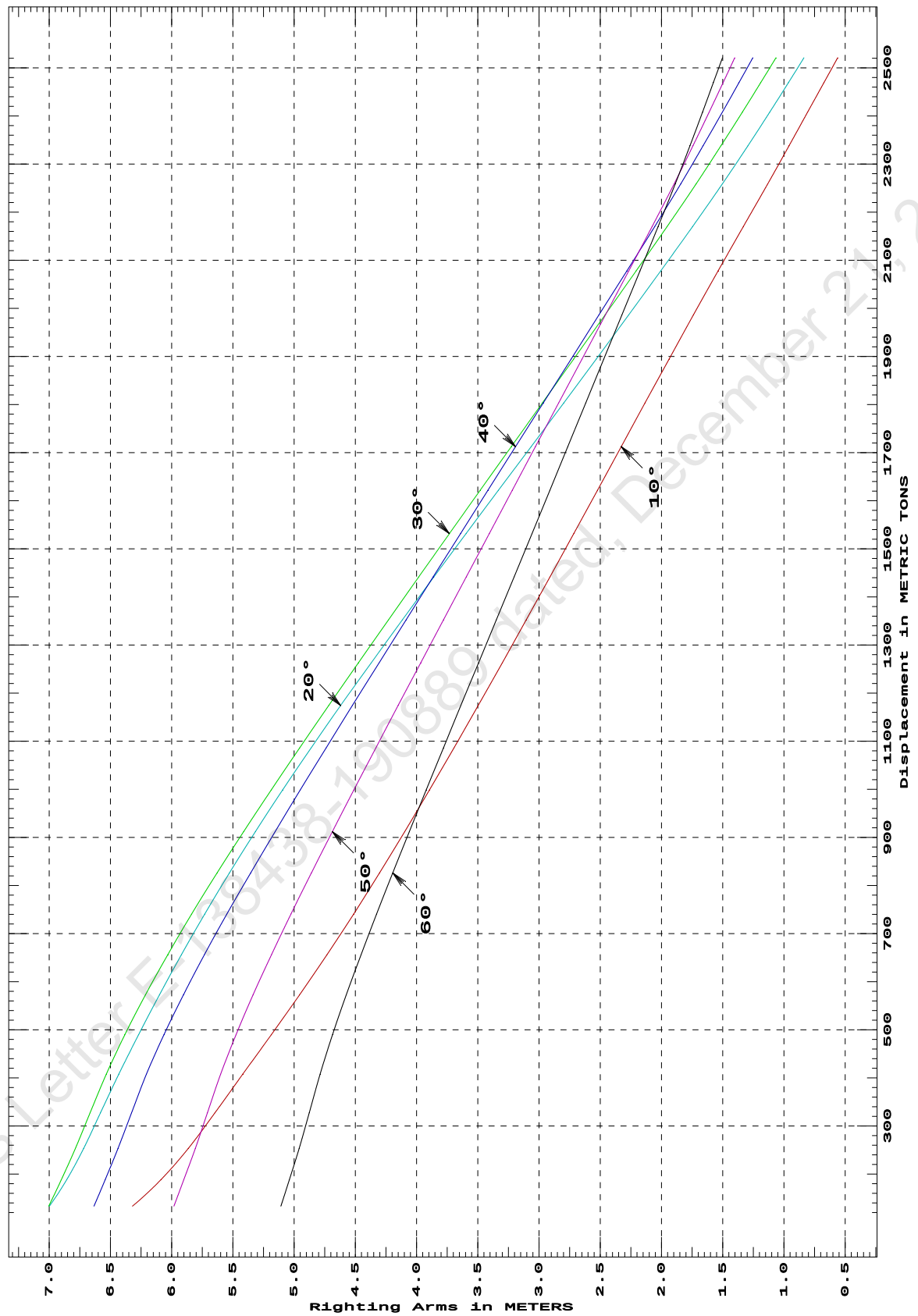
Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.320s	6.999s	7.003s	6.634s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.865s	6.719s	6.786s	6.438s	5.806s	4.959s
327.16	5.646s	6.581s	6.666s	6.327s	5.707s	4.876s
407.76	5.419s	6.431s	6.534s	6.205s	5.600s	4.788s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.859s	4.194s
912.01	4.097s	5.316s	5.412s	5.155s	4.687s	4.059s
998.69	3.895s	5.094s	5.186s	4.949s	4.512s	3.921s
1,086.07	3.695s	4.861s	4.955s	4.739s	4.333s	3.781s
1,174.15	3.497s	4.620s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.371s	4.475s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.086s	3.781s	3.349s
1,441.67	2.910s	3.858s	3.982s	3.864s	3.594s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.641s	3.406s	3.057s
1,621.83	2.523s	3.331s	3.480s	3.417s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.765s
1,803.32	2.134s	2.798s	2.972s	2.965s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.649s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.459s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.897s	1.888s
2,338.10	0.958s	1.294s	1.514s	1.660s	1.743s	1.767s
2,408.27	0.806s	1.115s	1.338s	1.502s	1.610s	1.664s
2,469.20	0.674s	0.963s	1.190s	1.368s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.066s	1.255s	1.402s	1.502s

Distances in METERS.---Specific Gravity = 1.025.-----

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

**CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. FWD TRIM (initial)**



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

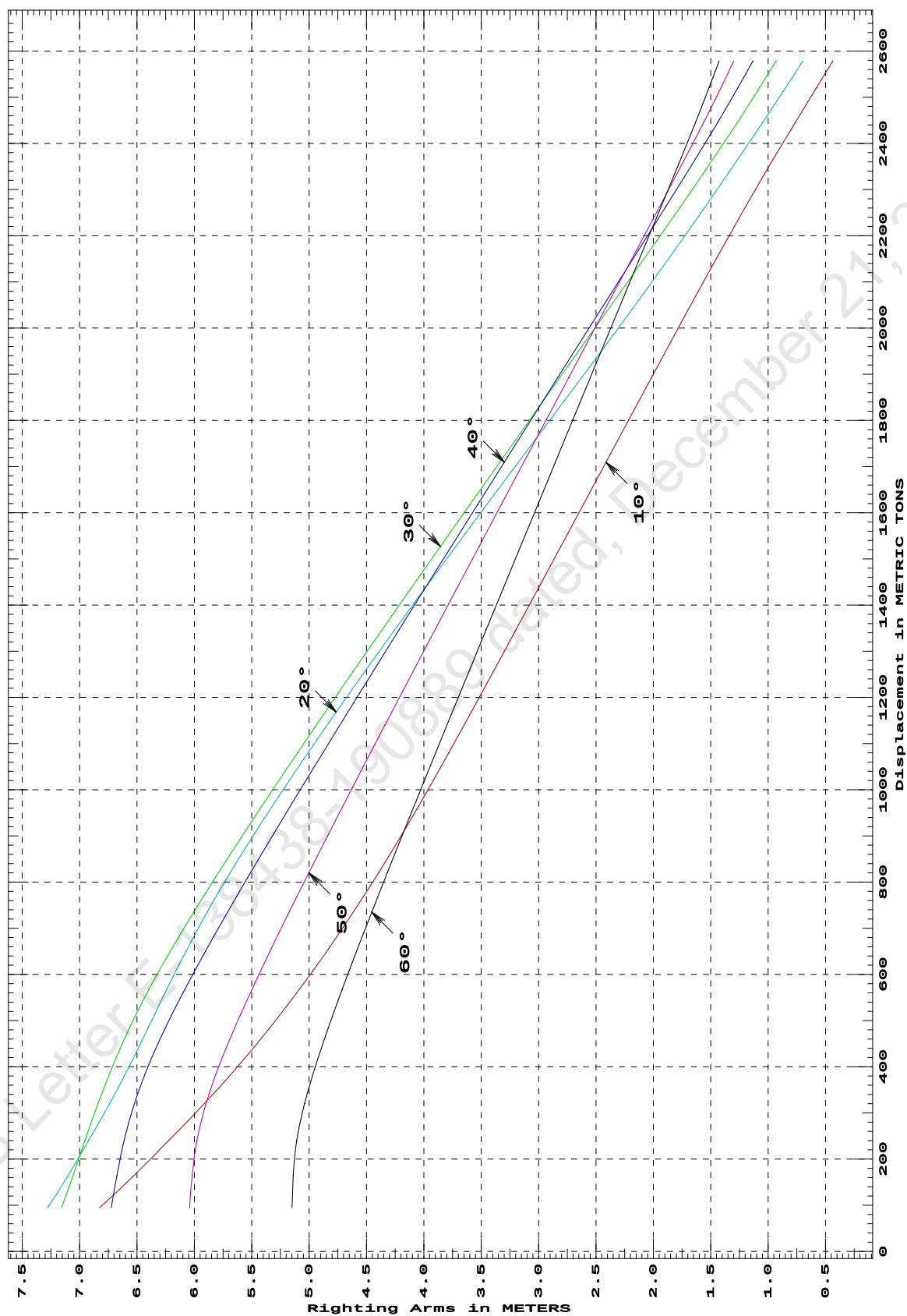
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.827s	7.278s	7.158s	6.723s	6.041s	5.149s
162.65	6.533s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.221s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.039s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.404s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.512s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.217s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.856s	3.761s	3.519s	3.157s
1,618.11	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.199s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.699s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

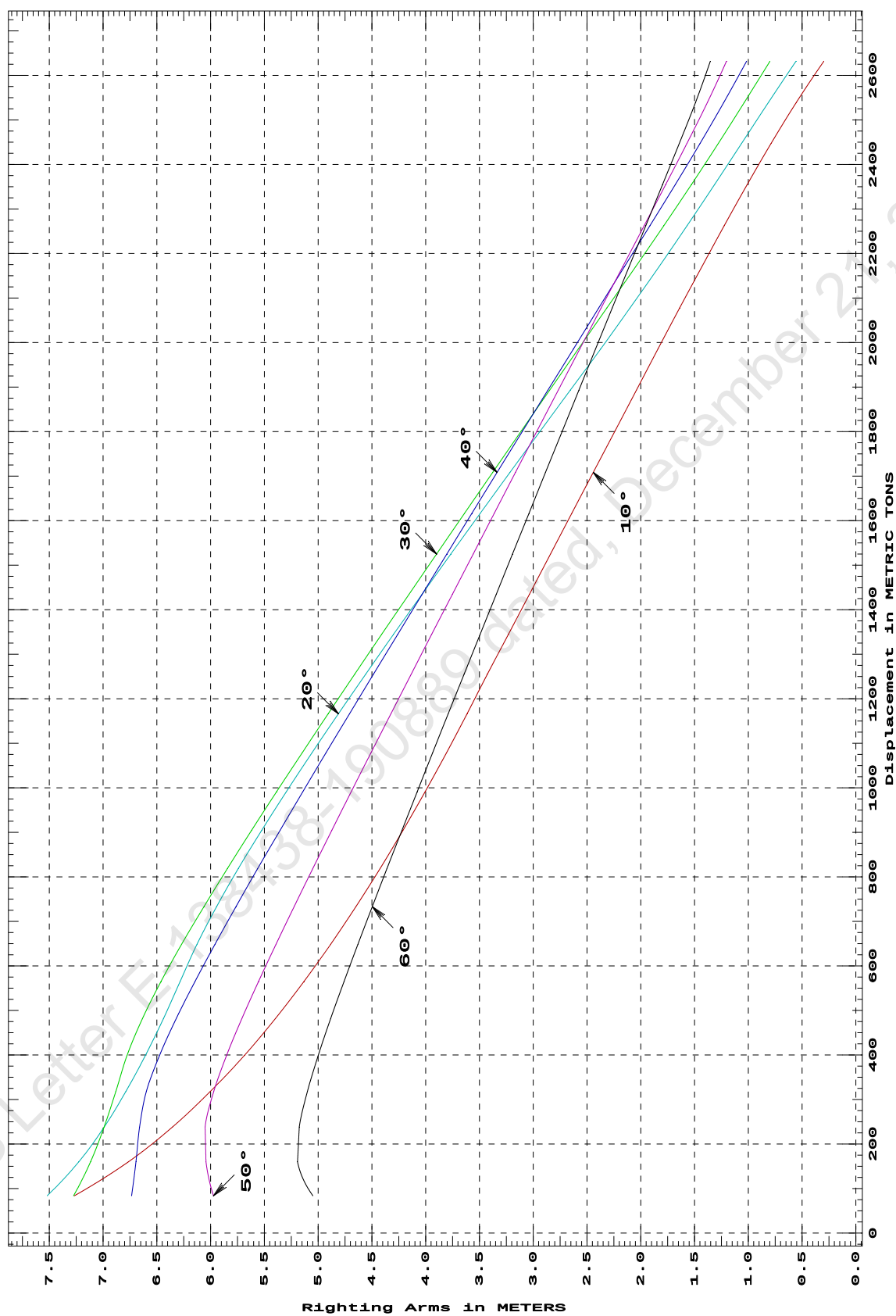
Showing righting arms in heel at VCG = 0.00

Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
83.79	7.268s	7.519s	7.274s	6.735s	5.973s	5.049s
160.29	6.763s	7.226s	7.116s	6.694s	6.043s	5.192s
238.38	6.354s	6.990s	6.990s	6.660s	6.051s	5.176s
318.09	6.003s	6.788s	6.881s	6.597s	5.971s	5.100s
399.40	5.685s	6.606s	6.776s	6.474s	5.851s	4.999s
481.78	5.397s	6.438s	6.634s	6.319s	5.709s	4.883s
564.87	5.132s	6.283s	6.462s	6.146s	5.555s	4.760s
648.69	4.884s	6.127s	6.269s	5.960s	5.392s	4.630s
733.24	4.648s	5.945s	6.061s	5.764s	5.224s	4.497s
818.51	4.425s	5.743s	5.844s	5.562s	5.050s	4.360s
904.44	4.211s	5.525s	5.619s	5.355s	4.873s	4.221s
991.06	4.006s	5.297s	5.387s	5.143s	4.693s	4.080s
1,078.37	3.807s	5.058s	5.149s	4.927s	4.509s	3.936s
1,166.40	3.615s	4.812s	4.907s	4.708s	4.324s	3.791s
1,255.11	3.427s	4.559s	4.660s	4.486s	4.135s	3.643s
1,344.46	3.231s	4.300s	4.410s	4.261s	3.945s	3.495s
1,434.46	3.036s	4.035s	4.156s	4.033s	3.753s	3.345s
1,525.13	2.840s	3.767s	3.899s	3.803s	3.559s	3.193s
1,616.48	2.642s	3.493s	3.640s	3.571s	3.363s	3.040s
1,708.51	2.442s	3.216s	3.377s	3.336s	3.165s	2.886s
1,800.77	2.241s	2.937s	3.113s	3.100s	2.966s	2.731s
1,893.06	2.041s	2.656s	2.848s	2.864s	2.767s	2.576s
1,985.36	1.840s	2.374s	2.582s	2.627s	2.568s	2.421s
2,077.68	1.638s	2.100s	2.316s	2.390s	2.369s	2.265s
2,170.03	1.433s	1.832s	2.049s	2.153s	2.169s	2.110s
2,262.38	1.225s	1.570s	1.786s	1.916s	1.969s	1.954s
2,354.73	1.010s	1.314s	1.529s	1.679s	1.769s	1.798s
2,447.09	0.789s	1.063s	1.280s	1.449s	1.571s	1.642s
2,539.44	0.554s	0.813s	1.038s	1.228s	1.380s	1.491s
2,631.79	0.298s	0.558s	0.799s	1.016s	1.204s	1.354s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00N

**CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)**



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

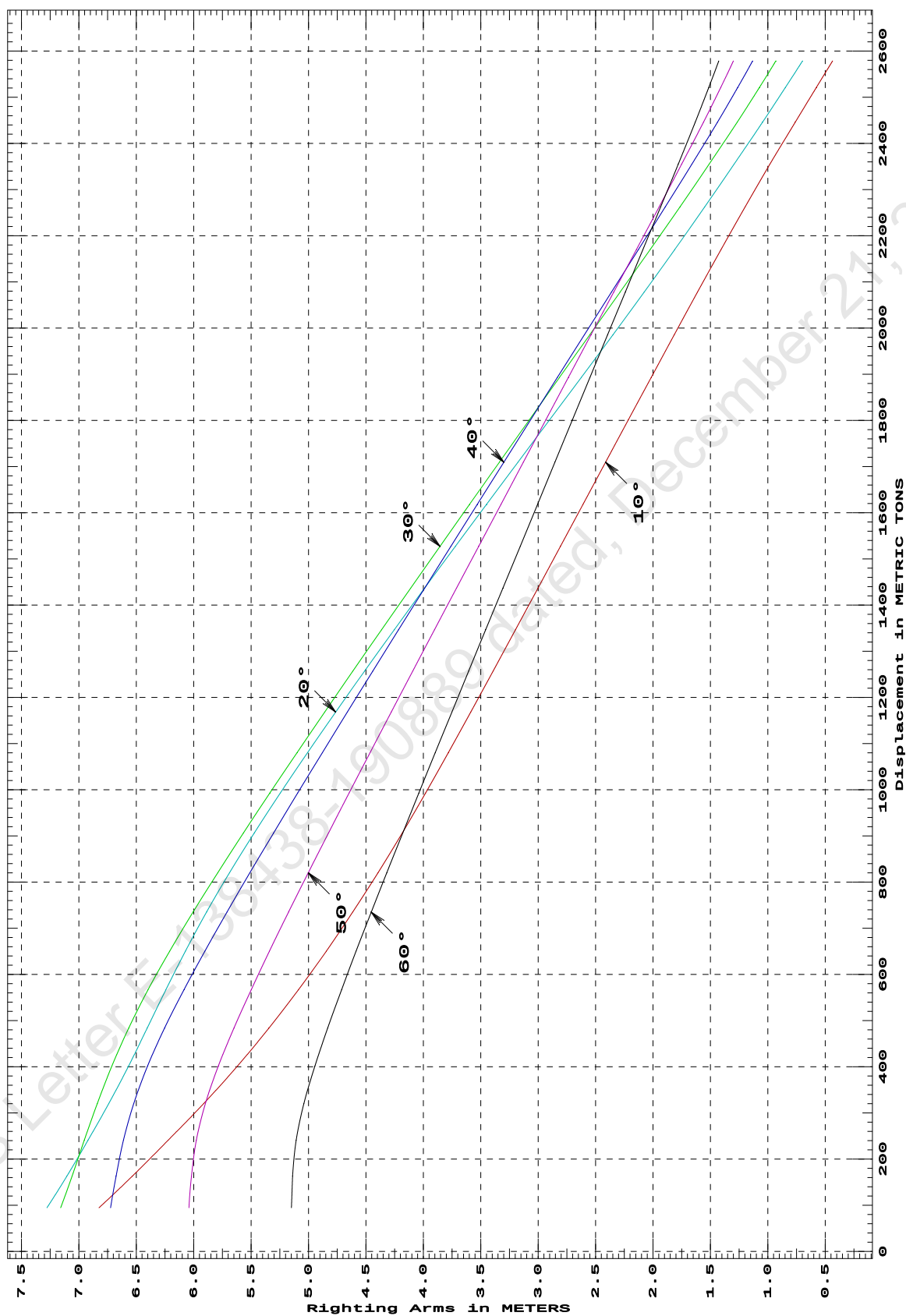
Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.824s	7.277s	7.158s	6.723s	6.041s	5.149s
162.65	6.535s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.222s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.040s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.405s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.511s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.216s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.855s	3.761s	3.519s	3.157s
1,618.10	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.198s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.698s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

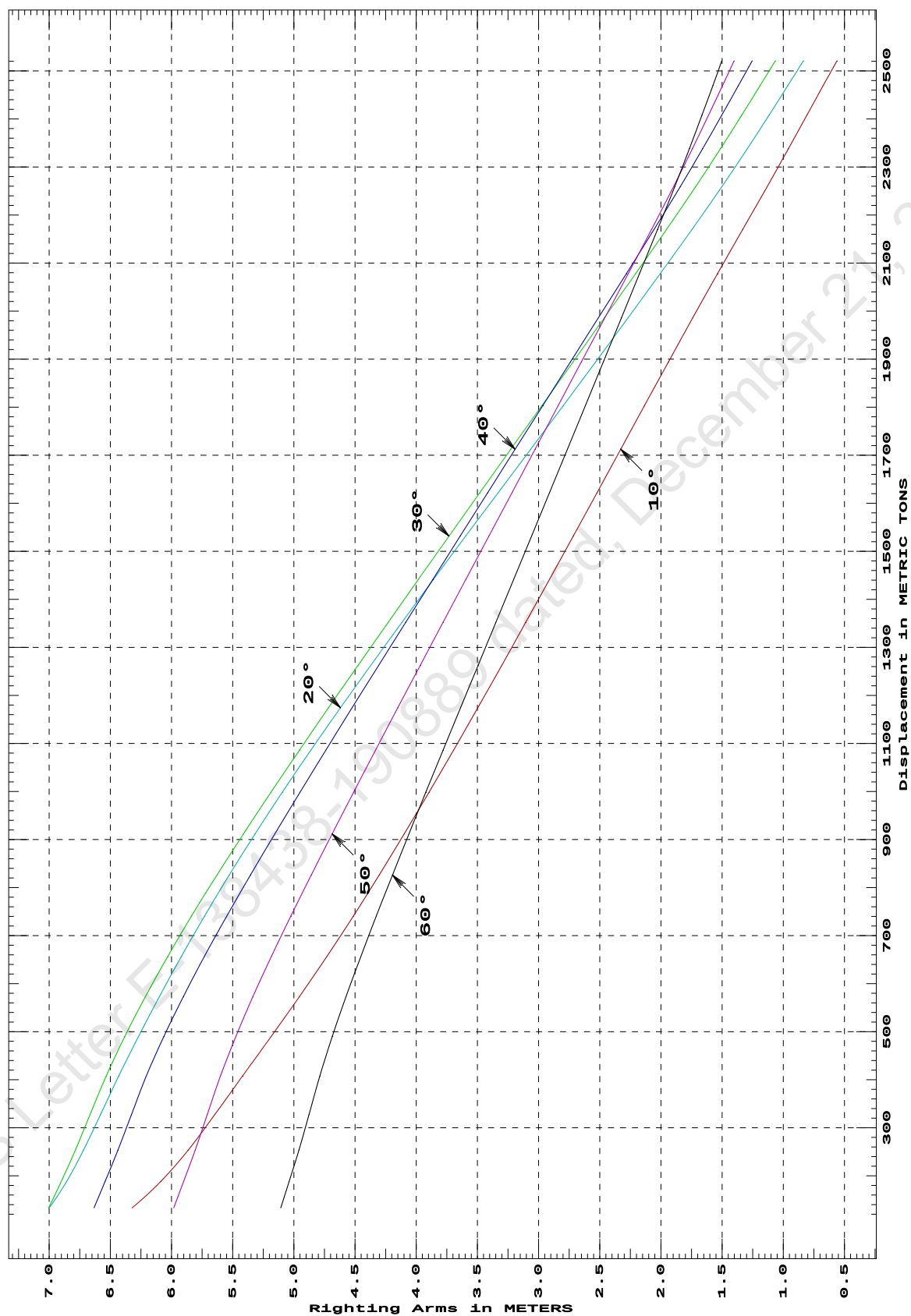
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.322s	6.999s	7.004s	6.633s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.864s	6.719s	6.786s	6.438s	5.807s	4.959s
327.16	5.646s	6.580s	6.666s	6.327s	5.708s	4.876s
407.76	5.419s	6.431s	6.533s	6.205s	5.600s	4.787s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.858s	4.194s
912.01	4.097s	5.316s	5.411s	5.155s	4.687s	4.059s
998.69	3.895s	5.093s	5.186s	4.949s	4.511s	3.921s
1,086.07	3.695s	4.861s	4.954s	4.738s	4.333s	3.780s
1,174.15	3.497s	4.619s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.370s	4.474s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.085s	3.780s	3.348s
1,441.67	2.910s	3.858s	3.981s	3.863s	3.593s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.640s	3.406s	3.057s
1,621.83	2.523s	3.330s	3.480s	3.416s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.764s
1,803.32	2.134s	2.798s	2.973s	2.966s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.650s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.458s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.898s	1.888s
2,338.10	0.958s	1.294s	1.513s	1.660s	1.743s	1.768s
2,408.27	0.806s	1.115s	1.339s	1.502s	1.610s	1.664s
2,469.21	0.674s	0.963s	1.189s	1.367s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.065s	1.254s	1.402s	1.503s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

**CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. AFT TRIM (initial)**



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

DEADWEIGHT TABLE

Draft	Disp	Dwt	
M	T	T	
0.612	481.78	75.81	
0.712	564.87	158.90	
0.812	648.69	242.72	
0.912	733.24	327.27	
1.012	818.51	412.54	
1.112	904.44	498.47	
1.212	991.06	585.09	
1.312	1078.37	672.40	
1.412	1166.40	760.43	
1.512	1255.11	849.14	
1.612	1344.46	938.49	
1.712	1434.46	1028.49	
1.812	1525.13	1119.16	
1.912	1616.48	1210.51	
2.012	1708.51	1302.54	
2.112	1800.77	1394.80	
2.212	1893.06	1437.72	Full Load Displacement

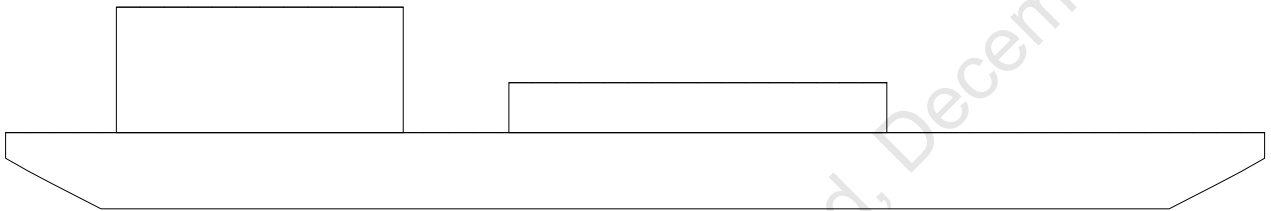
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GHS 17.22 PONT00N

USK draft refers to the line:

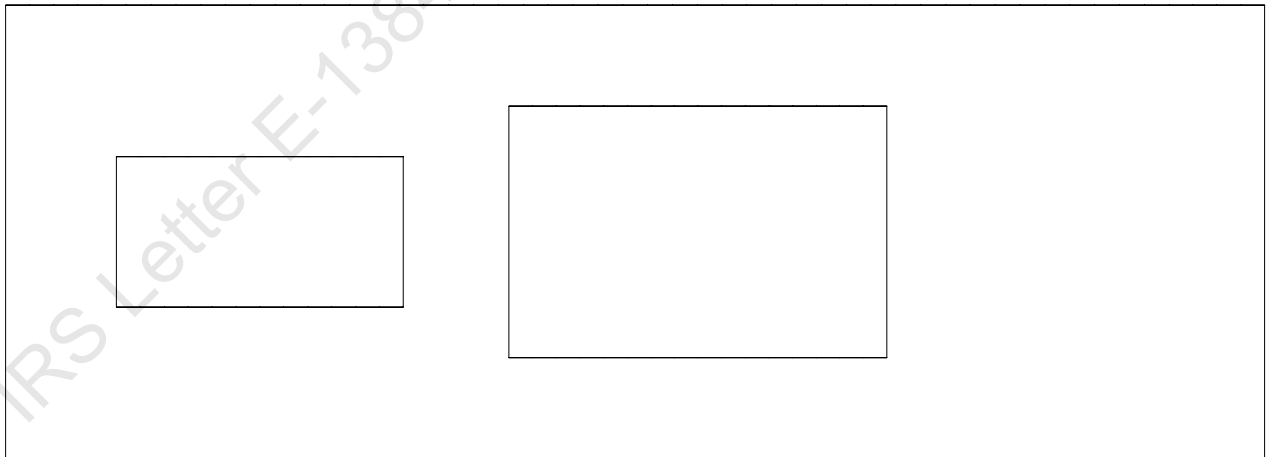
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic

Profile View



Plan View



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MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.489	122%	0d	7d	
600.00	17.349	115%	0d	6d	
800.00	16.087	112%	0d	5d	
1,000.00	14.631	73%	0d	4d	
1,200.00	13.079	51%	0d	3d	
1,400.00	11.429	30%	0d	3d	
1,600.00	9.697	12%	0d	2d	
1,800.00	7.619	0%	1d	1d	
2,000.00	5.293	0%	5d	1d	

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.792	138%	0d	10d	
600.00	17.834	103%	0d	7d	
800.00	16.573	83%	0d	6d	
1,000.00	15.128	70%	0d	5d	
1,200.00	13.510	57%	0d	4d	
1,400.00	11.824	35%	0d	3d	
1,600.00	10.137	7%	0d	3d	
1,800.00	7.959	0%	2d	2d	
2,000.00	5.353	3%	6d	1d	

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.899	137%	0d	14d	
600.00	17.990	117%	0d	10d	
800.00	16.745	80%	0d	7d	
1,000.00	15.270	69%	0d	6d	
1,200.00	13.683	59%	0d	5d	
1,400.00	11.943	37%	0d	4d	
1,600.00	10.176	8%	0d	3d	
1,800.00	8.007	0%	2d	3d	
2,000.00	5.351	0%	5d	2d	

Heeling moment is present from: wind

Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.793	138%	0d	10d	
600.00	17.835	103%	0d	7d	
800.00	16.573	83%	0d	6d	
1,000.00	15.129	70%	0d	5d	
1,200.00	13.510	57%	0d	4d	

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GHS 17.22 PONT00N

1,400.00	11.825	35%	0d	3d
1,600.00	10.136	7%	0d	3d
1,800.00	7.958	0%	2d	2d
2,000.00	5.363	4%	6d	1d

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement	--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	18.491	119%	0d	7d
600.00	17.392	114%	0d	6d
800.00	16.095	112%	0d	5d
1,000.00	14.632	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	11.430	31%	0d	3d
1,600.00	9.699	12%	0d	2d
1,800.00	7.661	0%	1d	1d
2,000.00	5.312	0%	5d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM-----IS CODE 2008 CRITERION-----Min/Max
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

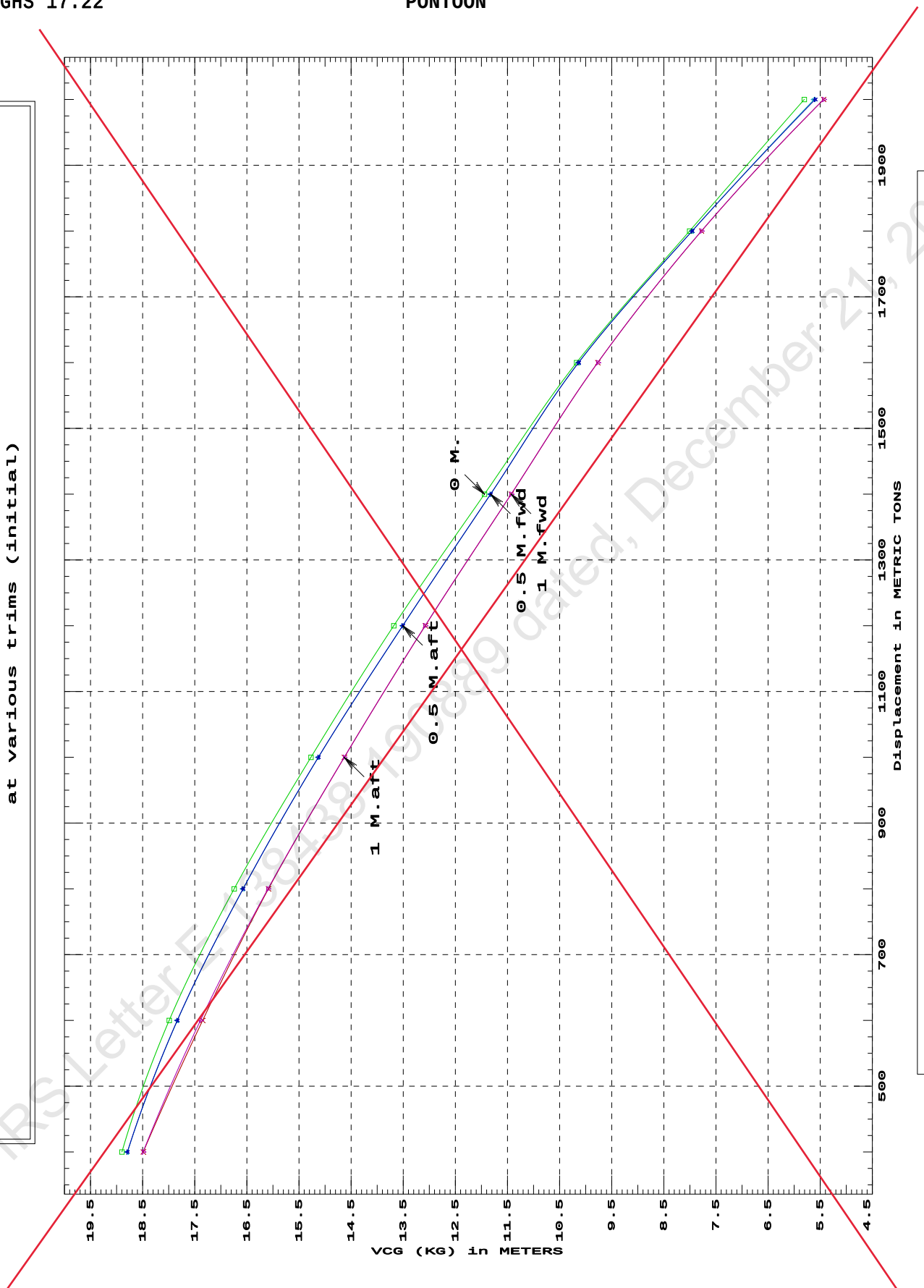
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MAX. VCG DOES NOT COVER CRANE OPERATION/LIFTING OF WEIGHTS AND VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION.

01-08-20 22:26:06
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

IS CODE 2008 criterion MAXIMUM VCG (KG)
at various trims (initial)



Specific Gravity = 1.025
Trim is per 50 M. "K" = BPL

LOADING CONDITIONS

SUMMARY OF CONDITIONS				
CONDITION NO.	1	2	3	4
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Crane + Deck Cargo + Ballast / Maximum Draft Condition	Condition 3 : Lightship + Deck Cargo / Maximum Deck Cargo Condition
TOTAL TANK LOAD (T)	0.000	234.870	234.870	0.000
DWT CONST. (T)	0.000	400.000	1202.850	1437.720
DEADWEIGHT (T)	0.000	634.870	1437.720	1437.720
LIGHT WEIGHT (T)	455.340	455.340	455.340	455.340
DISPLACEMENT (T)	455.340	1090.210	1893.060	1893.060
FWD DRAFT (M)	0.574	1.194	2.200	2.208
AFT DRAFT (M)	0.586	1.456	2.224	2.216
MEAN DRAFT (M)	0.580	1.325	2.212	2.212
TRIM (M)	0.012a	0.261a	0.024a	0.009a
L.C.G. (FROM AP) (M)	24.928f	24.225f	24.953f	24.983f
L.C.B. (FROM AP) (M)	24.928f	24.208f	24.951f	24.982f
M.C.T. (T-CM)	27.190	32.360	37.340	37.420
KG (M)	3.000	3.931	3.960	3.759
GM _T (M)	46.184	18.245	10.200	10.549
MAX. GZ (M)	5.443	3.619	1.363	1.452
Note:For details of deadweight constant refer respective loading condition				

SUMMARY OF CONDITIONS				
CONDITION NO.	5	6	7	8
CONDITION	Condition 4 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius aft	Condition 5 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius stbd	Condition 6 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius fwd	Condition 7 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius aft
TOTAL TANK LOAD (T)	284.980	284.980	284.980	422.400
DWT CONST. (T)	440.000	440.000	440.000	390.000
DEADWEIGHT (T)	724.980	724.980	724.980	812.400
LIGHT WEIGHT (T)	455.340	455.340	455.340	455.340
DISPLACEMENT (T)	1180.320	1180.320	1180.320	1267.740
FWD DRAFT (M)	1.199	1.390	1.578	1.296
AFT DRAFT (M)	1.652	1.466	1.276	1.753
MEAN DRAFT (M)	1.426	1.428	1.427	1.525
TRIM (M)	0.453a	0.076a	0.302f	0.457a
L.C.G. (FROM AP) (M)	23.739f	24.789f	25.840f	23.784f
L.C.B. (FROM AP) (M)	23.701f	24.783f	25.865f	23.755f
M.C.T. (T-CM)	32.810	32.920	32.830	33.750
KG (M)	4.883	4.883	4.883	4.033
GM _T (M)	15.773	15.826	15.776	15.360
MAX. GZ (M)	3.080	3.116	3.100	3.053

SUMMARY OF CONDITIONS				
CONDITION NO.	9	10	11	12
CONDITION	Condition 8 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius stbd	Condition 9 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius fwd	Condition 10 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius aft	Condition 11 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius stbd
TOTAL TANK LOAD (T)	422.400	422.400	422.400	422.400
DWT CONST. (T)	390.000	390.000	286.600	286.600
DEADWEIGHT (T)	812.400	812.400	709.000	709.000
LIGHT WEIGHT (T)	455.340	455.340	455.340	455.340
DISPLACEMENT (T)	1267.740	1267.740	1164.340	1164.340
FWD DRAFT (M)	1.505	1.711	1.325	1.426
AFT DRAFT (M)	1.547	1.339	1.494	1.393
MEAN DRAFT (M)	1.526	1.525	1.410	1.410
TRIM (M)	0.042a	0.373f	0.169a	0.034f
L.C.G. (FROM AP) (M)	24.889f	25.993f	24.527f	25.093f
L.C.B. (FROM AP) (M)	24.886f	26.016f	24.512f	25.097f
M.C.T. (T-CM)	33.750	33.760	32.620	32.680
KG (M)	4.033	4.033	5.289	5.289
GM _T (M)	15.347	15.360	15.513	15.547
MAX. GZ (M)	3.088	3.064	3.020	3.025

SUMMARY OF CONDITIONS				
CONDITION NO.	13	14	15	16
CONDITION	Condition 12 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius fwd	Condition 13 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius aft	Condition 14 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius stbd	Condition 15 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius fwd
TOTAL TANK LOAD (T)	422.400	323.200	323.200	323.200
DWT CONST. (T)	286.600	201.300	201.300	201.300
DEADWEIGHT (T)	709.000	524.500	524.500	524.500
LIGHT WEIGHT (T)	455.340	455.340	455.340	455.340
DISPLACEMENT (T)	1164.340	979.840	979.840	979.840
FWD DRAFT (M)	1.527	0.996	1.095	1.194
AFT DRAFT (M)	1.292	1.399	1.302	1.204
MEAN DRAFT (M)	1.410	1.198	1.199	1.199
TRIM (M)	0.235f	0.403a	0.206a	0.010a
L.C.G. (FROM AP) (M)	25.659f	23.712f	24.340f	24.968f
L.C.B. (FROM AP) (M)	25.680f	23.681f	24.325f	24.967f
M.C.T. (T-CM)	32.610	31.330	31.300	31.290
KG (M)	5.289	4.427	4.427	4.427
GM _T (M)	15.508	19.812	19.786	19.777
MAX. GZ (M)	3.015	3.721	3.744	3.752

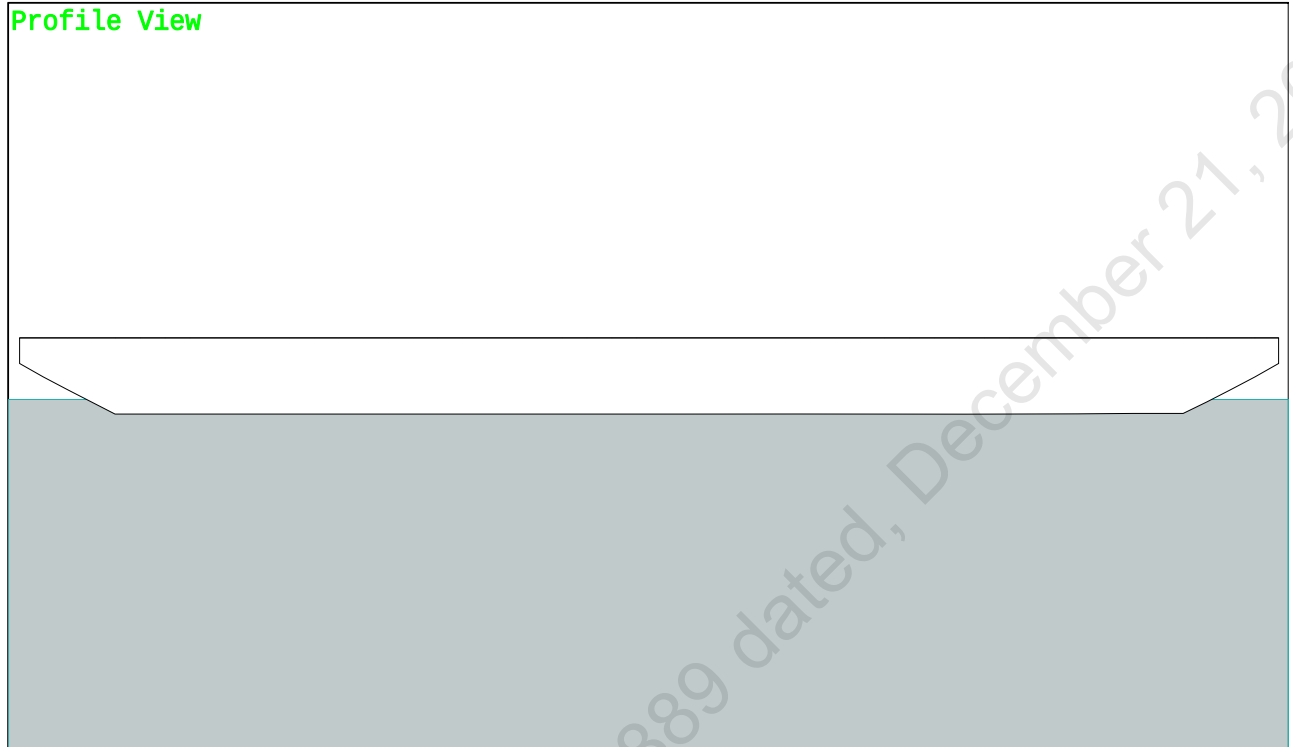
22-10-21 12:47:53 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 00 : Lightship
Lightship condition

Condition Graphic - Draft: 0.574 @ 50.000f, 0.586 @ 0.000 Heel: zero

Profile View



Plan View



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GHS 17.22

PONTOON

Condition 00 : Lightship
Lightship condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.574 @ 50.00f, 0.580 @ 25.00f, 0.586 @ 0.00

Trim: Aft 0.012/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WEIGHT	455.34	24.928f	0.000	3.000	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
	Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025	455.34	24.928f	0.000	0.283

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	804.9	24.990f	0.000	301.30 48.900
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		8.25	27.19	298.59	46.184

Distances in METERS.-----Moments in m.-MT.

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.574
DRAFT AT MS (M) = 0.580
DRAFT AT AP (M) = 0.586
DRAFT AT FWD MARK (M) = 0.575
DRAFT AT AFT MARK (M) = 0.585

HYDROSTATIC PROPERTIES

Trim: Aft 0.012/50.000, No Heel, VCG = 3.000

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----LCF-----cm trim-----GML-----GMT
0.580	455.34	24.928f	0.283	8.25 24.990f 27.19 298.59 46.184

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.574 @ 50.00f, 0.580 @ 25.00f, 0.586 @ 0.00

Trim: Aft 0.012/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment
HULL	118.3	1.246	1.532	0.05500	9.97

Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 24.928f TCG = 0.000 VCG = 3.000

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim---	Heel---	Weight(MT)---	in Trim--in Heel--> Area---Angle
0.574	0.01a	0.00	455.36	0.000 -0.022 0.0000 12.443
0.574	0.01a	0.03s	455.36	0.000 0.000 -0.0000 12.417
0.538	0.02a	5.03s	455.34	0.000 3.668 0.1600 7.417
0.282	0.02a	10.03s	455.36	0.000 4.947 0.5474 2.417
0.104	0.02a	12.44s	455.34	0.000 5.198 0.7622 0.000
-0.110	0.02a	15.03s	455.18	0.000 5.350 1.0007 -2.583
-0.562	0.03a	19.93s	455.09	0.000 5.443 1.4658 -7.485
-0.570	0.03a	20.03s	455.60	0.000 5.442 1.4752 -7.583

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GHS 17.22

PONTOON
Condition 00 : Lightship
Lightship condition

-1.075	0.03a	25.03s	455.34	0.000	5.372	1.9482	-12.583
-1.590	0.04a	30.03s	455.14	0.000	5.161	2.4088	-17.583
-2.091	0.04a	35.03s	455.34	0.000	4.830	2.8456	-22.583
-2.577	0.05a	40.03s	455.34	0.000	4.419	3.2497	-27.583
-3.043	0.05a	45.03s	455.34	0.000	3.949	3.6153	-32.583
-3.487	0.06a	50.03s	455.34	0.000	3.433	3.9377	-37.583
-3.903	0.06a	55.03s	455.34	0.000	2.880	4.2135	-42.583
-4.291	0.06a	60.03s	455.23	0.000	2.298	4.4396	-47.583

Distances in METERS.----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

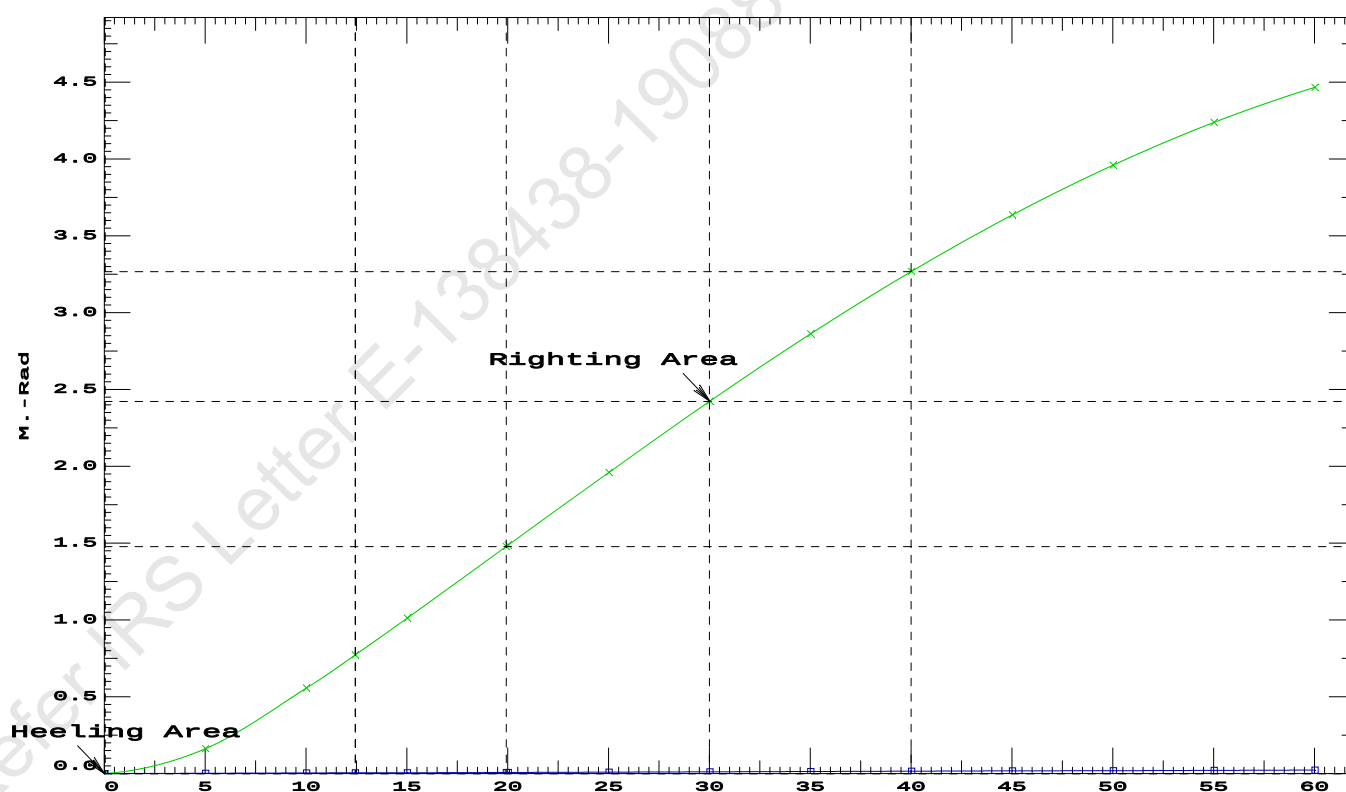
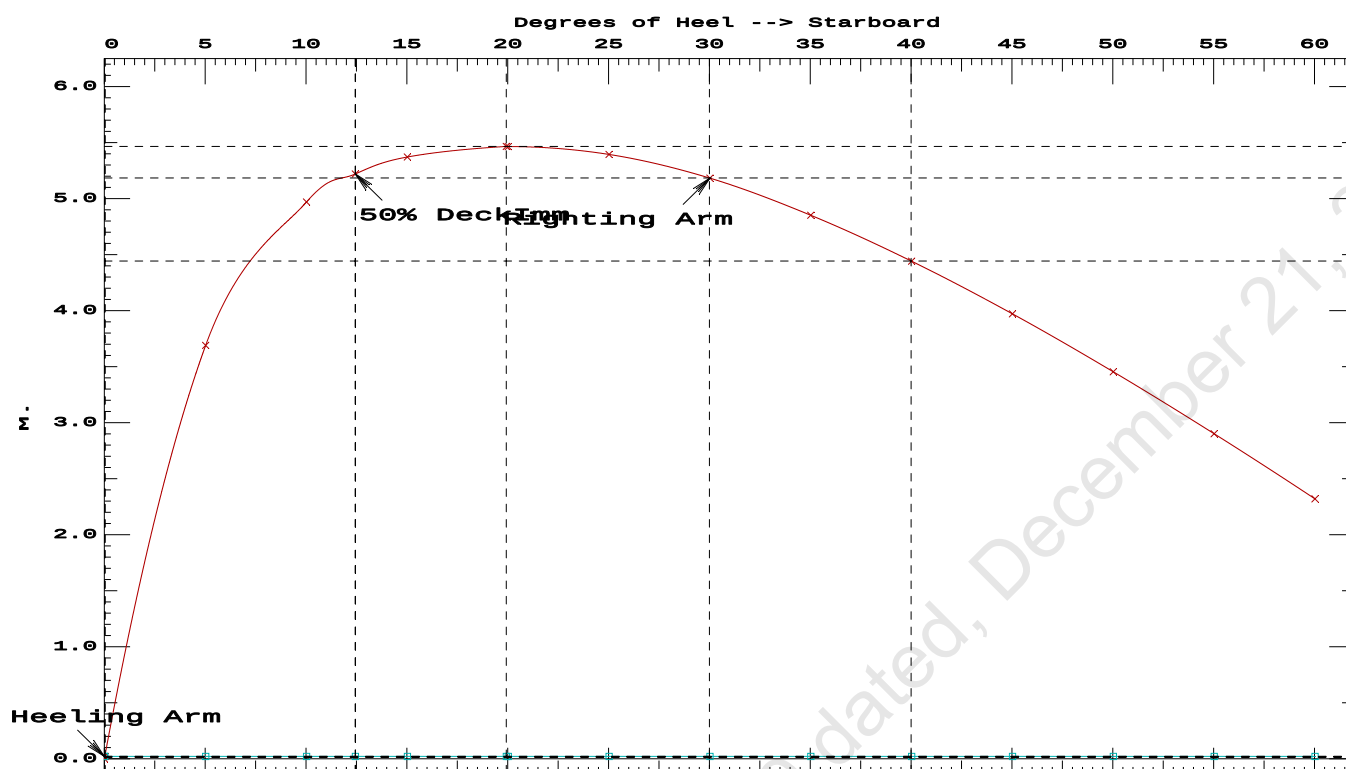
Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.97 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.4658 P
(2) Angle from Equilibrium to RZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 12.42 P

22-10-21 12:47:53
GHS 17.22

Cybermarine Technologies PTE. Ltd.

PONT00N

Condition 00 : LightshipLightship condition

22-10-21 12:47:55 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 01 : Lightship + Crane + Ballast

Condition Graphic - Draft: 1.194 @ 50.000f, 1.456 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C 19 07_WBTK.P 21 07_WBTK.S

22-10-21 12:47:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 01 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.194 @ 50.00f, 1.325 @ 25.00f, 1.456 @ 0.00

Trim: Aft 0.261/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	455.34	24.928f	0.000	3.000	
Crane	400.00	11.500f	0.000	6.325	
Total Fixed----->	855.34	18.648f	0.000	4.555	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->			234.87	44.536f	0.000
Total Weight----->			1,090.21	24.225f	0.000
			Displ(MT)----	LCB-----	TCB-----
HULL	1.025		1,090.21	24.208f	0.000
					0.670

Righting Arms:			0.000	0.000	
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025	856.6	24.756f	0.000	151.65
		MT/cm-----	m.-MT/cm-----	GML-----	BMT-----
		8.78	32.36	148.39	18.245
Distances in METERS.-----			Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

DRAFT AT FP (M) = 1.194
DRAFT AT MS (M) = 1.325
DRAFT AT AP (M) = 1.456
DRAFT AT FWD MARK (M) = 1.215
DRAFT AT AFT MARK (M) = 1.435

HYDROSTATIC PROPERTIES

Trim: Aft 0.261/50.000, No Heel, VCG = 3.931

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCG-----	cm trim-----
1.326	1,090.21	24.208f	0.670	8.78
				24.756f
				32.36
				148.39
				18.245
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.				
Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

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GHS 17.22

PONTOON

Condition 01 : Lightship + Crane + Ballast

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.194 @ 50.00f, 1.325 @ 25.00f, 1.456 @ 0.00

Trim: Aft 0.261/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	83.9	0.857	1.508	0.05500	6.96
CRANE	57.0	4.109	4.760	0.05500	14.92
Total wind heeling moment to starboard----->					21.88
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.225f TCG = 0.000 VCG = 3.931

Free Surface Adjustment: 0.257

Adjusted CG: LCG = 24.227f TCG = 0.000 VCG = 4.188

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle
1.443	0.30a	0.00	1,090.2	0.000 -0.020 0.0000 5.052
1.443	0.30a	0.05s	1,090.2	0.000 0.000 -0.0000 5.003
1.429	0.30a	5.05s	1,090.2	0.000 1.597 0.0697 0.003
1.384	0.33a	10.05s	1,090.2	0.000 3.039 0.2731 -4.997
1.287	0.47a	15.05s	1,090.2	0.000 3.596 0.5690 -9.997
1.251	0.53a	16.89s	1,090.2	0.000 3.619 0.6855 -11.843
1.187	0.63a	20.05s	1,090.2	0.000 3.559 0.8841 -14.997
1.081	0.79a	25.05s	1,090.2	0.000 3.320 1.1866 -19.997
0.970	0.96a	30.05s	1,090.2	0.000 2.986 1.4625 -24.997
0.854	1.12a	35.05s	1,090.2	0.000 2.597 1.7065 -29.997
0.736	1.29a	40.05s	1,090.2	0.000 2.169 1.9147 -34.997
0.614	1.45a	45.05s	1,090.2	0.000 1.714 2.0843 -39.997
0.489	1.60a	50.05s	1,090.2	0.000 1.240 2.2134 -44.997
0.361	1.74a	55.05s	1,090.2	0.000 0.752 2.3004 -49.997
0.230	1.87a	60.05s	1,090.2	0.000 0.255 2.3444 -54.997
Distances in METERS.-----Specific Gravity = 1.025.-----				Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

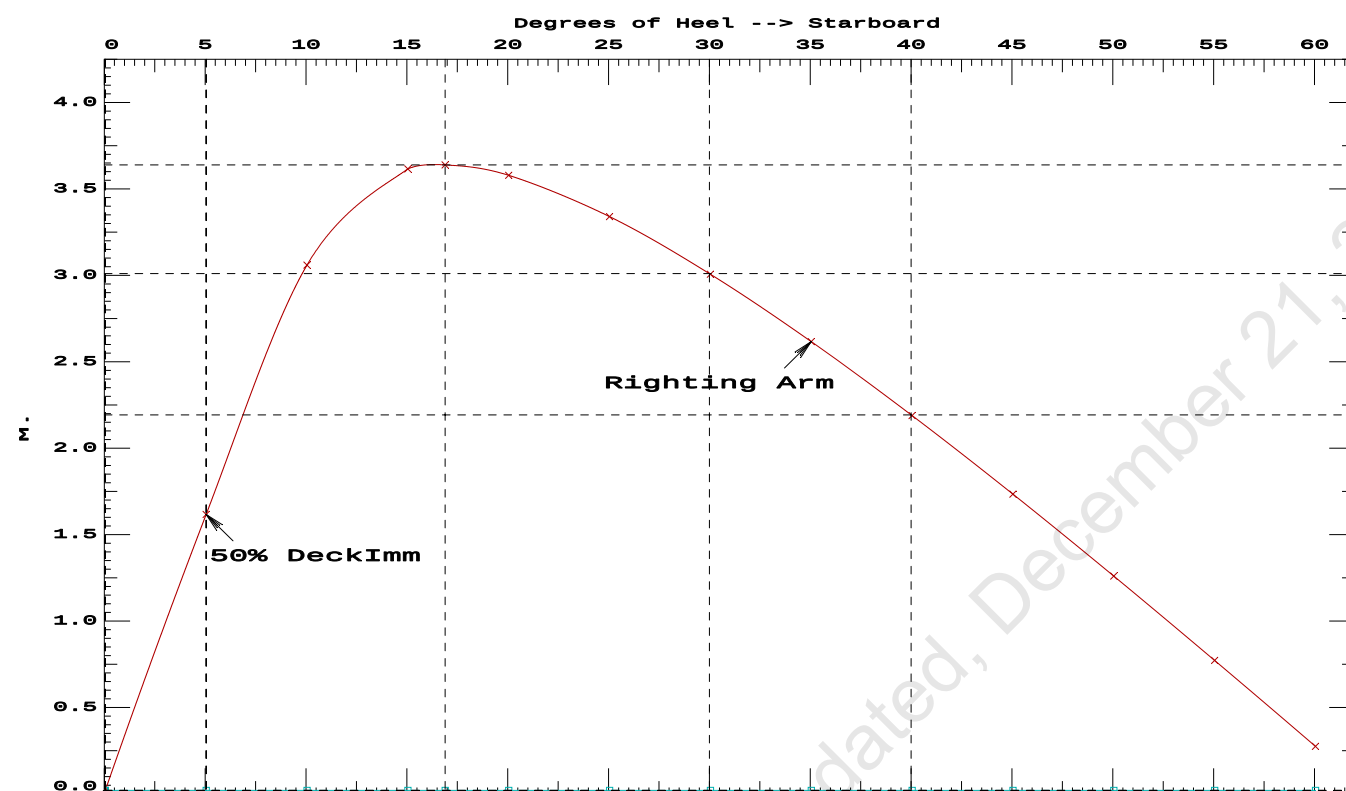
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.88 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.6855 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.00 P

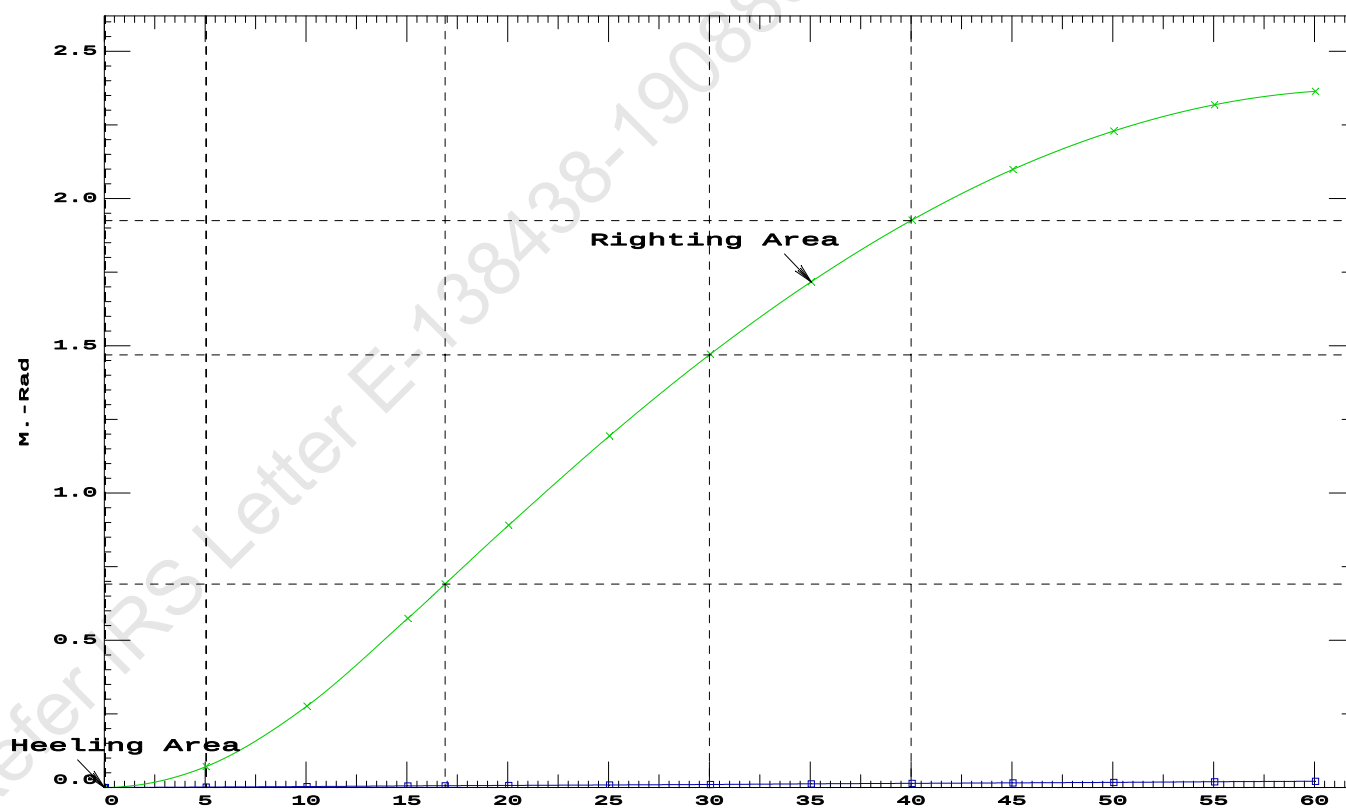
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GHS 17.22

PONT00N

Condition 01 : Lightship + Crane + Ballast

Heeling Arm



22-10-21 12:53:28 Cybermarine Technologies PTE. Ltd.

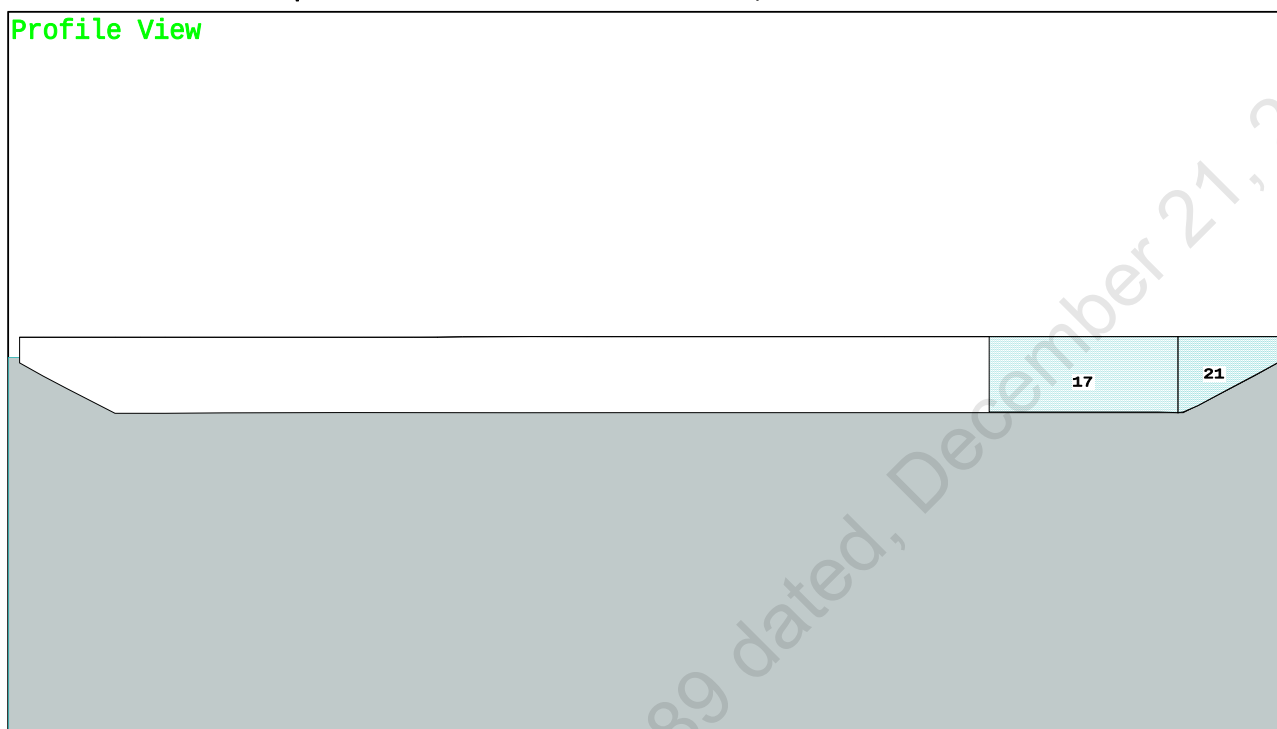
GHS 17.22

PONT00N

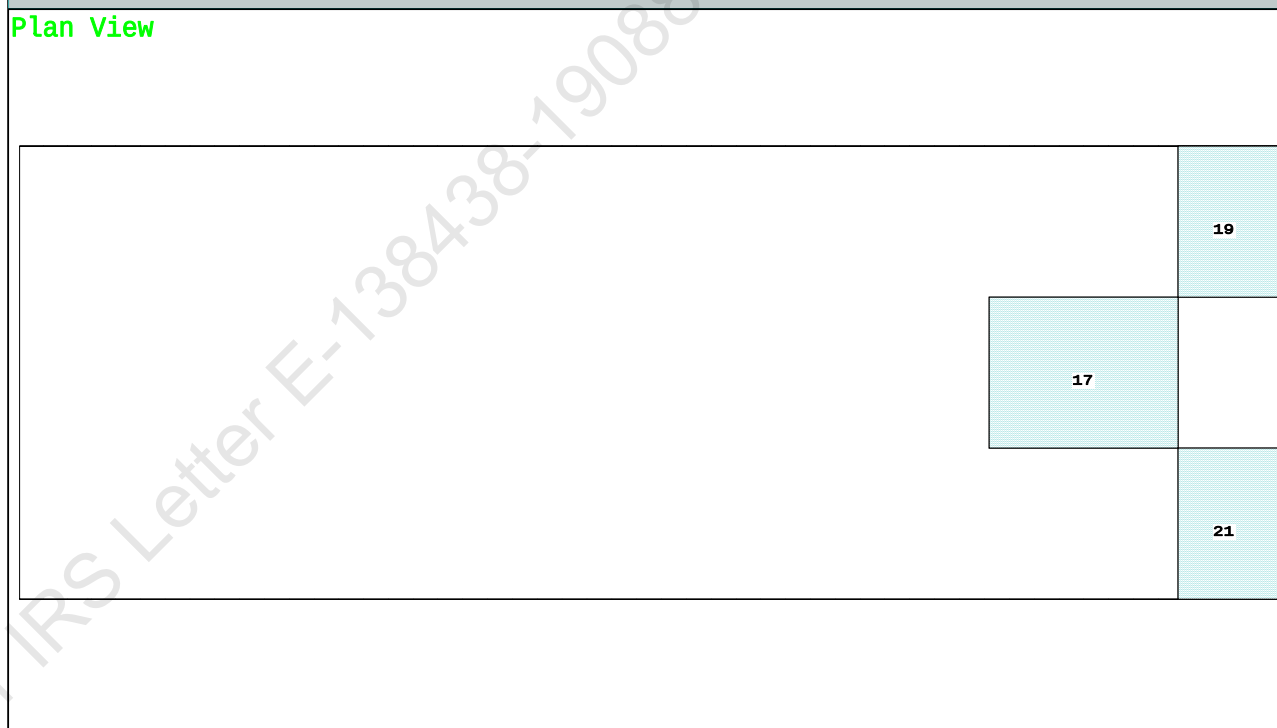
Condition 02 : Lightship + Crane + Deck Cargo + BallastMaximum Draft Condition

Condition Graphic - Draft: 2.200 @ 50.000f, 2.224 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C

19 07_WBTK.P

21 07_WBTK.S

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GHS 17.22

PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.200 @ 50.00f, 2.212 @ 25.00f, 2.224 @ 0.00

Trim: Aft 0.024/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Deck Cargo			802.85	25.940f	0.000	4.000	
Total Fixed----->			1,658.19	22.179f	0.000	4.286	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			234.87	44.536f	0.000	1.660	280.25
Total Weight----->			1,893.06	24.953f	0.000	3.960	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,893.06	24.951f	0.000	1.132	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	900.4	24.999f	0.000	101.45	13.029
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.23		37.34	98.62	10.200
Distances in METERS.-----				Moments in m.-MT-----			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

802.85 MT Deck Cargo

DRAFT AT FP (M) = 2.200
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.224
DRAFT AT FWD MARK (M) = 2.202
DRAFT AT AFT MARK (M) = 2.222

HYDROSTATIC PROPERTIES

Trim: Aft 0.024/50.000, No Heel, VCG = 3.960

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----
2.212	1,893.06	24.951f	1.132	9.23	24.999f	37.34	98.62	10.200
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

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GHS 17.22

PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.200 @ 50.00f, 2.212 @ 25.00f, 2.224 @ 0.00

Trim: Aft 0.024/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.293	4.366	0.05500	13.69
DECKCARGO.C	29.4	1.801	2.874	0.05500	4.65
Total wind heeling moment to starboard-->					21.63
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.953f TCG = 0.000 VCG = 3.960

Free Surface Adjustment: 0.148

Adjusted CG: LCG = 24.953f TCG = 0.000 VCG = 4.108

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight(MT)---	in Trim--in Heel---	Area---	Angle
2.212 0.03a 0.00	1,893.1	0.000 -0.011	0.0000	2.537
2.212 0.03a 0.06s	1,893.1	0.000 0.000	-0.0000	2.474
2.210 0.03a 2.54s	1,893.1	0.000 0.445	0.0096	-0.000
2.201 0.03a 5.06s	1,893.1	0.000 0.892	0.0391	-2.526
2.303 0.04a 10.06s	1,893.1	0.000 1.318	0.1406	-7.526
2.449 0.05a 13.39s	1,893.1	0.000 1.363	0.2194	-10.851
2.535 0.05a 15.06s	1,893.1	0.000 1.351	0.2590	-12.526
2.812 0.07a 20.06s	1,893.1	0.000 1.237	0.3736	-17.526
3.081 0.09a 25.06s	1,893.1	0.000 1.032	0.4733	-22.526
3.327 0.10a 30.06s	1,893.1	0.000 0.779	0.5526	-27.526
3.549 0.12a 35.06s	1,893.1	0.000 0.500	0.6086	-32.526
3.743 0.14a 40.06s	1,893.1	0.000 0.208	0.6396	-37.526
3.862 0.15a 43.54s	1,893.1	0.000 0.000	0.6460	-41.002
3.909 0.16a 45.06s	1,893.1	0.000 -0.092	0.6447	-42.526
4.046 0.17a 50.06s	1,893.1	0.000 -0.395	0.6235	-47.526
4.152 0.19a 55.06s	1,893.1	0.000 -0.698	0.5758	-52.526
4.225 0.20a 60.06s	1,893.1	0.000 -0.997	0.5018	-57.526
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.63 (constant)

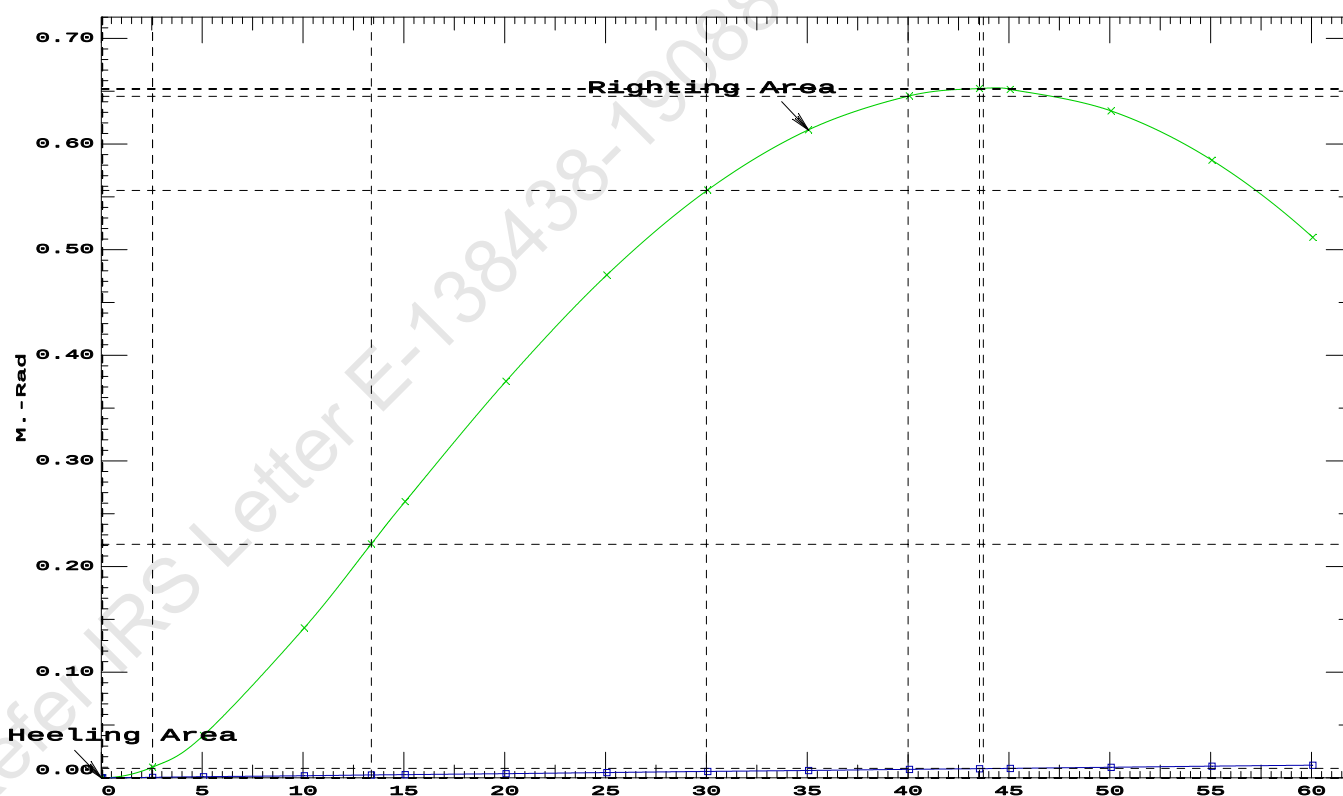
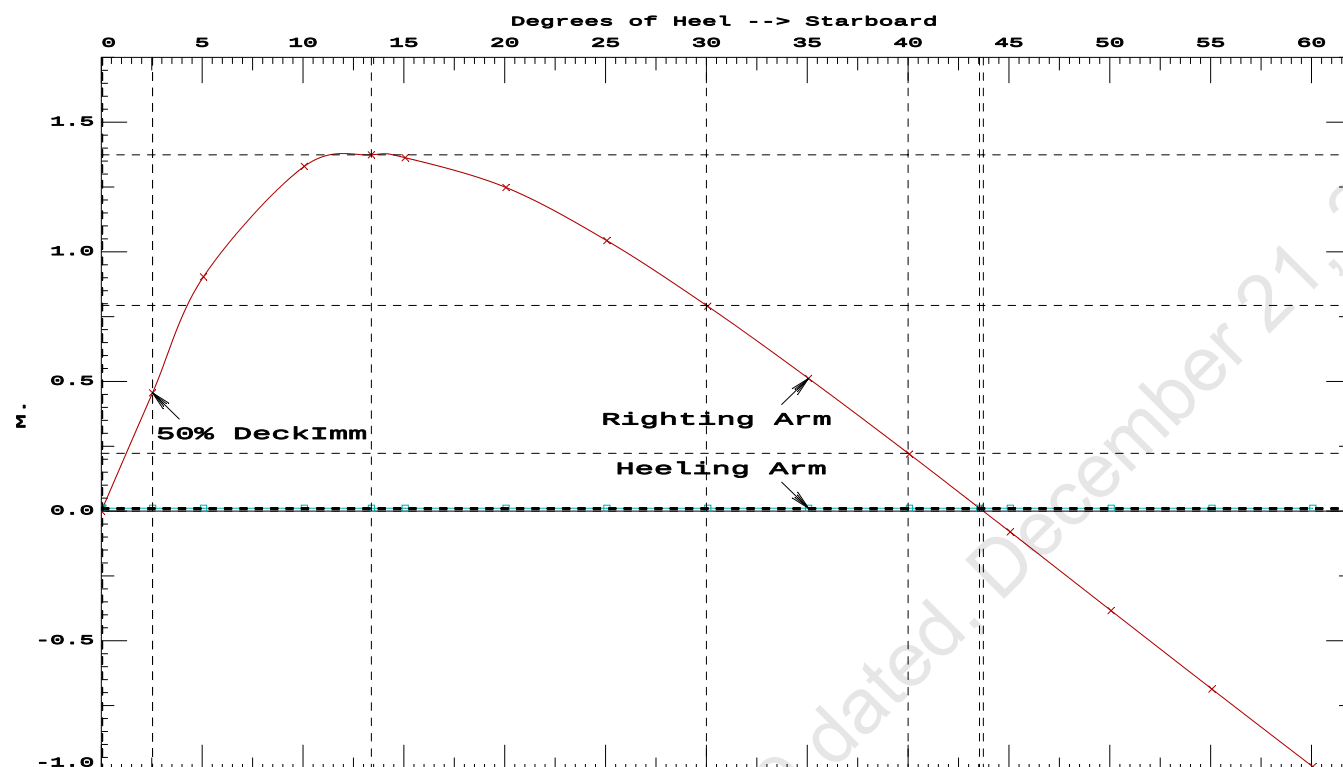
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2194 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	43.48 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.47 P

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PONTON

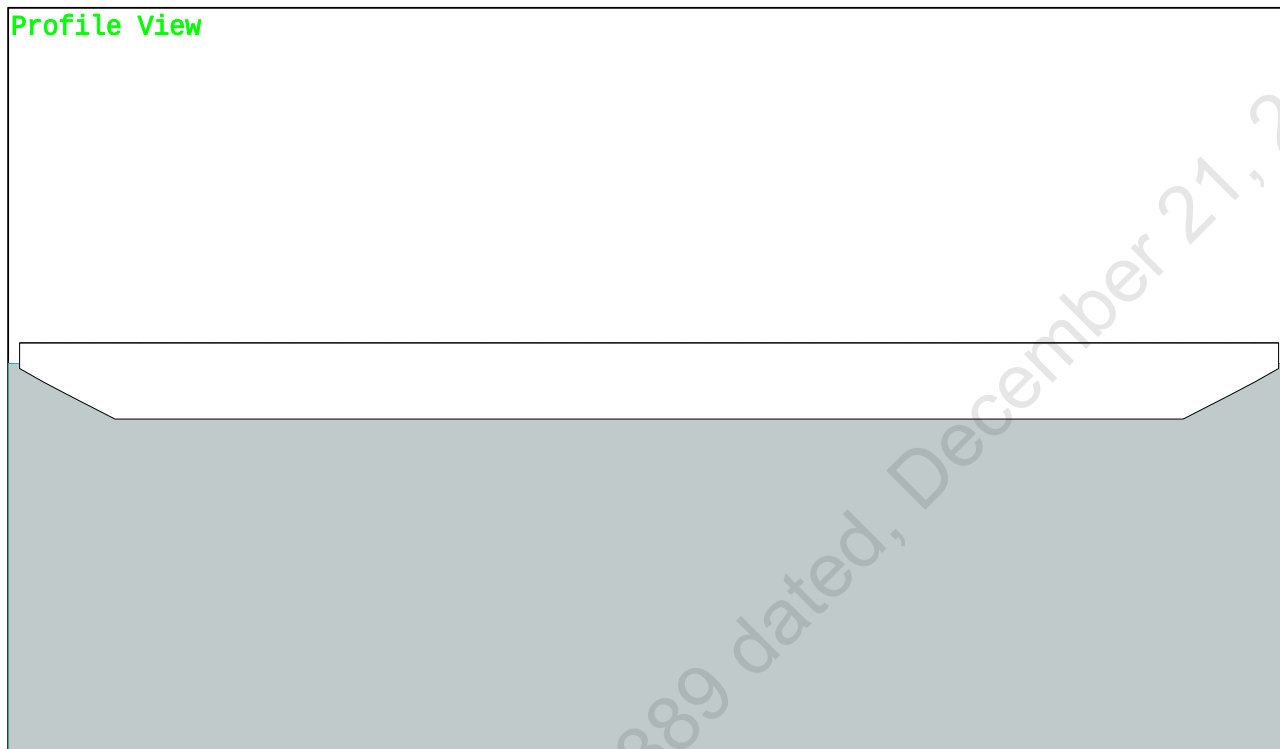
Condition 02 : Lightship + Crane + Deck Cargo + BallastMaximum Draft Condition

22-10-21 12:55:57 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

Condition Graphic - Draft: 2.208 @ 50.000f, 2.216 @ 0.000 Heel: zero

Profile View



Plan View



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GHS 17.22

PONTON

Condition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.208 @ 50.00f, 2.212 @ 25.00f, 2.216 @ 0.00

Trim: Aft 0.009/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	455.34	24.928f	0.000	3.000	
Deck Cargo	1,437.72	25.000f	0.000	4.000	
Total Weight----->	1,893.06	24.983f	0.000	3.759	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.06	24.982f	0.000	1.132

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.45 13.177
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		9.23	37.42	98.82	10.549

Distances in METERS.-----Moments in m.-MT.

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

1,437.72 MT Deck Cargo

DRAFT AT FP (M) = 2.208
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.216
DRAFT AT FWD MARK (M) = 2.208
DRAFT AT AFT MARK (M) = 2.216

HYDROSTATIC PROPERTIES

Trim: Aft 0.009/50.000, No Heel, VCG = 3.759

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----LCF-----cm trim-----GML-----GMT
2.212	1,893.06	24.982f	1.132	9.23 25.000f 37.42 98.82 10.549

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.208 @ 50.00f, 2.212 @ 25.00f, 2.216 @ 0.00

Trim: Aft 0.009/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment-----
HULL	40.5	0.405	1.478	0.05500	3.29
DECKCARGO.C	29.4	1.800	2.874	0.05500	4.65
Total wind heeling moment to starboard----->					7.94

Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

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GHS 17.22

PONTOON
Condition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 24.983f TCG = 0.000 VCG = 3.759

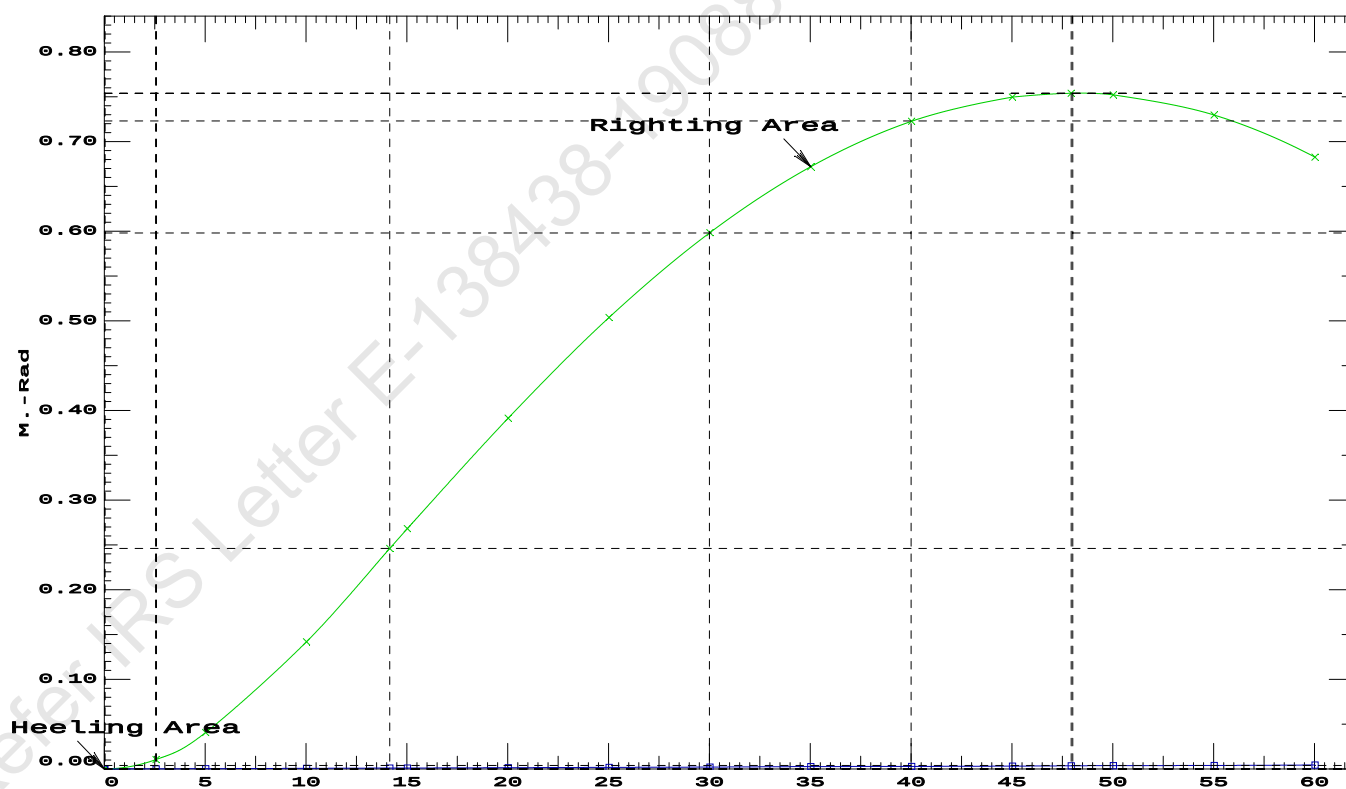
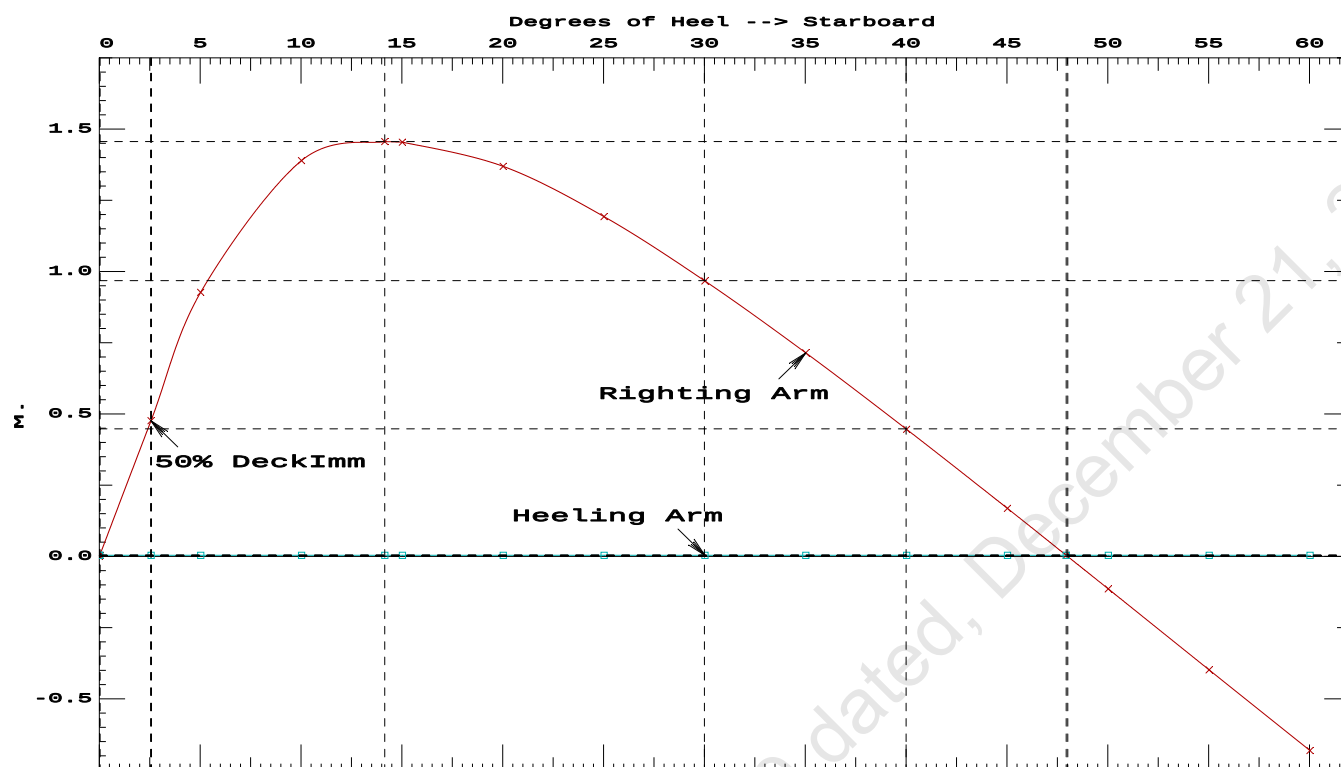
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
2.204	0.01a	0.00	1,893.1	0.000	-0.004	0.0000	2.562
2.204	0.01a	0.02s	1,893.1	0.000	0.000	-0.0000	2.539
2.202	0.01a	2.56s	1,893.1	0.000	0.472	0.0105	0.000
2.194	0.01a	5.02s	1,893.1	0.000	0.923	0.0404	-2.461
2.290	0.02a	10.02s	1,893.1	0.000	1.385	0.1463	-7.461
2.473	0.02a	14.15s	1,893.1	0.000	1.452	0.2502	-11.590
2.518	0.02a	15.02s	1,893.1	0.000	1.449	0.2723	-12.461
2.790	0.02a	20.02s	1,893.1	0.000	1.365	0.3968	-17.461
3.055	0.03a	25.02s	1,893.1	0.000	1.188	0.5088	-22.461
3.296	0.04a	30.02s	1,893.1	0.000	0.963	0.6030	-27.461
3.513	0.04a	35.02s	1,893.1	0.000	0.710	0.6762	-32.461
3.703	0.05a	40.02s	1,893.1	0.000	0.442	0.7266	-37.461
3.864	0.06a	45.02s	1,893.1	0.000	0.165	0.7532	-42.461
3.945	0.06a	47.94s	1,893.1	0.000	0.000	0.7574	-45.378
3.997	0.06a	50.02s	1,893.1	0.000	-0.118	0.7552	-47.461
4.099	0.07a	55.02s	1,893.1	0.000	-0.402	0.7325	-52.461
4.169	0.07a	60.02s	1,893.1	0.000	-0.685	0.6850	-57.461

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.94 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.2502 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg 47.92 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 2.54 P

22-10-21 12:55:57
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTOONCondition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

22-10-21 12:44:08 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

Condition Graphic - Draft: 1.199 @ 50.000f, 1.652 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

22-10-21 12:44:08 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.199 @ 50.00f, 1.426 @ 25.00f, 1.652 @ 0.00

Trim: Aft 0.453/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	19.500a	0.000	34.602	
Total Fixed----->			895.34	16.944f	0.000	5.897	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,180.32	23.739f	0.000	4.883	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,180.21	23.701f	0.000	0.727	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025		862.8	24.603f	0.000	143.17	19.930
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.84		32.81	139.02	15.773
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.199
DRAFT AT MS (M) = 1.426
DRAFT AT AP (M) = 1.652
DRAFT AT FWD MARK (M) = 1.235
DRAFT AT AFT MARK (M) = 1.616

HYDROSTATIC PROPERTIES

Trim: Aft 0.453/50.000, No Heel, VCG = 4.883

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/					
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT-----	
1.429	1,180.21	23.701f	0.727	8.84	24.603f	32.81	139.02	15.773	
Distances in METERS.-----					Specific Gravity = 1.025.-----				
					Moment in m.-MT.				
					Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

22-10-21 12:44:08 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.199 @ 50.00f, 1.426 @ 25.00f, 1.652 @ 0.00

Trim: Aft 0.453/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.0	0.810	1.511	0.05500	6.57
CRANE	57.0	3.951	4.652	0.05500	14.59
Total wind heeling moment to starboard-->					21.15
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.739f TCG = 0.000 VCG = 4.883

Free Surface Adjustment: 0.299

Adjusted CG: LCG = 23.741f TCG = 0.000 VCG = 5.182

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.641	0.52a	0.00	1,180.3	0.000	-0.018	0.0000	4.376
1.641	0.52a	0.06s	1,180.3	0.000	0.000	-0.0000	4.314
1.629	0.52a	4.38s	1,180.3	0.000	1.196	0.0450	0.000
1.627	0.52a	5.06s	1,180.3	0.000	1.386	0.0605	-0.686
1.596	0.57a	10.06s	1,180.3	0.000	2.660	0.2378	-5.686
1.606	0.84a	15.06s	1,180.3	0.000	3.079	0.4952	-10.686
1.607	0.85a	15.32s	1,180.3	0.000	3.080	0.5093	-10.947
1.614	1.13a	20.06s	1,180.3	0.000	2.942	0.7617	-15.686
1.616	1.43a	25.06s	1,180.3	0.000	2.621	1.0057	-20.686
1.613	1.73a	30.06s	1,180.3	0.000	2.216	1.2174	-25.686
1.603	2.04a	35.06s	1,180.3	0.000	1.763	1.3914	-30.686
1.587	2.34a	40.06s	1,180.3	0.000	1.281	1.5244	-35.686
1.562	2.63a	45.06s	1,180.3	0.000	0.780	1.6145	-40.686
1.528	2.91a	50.06s	1,180.3	0.000	0.267	1.6603	-45.686
1.506	3.04a	52.64s	1,180.3	0.000	0.000	1.6663	-48.261
1.482	3.16a	55.06s	1,180.3	0.000	-0.252	1.6609	-50.686
1.422	3.39a	60.06s	1,180.3	0.000	-0.771	1.6163	-55.686
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.15 (constant)

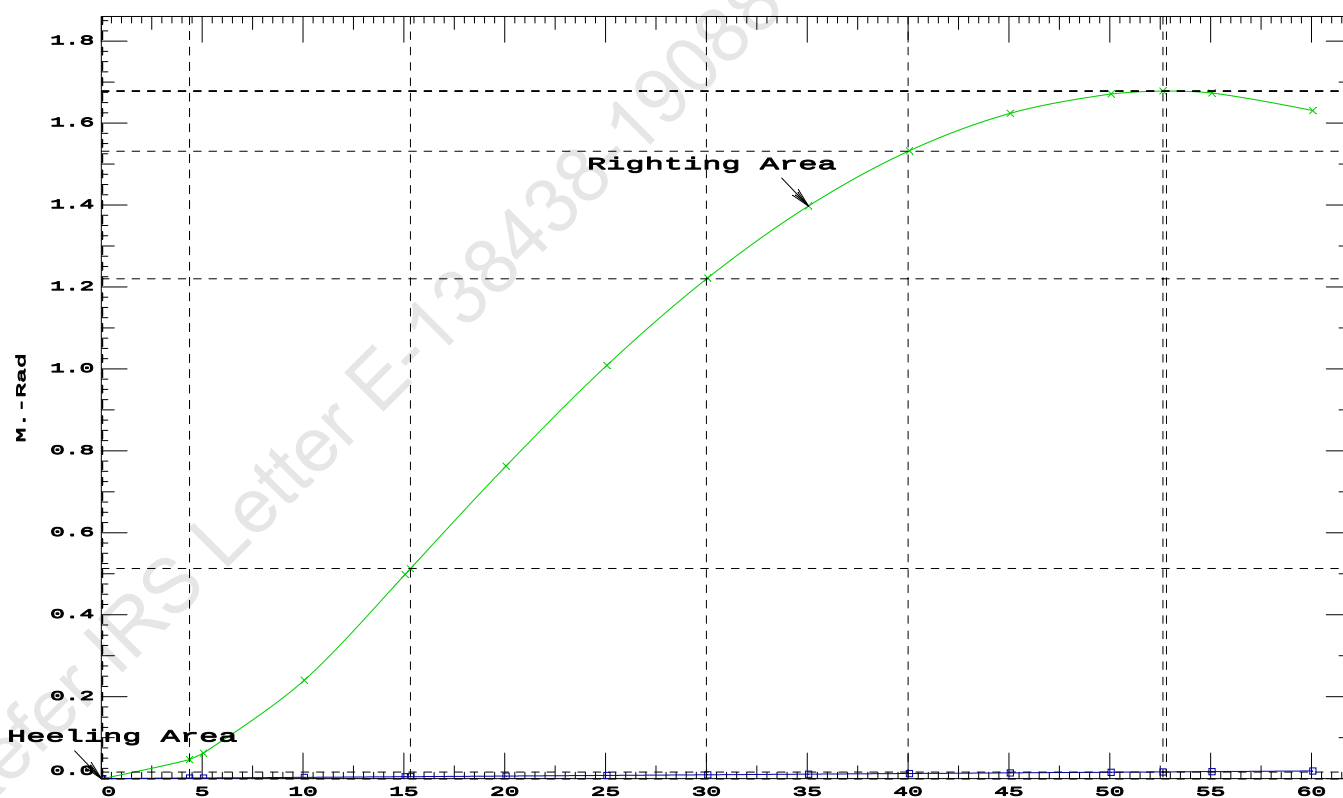
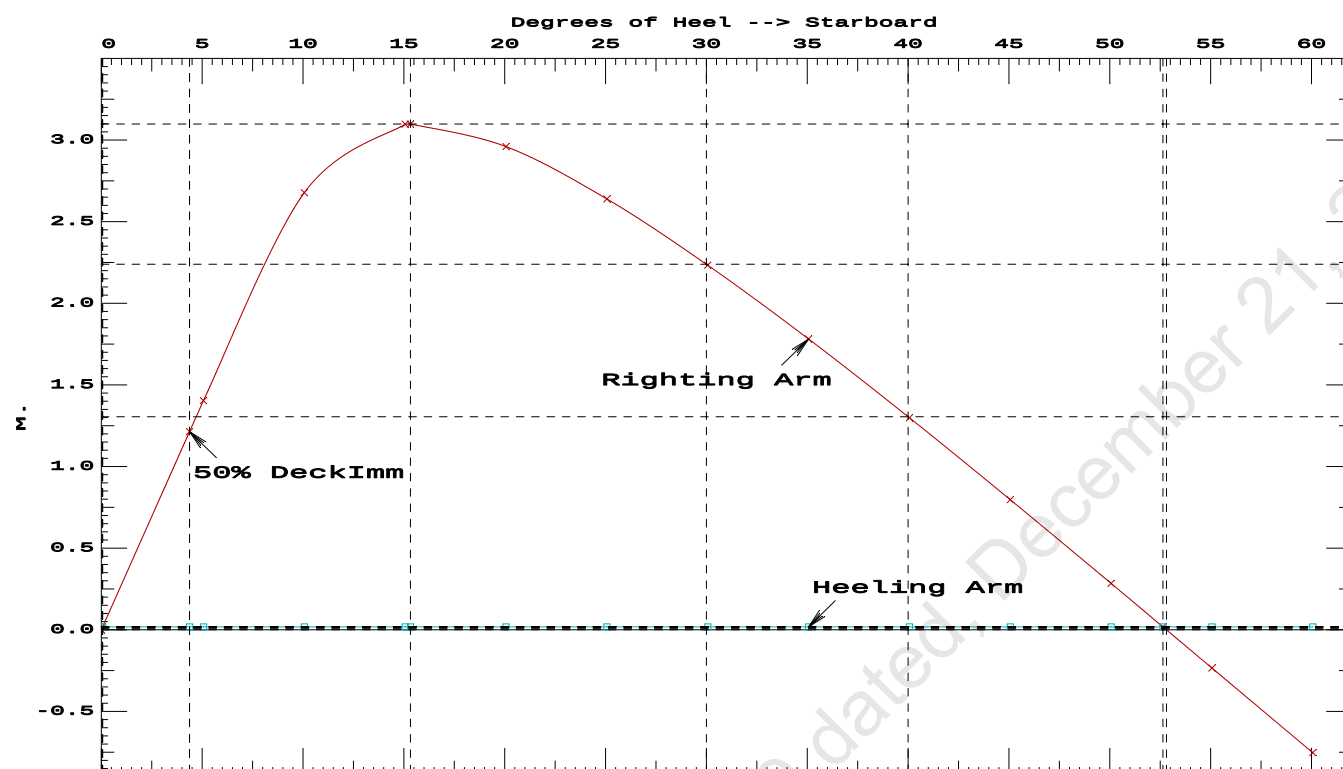
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.5093 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	52.57 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	4.31 P

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GHS 17.22

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft



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GHS 17.22

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.739f TCG = 0.000 VCG = 4.883
Free Surface Adjustment: 0.299
Adjusted CG: LCG = 23.741f TCG = 0.000 VCG = 5.182

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
1.641	0.52a	0.01s	1,180.3	0.000	0.000	0.0000
1.627	0.52a	5.01s	1,180.3	0.000	1.386	0.0605
1.596	0.57a	10.01s	1,180.3	0.000	2.664	0.2380
1.606	0.83a	15.01s	1,180.3	0.000	3.093	0.4962
1.607	0.85a	15.33s	1,180.3	0.000	3.094	0.5135
1.614	1.13a	20.01s	1,180.3	0.000	2.958	0.7639
1.616	1.43a	25.01s	1,180.3	0.000	2.638	1.0094
1.613	1.73a	30.01s	1,180.3	0.000	2.234	1.2226
1.603	2.03a	35.01s	1,180.3	0.000	1.781	1.3981
1.587	2.33a	40.01s	1,180.3	0.000	1.299	1.5328
1.562	2.63a	45.01s	1,180.3	0.000	0.798	1.6244
1.528	2.90a	50.01s	1,180.3	0.000	0.285	1.6718
1.504	3.05a	52.77s	1,180.3	0.000	0.000	1.6786
1.482	3.16a	55.01s	1,180.3	0.000	-0.233	1.6741
1.423	3.39a	60.01s	1,180.3	0.000	-0.752	1.6311

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

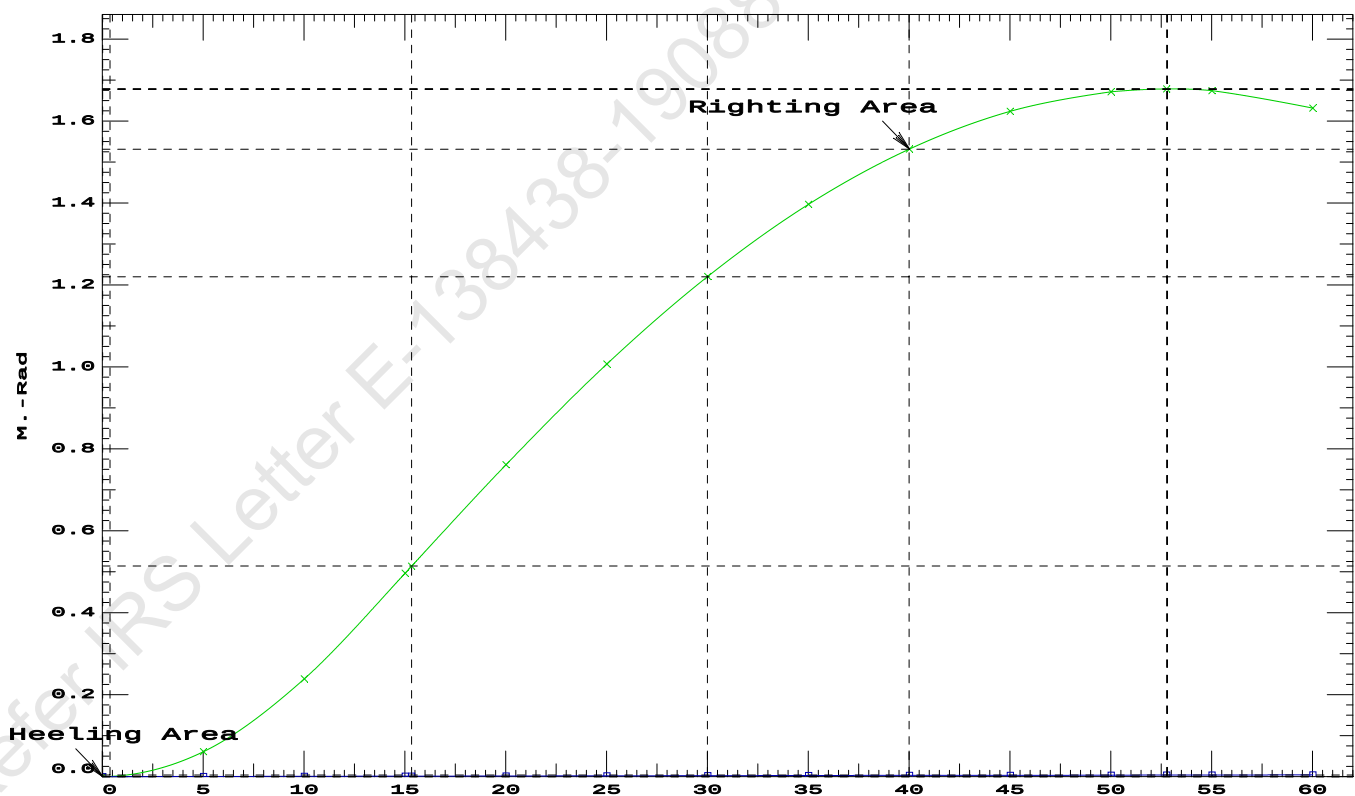
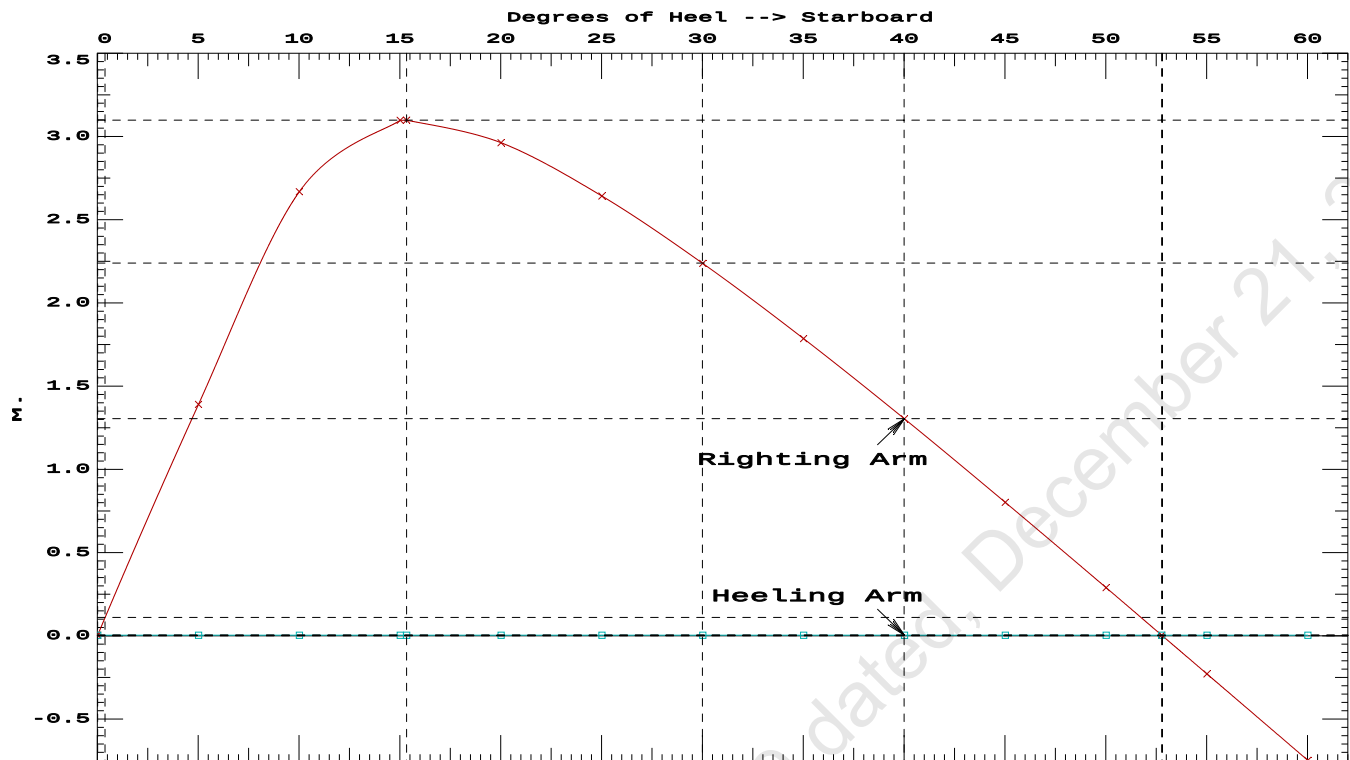
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----STABILITY CRITERIA FOR CRANE OPERAT-----Min/Max-----Attained

(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	73036.0 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667		431.383 P

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GHS 17.22 PONT00N

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft



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GHS 17.22 PONTON

Trim: Aft 0.453/50.000, Heel: 0.00 deg.

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----	> 1.025	859.9	24.573f	0.023s	141.74	19.820
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		8.81		32.48	137.58	15.664
Distances in METERS.-----				Moments in m.-MT.		

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.427 @ 50.00f, 1.382 @ 25.00f, 1.338 @ 0.00
Trim: Fwd 0.089/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----		
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Total Fixed----->	855.34	18.648f	0.000	4.555			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,140.32	25.255f	0.000	3.841	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,140.33	25.261f	0.005s	0.698	

Righting Arms:			0.000	0.005s			
External Arms:			0.000	0.004s			
Residual Righting Arms:			0.000	0.000s			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane----->	1.025	849.2	25.023f	0.070s	141.32	20.329	
		MT/cm					
		8.70					
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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GHS 17.22 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.648f TCG = 0.000 VCG = 4.555

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.326	0.10f	0.00	1,140.3	0.000	-0.000	0.0000
1.326	0.10f	0.07p	1,140.5	0.000	0.031	0.0000
1.311	0.10f	5.00p	1,140.3	0.000	1.563	0.0686
1.256	0.11f	10.00p	1,140.3	0.000	3.008	0.2688
1.102	0.16f	15.00p	1,140.3	0.000	3.589	0.5630
1.029	0.19f	17.23p	1,140.3	0.000	3.620	0.7038
0.936	0.22f	20.00p	1,140.3	0.000	3.575	0.8782
0.761	0.27f	25.00p	1,140.3	0.000	3.364	1.1834
0.580	0.33f	30.00p	1,140.3	0.000	3.062	1.4645

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 855.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.648f TCG = 0.000 VCG = 4.555

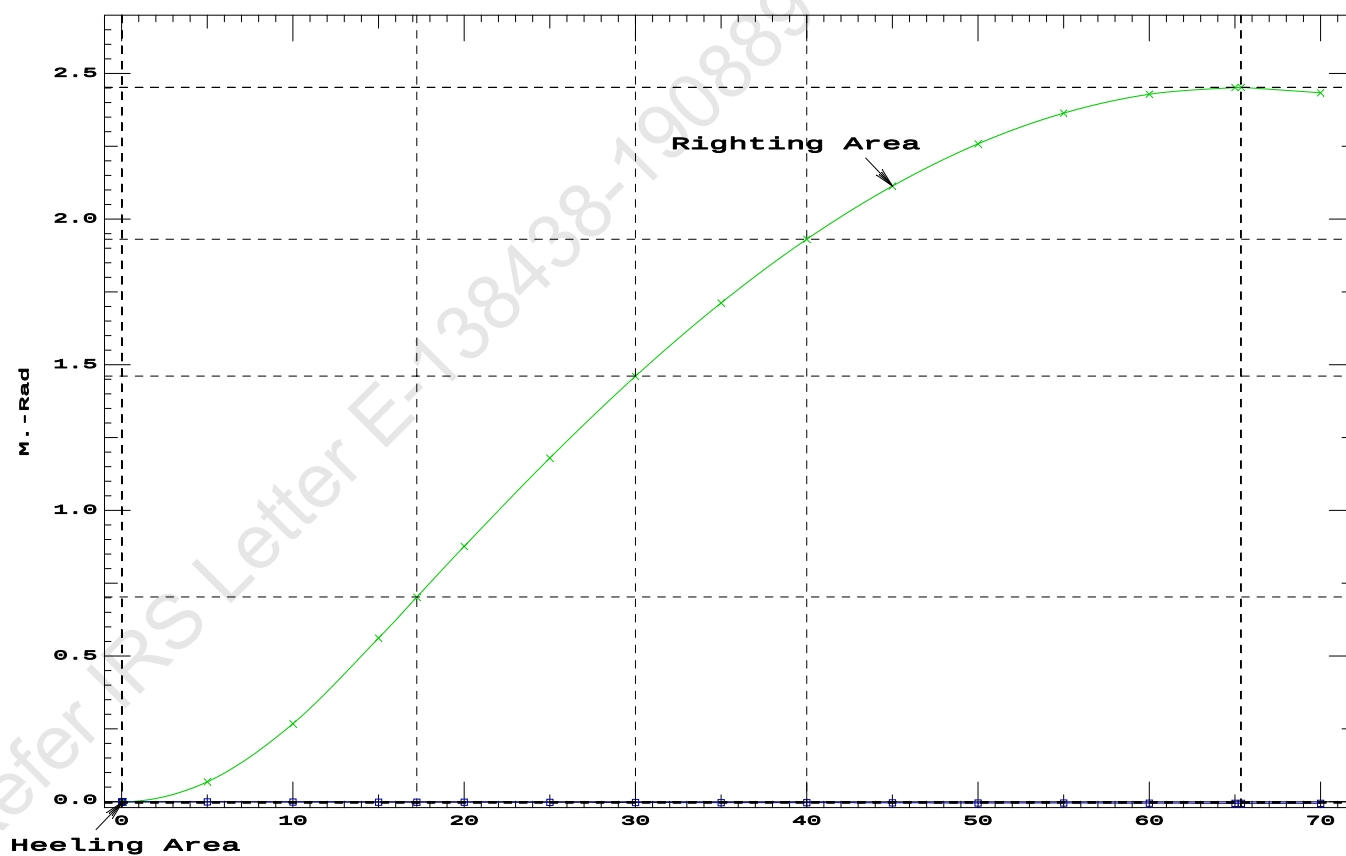
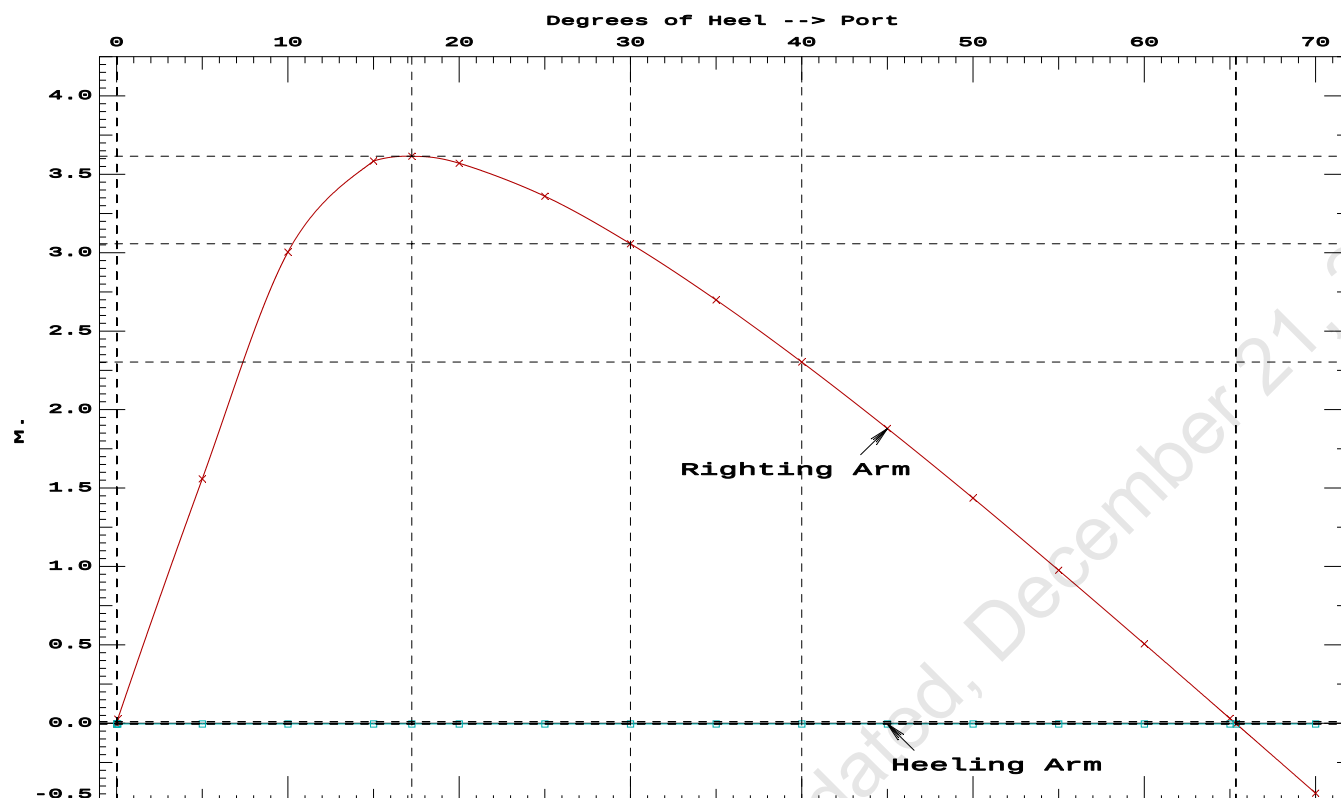
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.326	0.10f	0.00	1,140.3	0.000	-0.000	0.0000
1.326	0.10f	0.07p	1,140.5	0.000	0.031	0.0000
1.311	0.10f	5.00p	1,140.3	0.000	1.563	0.0686
1.256	0.11f	10.00p	1,140.3	0.000	3.008	0.2688
1.102	0.16f	15.00p	1,140.3	0.000	3.589	0.5630
1.029	0.19f	17.23p	1,140.3	0.000	3.620	0.7038
0.936	0.22f	20.00p	1,140.3	0.000	3.575	0.8782
0.761	0.27f	25.00p	1,140.3	0.000	3.364	1.1834
0.580	0.33f	30.00p	1,140.3	0.000	3.062	1.4645
0.393	0.39f	35.00p	1,140.3	0.000	2.703	1.7164
0.202	0.44f	40.00p	1,140.3	0.000	2.307	1.9353
0.008	0.50f	45.00p	1,140.3	0.000	1.884	2.1184
-0.186	0.55f	50.00p	1,140.3	0.000	1.440	2.2636
-0.379	0.60f	55.00p	1,140.3	0.000	0.980	2.3693
-0.569	0.64f	60.00p	1,140.3	0.000	0.511	2.4344
-0.755	0.68f	65.00p	1,140.3	0.000	0.035	2.4583
-0.768	0.68f	65.37p	1,140.3	0.000	0.000	2.4584
-0.933	0.71f	70.00p	1,140.3	0.000	-0.441	2.4406

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 855.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 65.30 P

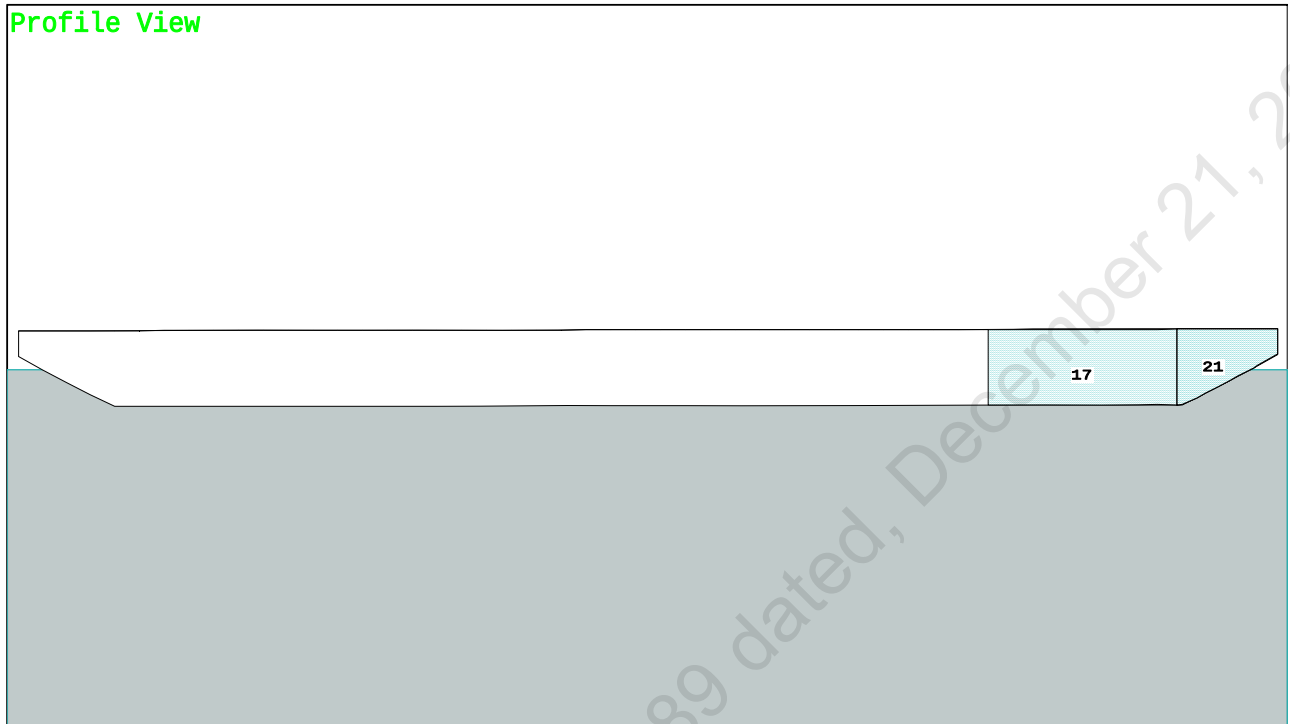
22-10-21 12:44:06
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00N

22-10-21 12:44:11 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

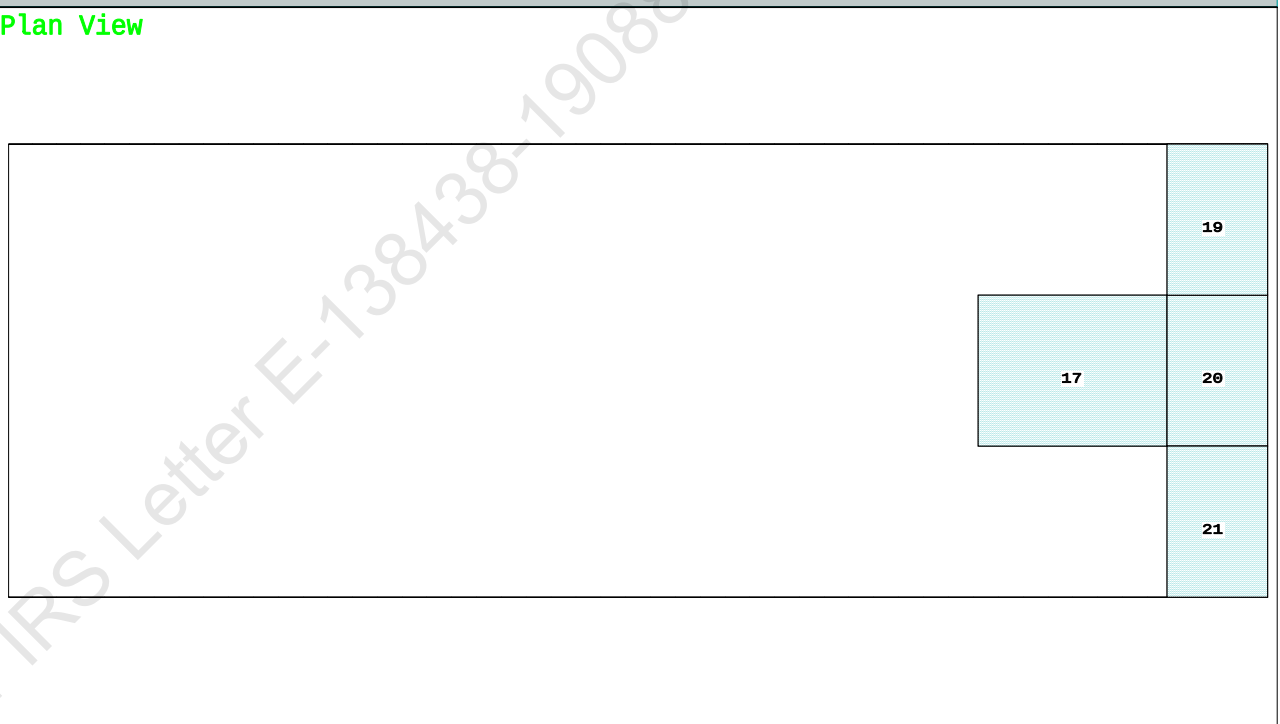
Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

Condition Graphic - Draft: 1.390 @ 50.000f, 1.466 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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GHS 17.22

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.390 @ 50.00f, 1.428 @ 25.00f, 1.466 @ 0.00

Trim: Aft 0.076/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed----->			895.34	18.329f	0.000	5.897	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,180.32	24.789f	0.000	4.883	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,180.33	24.783f	0.000	0.722	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		863.8	24.926f	0.000	143.62	19.988
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.85		32.92	139.46	15.826
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.390
DRAFT AT MS (M) = 1.428
DRAFT AT AP (M) = 1.466
DRAFT AT FWD MARK (M) = 1.396
DRAFT AT AFT MARK (M) = 1.459

HYDROSTATIC PROPERTIES

Trim: Aft 0.076/50.000, No Heel, VCG = 4.883

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT-----
1.428	1,180.33	24.783f	0.722	8.85	24.926f	32.92	139.46	15.826
Distances in METERS.-----					Specific Gravity = 1.025.-----			
					Moment in m.-MT.			
					Trim is per 50.00m.			

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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GHS 17.22

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.390 @ 50.00f, 1.428 @ 25.00f, 1.466 @ 0.00

Trim: Aft 0.076/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.1	0.802	1.502	0.05500	6.53
CRANE	57.0	4.062	4.761	0.05500	14.93
Total wind heeling moment to starboard-->					21.46
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.789f TCG = 0.000 VCG = 4.883

Free Surface Adjustment: 0.299

Adjusted CG: LCG = 24.790f TCG = 0.000 VCG = 5.182

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.454	0.09a	0.00	1,180.3	0.000 -0.018	0.0000	5.022
1.453	0.09a	0.04s	1,180.3	0.000 0.000	-0.0000	4.978
1.439	0.09a	5.04s	1,180.3	0.000 1.382	0.0603	-0.022
1.393	0.09a	10.04s	1,180.3	0.000 2.677	0.2380	-5.022
1.303	0.14a	15.04s	1,180.3	0.000 3.114	0.4981	-10.022
1.297	0.14a	15.34s	1,180.3	0.000 3.116	0.5140	-10.315
1.204	0.19a	20.04s	1,180.3	0.000 2.979	0.7677	-15.022
1.098	0.24a	25.04s	1,180.3	0.000 2.659	1.0150	-20.022
0.984	0.29a	30.04s	1,180.3	0.000 2.254	1.2300	-25.022
0.864	0.34a	35.04s	1,180.3	0.000 1.802	1.4074	-30.022
0.739	0.39a	40.04s	1,180.3	0.000 1.319	1.5438	-35.022
0.608	0.44a	45.04s	1,180.3	0.000 0.816	1.6371	-40.022
0.473	0.49a	50.04s	1,180.3	0.000 0.301	1.6859	-45.022
0.393	0.51a	52.93s	1,180.3	0.000 0.000	1.6935	-47.910
0.334	0.53a	55.04s	1,180.3	0.000 -0.221	1.6895	-50.022
0.193	0.57a	60.04s	1,180.3	0.000 -0.744	1.6474	-55.022
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

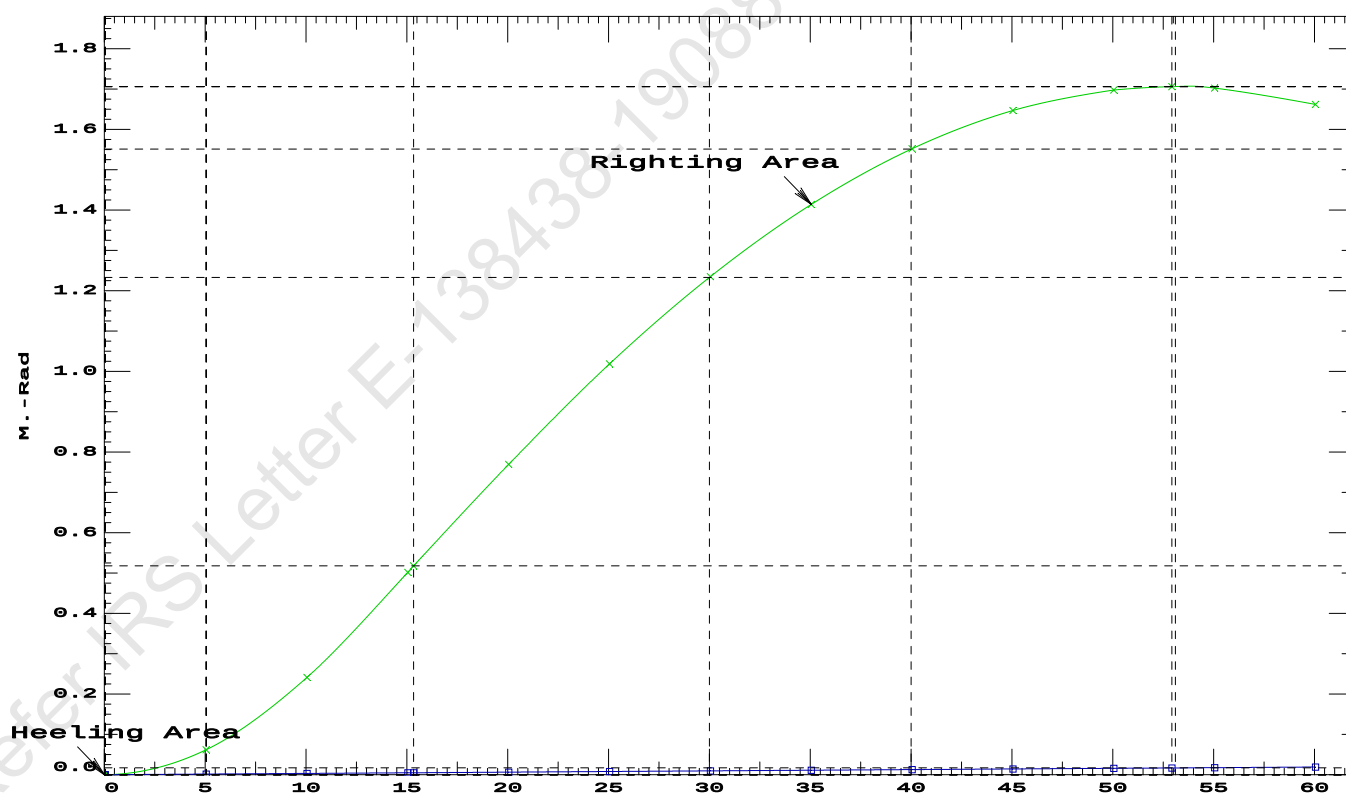
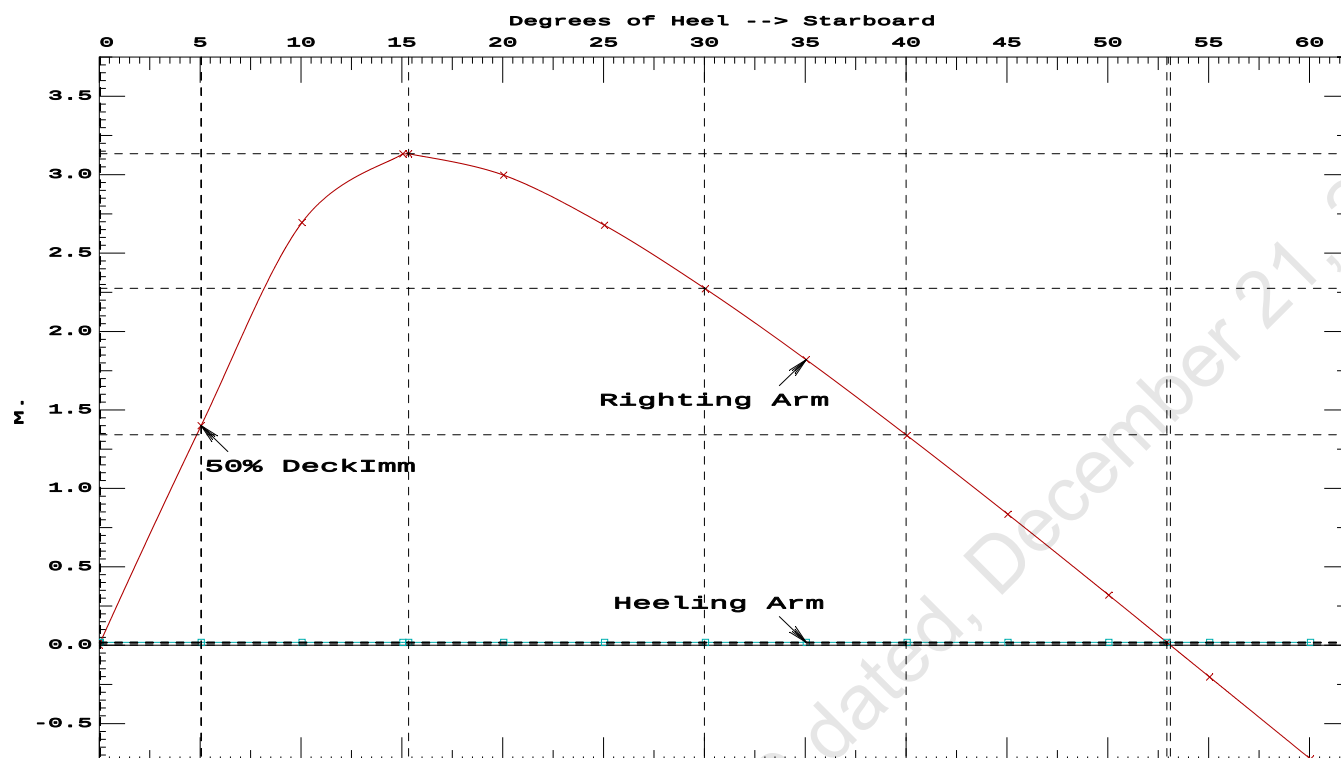
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.46 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5140 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	52.89 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	5.00 P

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GHS 17.22

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd



22-10-21 12:44:11 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.789f TCG = 0.000 VCG = 4.883
Free Surface Adjustment: 0.299
Adjusted CG: LCG = 24.790f TCG = 0.019p VCG = 5.181

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
1.445	0.09a	3.59s	1,180.3	0.000	0.000	0.0000
1.412	0.09a	8.59s	1,180.3	0.000	1.346	0.0587
1.330	0.13a	13.59s	1,180.3	0.000	2.101	0.2186
1.293	0.14a	15.56s	1,180.3	0.000	2.135	0.2920
1.234	0.17a	18.59s	1,180.3	0.000	2.064	0.4040
1.130	0.22a	23.59s	1,180.3	0.000	1.782	0.5749
1.018	0.27a	28.59s	1,180.3	0.000	1.396	0.7144
0.900	0.33a	33.59s	1,180.3	0.000	0.955	0.8174
0.776	0.38a	38.59s	1,180.3	0.000	0.479	0.8801
0.652	0.42a	43.39s	1,180.3	0.000	0.000	0.9003
0.646	0.43a	43.59s	1,180.3	0.000	-0.020	0.9003
0.513	0.47a	48.59s	1,180.3	0.000	-0.534	0.8762
0.375	0.52a	53.59s	1,180.3	0.000	-1.055	0.8069
0.235	0.56a	58.59s	1,180.3	0.000	-1.580	0.6920
0.091	0.59a	63.59s	1,180.3	0.000	-2.102	0.5313

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

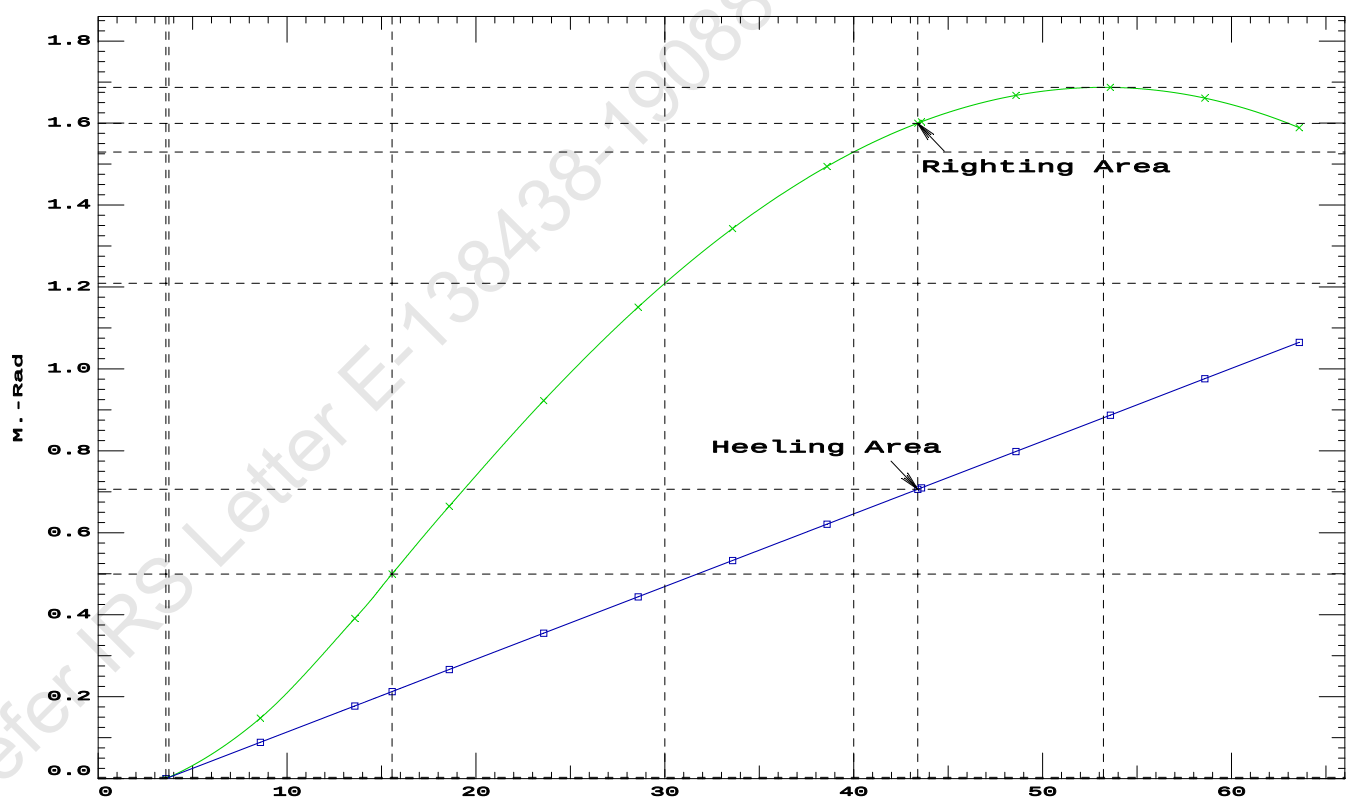
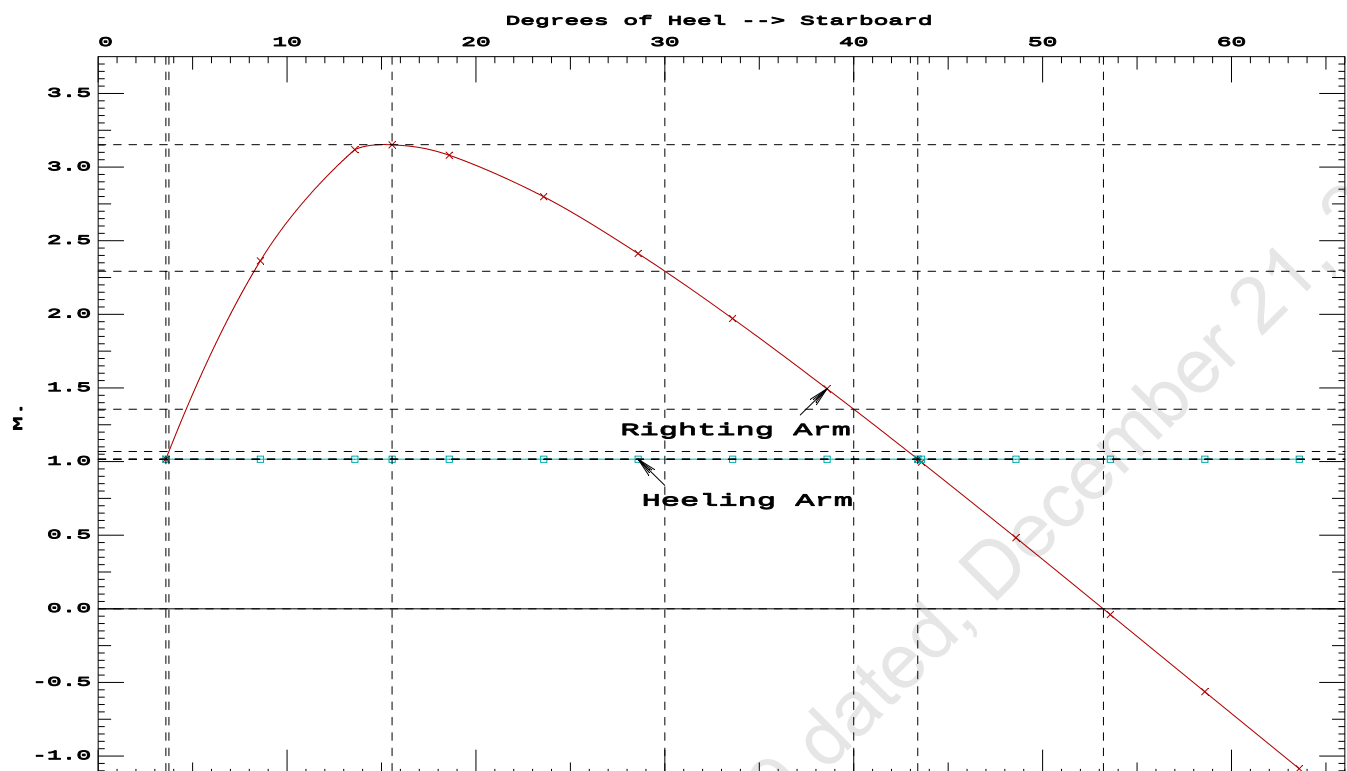
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1200.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.59 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	210.0 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.275 P

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GHS 17.22 PONT00N

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd



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GHS 17.90 PONTON

Condition 05 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.385 @ 50.00f, 1.422 @ 25.00f, 1.460 @ 0.00

Trim: Aft 0.075/50.000, Heel: Stbd 3.59 deg.

Part-----	Weight (MT)	LCG	TCG	VCG			
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Crane Load	40.00	11.500f	0.000	34.602			
Total Fixed----->	895.34	18.329f	0.000	5.897			
Load-----	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,180.32	24.789f	0.000	4.883	
			Displ (MT)	LCB	TCB	VCB	
HULL	1.025		1,180.33	24.783f	1.277s	0.762	

Righting Arms:			0.000	1.017s			
External Arms:			0.000	1.017s			
Residual Righting Arms:			0.000	0.000s			
Part-----	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane---->	1.025	863.3	24.917f	0.151s	143.18	20.005*	
		MT/cm					
		8.85					
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.59

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GHS 17.90 PONTON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.427 @ 50.00f, 1.382 @ 25.00f, 1.338 @ 0.00

Trim: Fwd 0.089/50.000, Heel: zero

Part	Weight (MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Total Fixed	855.34	18.648f	0.000	4.555			
Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks			284.98	45.086f	0.000	1.697	352.57
Total Weight			1,140.32	25.255f	0.000	3.841	
			Displ (MT)	LCB	TCB	VCB	
HULL	1.025		1,140.30	25.261f	0.000	0.698	
Righting Arms:			0.000	0.000			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	-0.001s			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	860.5	25.085f	0.000	146.97	20.597*	
		MT/cm	m.	MT/cm	GML	GMT	
		8.82		32.80	143.83	17.454	
Distances in METERS.						Moments in m.-MT.	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 4076.95; t3: 30.0; t: 15.0
limmarg: 1504.99; t3: 15.0; t: 6.0
limmarg: 524.86; t3: 9.0; t: 2.4
limmarg: 238.47; t3: 6.6; t: 1.0
limmarg: 143.94; t3: 5.6; t: 0.4
limmarg: 110.38; t3: 5.2; t: 0.2
limmarg: 94.52; t3: 5.0; t: 0.1
limmarg: 86.84; t3: 4.9; t: 0.0
limmarg: 86.84; t3: 4.9; t: 0.0

Theta3 is 4.90 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.648f TCG = 0.000 VCG = 4.555

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth	Trim	Heel	Weight (MT)	in Trim--in Heel--> Area
1.318	0.10f	3.59s	1,140.3	0.000 -1.119 0.0000
1.326	0.10f	0.00	1,140.3	0.000 0.001 -0.0350
1.312	0.10f	4.90p	1,140.3	0.000 1.528 0.0304
1.311	0.10f	5.00p	1,140.3	0.000 1.559 0.0331
1.256	0.11f	10.00p	1,140.3	0.000 3.005 0.2331
1.102	0.16f	15.00p	1,140.3	0.000 3.586 0.5269
1.029	0.19f	17.23p	1,140.3	0.000 3.616 0.6676

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GHS 17.90 PONTON

0.936	0.22f	20.00p	1,140.3	0.000	3.571	0.8419
0.761	0.27f	25.00p	1,140.3	0.000	3.361	1.1468
0.580	0.33f	30.00p	1,140.3	0.000	3.058	1.4275

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 855.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.6 deg to abs 4.9 > 1.000 1.868 P

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GHS 17.90 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.648f TCG = 0.000 VCG = 4.555

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.318	0.10f	3.59s	1,140.3	0.000 -1.119 0.0000
1.326	0.10f	0.00	1,140.3	0.000 0.001 -0.0350
1.312	0.10f	4.90p	1,140.3	0.000 1.528 0.0304
1.311	0.10f	5.00p	1,140.3	0.000 1.559 0.0331
1.256	0.11f	10.00p	1,140.3	0.000 3.005 0.2331
1.102	0.16f	15.00p	1,140.3	0.000 3.586 0.5269
1.029	0.19f	17.23p	1,140.3	0.000 3.616 0.6676
0.936	0.22f	20.00p	1,140.3	0.000 3.571 0.8419
0.761	0.27f	25.00p	1,140.3	0.000 3.361 1.1468
0.580	0.33f	30.00p	1,140.3	0.000 3.058 1.4275
0.393	0.39f	35.00p	1,140.3	0.000 2.700 1.6791
0.202	0.44f	40.00p	1,140.3	0.000 2.304 1.8977
0.008	0.50f	45.00p	1,140.3	0.000 1.880 2.0805
-0.186	0.55f	50.00p	1,140.3	0.000 1.436 2.2254
-0.379	0.60f	55.00p	1,140.3	0.000 0.977 2.3308
-0.569	0.64f	60.00p	1,140.3	0.000 0.507 2.3956
-0.755	0.68f	65.00p	1,140.3	0.000 0.032 2.4192
-0.767	0.68f	65.33p	1,140.3	0.000 0.000 2.4193
-0.933	0.71f	70.00p	1,140.3	0.000 -0.445 2.4012

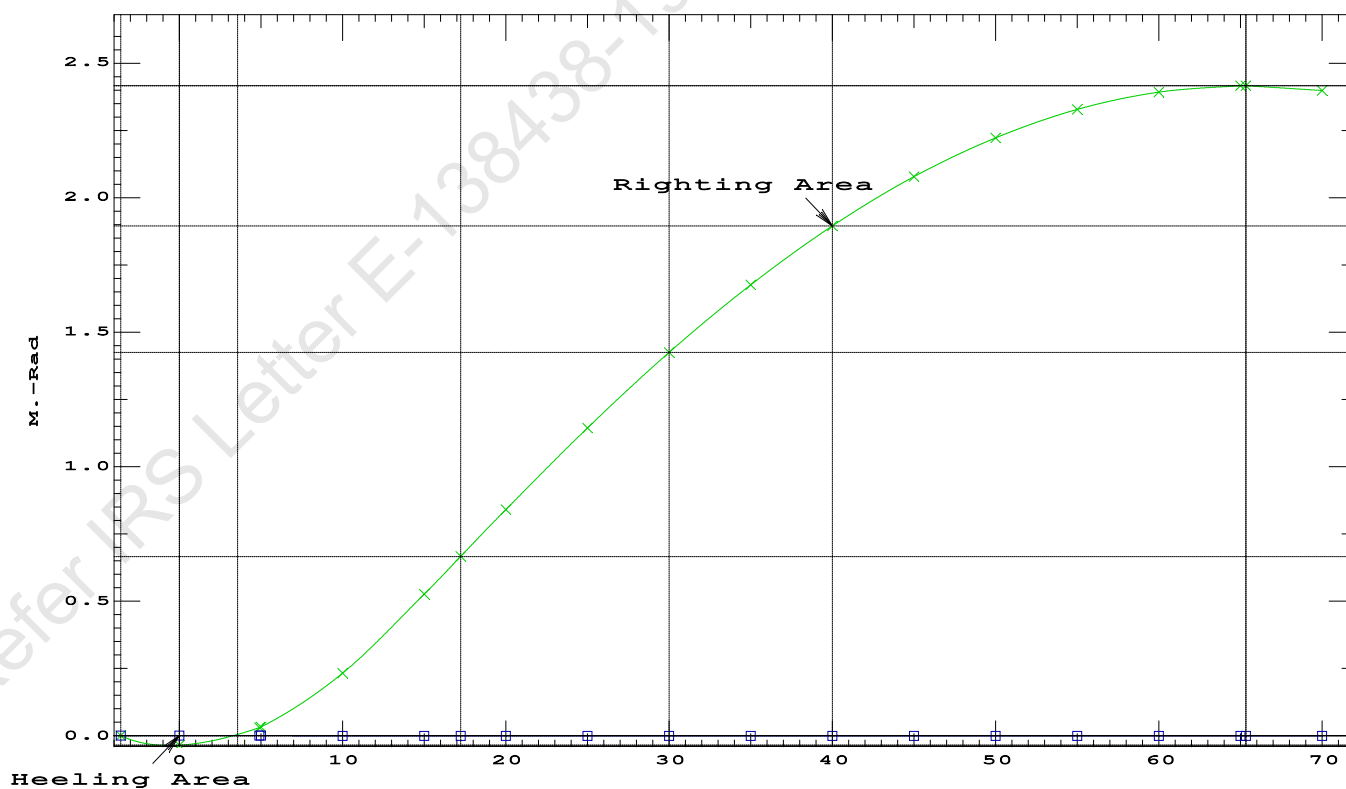
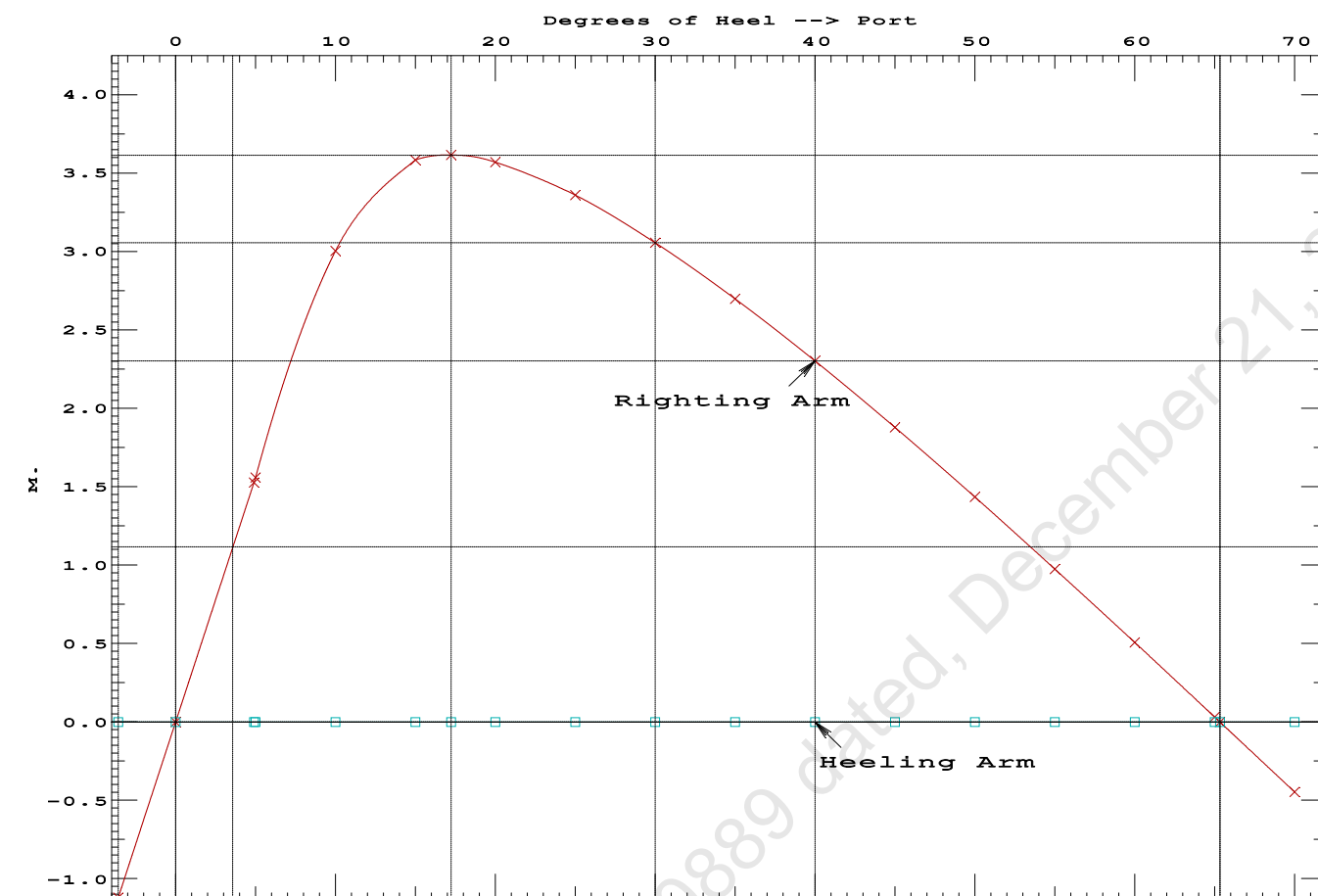
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 855.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.6 deg to RAZero > 1.000 70.100 P
(2) Angle from abs 4.9 deg to RAZero > 20.00 deg 60.43 P

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GHS 17.90 PONTON



22-10-21 12:44:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

Condition Graphic - Draft: 1.578 @ 50.000f, 1.276 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C 19 07_WBTK.P 20 07_WBTK.C 21 07_WBTK.S

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GHS 17.22

PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.578 @ 50.00f, 1.427 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.302/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	42.500f	0.000	34.602	
Total Fixed----->			895.34	19.714f	0.000	5.897	
	Load----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,180.32	25.840f	0.000	4.883	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,180.26	25.865f	0.000	0.724	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	862.9	25.263f	0.000	143.22	19.935
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.84		32.83	139.06	15.776
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.578
DRAFT AT MS (M) = 1.427
DRAFT AT AP (M) = 1.276
DRAFT AT FWD MARK (M) = 1.554
DRAFT AT AFT MARK (M) = 1.300

HYDROSTATIC PROPERTIES

Trim: Fwd 0.302/50.000, No Heel, VCG = 4.883

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/					
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT-----	
1.428	1,180.26	25.865f	0.724	8.84	25.263f	32.83	139.06	15.776	
Distances in METERS.-----					Specific Gravity = 1.025.-----				
					Moment in m.-MT.				
					Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

22-10-21 12:44:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.578 @ 50.00f, 1.427 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.302/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.0	0.806	1.505	0.05500	6.54
CRANE	57.0	4.175	4.875	0.05500	15.28
Total wind heeling moment to starboard-->					21.82
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.840f TCG = 0.000 VCG = 4.883

Free Surface Adjustment: 0.299

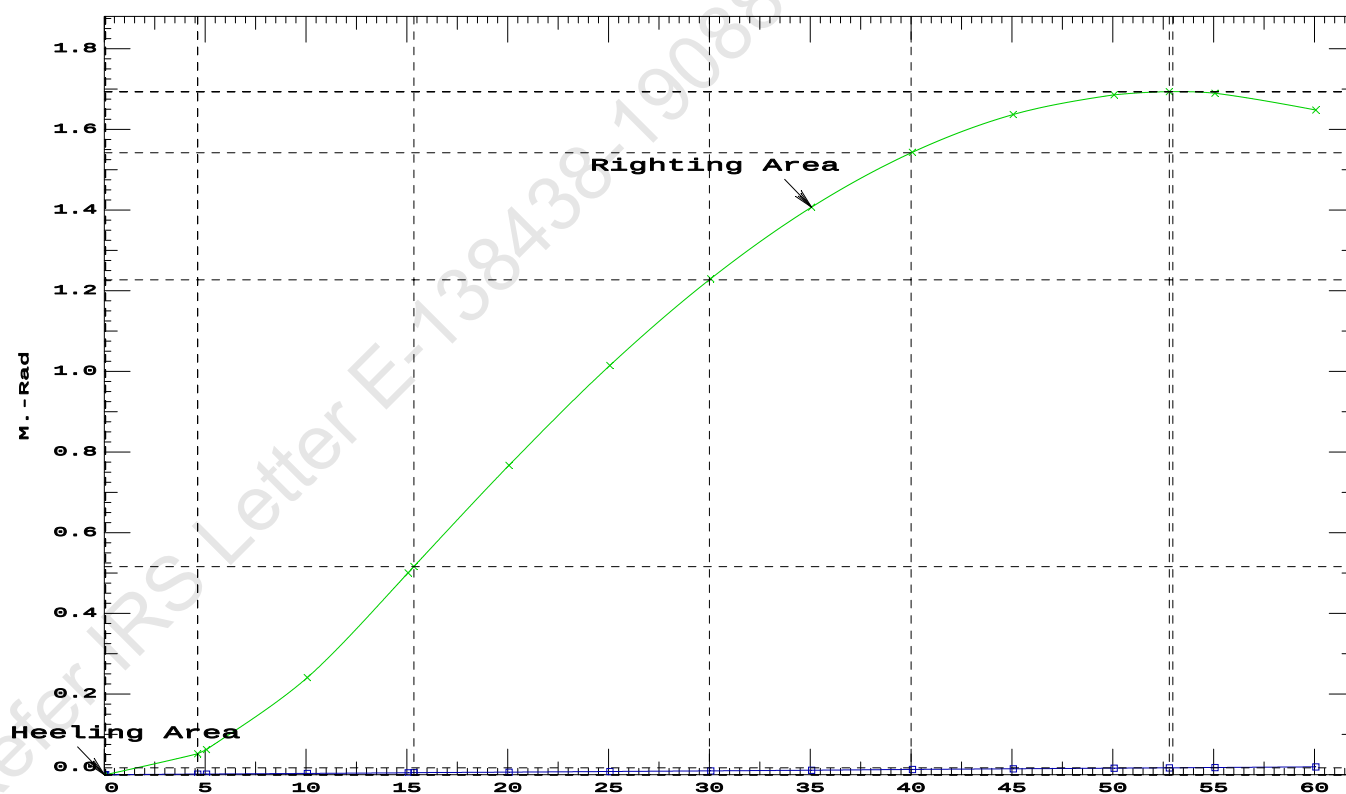
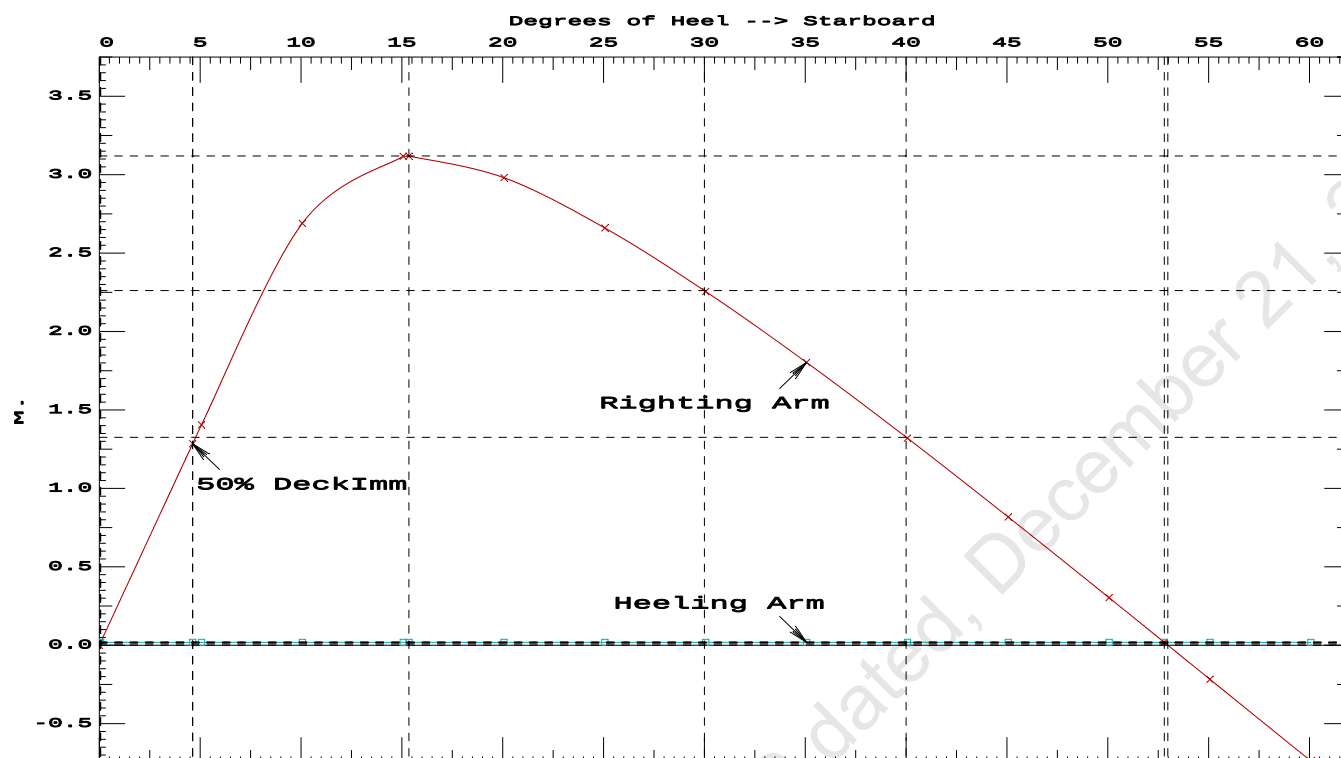
Adjusted CG: LCG = 25.838f TCG = 0.000 VCG = 5.182

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.264	0.35f	0.00	1,180.3	0.000 -0.018	0.0000	4.629
1.264	0.35f	0.06s	1,180.3	0.000 0.000	-0.0000	4.567
1.251	0.35f	4.63s	1,180.3	0.000 1.267	0.0505	0.000
1.248	0.35f	5.06s	1,180.3	0.000 1.386	0.0605	-0.433
1.187	0.37f	10.06s	1,180.3	0.000 2.672	0.2383	-5.433
0.998	0.56f	15.06s	1,180.4	0.000 3.099	0.4973	-10.433
0.987	0.57f	15.35s	1,180.3	0.000 3.100	0.5130	-10.724
0.794	0.75f	20.06s	1,180.3	0.000 2.962	0.7655	-15.433
0.578	0.95f	25.06s	1,180.3	0.000 2.642	1.0113	-20.433
0.354	1.15f	30.06s	1,180.3	0.000 2.237	1.2248	-25.433
0.123	1.36f	35.06s	1,180.3	0.000 1.784	1.4006	-30.433
-0.113	1.56f	40.06s	1,180.3	0.000 1.301	1.5354	-35.433
-0.350	1.75f	45.06s	1,180.3	0.000 0.799	1.6272	-40.433
-0.586	1.94f	50.06s	1,180.3	0.000 0.285	1.6746	-45.433
-0.714	2.03f	52.80s	1,180.3	0.000 0.000	1.6814	-48.174
-0.817	2.11f	55.06s	1,180.3	0.000 -0.236	1.6768	-50.433
-1.041	2.26f	60.06s	1,180.3	0.000 -0.757	1.6335	-55.433
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.82 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5130 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	52.74 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.57 P

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GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00NCondition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

22-10-21 12:44:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.840f TCG = 0.000 VCG = 4.883
Free Surface Adjustment: 0.299
Adjusted CG: LCG = 25.838f TCG = 0.000 VCG = 5.182

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.264	0.35f	0.00	1,180.3	0.000	0.000	0.0000
1.249	0.35f	5.00s	1,180.3	0.000	1.387	0.0605
1.188	0.37f	10.00s	1,180.3	0.000	2.676	0.2385
1.001	0.56f	15.00s	1,180.3	0.000	3.116	0.4985
0.987	0.57f	15.35s	1,180.3	0.000	3.118	0.5173
0.796	0.75f	20.00s	1,180.3	0.000	2.983	0.7684
0.581	0.95f	25.00s	1,180.3	0.000	2.664	1.0161
0.357	1.15f	30.00s	1,180.3	0.000	2.259	1.2315
0.126	1.35f	35.00s	1,180.3	0.000	1.807	1.4093
-0.110	1.56f	40.00s	1,180.3	0.000	1.325	1.5462
-0.347	1.75f	45.00s	1,180.3	0.000	0.823	1.6400
-0.583	1.94f	50.00s	1,180.3	0.000	0.309	1.6895
-0.721	2.04f	52.97s	1,180.3	0.000	0.000	1.6975
-0.815	2.11f	55.00s	1,180.3	0.000	-0.212	1.6937
-1.038	2.26f	60.00s	1,180.3	0.000	-0.733	1.6525

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

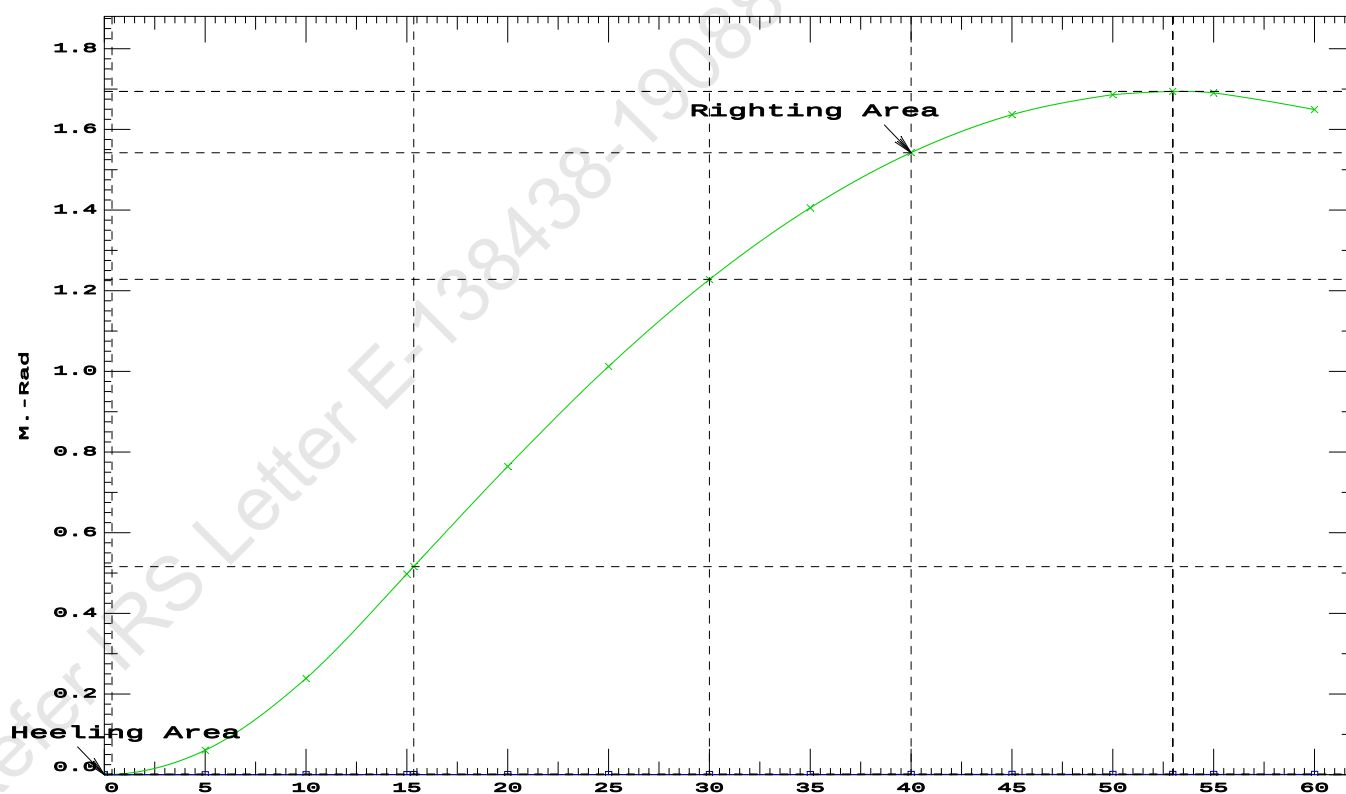
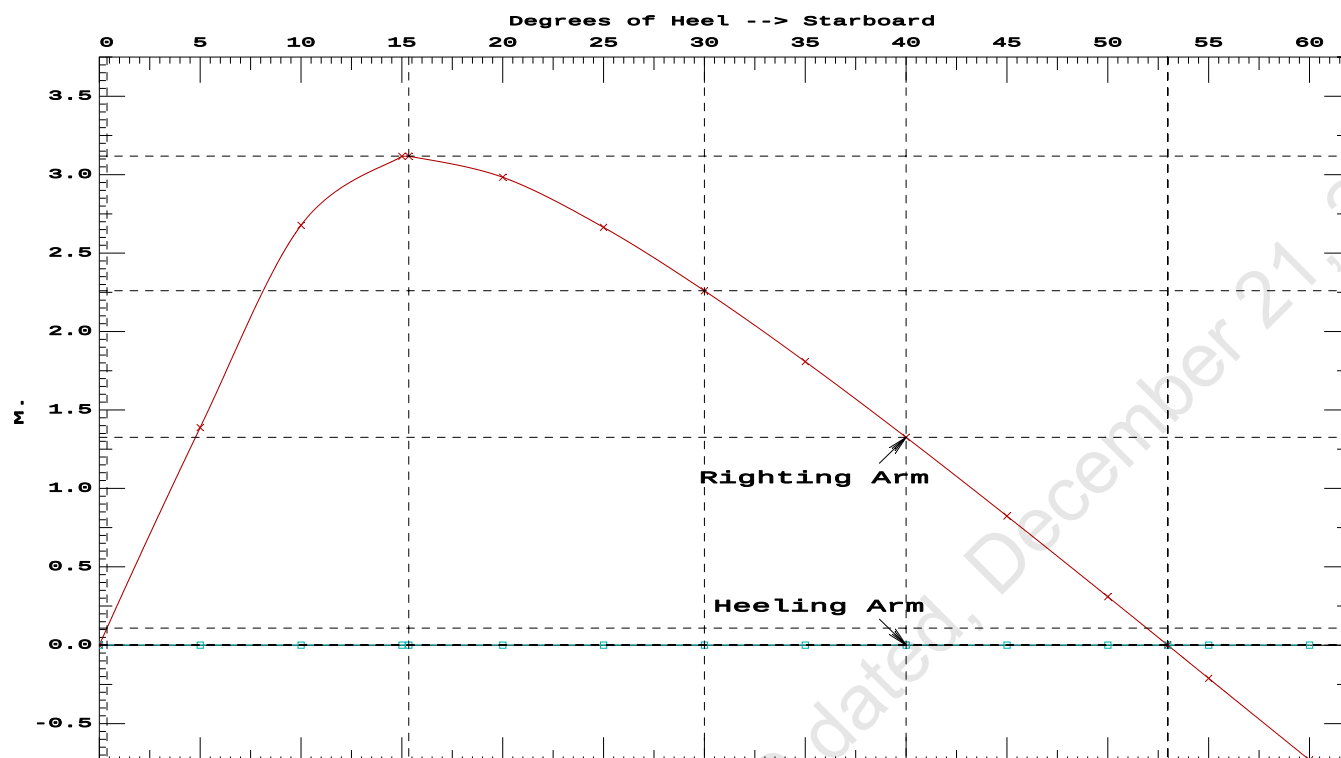
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.00 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	367969 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	2168.21 P

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GHS 17.22 PONT00N

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd



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GHS 17.22 PONTON

Condition 06 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.578 @ 50.00f, 1.427 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.302/50.000, Heel: Stbd 0.01 deg.

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	455.34	24.928f	0.000	3.000	
Crane	400.00	11.500f	0.000	6.325	
Crane Load	40.00	42.500f	0.000	34.602	
Total Fixed----->	895.34	19.714f	0.000	5.897	
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		284.98	45.086f	0.000	1.697
Total Weight----->		1,180.32	25.840f	0.000	4.883
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,180.32	25.865f	0.005s	0.724

Righting Arms:		0.000	0.004s		
External Arms:		0.000	0.004s		
Residual Righting Arms:		0.000	0.000		
Part-----	SpGr	WPA	LCF	TCF	BML
Total Waterplane----->	1.025	859.4	25.345f	0.026s	141.48
		MT/cm			
		8.81			

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

PONTOON

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limmarg: ; t3: 0; t: 0.02
limmarg: 469.41; t3: 0; t: 0.02
limmarg: 0.00; t3: 0; t: 0.02

Theta3 is -0.02 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.648f TCG = 0.000 VCG = 4.555

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.326	0.10f	0.01s	1,140.2	0.000	-0.004	0.0000
1.326	0.10f	0.00	1,140.2	0.000	-0.000	-0.0000
1.311	0.10f	5.00p	1,140.3	0.000	1.563	0.0682
1.256	0.11f	10.00p	1,140.3	0.000	3.008	0.2685
1.102	0.16f	15.00p	1,140.3	0.000	3.589	0.5626
1.029	0.19f	17.23p	1,140.3	0.000	3.620	0.7035
0.936	0.22f	20.00p	1,140.3	0.000	3.575	0.8779
0.761	0.27f	25.00p	1,140.3	0.000	3.364	1.1831
0.580	0.33f	30.00p	1,140.3	0.000	3.061	1.4641

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 855.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0 > 1.000 1.000 P

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.648f TCG = 0.000 VCG = 4.555

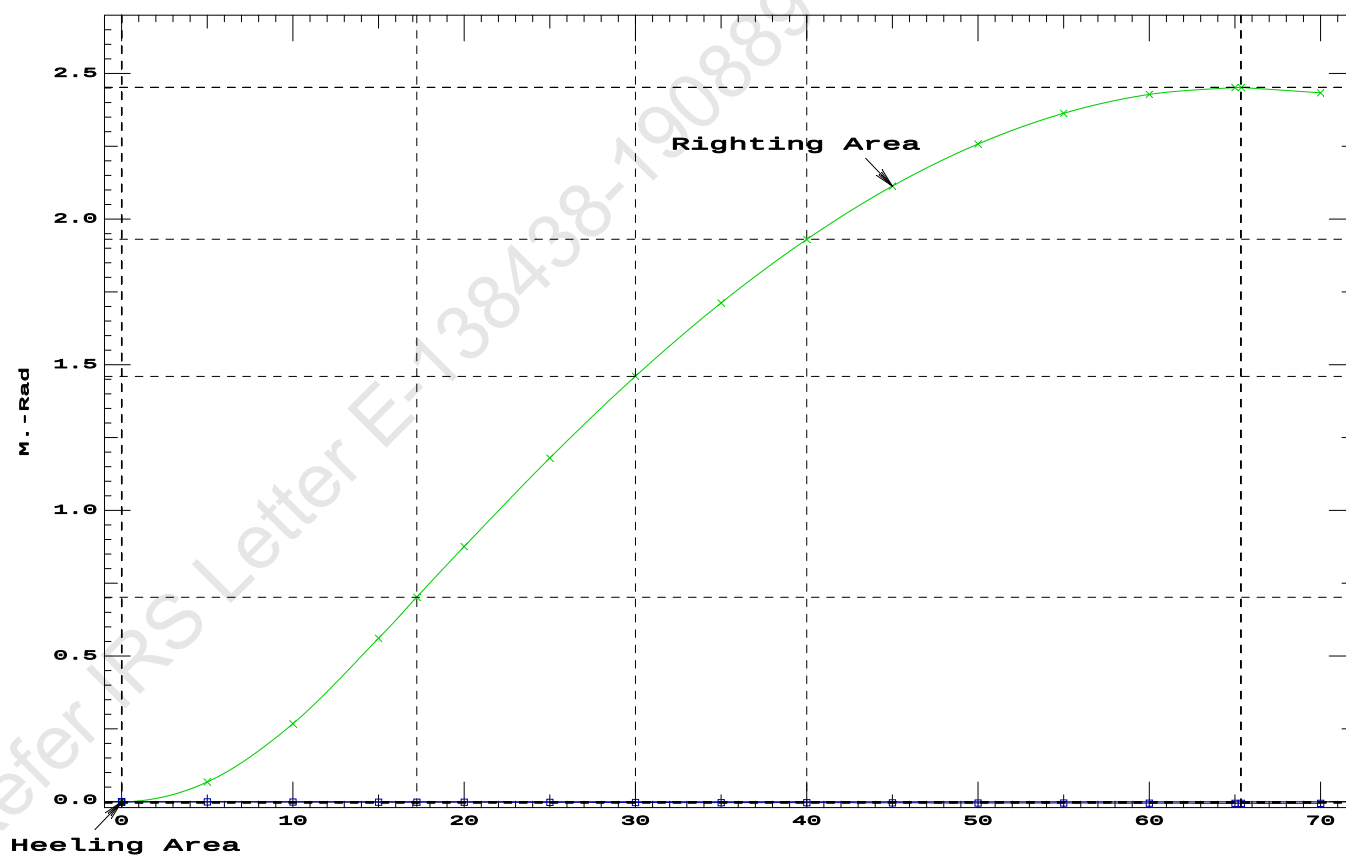
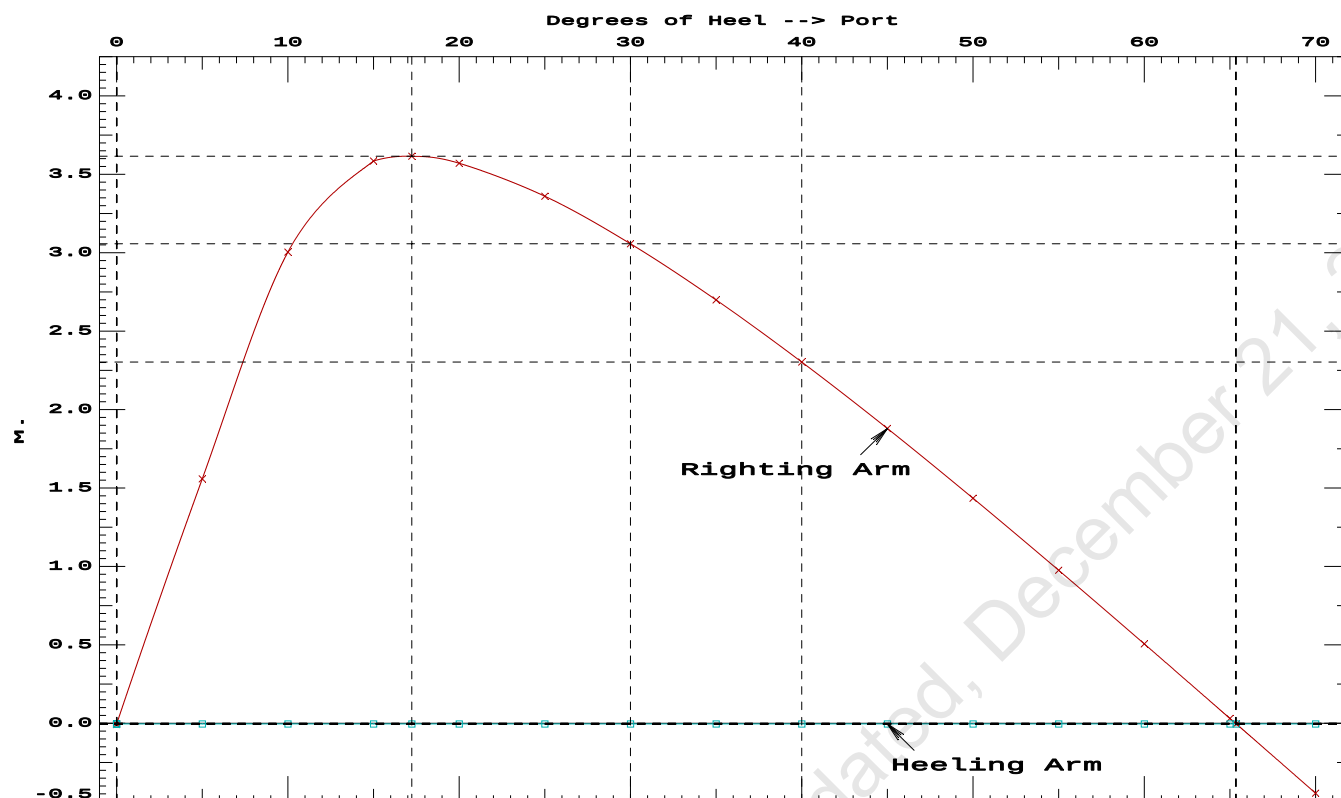
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.326	0.10f	0.01s	1,140.2	0.000	-0.004	0.0000
1.326	0.10f	0.00	1,140.2	0.000	-0.000	-0.0000
1.311	0.10f	5.00p	1,140.3	0.000	1.563	0.0682
1.256	0.11f	10.00p	1,140.3	0.000	3.008	0.2685
1.102	0.16f	15.00p	1,140.3	0.000	3.589	0.5626
1.029	0.19f	17.23p	1,140.3	0.000	3.620	0.7035
0.936	0.22f	20.00p	1,140.3	0.000	3.575	0.8779
0.761	0.27f	25.00p	1,140.3	0.000	3.364	1.1831
0.580	0.33f	30.00p	1,140.3	0.000	3.061	1.4641
0.393	0.39f	35.00p	1,140.3	0.000	2.703	1.7161
0.202	0.44f	40.00p	1,140.3	0.000	2.307	1.9350
0.008	0.50f	45.00p	1,140.3	0.000	1.884	2.1181
-0.186	0.55f	50.00p	1,140.3	0.000	1.440	2.2632
-0.379	0.60f	55.00p	1,140.3	0.000	0.980	2.3689
-0.569	0.64f	60.00p	1,140.3	0.000	0.511	2.4341
-0.755	0.68f	65.00p	1,140.3	0.000	0.035	2.4579
-0.768	0.68f	65.37p	1,140.3	0.000	0.000	2.4581
-0.933	0.71f	70.00p	1,140.3	0.000	-0.441	2.4402

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 855.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 6202837 P
(2) Angle from abs 0 deg to RAzero > 20.00 deg 65.38 P

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GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00N

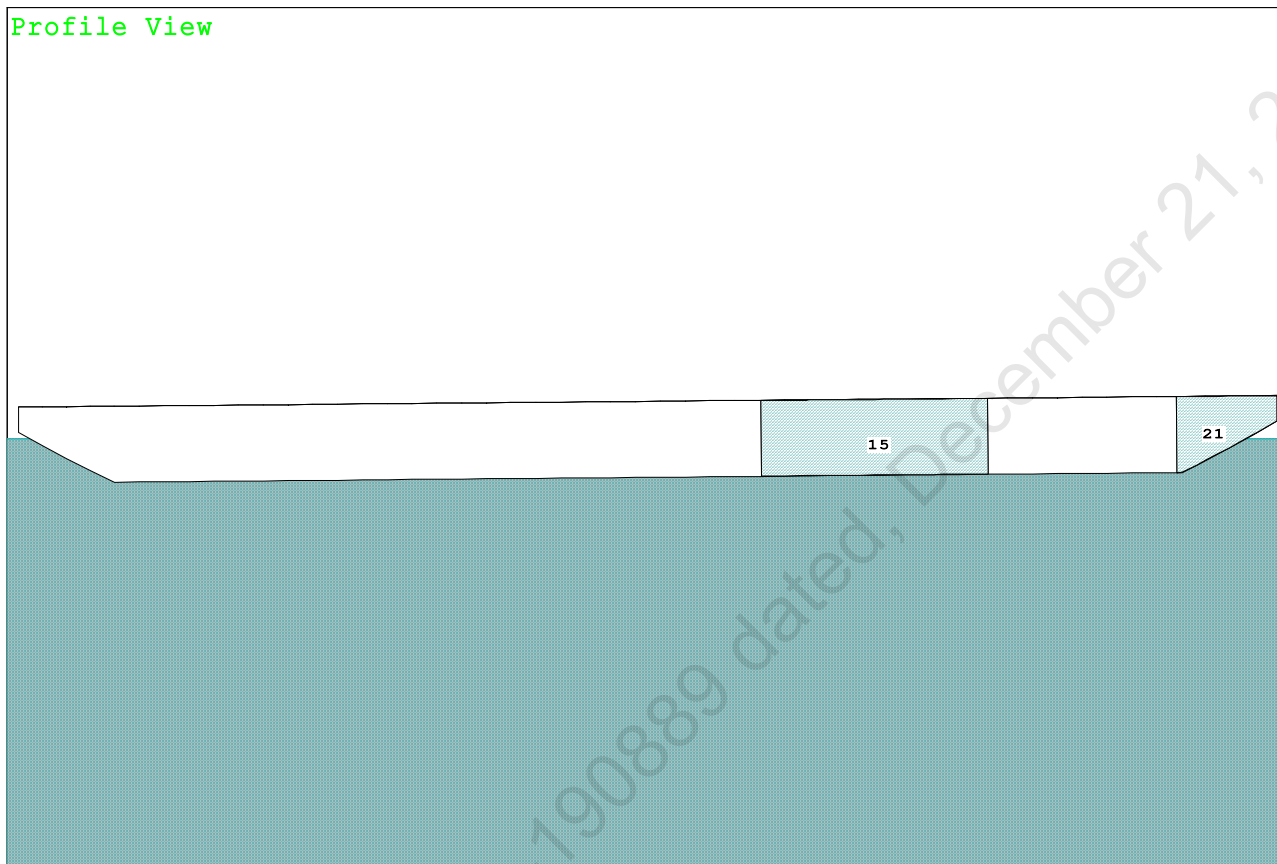
13-12-21 20:33:46 Cybermarine Technologies PTE. Ltd.
GHS 17.90

PONTOON

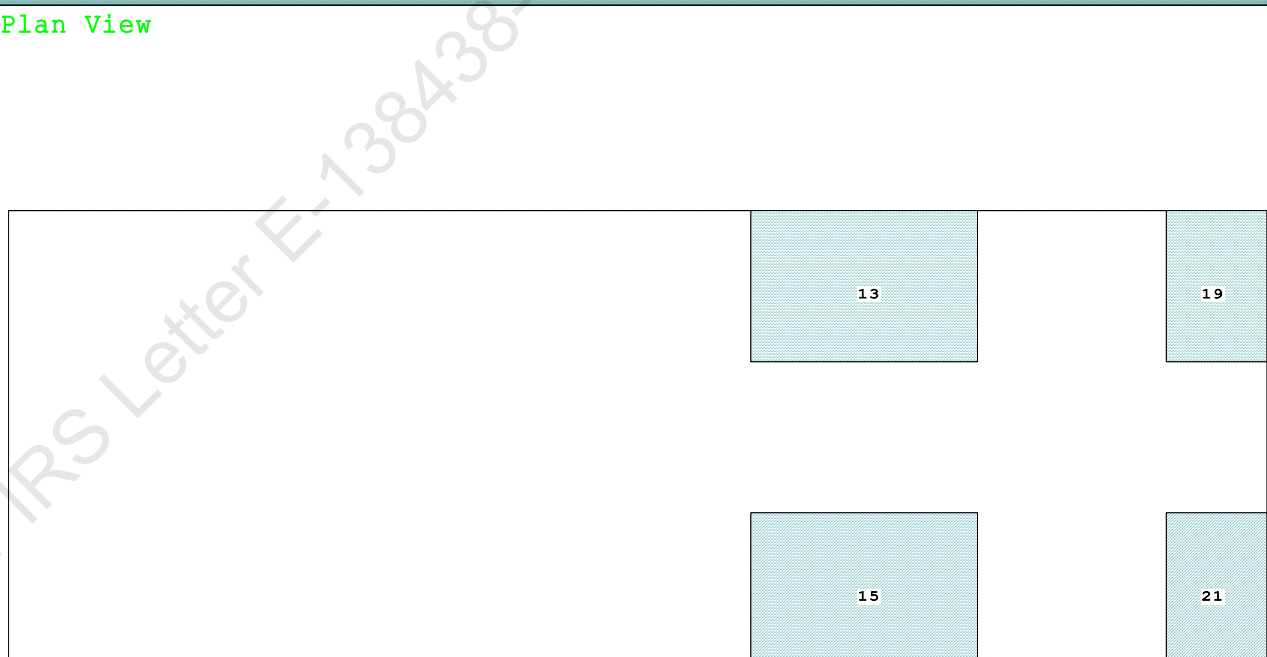
Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Condition Graphic - Draft: 1.296 @ 50.000f, 1.753 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.296 @ 50.00f, 1.524 @ 25.00f, 1.753 @ 0.00

Trim: Aft 0.457/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed----->			845.34	17.077f	0.000	5.251	
	Load----	SpGr----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,267.74	23.784f	0.000	4.033	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,267.62	23.755f	0.000	0.778	

Righting Arms:				0.000	0.000		
Part-----			SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025		869.4	24.603f	0.000	136.40	18.615*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.91		33.75	133.14	15.360
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.296
DRAFT AT MS (M) = 1.524
DRAFT AT AP (M) = 1.753
DRAFT AT FWD MARK (M) = 1.333
DRAFT AT AFT MARK (M) = 1.716

HYDROSTATIC PROPERTIES

Trim: Aft 0.457/50.000, No Heel, VCG = 4.033

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm trim-----
1.528	1,267.62	23.755f	0.778	8.91
				24.603f
				33.75
				133.14
				15.360

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.90

PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.296 @ 50.00f, 1.524 @ 25.00f, 1.753 @ 0.00

Trim: Aft 0.457/50.000, Heel: zero

Part	LPA*Sf	HCP	Arm	Pressure	Moment
HULL	74.3	0.759	1.508	0.05500	6.16
CRANE	57.0	3.852	4.600	0.05500	14.42
Total wind heeling moment to starboard-->					20.58
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.784f TCG = 0.000 VCG = 4.033

Free Surface Adjustment: 0.371

Adjusted CG: LCG = 23.788f TCG = 0.000 VCG = 4.404

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area---Angle	
1.741 0.52a 0.00	1,267.7	0.000 -0.016	0.0000	4.046
1.741 0.52a 0.05s	1,267.7	0.000 0.000	-0.0000	3.993
1.731 0.52a 4.05s	1,267.7	0.000 1.076	0.0375	0.000
1.728 0.53a 5.05s	1,267.7	0.000 1.346	0.0588	-1.007
1.707 0.58a 10.05s	1,267.7	0.000 2.604	0.2318	-6.007
1.771 0.86a 15.05s	1,267.7	0.000 3.046	0.4850	-11.007
1.783 0.92a 16.00s	1,267.7	0.000 3.053	0.5355	-11.957
1.831 1.16a 20.05s	1,267.7	0.000 2.961	0.7500	-16.007
1.883 1.46a 25.05s	1,267.7	0.000 2.704	0.9993	-21.007
1.927 1.77a 30.05s	1,267.7	0.000 2.366	1.2210	-26.007
1.963 2.08a 35.05s	1,267.7	0.000 1.981	1.4110	-31.007
1.990 2.39a 40.05s	1,267.7	0.000 1.566	1.5660	-36.007
2.006 2.68a 45.05s	1,267.7	0.000 1.130	1.6838	-41.007
2.009 2.97a 50.05s	1,267.7	0.000 0.681	1.7629	-46.007
1.997 3.23a 55.05s	1,267.7	0.000 0.223	1.8024	-51.007
1.986 3.35a 57.47s	1,267.7	0.000 0.000	1.8071	-53.420
1.969 3.46a 60.05s	1,267.7	0.000 -0.239	1.8017	-56.007
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.58 (constant)

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PONTOON

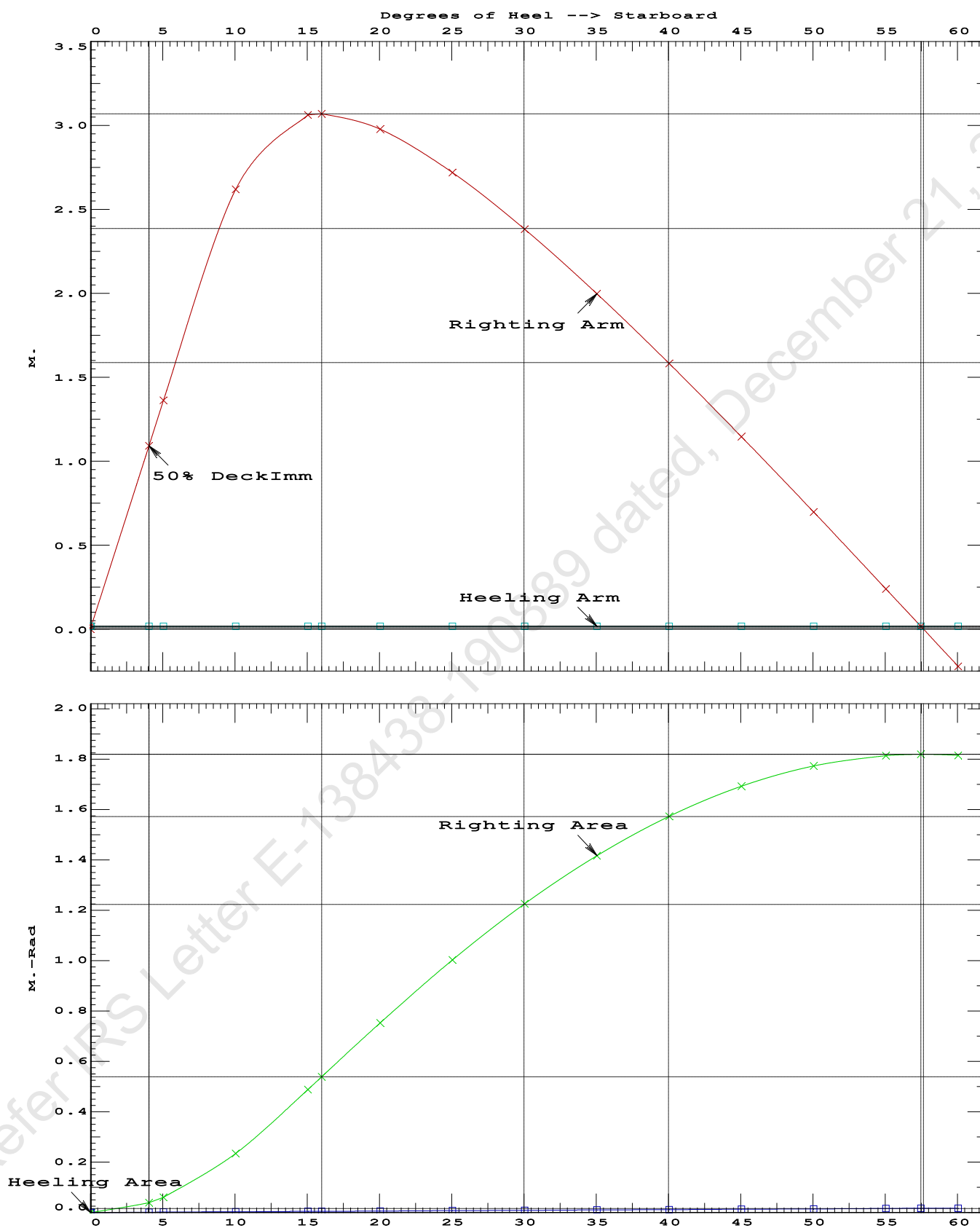
Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5355 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 57.41 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 3.99 P

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PONTOON

Condition 07 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Aft

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PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.784f TCG = 0.000 VCG = 4.033
Free Surface Adjustment: 0.371
Adjusted CG: LCG = 23.788f TCG = 0.000 VCG = 4.404

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.741 0.52a 0.00	1,267.7	0.000 -0.001	0.0000	
1.728 0.53a 5.00s	1,267.7	0.000 1.347	0.0587	
1.706 0.57a 10.00s	1,267.7	0.000 2.608	0.2320	
1.770 0.86a 15.00s	1,267.7	0.000 3.061	0.4861	
1.783 0.92a 16.01s	1,267.7	0.000 3.068	0.5402	
1.830 1.16a 20.00s	1,267.7	0.000 2.979	0.7524	
1.882 1.46a 25.00s	1,267.7	0.000 2.722	1.0033	
1.927 1.77a 30.00s	1,267.7	0.000 2.385	1.2267	
1.963 2.08a 35.00s	1,267.7	0.000 2.000	1.4184	
1.990 2.38a 40.00s	1,267.7	0.000 1.586	1.5751	
2.006 2.68a 45.00s	1,267.7	0.000 1.150	1.6946	
2.009 2.97a 50.00s	1,267.7	0.000 0.701	1.7754	
1.998 3.23a 55.00s	1,267.7	0.000 0.243	1.8167	
1.985 3.35a 57.63s	1,267.7	0.000 0.000	1.8223	
1.969 3.46a 60.00s	1,267.7	0.000 -0.219	1.8177	

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

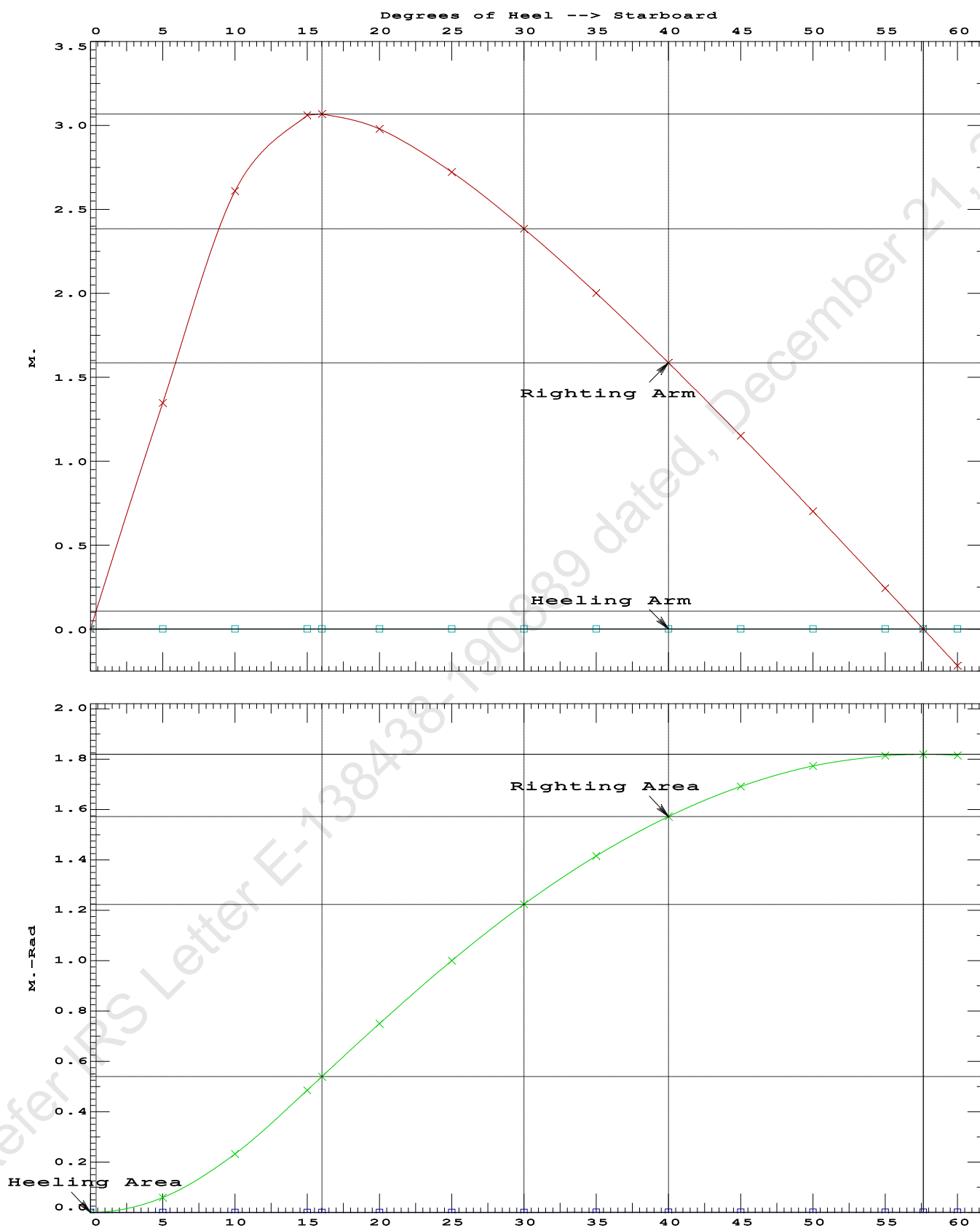
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	388896 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2297.70 P

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GHS 17.90

PONTOON

Condition 07 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Aft

PONTOON

Crane in operation with load in place

USK draft: 1.296 @ 50.00f, 1.524 @ 25.00f, 1.753 @ 0.00

Trim: Aft 0.457/50.000, Heel: zero

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 17.90 PONTON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.540 @ 50.00f, 1.481 @ 25.00f, 1.422 @ 0.00

Trim: Fwd 0.119/50.000, Heel: zero

Part	Weight (MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	350.00	11.500f	0.000	6.325			
Total Fixed	805.34	19.092f	0.000	4.445			
Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks			422.40	37.208f	0.000	1.596	470.09
Total Weight			1,227.74	25.325f	0.000	3.465	
			Displ (MT)	LCB	TCB	VCB	
HULL	1.025		1,227.73	25.331f	0.000	0.750	
Righting Arms:			0.000	0.000			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	-0.001s			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	866.8	25.093f	0.000	139.56	19.171*	
		MT/cm	m.	MT/cm	GML	GMT	
		8.88		33.60	136.84	16.456	
Distances in METERS.						Moments in m.-MT.	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 0.9; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.2; t: 0.12
limmarg: INFINITY; t3: 0.1; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.0; t: 0.01

Theta3 is 0.03 degrees, as demonstrated below.

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GHS 17.90 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.092f TCG = 0.000 VCG = 4.445

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.410	0.14f	0.00	1,227.7	0.000	-0.001	0.0000
1.409	0.14f	0.03s	1,227.8	0.000	0.011	0.0000
1.394	0.14f	5.00s	1,227.7	0.000	1.477	0.0645
1.345	0.15f	10.00s	1,227.7	0.000	2.881	0.2552
1.230	0.22f	15.00s	1,227.7	0.000	3.446	0.5374
1.168	0.26f	17.54s	1,227.7	0.000	3.482	0.6918
1.105	0.30f	20.00s	1,227.7	0.000	3.449	0.8404
0.969	0.37f	25.00s	1,227.7	0.000	3.268	1.1357
0.824	0.45f	30.00s	1,227.7	0.000	3.001	1.4099

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0 > 1.000 LARGE

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GHS 17.90 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.092f TCG = 0.000 VCG = 4.445

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.410	0.14f	0.00	1,227.7	0.000	-0.001	0.0000
1.409	0.14f	0.03s	1,227.8	0.000	0.011	0.0000
1.394	0.14f	5.00s	1,227.7	0.000	1.477	0.0645
1.345	0.15f	10.00s	1,227.7	0.000	2.881	0.2552
1.230	0.22f	15.00s	1,227.7	0.000	3.446	0.5374
1.168	0.26f	17.54s	1,227.7	0.000	3.482	0.6918
1.105	0.30f	20.00s	1,227.7	0.000	3.449	0.8404
0.969	0.37f	25.00s	1,227.7	0.000	3.268	1.1357
0.824	0.45f	30.00s	1,227.7	0.000	3.001	1.4099
0.672	0.53f	35.00s	1,227.7	0.000	2.681	1.6582
0.513	0.60f	40.00s	1,227.7	0.000	2.324	1.8768
0.349	0.68f	45.00s	1,227.7	0.000	1.940	2.0631
0.181	0.75f	50.00s	1,227.7	0.000	1.536	2.2149
0.012	0.82f	55.00s	1,227.7	0.000	1.116	2.3308
-0.158	0.88f	60.00s	1,227.7	0.000	0.685	2.4095
-0.326	0.93f	65.00s	1,227.7	0.000	0.248	2.4502
-0.419	0.96f	67.81s	1,227.7	0.000	0.000	2.4563
-0.490	0.98f	70.00s	1,227.7	0.000	-0.194	2.4526

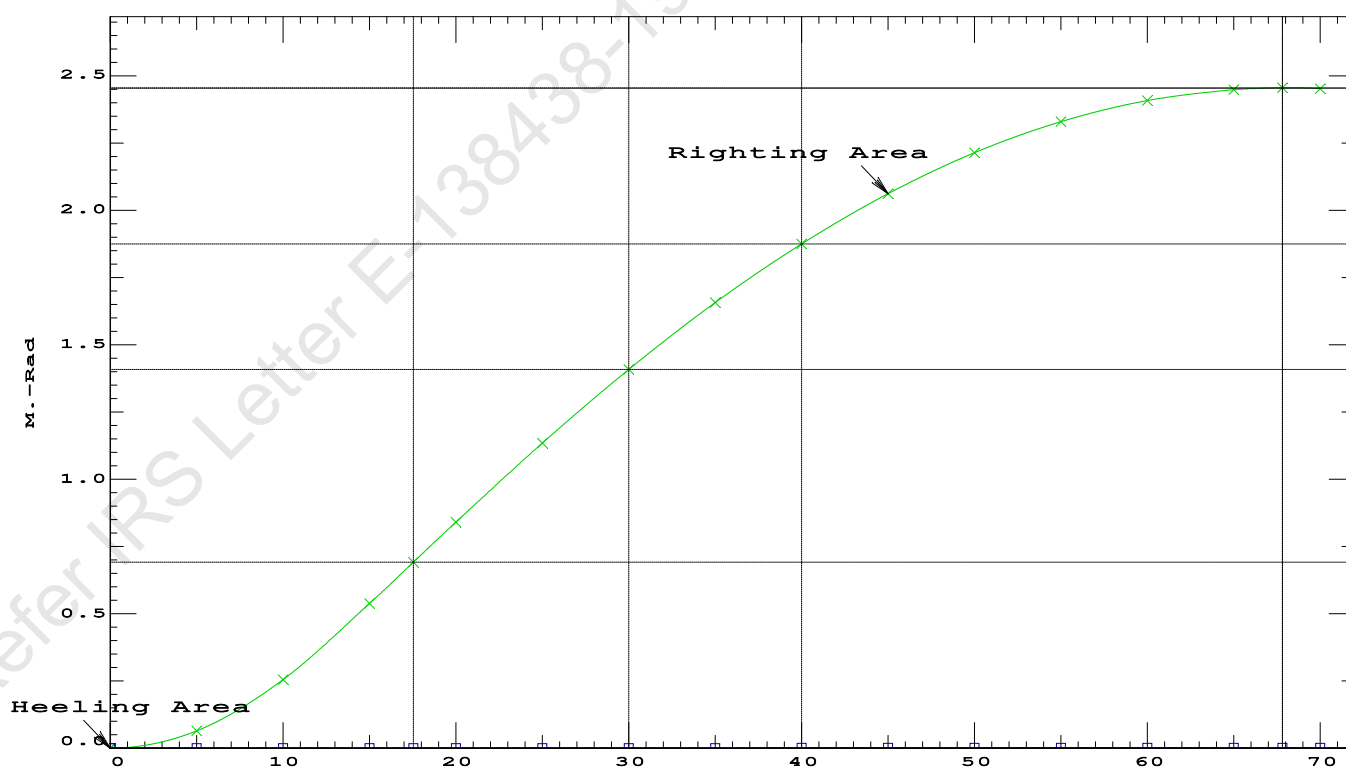
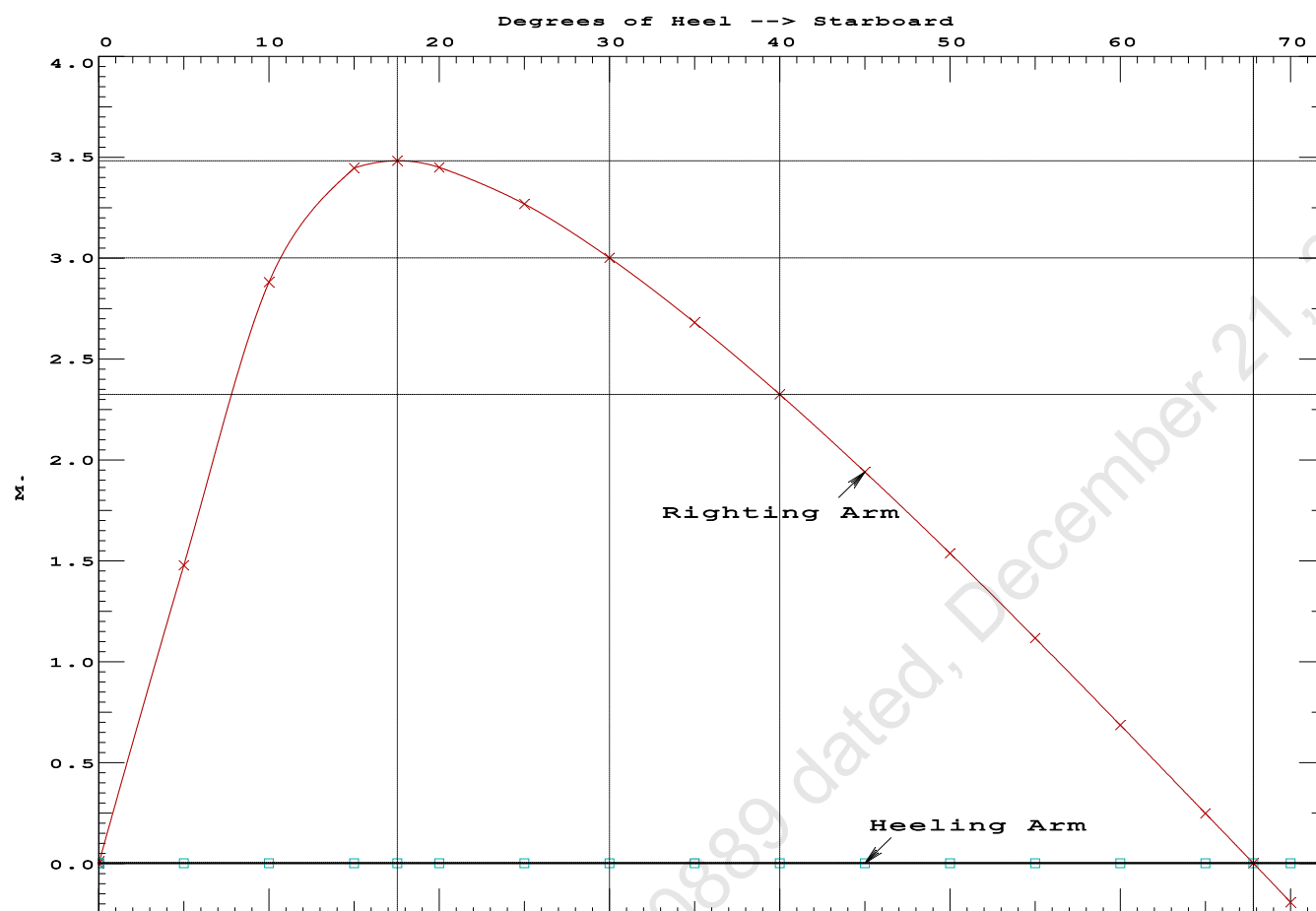
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0 deg to RAzero > 20.00 deg 67.78 P

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GHS 17.90 PONTOON

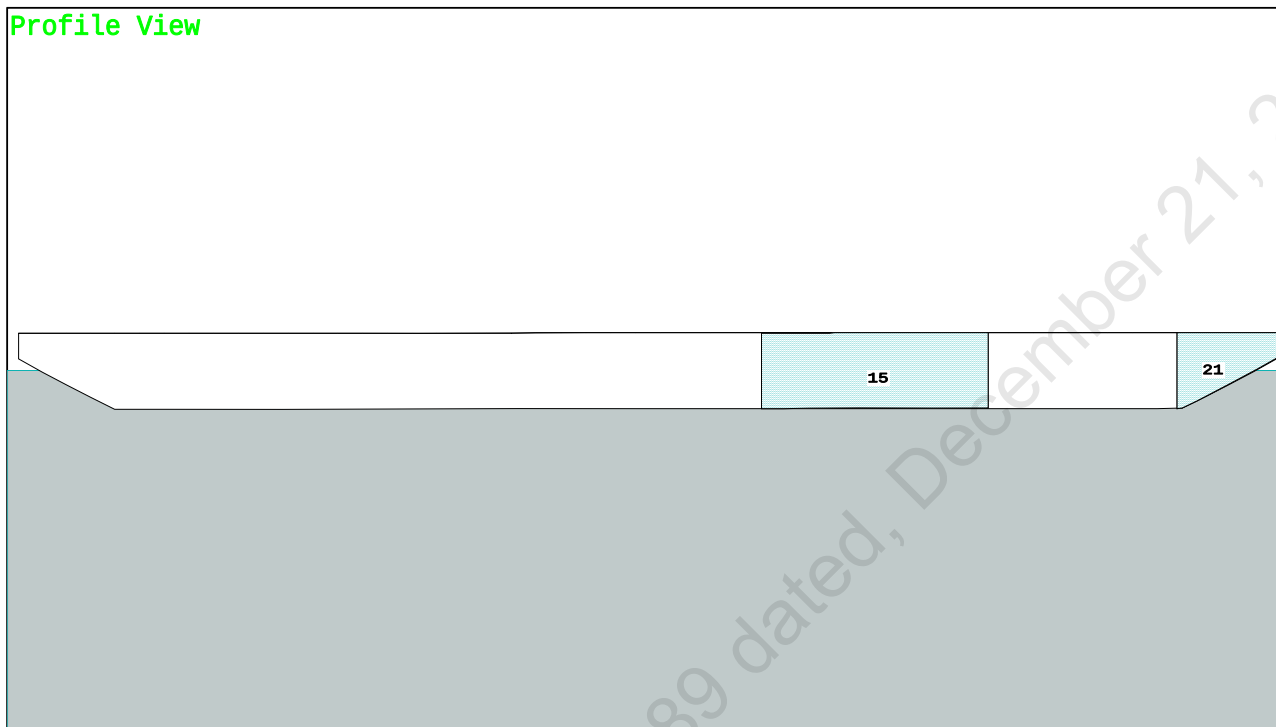


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GHS 17.22 PONT00N

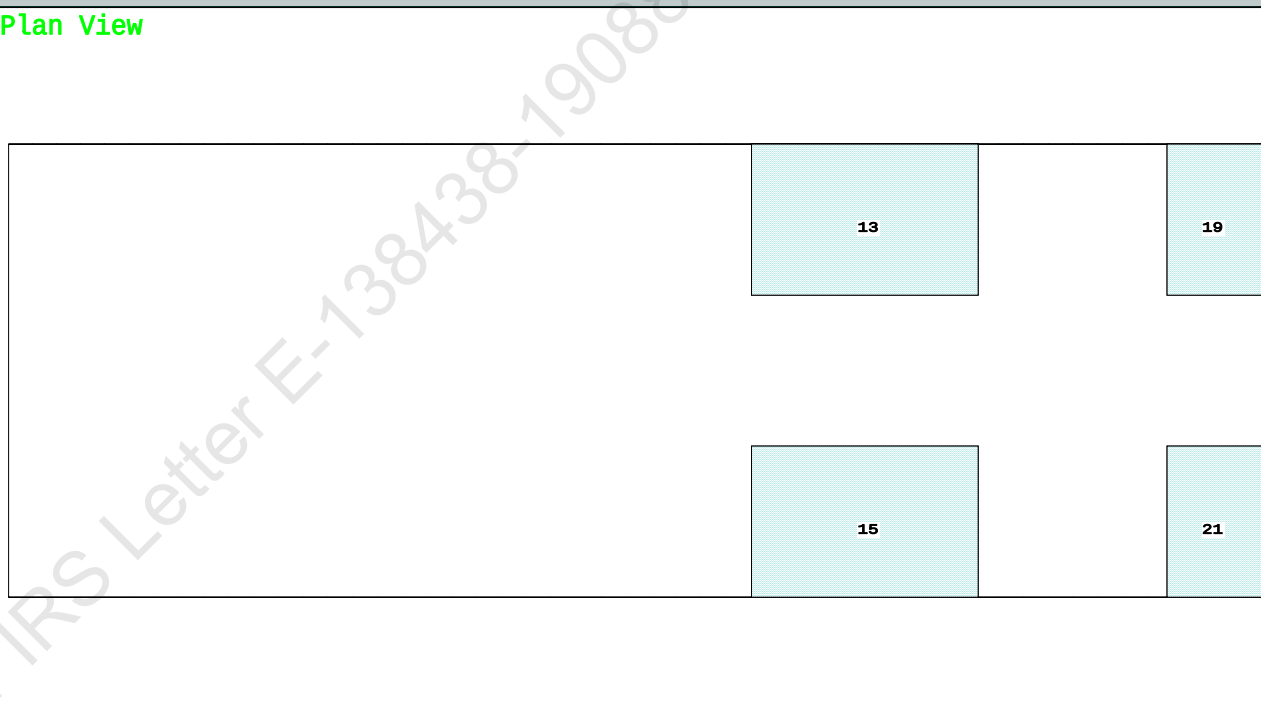
Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

Condition Graphic - Draft: 1.505 @ 50.000f, 1.547 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S 19 07_WBTK.P 21 07_WBTK.S

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GHS 17.22

PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.505 @ 50.00f, 1.526 @ 25.00f, 1.547 @ 0.00

Trim: Aft 0.042/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	21.478	
Total Fixed----->			845.34	18.733f	0.000	5.251	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,267.74	24.889f	0.000	4.033	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,267.74	24.886f	0.000	0.773	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	869.4	24.965f	0.000	136.36	18.607
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.91		33.75	133.10	15.347
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.505
DRAFT AT MS (M) = 1.526
DRAFT AT AP (M) = 1.547
DRAFT AT FWD MARK (M) = 1.509
DRAFT AT AFT MARK (M) = 1.544

HYDROSTATIC PROPERTIES

Trim: Aft 0.042/50.000, No Heel, VCG = 4.033

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----
1.526	1,267.74	24.886f	0.773	8.91	24.965f	33.75	133.10	15.347
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.505 @ 50.00f, 1.526 @ 25.00f, 1.547 @ 0.00

Trim: Aft 0.042/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	74.3	0.752	1.497	0.05500	6.12
CRANE	57.0	3.973	4.719	0.05500	14.79
Total wind heeling moment to starboard-->					20.91
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.889f TCG = 0.000 VCG = 4.033

Free Surface Adjustment: 0.371

Adjusted CG: LCG = 24.889f TCG = 0.000 VCG = 4.404

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.535	0.05a	0.00	1,267.7	0.000	-0.016	4.742
1.535	0.05a	0.06s	1,267.7	0.000	0.000	4.684
1.522	0.05a	4.74s	1,267.7	0.000	1.264	0.000
1.520	0.05a	5.06s	1,267.7	0.000	1.348	-0.316
1.478	0.05a	10.06s	1,267.7	0.000	2.631	-5.316
1.430	0.08a	15.06s	1,267.7	0.000	3.082	-10.316
1.420	0.08a	16.01s	1,267.7	0.000	3.088	-11.272
1.372	0.11a	20.06s	1,267.7	0.000	2.997	-15.316
1.303	0.13a	25.06s	1,267.7	0.000	2.740	-20.316
1.226	0.16a	30.06s	1,267.7	0.000	2.402	-25.316
1.140	0.19a	35.06s	1,267.7	0.000	2.017	-30.316
1.045	0.22a	40.06s	1,267.7	0.000	1.602	-35.316
0.943	0.25a	45.06s	1,267.7	0.000	1.165	-40.316
0.834	0.27a	50.06s	1,267.7	0.000	0.713	-45.316
0.719	0.30a	55.06s	1,267.7	0.000	0.252	-50.316
0.654	0.31a	57.77s	1,267.7	0.000	0.000	-53.025
0.598	0.32a	60.06s	1,267.7	0.000	-0.214	-55.316
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

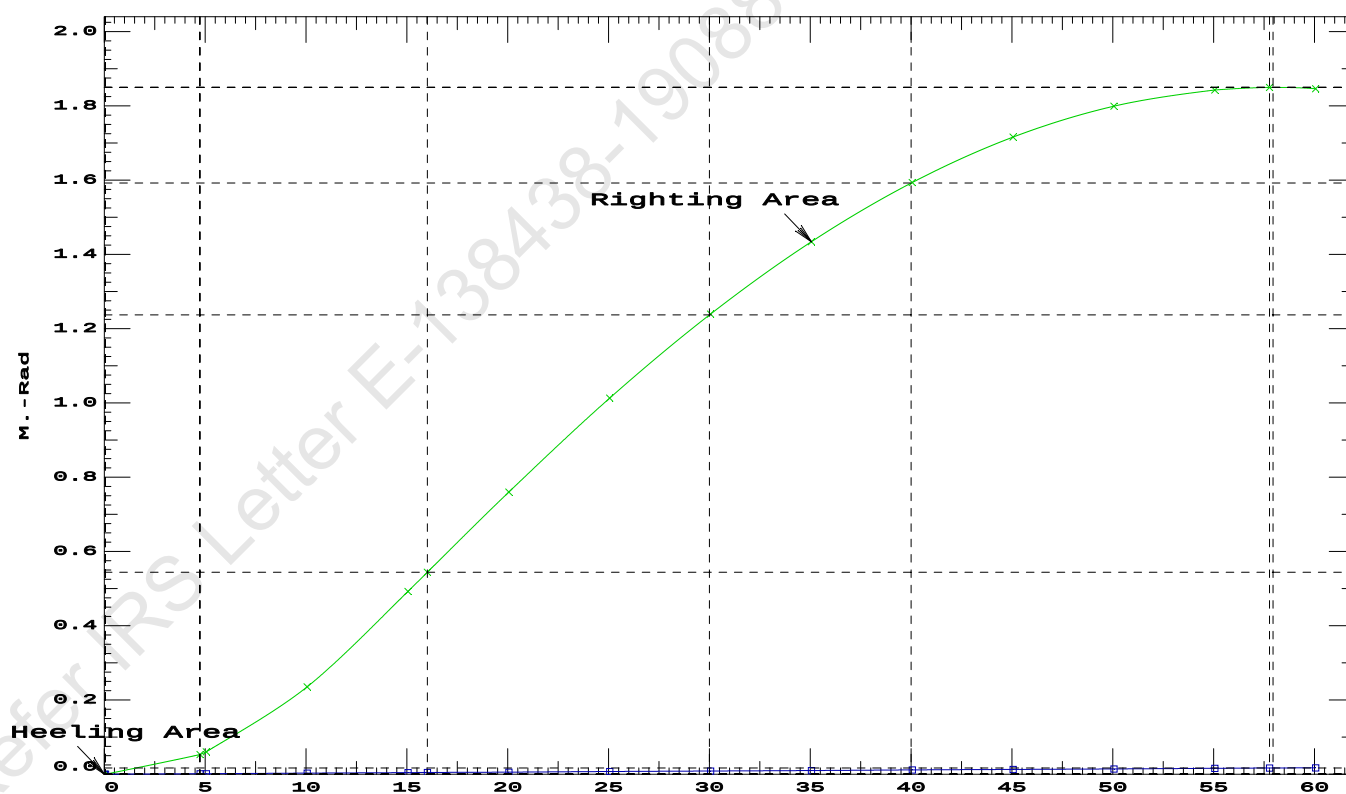
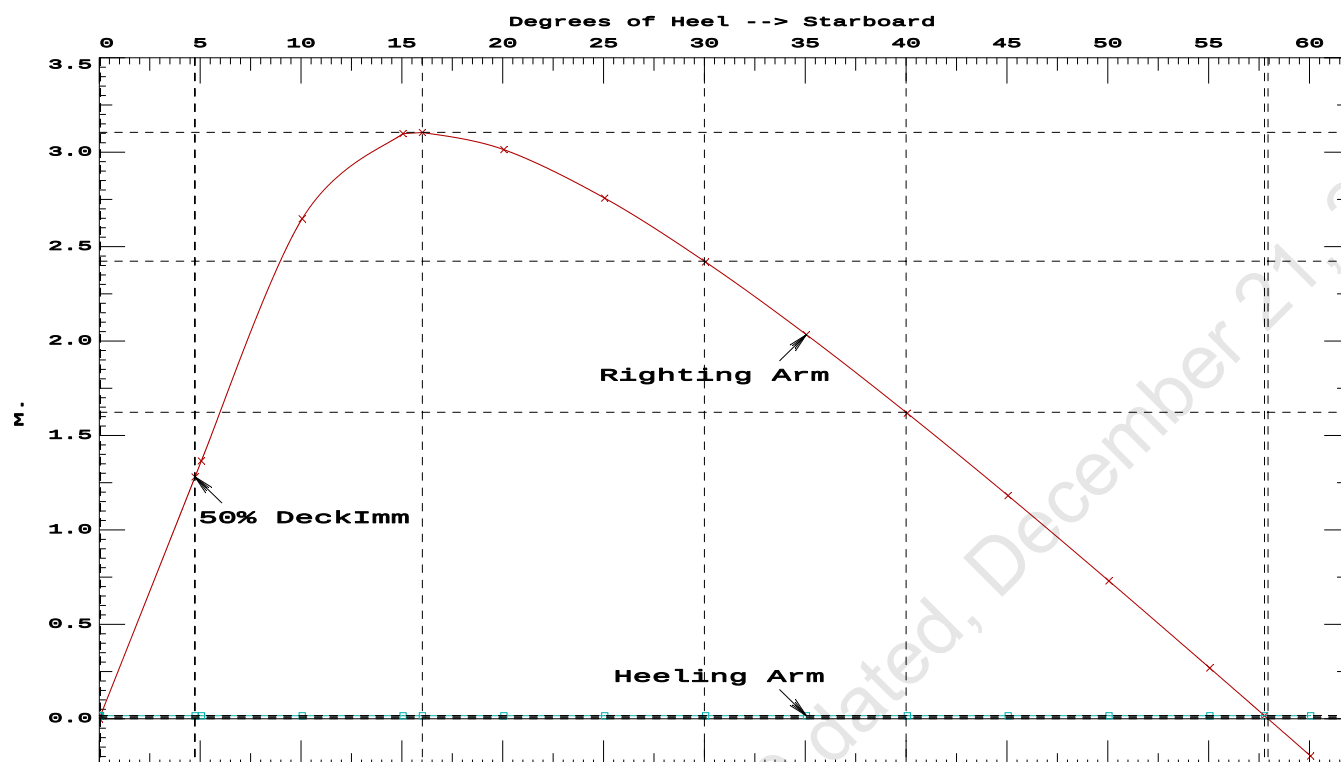
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.91 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5406 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	57.71 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.68 P

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PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd



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PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.889f TCG = 0.000 VCG = 4.033
Free Surface Adjustment: 0.371
Adjusted CG: LCG = 24.889f TCG = 0.026p VCG = 4.403

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
1.525	0.05a	3.99s	1,267.7	0.000	0.000	0.0000
1.490	0.05a	8.99s	1,267.7	0.000	1.312	0.0572
1.441	0.07a	13.99s	1,267.7	0.000	1.992	0.2104
1.418	0.08a	16.15s	1,267.7	0.000	2.025	0.2866
1.385	0.10a	18.99s	1,267.7	0.000	1.972	0.3864
1.319	0.13a	23.99s	1,267.7	0.000	1.739	0.5511
1.243	0.16a	28.99s	1,267.7	0.000	1.414	0.6893
1.159	0.18a	33.99s	1,267.7	0.000	1.036	0.7966
1.066	0.21a	38.99s	1,267.7	0.000	0.625	0.8694
0.965	0.24a	43.99s	1,267.7	0.000	0.191	0.9051
0.920	0.25a	46.12s	1,267.7	0.000	0.000	0.9087
0.858	0.27a	48.99s	1,267.7	0.000	-0.260	0.9022
0.744	0.29a	53.99s	1,267.7	0.000	-0.721	0.8595
0.624	0.31a	58.99s	1,267.7	0.000	-1.188	0.7762
0.500	0.33a	63.99s	1,267.7	0.000	-1.656	0.6522

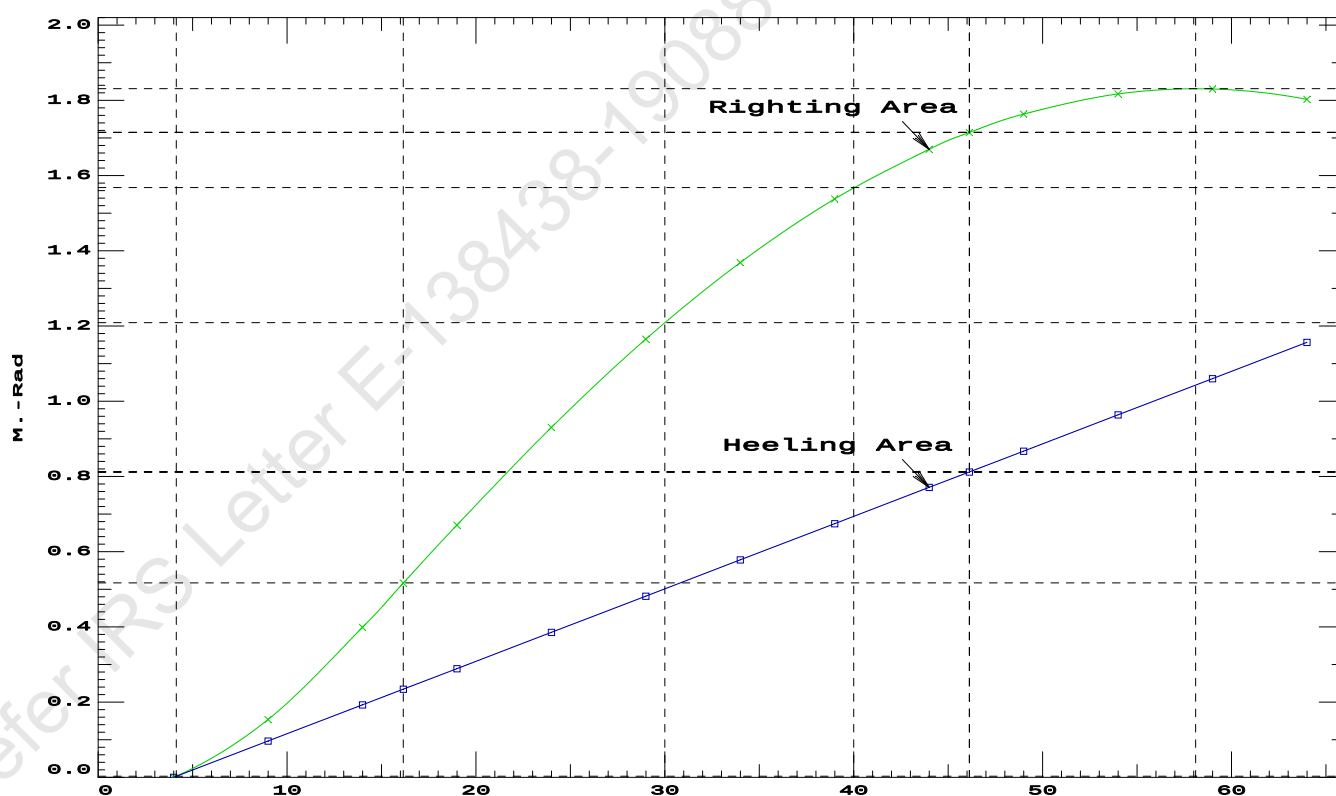
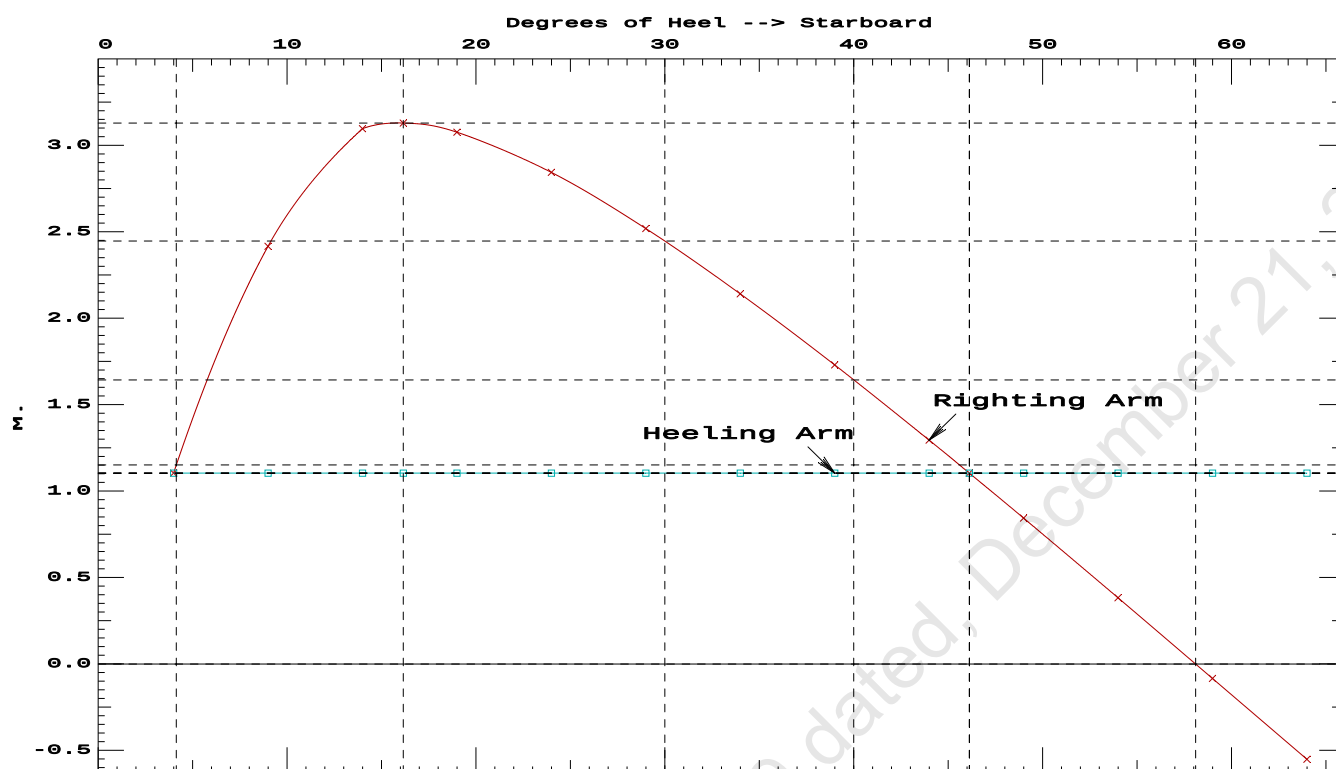
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1400.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.99 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	183.3 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.119 P

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GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTOONCondition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

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GHS 17.22 PONTON

Condition 08 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.499 @ 50.00f, 1.520 @ 25.00f, 1.541 @ 0.00

Trim: Aft 0.042/50.000, Heel: Stbd 3.99 deg.

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	455.34	24.928f	0.000	3.000	
Crane	350.00	11.500f	0.000	6.325	
Crane Load	40.00	11.500f	0.000	21.478	
Total Fixed----->	845.34	18.733f	0.000	5.251	
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		422.40	37.208f	0.000	1.596
Total Weight----->		1,267.74	24.889f	0.000	4.033
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,267.74	24.886f	1.331s	0.819

Righting Arms:		0.000	1.104s		
External Arms:		0.000	1.104s		
Residual Righting Arms:		0.000	0.000s		
Part-----	SpGr	WPA	LCF	TCF	BML
Total Waterplane----->	1.025	869.9	24.961f	0.153s	136.25
		MT/cm			
		8.92			

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.99

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.540 @ 50.00f, 1.481 @ 25.00f, 1.422 @ 0.00
Trim: Fwd 0.119/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----		
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	350.00	11.500f	0.000	6.325			
Total Fixed----->	805.34	19.092f	0.000	4.445			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,227.74	25.325f	0.000	3.465	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,227.74	25.331f	0.003s	0.750	

Righting Arms:			0.000	0.003s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.002s			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	860.0	25.275f	0.041s	136.30	19.030	
		MT/cm					
		8.81					
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3326.77; t3: 30.0; t: 15.0
limmarg: 1206.09; t3: 15.0; t: 7.5
limmarg: 252.08; t3: 7.5; t: 3.8
limmarg: -13.91; t3: 3.7; t: 1.9
limmarg: 96.70; t3: 5.6; t: 0.9
limmarg: 38.66; t3: 4.7; t: 0.5
limmarg: 10.84; t3: 4.2; t: 0.2
limmarg: 0.58; t3: 4.0; t: 0.1
limmarg: -4.37; t3: 3.9; t: 0.1
limmarg: 0.58; t3: 4.0; t: 0.0
limmarg: 0.58; t3: 4.0; t: 0.0

Theta3 is 4.00 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.092f TCG = 0.000 VCG = 4.445

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.399	0.14f	3.99s	1,227.7	0.000	-1.181	0.0000
1.410	0.14f	0.00	1,227.7	0.000	-0.002	-0.0412
1.410	0.14f	0.00	1,227.7	0.000	0.001	-0.0412
1.399	0.14f	4.00p	1,227.7	0.000	1.185	0.0002
1.394	0.14f	5.00p	1,227.7	0.000	1.478	0.0235
1.345	0.15f	10.00p	1,227.7	0.000	2.882	0.2141
1.230	0.22f	15.00p	1,227.7	0.000	3.448	0.4964
1.168	0.26f	17.54p	1,227.7	0.000	3.484	0.6510
1.105	0.30f	20.00p	1,227.7	0.000	3.450	0.7996
0.969	0.37f	25.00p	1,227.7	0.000	3.270	1.0951
0.824	0.45f	30.00p	1,227.7	0.000	3.002	1.3694

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -4 deg to abs 4 > 1.000 1.006 P

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.092f TCG = 0.000 VCG = 4.445

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.399	0.14f	3.99s	1,227.7	0.000	-1.181	0.0000
1.410	0.14f	0.00	1,227.7	0.000	-0.002	-0.0412
1.410	0.14f	0.00	1,227.7	0.000	0.001	-0.0412
1.399	0.14f	4.00p	1,227.7	0.000	1.185	0.0002
1.394	0.14f	5.00p	1,227.7	0.000	1.478	0.0235
1.345	0.15f	10.00p	1,227.7	0.000	2.882	0.2141
1.230	0.22f	15.00p	1,227.7	0.000	3.448	0.4964
1.168	0.26f	17.54p	1,227.7	0.000	3.484	0.6510
1.105	0.30f	20.00p	1,227.7	0.000	3.450	0.7996
0.969	0.37f	25.00p	1,227.7	0.000	3.270	1.0951
0.824	0.45f	30.00p	1,227.7	0.000	3.002	1.3694
0.672	0.53f	35.00p	1,227.7	0.000	2.682	1.6178
0.513	0.60f	40.00p	1,227.7	0.000	2.326	1.8366
0.349	0.68f	45.00p	1,227.7	0.000	1.942	2.0230
0.181	0.75f	50.00p	1,227.7	0.000	1.538	2.1750
0.012	0.82f	55.00p	1,227.7	0.000	1.118	2.2910
-0.158	0.88f	60.00p	1,227.7	0.000	0.687	2.3698
-0.326	0.93f	65.00p	1,227.7	0.000	0.249	2.4107
-0.419	0.96f	67.83p	1,227.7	0.000	0.000	2.4169
-0.490	0.98f	70.00p	1,227.7	0.000	-0.192	2.4133

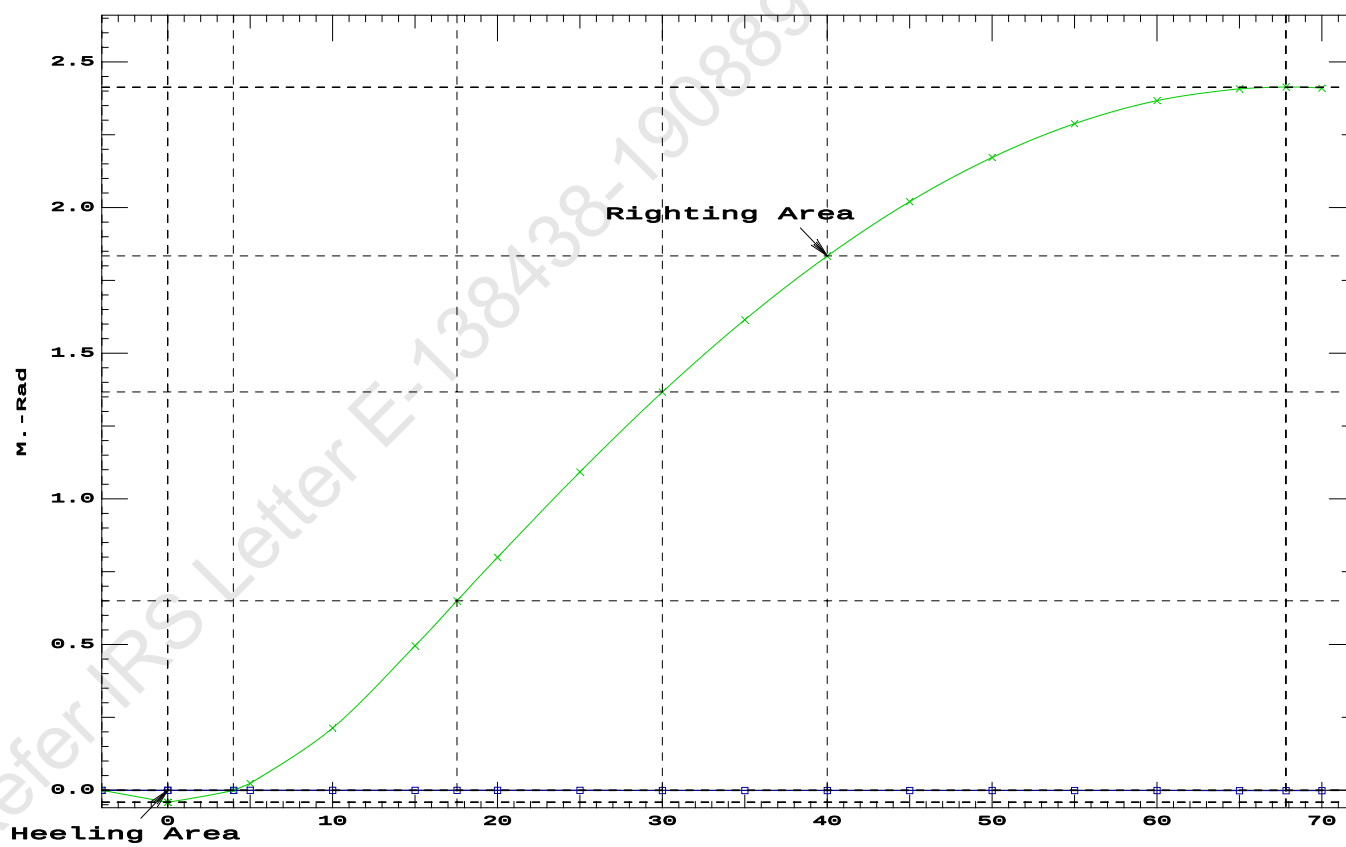
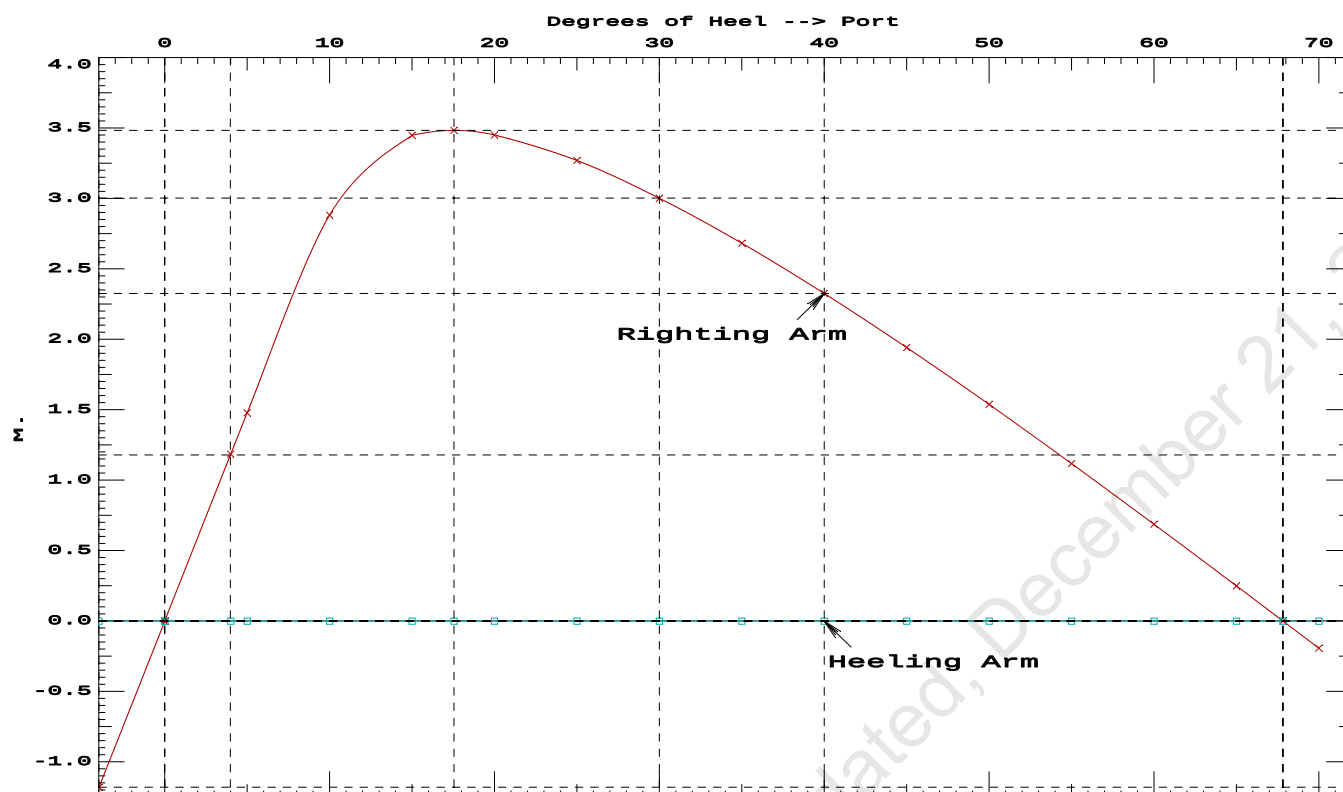
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -4 deg to RAZero > 1.000 59.710 P
(2) Angle from abs 4 deg to RAZero > 20.00 deg 63.83 P

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GHS 17.22 PONT00N



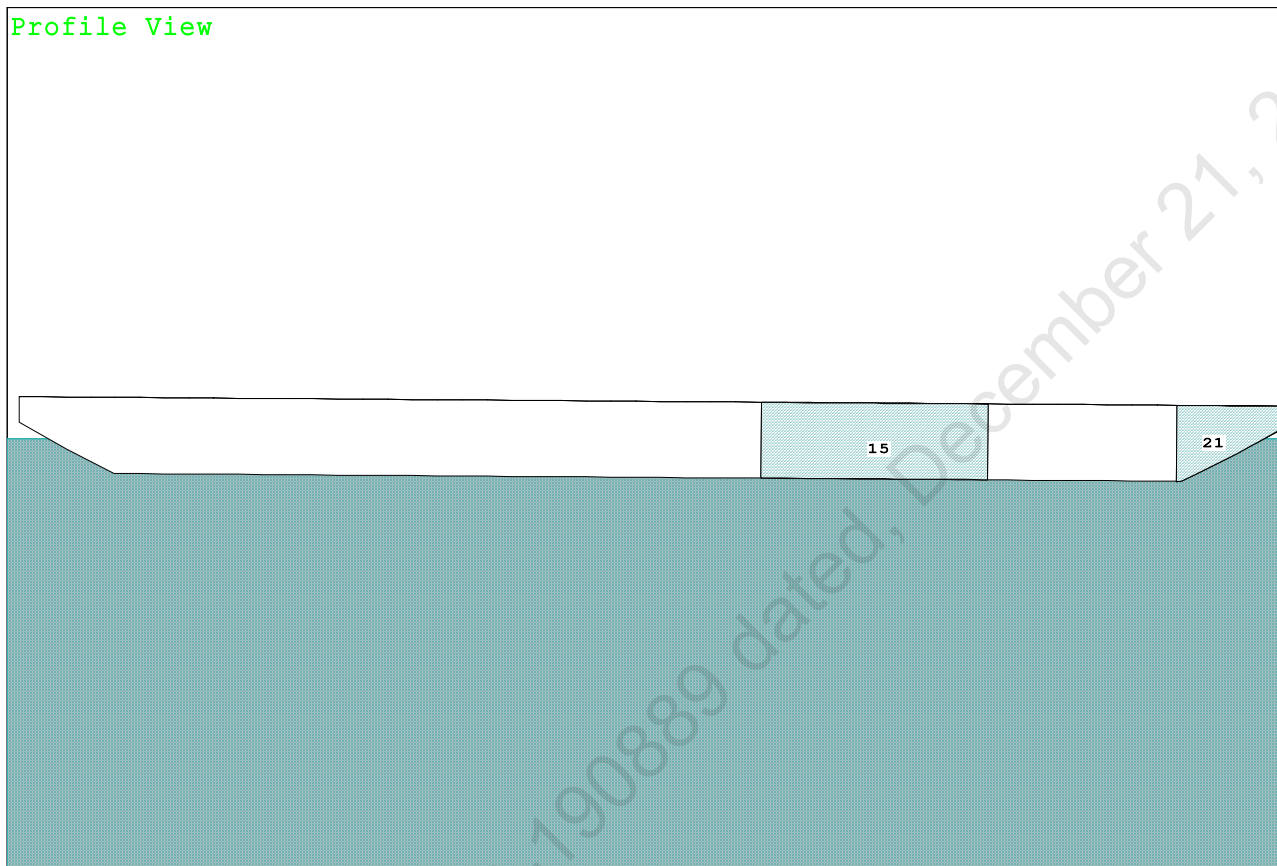
13-12-21 20:33:51 Cybermarine Technologies PTE. Ltd.
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PONTOON

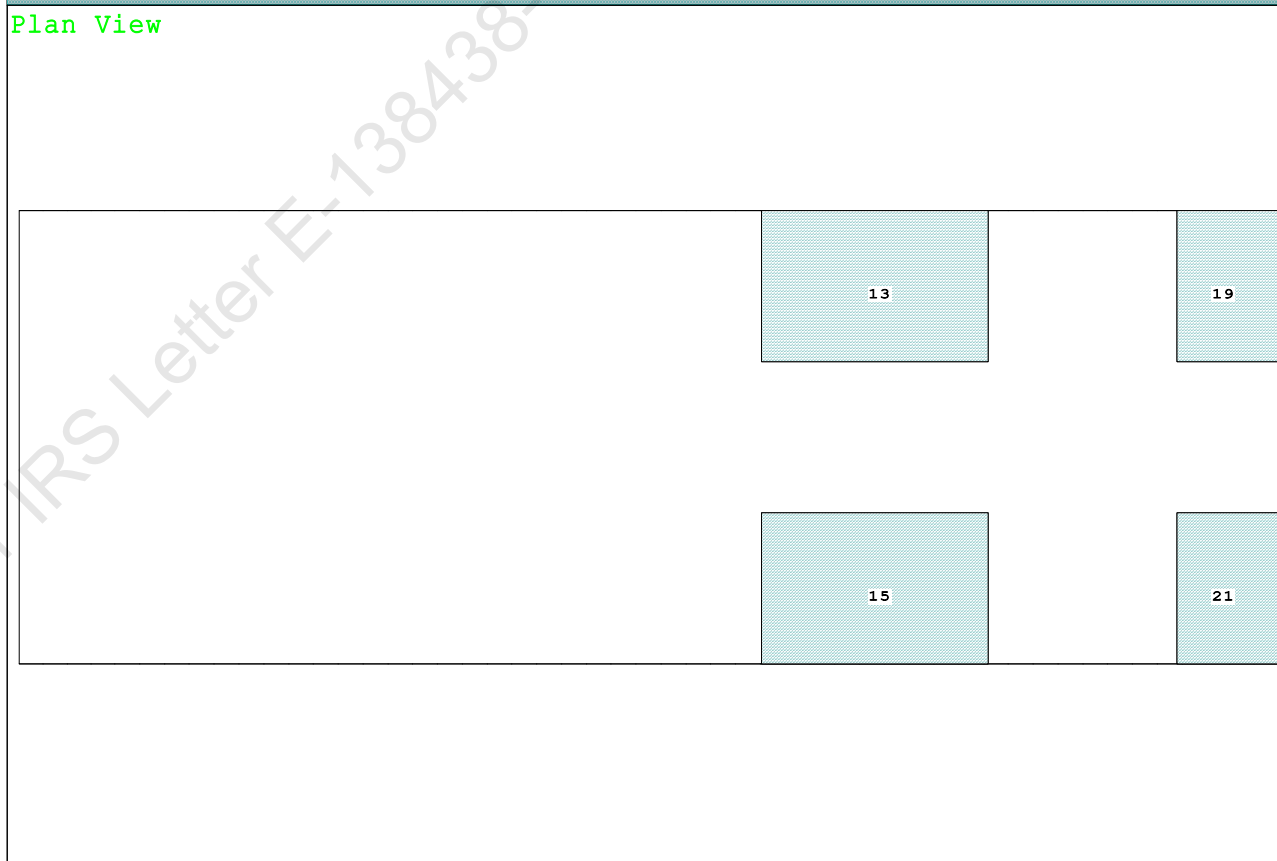
Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

Condition Graphic - Draft: 1.711 @ 50.000f, 1.339 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.711 @ 50.00f, 1.525 @ 25.00f, 1.339 @ 0.00

Trim: Fwd 0.373/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed----->			845.34	20.389f	0.000	5.251	
	Load----	SpGr----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,267.74	25.993f	0.000	4.033	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,267.65	26.016f	0.000	0.777	

	Righting Arms:			0.001a	0.000		
Part-----		SpGr----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025		869.5	25.320f	0.000	136.42	18.617*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.91		33.76	133.16	15.360
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.711
DRAFT AT MS (M) = 1.525
DRAFT AT AP (M) = 1.339
DRAFT AT FWD MARK (M) = 1.681
DRAFT AT AFT MARK (M) = 1.368

HYDROSTATIC PROPERTIES

Trim: Fwd 0.373/50.000, No Heel, VCG = 4.033

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm trim-----
1.527	1,267.65	26.016f	0.777	8.91
				25.320f
				33.76
				133.16
				15.360

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.711 @ 50.00f, 1.525 @ 25.00f, 1.339 @ 0.00

Trim: Fwd 0.373/50.000, Heel: zero

Part	LPA*Sf	HCP	Arm	Pressure	Moment
HULL	74.3	0.757	1.505	0.05500	6.15
CRANE	57.0	4.098	4.846	0.05500	15.19
Total wind heeling moment to starboard-->					21.34
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.993f TCG = 0.000 VCG = 4.033

Free Surface Adjustment: 0.371

Adjusted CG: LCG = 25.990f TCG = 0.000 VCG = 4.404

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth	Trim	Heel	Weight (MT)	Area
1.327	0.43f	0.00	1,267.7	0.0000
1.326	0.43f	0.05s	1,267.8	-0.0000
1.315	0.43f	4.18s	1,267.7	0.0401
1.310	0.43f	5.05s	1,267.7	0.0588
1.251	0.47f	10.05s	1,267.8	0.2321
1.089	0.70f	15.05s	1,267.7	0.4862
1.057	0.75f	16.01s	1,267.7	0.5371
0.912	0.95f	20.05s	1,267.7	0.7522
0.723	1.20f	25.05s	1,267.7	1.0025
0.523	1.45f	30.05s	1,267.7	1.2253
0.314	1.70f	35.05s	1,267.7	1.4164
0.098	1.95f	40.05s	1,267.7	1.5724
-0.122	2.19f	45.05s	1,267.7	1.6911
-0.344	2.43f	50.05s	1,267.7	1.7712
-0.563	2.64f	55.05s	1,267.7	1.8115
-0.671	2.74f	57.56s	1,267.7	1.8166
-0.777	2.83f	60.05s	1,267.7	1.8116
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.34 (constant)

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PONTOON

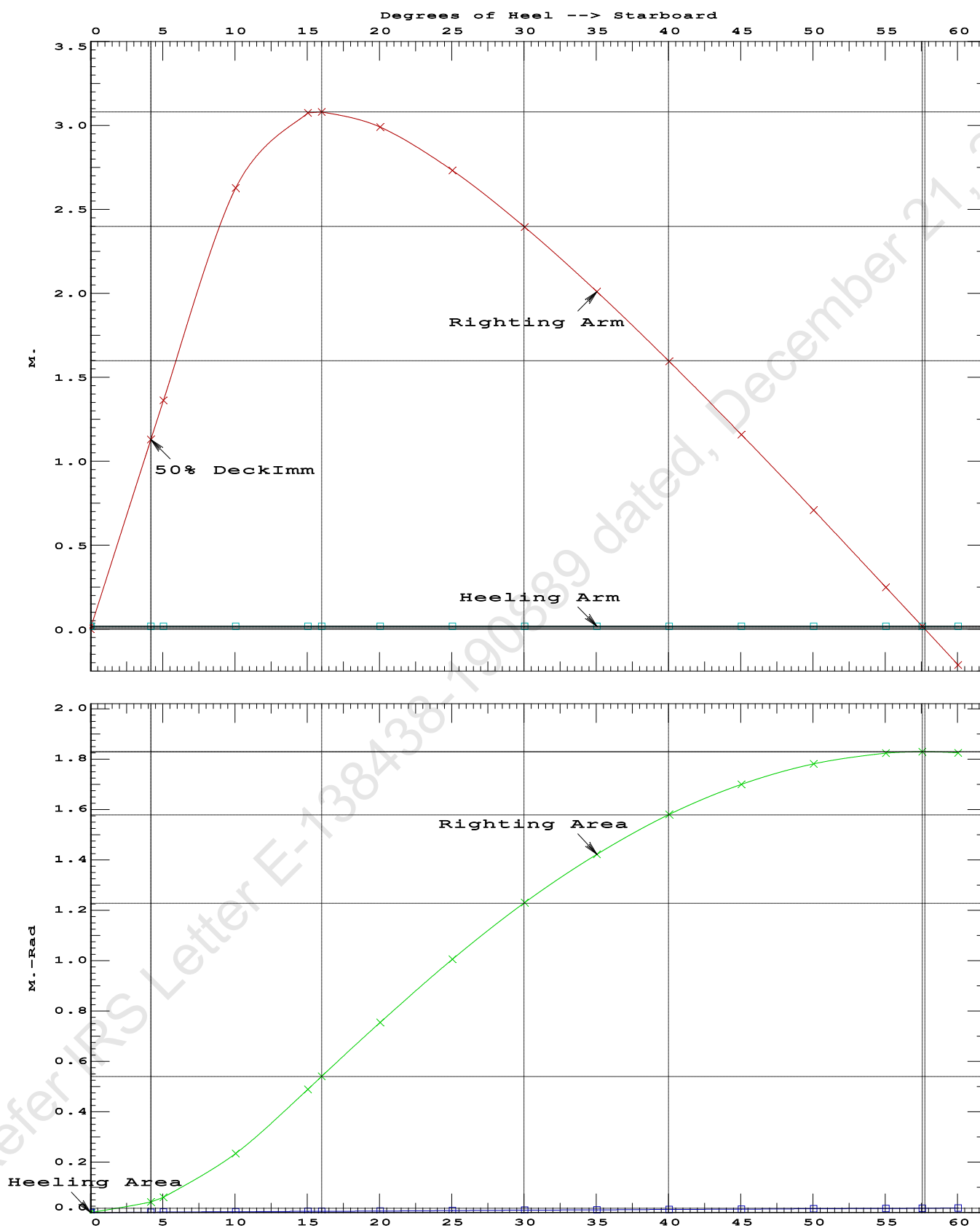
Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5371 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 57.51 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.13 P

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PONTOON

Condition 09 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Fwd

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.993f TCG = 0.000 VCG = 4.033
Free Surface Adjustment: 0.371
Adjusted CG: LCG = 25.990f TCG = 0.000 VCG = 4.404

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim----	Heel----	Weight (MT)----	in Trim--	in Heel---	Area
1.326	0.43f	0.00	1,267.8	0.000	0.001	0.0000
1.310	0.43f	5.00s	1,267.7	0.000	1.347	0.0588
1.253	0.47f	10.00s	1,267.8	0.000	2.615	0.2322
1.091	0.70f	15.00s	1,267.7	0.000	3.072	0.4873
1.056	0.75f	16.02s	1,267.7	0.000	3.079	0.5417
0.914	0.94f	20.00s	1,267.7	0.000	2.990	0.7546
0.725	1.19f	25.00s	1,267.7	0.000	2.734	1.0065
0.525	1.44f	30.00s	1,267.7	0.000	2.396	1.2309
0.316	1.70f	35.00s	1,267.7	0.000	2.012	1.4236
0.100	1.95f	40.00s	1,267.7	0.000	1.597	1.5813
-0.120	2.19f	45.00s	1,267.7	0.000	1.161	1.7018
-0.342	2.42f	50.00s	1,267.7	0.000	0.711	1.7836
-0.561	2.64f	55.00s	1,267.7	0.000	0.252	1.8257
-0.678	2.75f	57.72s	1,267.7	0.000	0.000	1.8317
-0.775	2.83f	60.00s	1,267.7	0.000	-0.211	1.8275
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 2.00

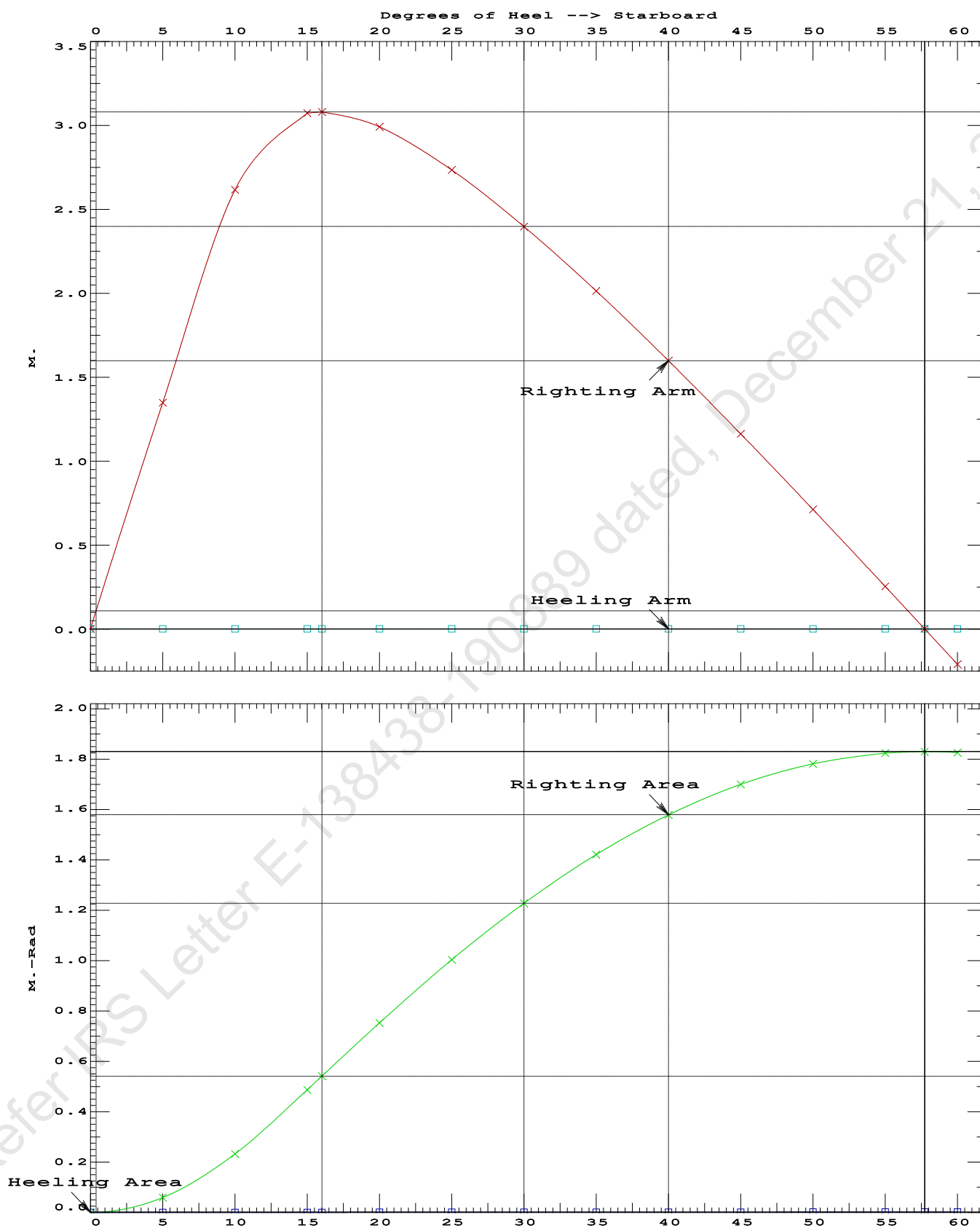
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	195160 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1153.44 P

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PONTOON

Condition 09 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Fwd

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Condition 09 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.712 @ 50.00f, 1.525 @ 25.00f, 1.338 @ 0.00

Trim: Fwd 0.373/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed----->			845.34	20.389f	0.000	5.251	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,267.74	25.993f	0.000	4.033	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,267.74	26.018f	0.003p	0.777	

	Righting Arms:			0.001f	-0.003s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.001f	-0.004s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		863.2	25.405f	0.043p	133.50	18.452*
			MT/cm				
			8.85				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.540 @ 50.00f, 1.481 @ 25.00f, 1.422 @ 0.00
Trim: Fwd 0.119/50.000, Heel: zero

Part-----	Weight (MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	455.34	24.928f	0.000	3.000
Crane	350.00	11.500f	0.000	6.325
Total Fixed----->	805.34	19.092f	0.000	4.445

Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880
Total Tanks----->			422.40	37.208f	0.000	1.596
Total Weight----->			1,227.74	25.325f	0.000	3.465

Displ (MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,227.73	25.331f
		0.000	0.750

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	-0.001s

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	866.8	25.093f	0.000	139.56	19.171*
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		8.88	33.60		136.84	16.456

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.092f TCG = 0.000 VCG = 4.445

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.410	0.14f	0.00	1,227.7	0.000	-0.001	0.0000
1.407	0.14f	0.97s	1,227.7	0.000	0.289	0.0024
1.395	0.14f	5.00s	1,227.7	0.000	1.477	0.0645
1.345	0.15f	10.00s	1,227.7	0.000	2.881	0.2552
1.230	0.22f	15.00s	1,227.7	0.000	3.446	0.5374
1.168	0.26f	17.54s	1,227.7	0.000	3.482	0.6918
1.105	0.30f	20.00s	1,227.7	0.000	3.449	0.8404
0.969	0.37f	25.00s	1,227.7	0.000	3.268	1.1357
0.824	0.45f	30.00s	1,227.7	0.000	3.001	1.4099

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.092f TCG = 0.000 VCG = 4.445

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.410	0.14f	0.00	1,227.7	0.000 -0.001 0.0000
1.407	0.14f	0.97s	1,227.7	0.000 0.289 0.0024
1.395	0.14f	5.00s	1,227.7	0.000 1.477 0.0645
1.345	0.15f	10.00s	1,227.7	0.000 2.881 0.2552
1.230	0.22f	15.00s	1,227.7	0.000 3.446 0.5374
1.168	0.26f	17.54s	1,227.7	0.000 3.482 0.6918
1.105	0.30f	20.00s	1,227.7	0.000 3.449 0.8404
0.969	0.37f	25.00s	1,227.7	0.000 3.268 1.1357
0.824	0.45f	30.00s	1,227.7	0.000 3.001 1.4099
0.672	0.53f	35.00s	1,227.7	0.000 2.681 1.6582
0.513	0.60f	40.00s	1,227.7	0.000 2.324 1.8768
0.349	0.68f	45.00s	1,227.7	0.000 1.940 2.0631
0.181	0.75f	50.00s	1,227.7	0.000 1.536 2.2149
0.012	0.82f	55.00s	1,227.7	0.000 1.116 2.3308
-0.158	0.88f	60.00s	1,227.7	0.000 0.685 2.4095
-0.326	0.93f	65.00s	1,227.7	0.000 0.248 2.4502
-0.419	0.96f	67.81s	1,227.7	0.000 0.000 2.4563
-0.490	0.98f	70.00s	1,227.7	0.000 -0.194 2.4526

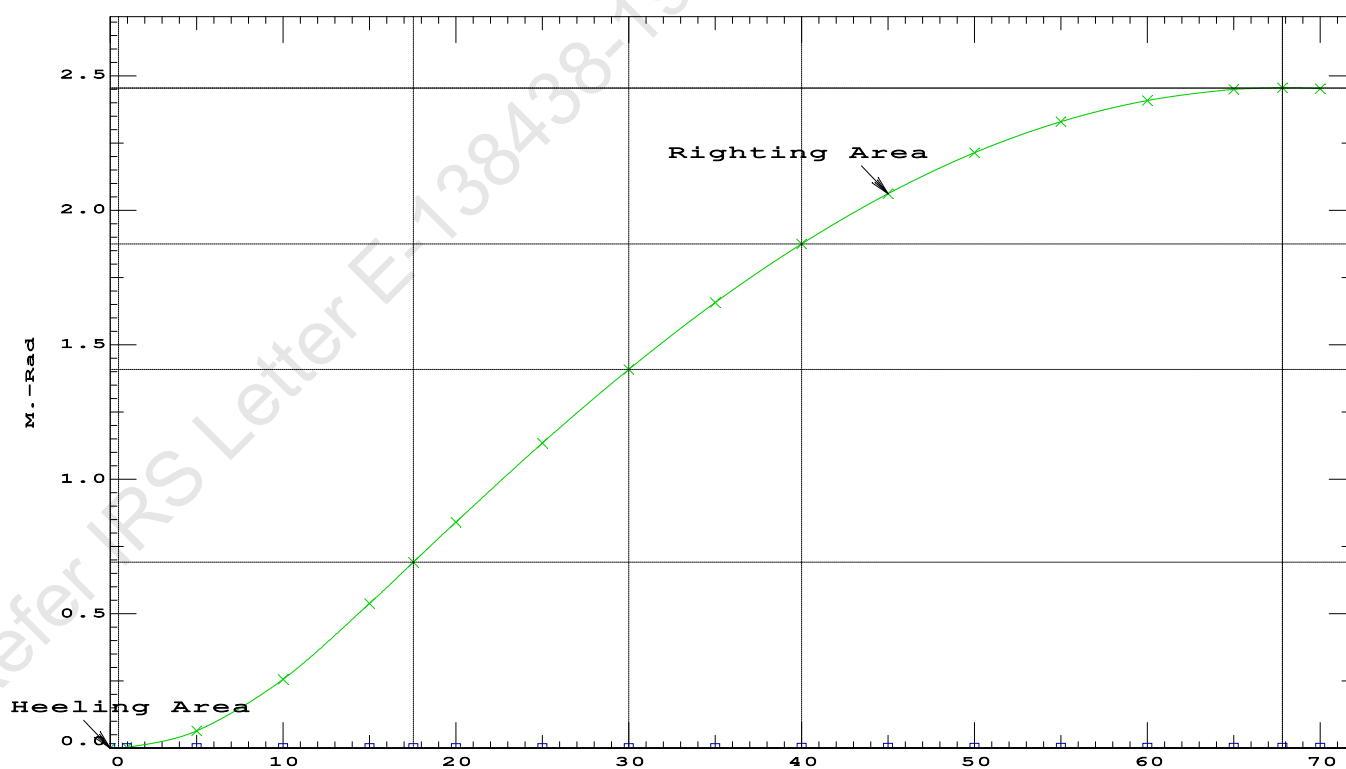
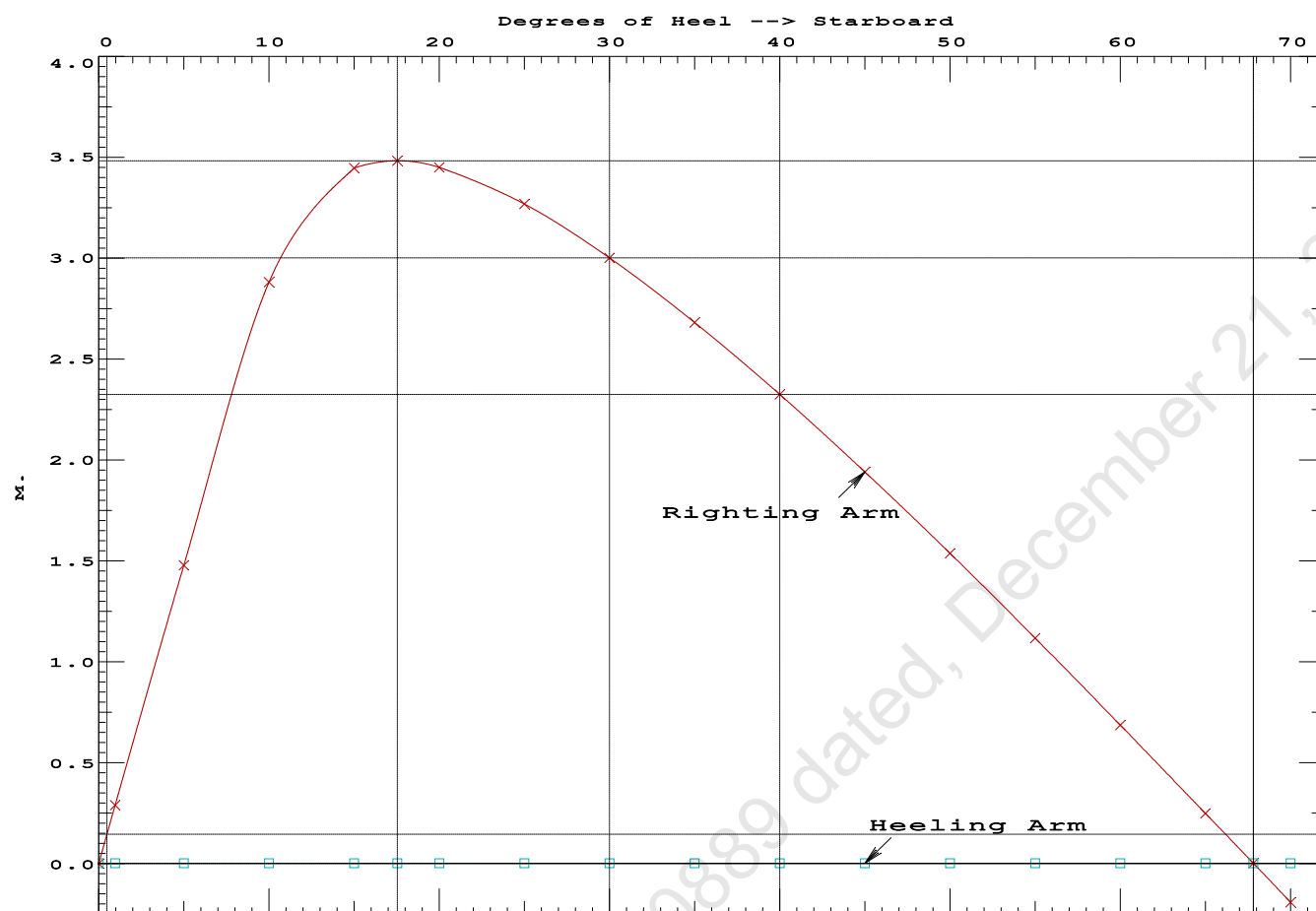
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 1 deg to RAzero > 20.00 deg 66.84 P

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GHS 17.90 PONTOON

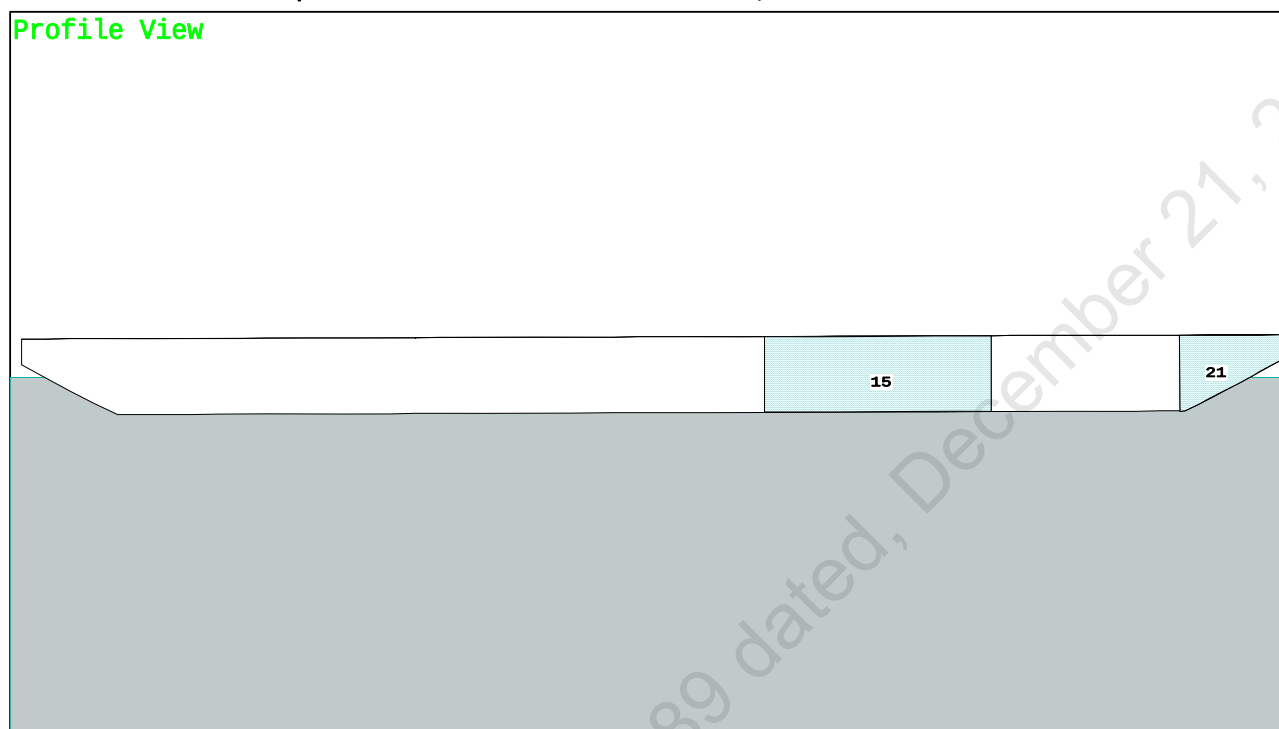


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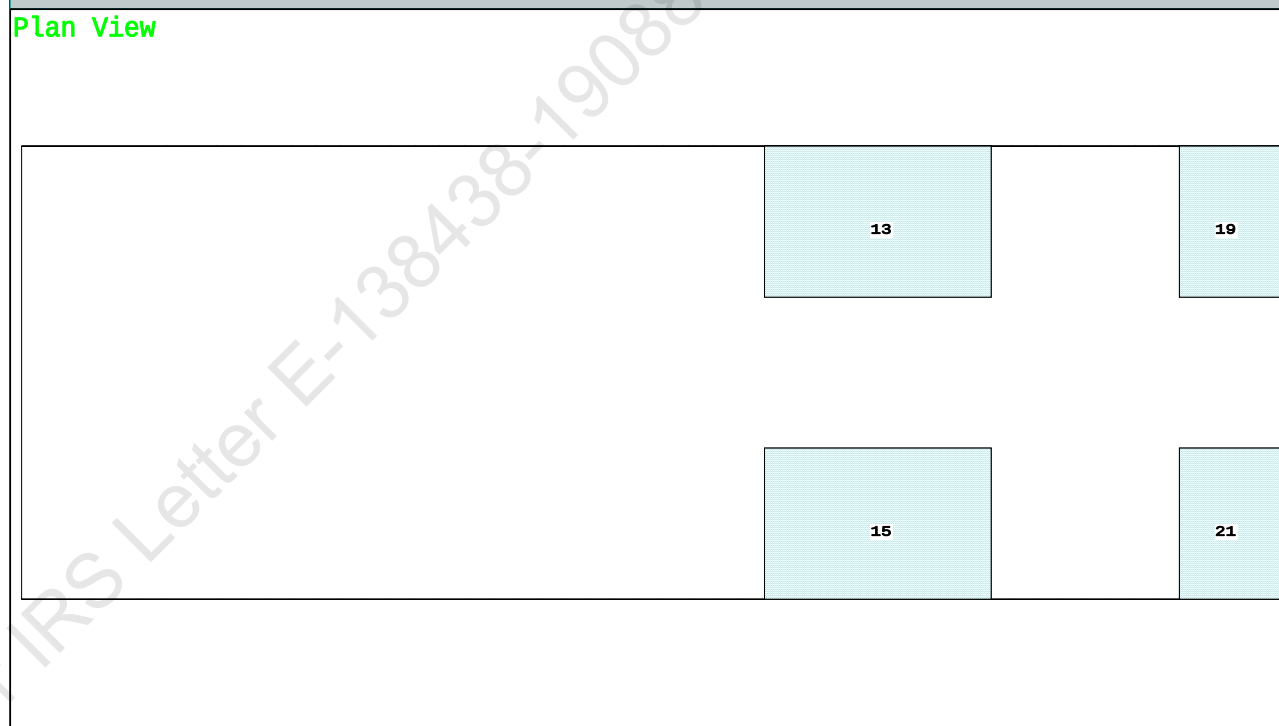
Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.325 @ 50.000f, 1.494 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S 19 07_WBTK.P 21 07_WBTK.S

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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.325 @ 50.00f, 1.409 @ 25.00f, 1.494 @ 0.00

Trim: Aft 0.169/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Total Fixed----->			741.94	17.308f	0.000	7.391	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,164.34	24.527f	0.000	5.289	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,164.31	24.512f	0.000	0.713	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	861.9	24.854f	0.000	144.66	20.089
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.83		32.62	140.09	15.513
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.325
DRAFT AT MS (M) = 1.409
DRAFT AT AP (M) = 1.494
DRAFT AT FWD MARK (M) = 1.339
DRAFT AT AFT MARK (M) = 1.480

HYDROSTATIC PROPERTIES

Trim: Aft 0.169/50.000, No Heel, VCG = 5.289

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/					
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT-----	
1.410	1,164.31	24.512f	0.713	8.83	24.854f	32.62	140.09	15.513	
Distances in METERS.-----					Specific Gravity = 1.025.-----				
					Moment in m.-MT.				
					Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.325 @ 50.00f, 1.409 @ 25.00f, 1.494 @ 0.00

Trim: Aft 0.169/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.9	0.813	1.503	0.05500	6.61
CRANE	57.0	4.052	4.742	0.05500	14.87
Total wind heeling moment to starboard-->					21.47
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.527f TCG = 0.000 VCG = 5.289

Free Surface Adjustment: 0.404

Adjusted CG: LCG = 24.529f TCG = 0.000 VCG = 5.692

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.482	0.19a	0.00	1,164.3	0.000 -0.018	0.0000	4.919
1.482	0.19a	0.06s	1,164.3	0.000 0.000	-0.0000	4.860
1.468	0.19a	4.92s	1,164.3	0.000 1.324	0.0561	0.000
1.467	0.19a	5.06s	1,164.3	0.000 1.362	0.0594	-0.140
1.423	0.21a	10.06s	1,164.3	0.000 2.623	0.2340	-5.140
1.349	0.31a	15.06s	1,164.3	0.000 3.020	0.4876	-10.140
1.268	0.42a	20.06s	1,164.3	0.000 2.844	0.7477	-15.140
1.180	0.53a	25.06s	1,164.3	0.000 2.483	0.9814	-20.140
1.085	0.65a	30.06s	1,164.3	0.000 2.037	1.1793	-25.140
0.985	0.76a	35.06s	1,164.3	0.000 1.545	1.3359	-30.140
0.879	0.87a	40.06s	1,164.3	0.000 1.025	1.4483	-35.140
0.768	0.98a	45.06s	1,164.3	0.000 0.486	1.5144	-40.140
0.665	1.08a	49.50s	1,164.3	0.000 0.000	1.5332	-44.578
0.652	1.09a	50.06s	1,164.3	0.000 -0.062	1.5329	-45.140
0.531	1.18a	55.06s	1,164.3	0.000 -0.614	1.5035	-50.140
0.404	1.27a	60.06s	1,164.3	0.000 -1.164	1.4259	-55.140
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

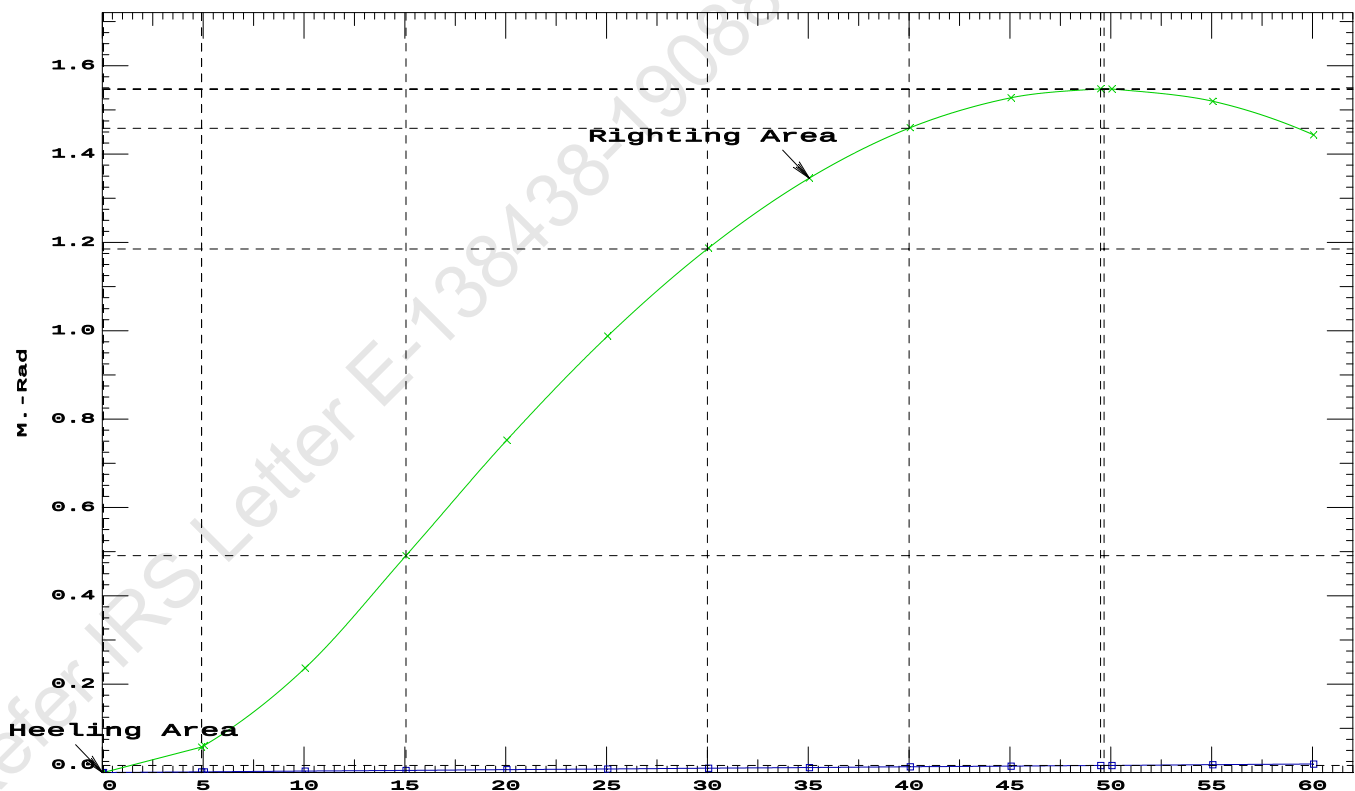
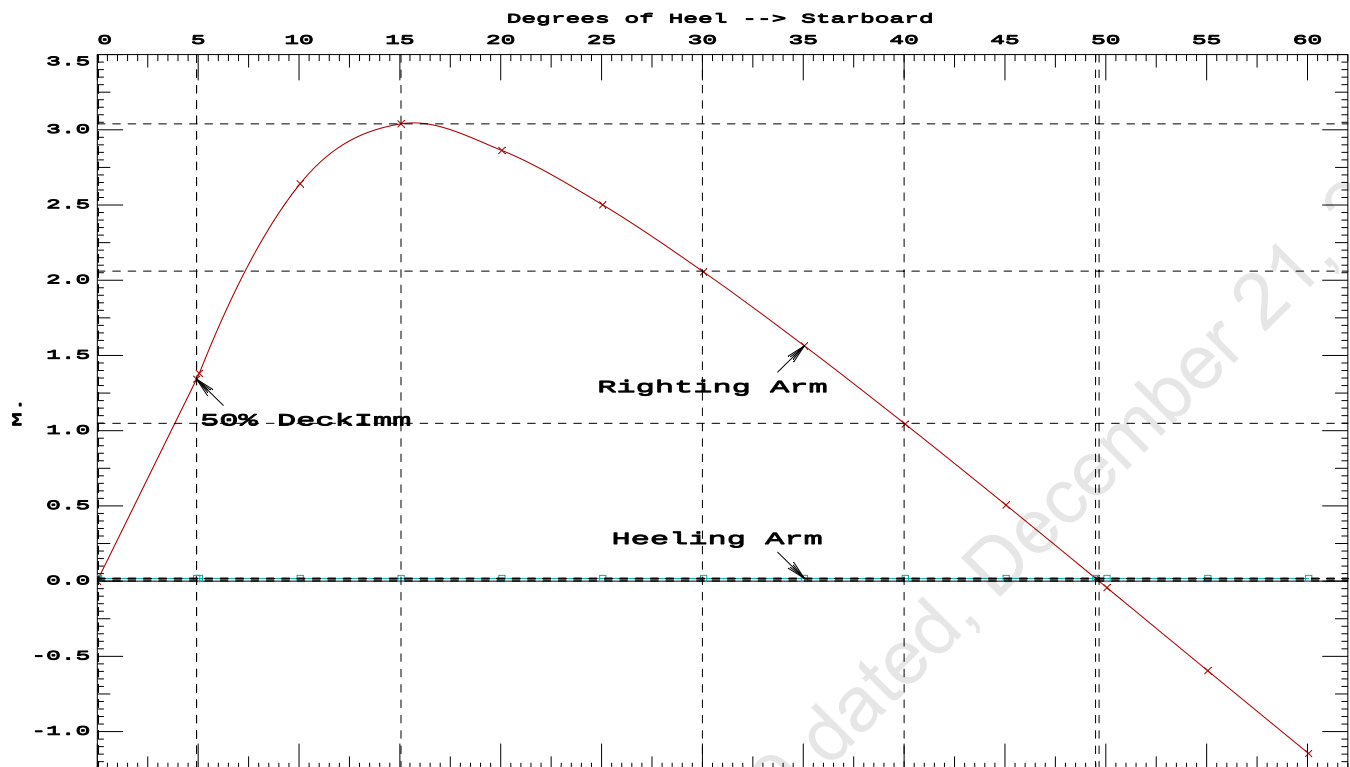
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.47 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.4876 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	49.44 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.86 P

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**Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft**



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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.527f TCG = 0.000 VCG = 5.289
Free Surface Adjustment: 0.404
Adjusted CG: LCG = 24.529f TCG = 0.000 VCG = 5.692

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.482	0.19a	0.01s	1,164.3	0.000	0.000	0.0000
1.467	0.19a	5.01s	1,164.3	0.000	1.363	0.0595
1.424	0.21a	10.01s	1,164.3	0.000	2.627	0.2342
1.350	0.31a	15.01s	1,164.3	0.000	3.034	0.4887
1.269	0.42a	20.01s	1,164.3	0.000	2.861	0.7501
1.181	0.53a	25.01s	1,164.3	0.000	2.501	0.9854
1.086	0.65a	30.01s	1,164.3	0.000	2.056	1.1848
0.986	0.76a	35.01s	1,164.3	0.000	1.564	1.3431
0.880	0.87a	40.01s	1,164.3	0.000	1.044	1.4571
0.769	0.98a	45.01s	1,164.3	0.000	0.506	1.5249
0.662	1.08a	49.62s	1,164.3	0.000	0.000	1.5453
0.653	1.09a	50.01s	1,164.3	0.000	-0.043	1.5451
0.532	1.18a	55.01s	1,164.3	0.000	-0.595	1.5174
0.405	1.27a	60.01s	1,164.3	0.000	-1.145	1.4415

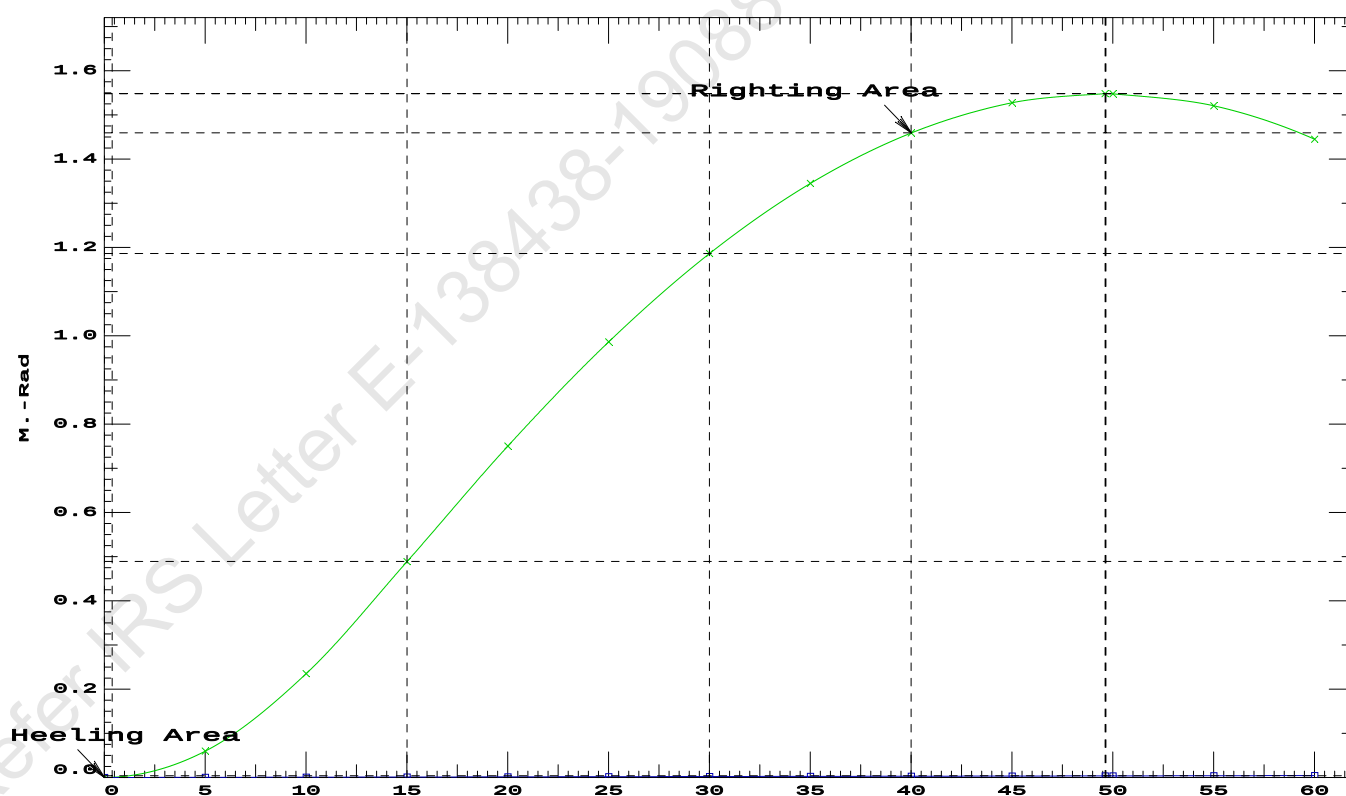
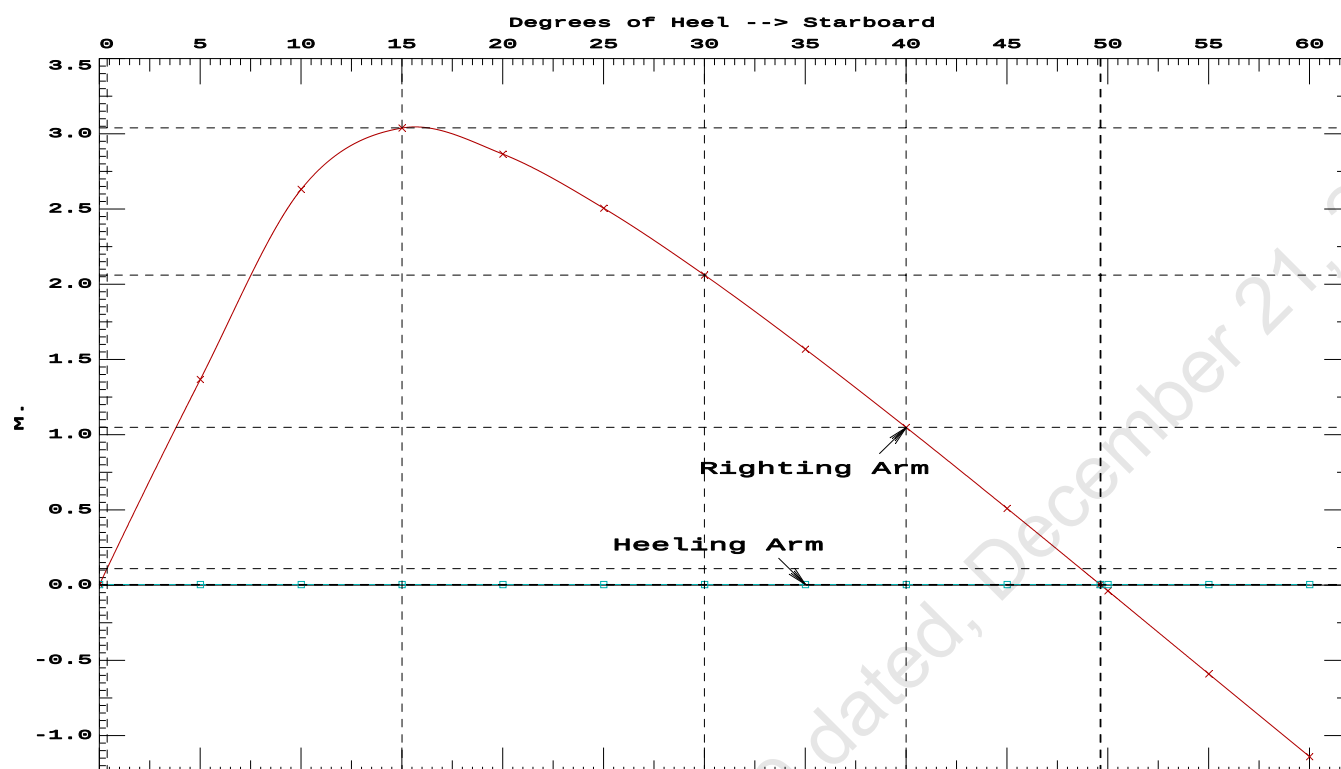
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.01 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	70659.3 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	416.572 P

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PONT00NCondition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

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Condition 10 : Operating Condition of 250T Crane**Crane Lifting 36.6T @ 18m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.325 @ 50.00f, 1.409 @ 25.00f, 1.494 @ 0.00

Trim: Aft 0.169/50.000, Heel: Stbd 0.01 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Total Fixed----->			741.94	17.308f	0.000	7.391	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,164.34	24.527f	0.000	5.289	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,164.34	24.512f	0.005s	0.713	

	Righting Arms:			0.000	0.004s		
	External Arms:			0.000	0.004s		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		857.7	24.736f	0.029s	142.57	19.975
			MT/cm				
			8.79				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.482 @ 50.00f, 1.368 @ 25.00f, 1.254 @ 0.00
Trim: Fwd 0.228/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	250.00	7.500f	0.000	5.200			
Total Fixed----->	705.34	18.751f	0.000	3.780			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,127.74	25.664f	0.000	2.962	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,127.74	25.675f	0.001s	0.692	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane----->	1.025	856.2	25.280f	0.021s	146.45	20.576	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.78	32.52	144.18	18.306		
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 530154496.00; t3: 30.0; t: 15.00
limmarg: 198642368.00; t3: 15.0; t: 7.50
limmarg: 53900080.00; t3: 7.5; t: 3.75
limmarg: 13535362.00; t3: 3.8; t: 1.88
limmarg: 3558001.25; t3: 1.9; t: 0.94
limmarg: 894219.25; t3: 1.0; t: 0.47
limmarg: 275135.97; t3: 0.5; t: 0.23
limmarg: 72902.02; t3: 0.3; t: 0.12
limmarg: 33026.57; t3: 0.2; t: 0.06
limmarg: 20272.39; t3: 0.1; t: 0.03
limmarg: 5296.18; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.751f TCG = 0.000 VCG = 3.780

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.242	0.26f	0.01s	1,127.7	0.000	-0.003	0.0000
1.242	0.26f	0.00	1,127.7	0.000	0.000	-0.0000
1.241	0.26f	0.07p	1,127.8	0.000	0.026	0.0000
1.227	0.26f	5.00p	1,127.7	0.000	1.654	0.0723
1.164	0.28f	10.00p	1,127.7	0.000	3.181	0.2842
0.972	0.41f	15.00p	1,127.7	0.000	3.838	0.5968
0.822	0.51f	18.59p	1,128.0	0.000	3.910	0.8415
0.762	0.55f	20.00p	1,127.7	0.000	3.899	0.9373
0.543	0.69f	25.00p	1,127.7	0.000	3.760	1.2739
0.318	0.83f	30.00p	1,127.7	0.000	3.524	1.5925

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 705.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 53.962 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.751f TCG = 0.000 VCG = 3.780

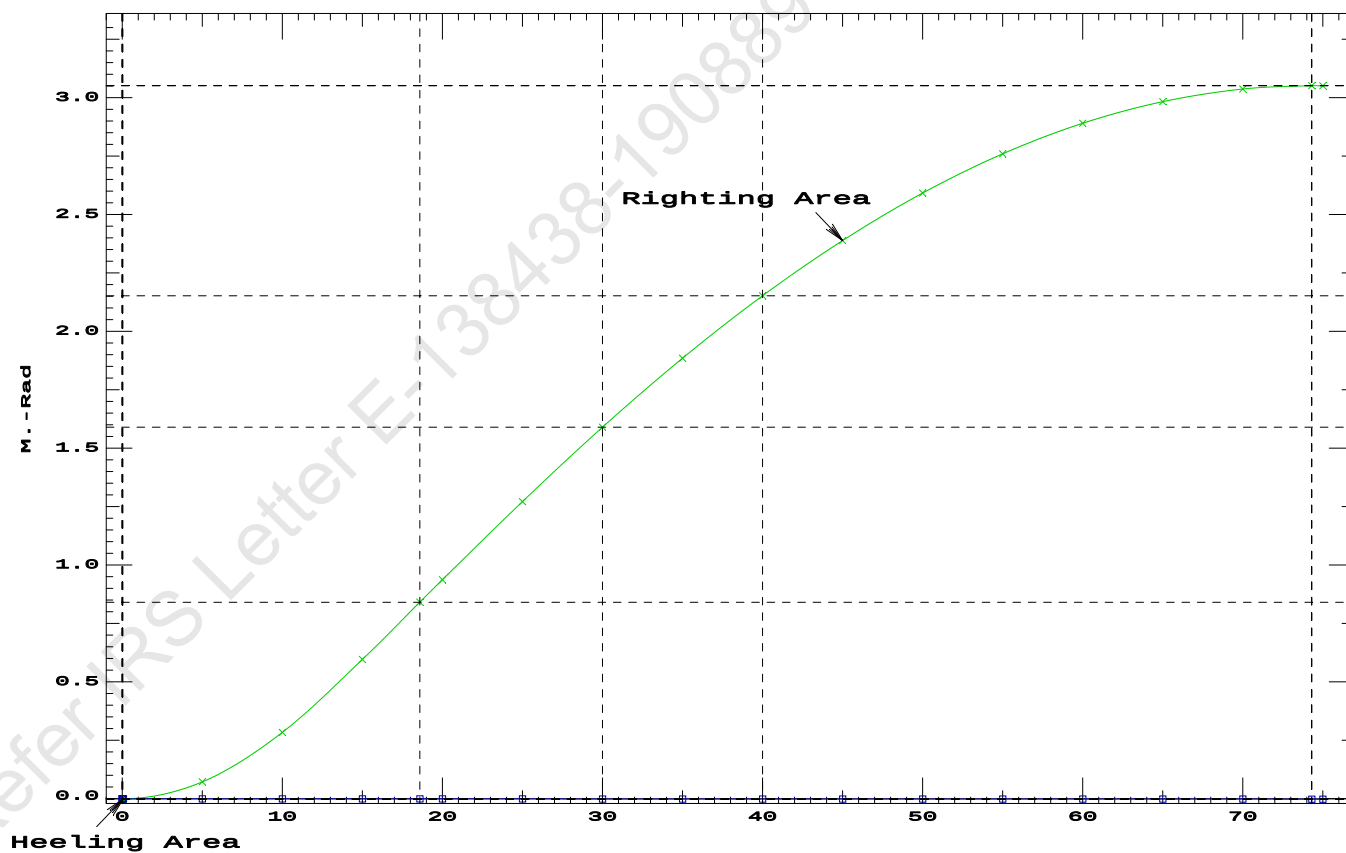
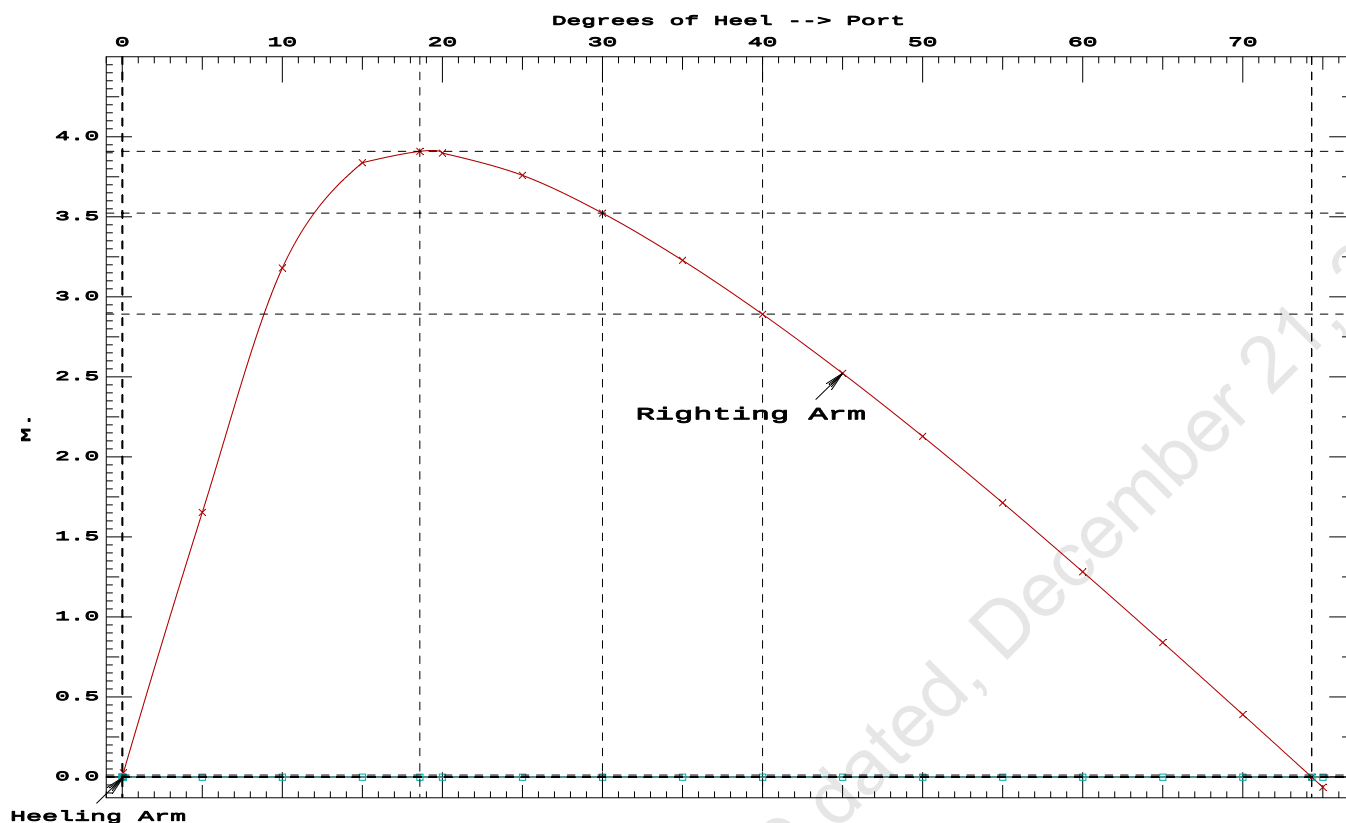
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.242	0.26f	0.01s	1,127.7	0.000	-0.003	0.0000
1.242	0.26f	0.00	1,127.7	0.000	0.000	-0.0000
1.241	0.26f	0.07p	1,127.8	0.000	0.026	0.0000
1.227	0.26f	5.00p	1,127.7	0.000	1.654	0.0723
1.164	0.28f	10.00p	1,127.7	0.000	3.181	0.2842
0.972	0.41f	15.00p	1,127.7	0.000	3.838	0.5968
0.822	0.51f	18.59p	1,128.0	0.000	3.910	0.8415
0.762	0.55f	20.00p	1,127.7	0.000	3.899	0.9373
0.543	0.69f	25.00p	1,127.7	0.000	3.760	1.2739
0.318	0.83f	30.00p	1,127.7	0.000	3.524	1.5925
0.087	0.97f	35.00p	1,127.7	0.000	3.229	1.8876
-0.146	1.11f	40.00p	1,127.7	0.000	2.892	2.1550
-0.381	1.25f	45.00p	1,127.7	0.000	2.523	2.3915
-0.615	1.38f	50.00p	1,127.7	0.000	2.129	2.5947
-0.845	1.51f	55.00p	1,127.7	0.000	1.714	2.7625
-1.069	1.62f	60.00p	1,127.7	0.000	1.283	2.8934
-1.285	1.72f	65.00p	1,127.7	0.000	0.841	2.9862
-1.489	1.80f	70.00p	1,127.7	0.000	0.391	3.0400
-1.654	1.86f	74.31p	1,127.7	0.000	0.000	3.0547
-1.680	1.87f	75.00p	1,127.7	0.000	-0.063	3.0543

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 705.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 74.24 P

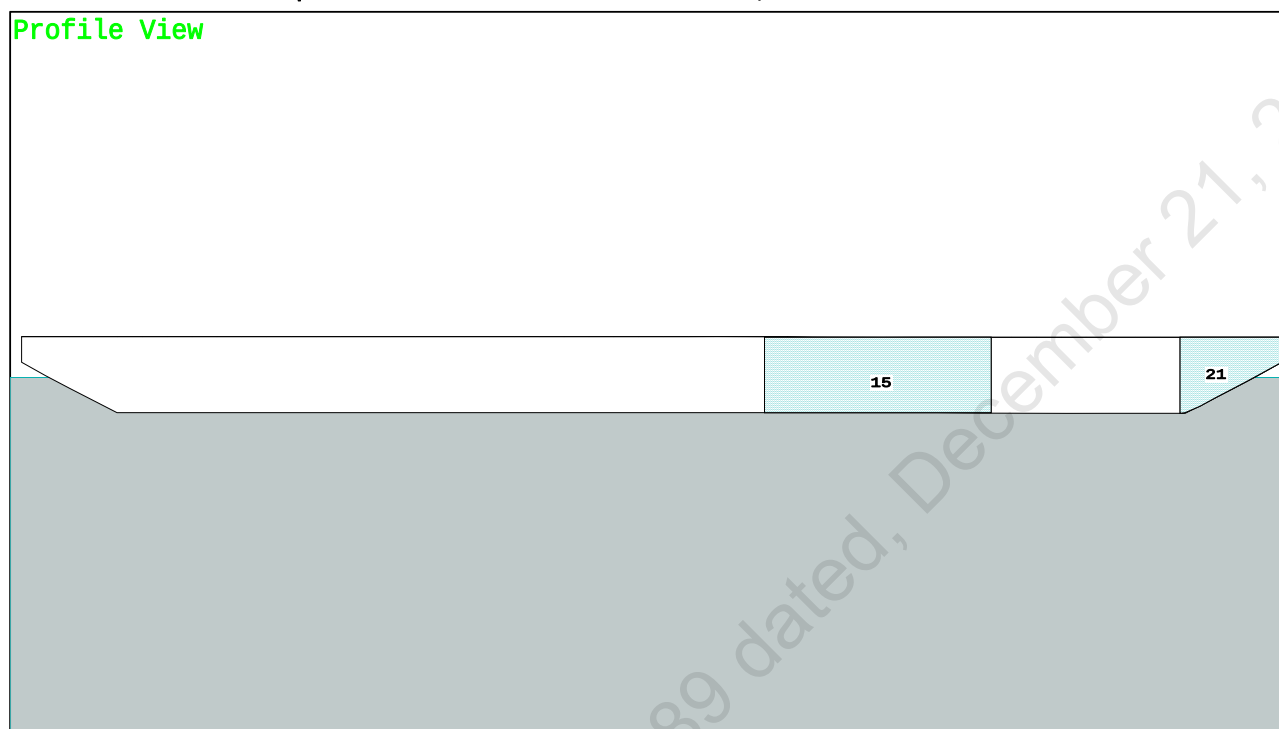
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PONTOON

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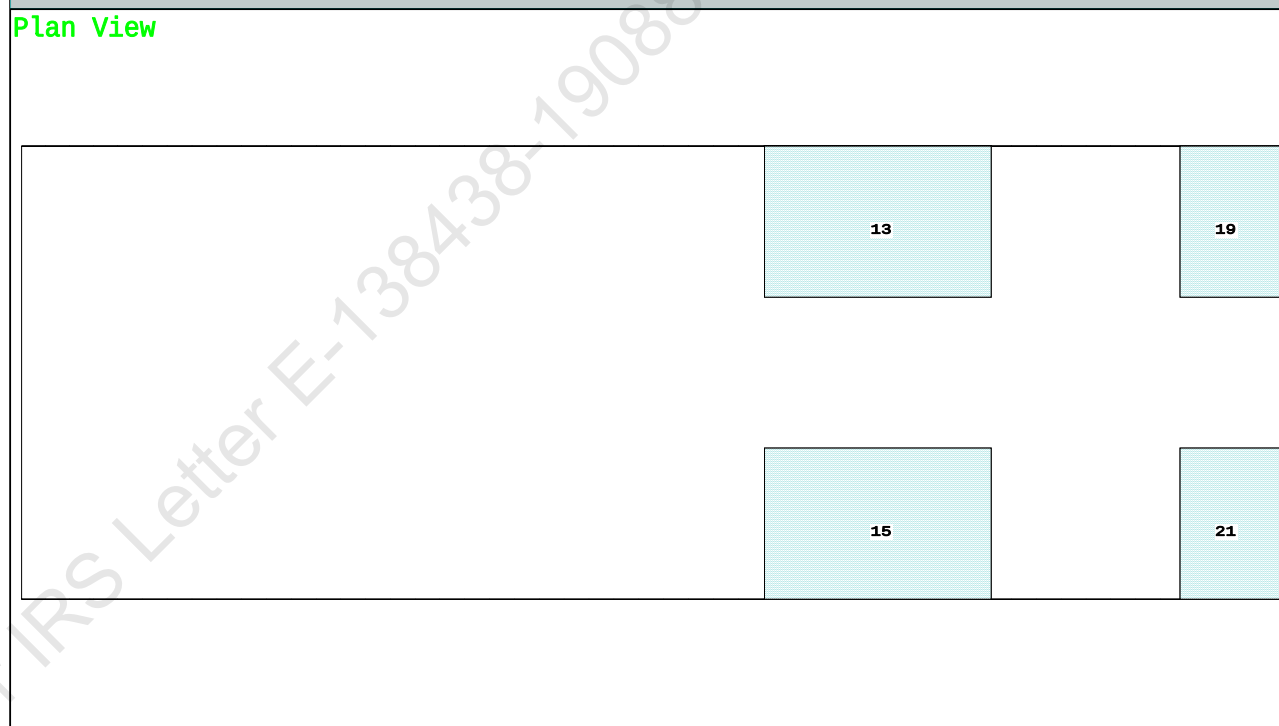
Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

Condition Graphic - Draft: 1.426 @ 50.000f, 1.393 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S 19 07_WBTK.P 21 07_WBTK.S

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PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.426 @ 50.00f, 1.410 @ 25.00f, 1.393 @ 0.00

Trim: Fwd 0.034/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed----->			741.94	18.196f	0.000	7.391	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,164.34	25.093f	0.000	5.289	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,164.32	25.097f	0.000	0.712	

	Righting Arms:			0.001f	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	862.4	25.032f	0.000	144.92	20.124
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.84		32.68	140.34	15.547
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.426
DRAFT AT MS (M) = 1.410
DRAFT AT AP (M) = 1.393
DRAFT AT FWD MARK (M) = 1.424
DRAFT AT AFT MARK (M) = 1.396

HYDROSTATIC PROPERTIES

Trim: Fwd 0.034/50.000, No Heel, VCG = 5.289

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
1.410	1,164.32	25.097f	0.712	8.84	25.032f	32.68	140.34	15.547
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.22

PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.426 @ 50.00f, 1.410 @ 25.00f, 1.393 @ 0.00

Trim: Fwd 0.034/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.9	0.812	1.502	0.05500	6.60
CRANE	57.0	4.112	4.802	0.05500	15.06
Total wind heeling moment to starboard-->					21.66
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.093f TCG = 0.000 VCG = 5.289

Free Surface Adjustment: 0.404

Adjusted CG: LCG = 25.093f TCG = 0.000 VCG = 5.692

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
1.381	0.04f	0.00	1,164.2	0.000	-0.018	0.0000	5.169
1.381	0.04f	0.05s	1,164.3	0.000	0.000	-0.0000	5.121
1.366	0.04f	5.05s	1,164.3	0.000	1.359	0.0593	0.121
1.365	0.04f	5.17s	1,164.3	0.000	1.392	0.0622	0.000
1.315	0.04f	10.05s	1,164.3	0.000	2.622	0.2337	-4.879
1.187	0.06f	15.05s	1,164.3	0.000	3.025	0.4876	-9.879
1.049	0.08f	20.05s	1,164.3	0.000	2.850	0.7481	-14.879
0.902	0.11f	25.05s	1,164.3	0.000	2.489	0.9824	-19.879
0.748	0.13f	30.05s	1,164.3	0.000	2.043	1.1808	-24.879
0.588	0.15f	35.05s	1,164.3	0.000	1.551	1.3379	-29.879
0.423	0.17f	40.05s	1,164.3	0.000	1.031	1.4508	-34.879
0.254	0.19f	45.05s	1,164.3	0.000	0.492	1.5174	-39.879
0.101	0.21f	49.54s	1,164.3	0.000	0.000	1.5367	-44.367
0.084	0.21f	50.05s	1,164.3	0.000	-0.056	1.5365	-44.879
-0.088	0.23f	55.05s	1,164.3	0.000	-0.609	1.5075	-49.879
-0.258	0.25f	60.05s	1,164.3	0.000	-1.159	1.4303	-54.879
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

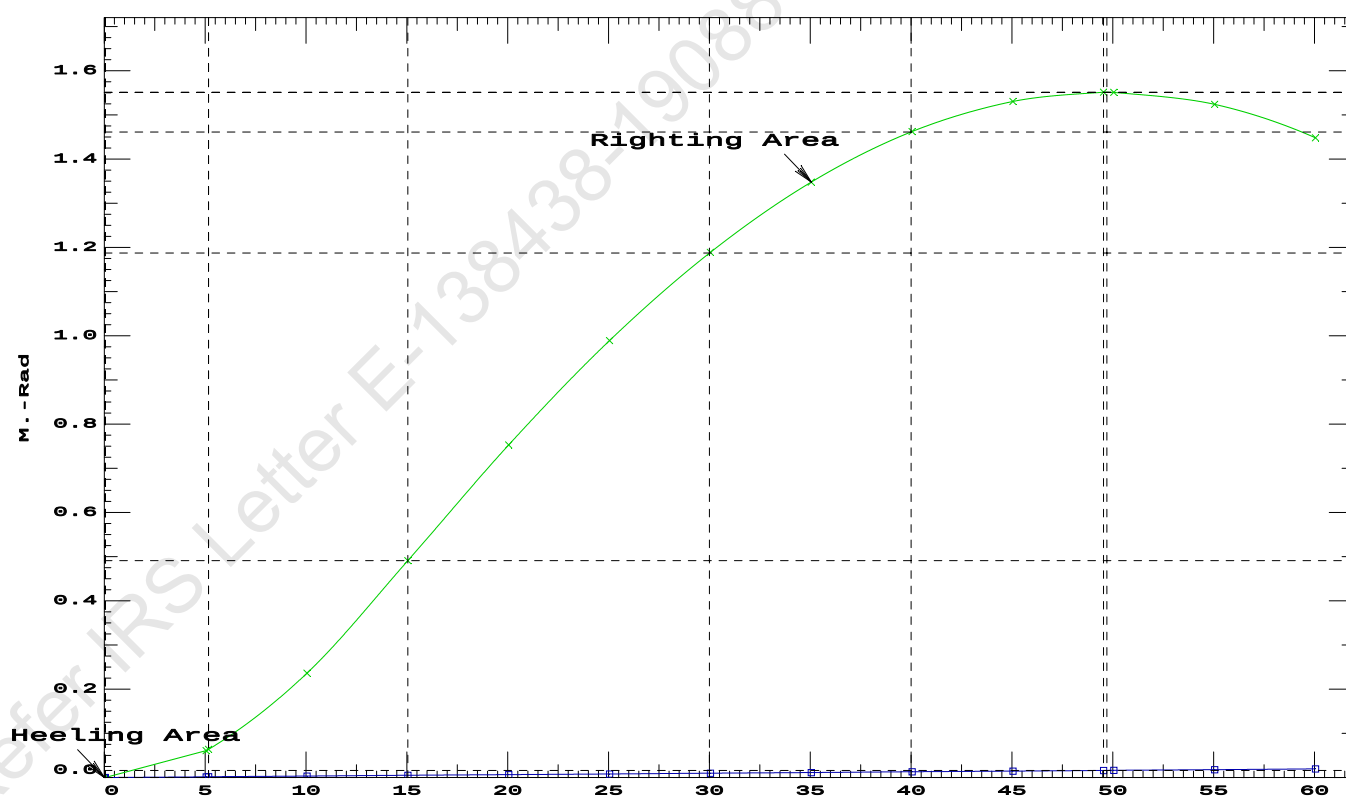
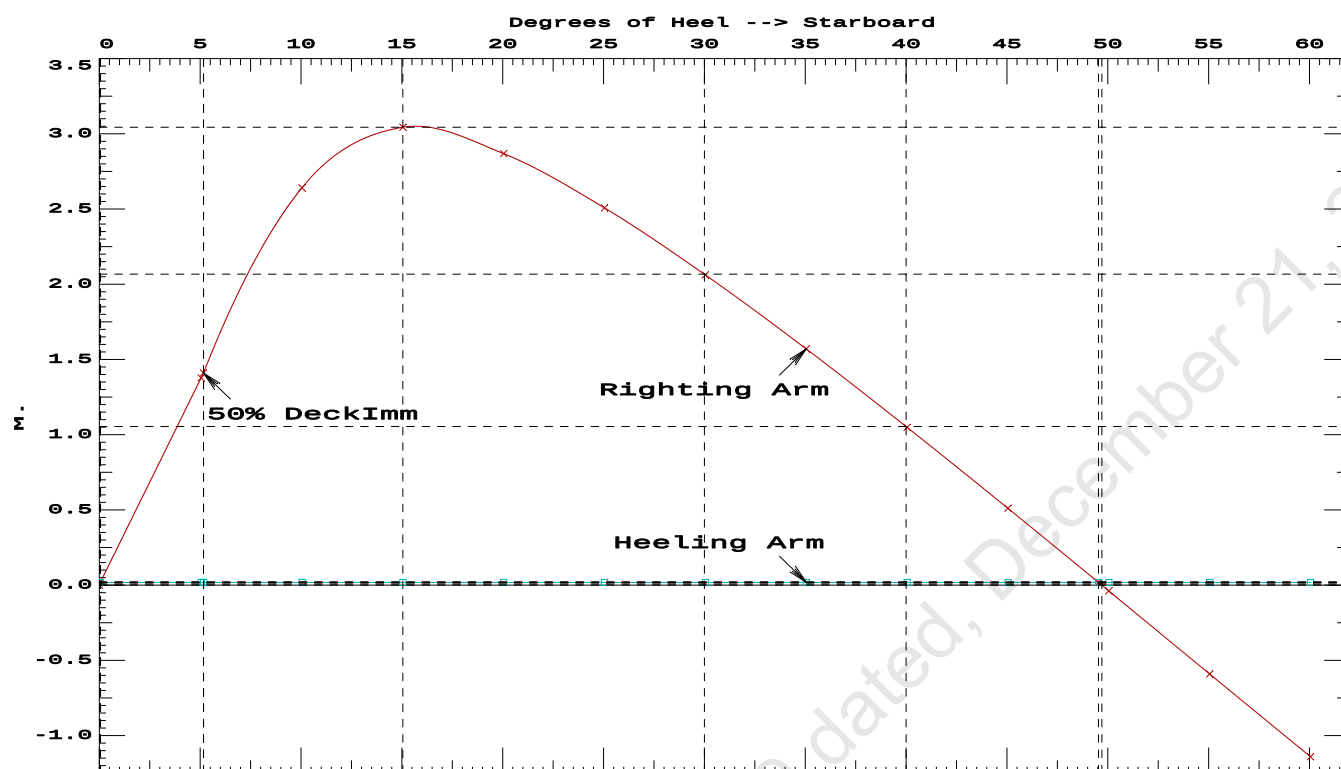
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.66 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.4876 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	49.49 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	5.12 P

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PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd



22-10-21 12:43:42 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.093f TCG = 0.000 VCG = 5.289

Free Surface Adjustment: 0.404

Adjusted CG: LCG = 25.093f TCG = 0.014p VCG = 5.692

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---	Trim----	Heel----	Weight(MT)---	in Trim--in Heel---> Area
1.377	0.04f	2.02s	1,164.3	0.000 0.000 0.0000
1.353	0.04f	7.02s	1,164.4	0.000 1.352 0.0590
1.266	0.05f	12.02s	1,164.3	0.000 2.376 0.2241
1.181	0.06f	15.27s	1,164.5	0.000 2.490 0.3640
1.134	0.07f	17.02s	1,164.3	0.000 2.457 0.4397
0.992	0.09f	22.02s	1,164.3	0.000 2.187 0.6463
0.842	0.11f	27.02s	1,164.3	0.000 1.786 0.8206
0.686	0.14f	32.02s	1,164.3	0.000 1.319 0.9566
0.523	0.16f	37.02s	1,164.3	0.000 0.813 1.0498
0.357	0.18f	42.02s	1,164.3	0.000 0.283 1.0979
0.268	0.19f	44.63s	1,164.3	0.000 0.000 1.1043
0.187	0.20f	47.02s	1,164.3	0.000 -0.261 1.0989
0.016	0.22f	52.02s	1,164.3	0.000 -0.813 1.0522
-0.155	0.24f	57.02s	1,164.3	0.000 -1.366 0.9571
-0.325	0.26f	62.02s	1,164.3	0.000 -1.915 0.8139

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

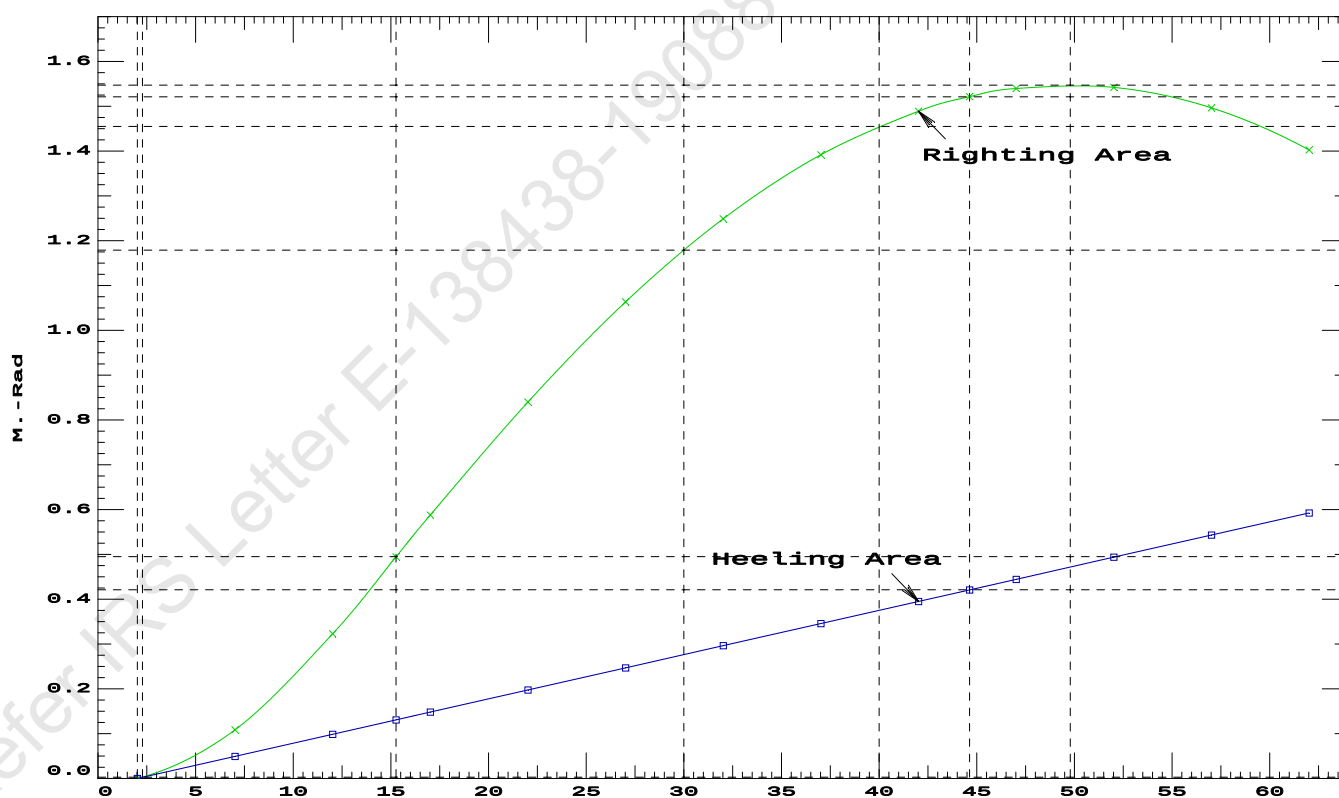
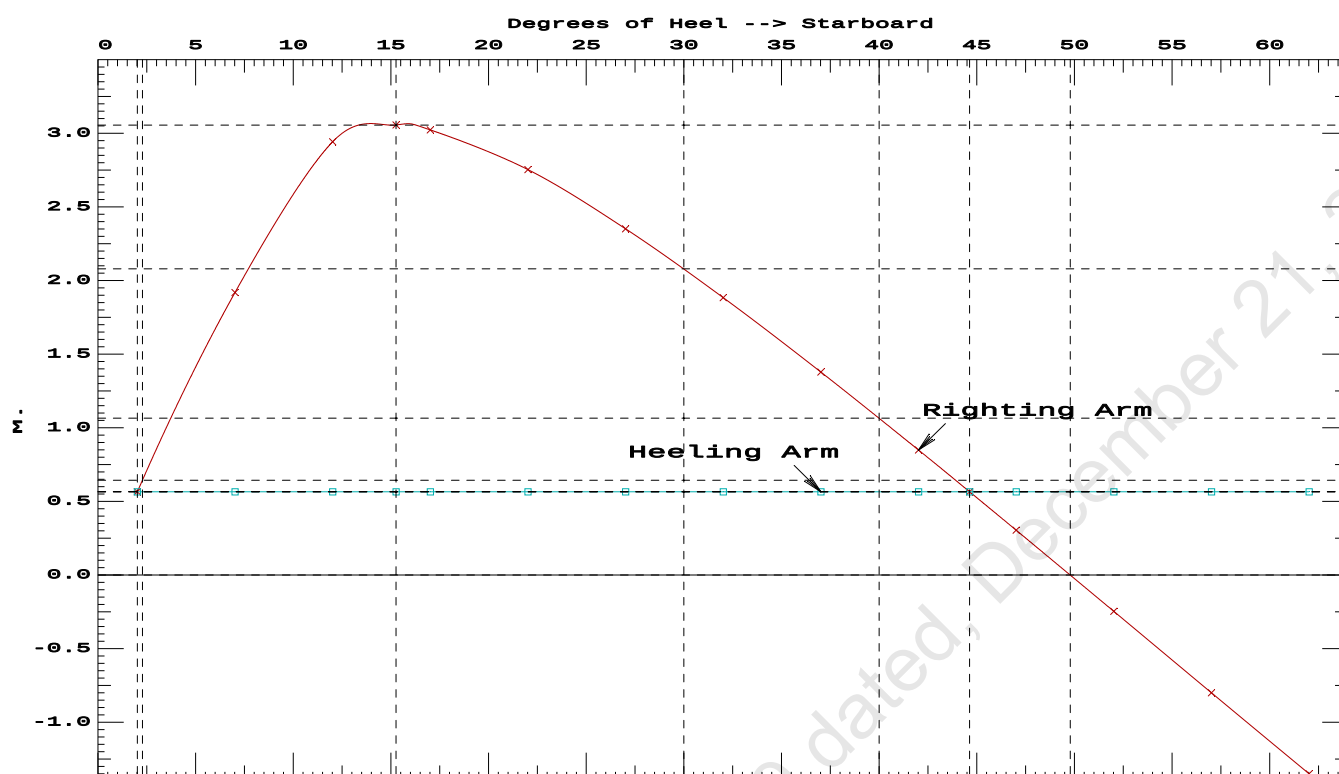
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	2.02 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	440.1 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	3.624 P

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GHS 17.22 PONT00N

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd



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GHS 17.22 PONTON

Condition 11 : Operating Condition of 250T Crane**Crane Lifting 36.6T @ 18m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.424 @ 50.00f, 1.407 @ 25.00f, 1.390 @ 0.00

Trim: Fwd 0.033/50.000, Heel: Stbd 2.02 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed----->			741.94	18.196f	0.000	7.391	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,164.34	25.093f	0.000	5.289	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,164.34	25.096f	0.727s	0.725	

	Righting Arms:			0.000	0.566s		
	External Arms:			0.000	0.566s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		859.4	25.033f	0.100s	143.37	20.007
			MT/cm				
			8.81				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.02

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After hook load drop									
WEIGHT and DISPLACEMENT and WATERPLANE STATUS									
USK draft: 1.482 @ 50.00f, 1.368 @ 25.00f, 1.254 @ 0.00									
Trim: Fwd 0.228/50.000, Heel: 0.00 deg.									
Part-----	Weight(MT)----			LCG-----	TCG-----	VCG-----			
LIGHT SHIP	455.34			24.928f	0.000	3.000			
Crane	250.00			7.500f	0.000	5.200			
Total Fixed----->	705.34			18.751f	0.000	3.780			
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----		
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*		
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*		
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*		
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*		
Total Tanks----->	422.40			37.208f	0.000	1.596	470.09		
Total Weight----->	1,127.74			25.664f	0.000	2.962			
	Displ(MT)----			LCB-----	TCB-----	VCB-----			
HULL	1.025		1,127.74	25.674f	0.001s	0.692			

	Righting Arms:			0.000	0.001s				
	External Arms:			0.000	0.001				
	Residual Righting Arms:			0.000	0.000				
Part-----	SpGr-----			WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	856.2			25.280f	0.021s	146.45	20.576	
	MT/cm-----			m.-	MT/cm-----	GML-----	GMT-----		
	8.78				32.52	144.18	18.306		
Distances in METERS.-----Moments in m.-MT.									

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 13415.21; t3: 30.00; t: 15.00
limmarg: 4964.20; t3: 15.00; t: 7.35
limmarg: 1329.18; t3: 7.65; t: 3.60
limmarg: 302.29; t3: 4.05; t: 1.76
limmarg: 28.89; t3: 2.29; t: 0.86
limmarg: -49.60; t3: 1.43; t: 0.42
limmarg: -15.81; t3: 1.85; t: 0.21
limmarg: 4.34; t3: 2.06; t: 0.10
limmarg: -5.53; t3: 1.96; t: 0.05
limmarg: -0.65; t3: 2.01; t: 0.02
limmarg: 1.33; t3: 2.03; t: 0.01

Theta3 is 2.03 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.751f TCG = 0.000 VCG = 3.780

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.238	0.26f	2.02s	1,127.7	0.000	-0.669	0.0000
1.242	0.26f	0.00	1,127.7	0.000	0.000	-0.0118
1.238	0.26f	2.03p	1,127.7	0.000	0.674	0.0002
1.227	0.26f	5.00p	1,127.7	0.000	1.654	0.0605
1.164	0.28f	10.00p	1,127.7	0.000	3.181	0.2722
0.972	0.41f	15.00p	1,127.7	0.000	3.838	0.5849
0.822	0.51f	18.59p	1,128.0	0.000	3.910	0.8295
0.762	0.55f	20.00p	1,127.7	0.000	3.899	0.9254
0.543	0.69f	25.00p	1,127.7	0.000	3.760	1.2620
0.318	0.83f	30.00p	1,127.7	0.000	3.524	1.5806

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 705.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to abs 2 > 1.000 1.013 P

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.751f TCG = 0.000 VCG = 3.780

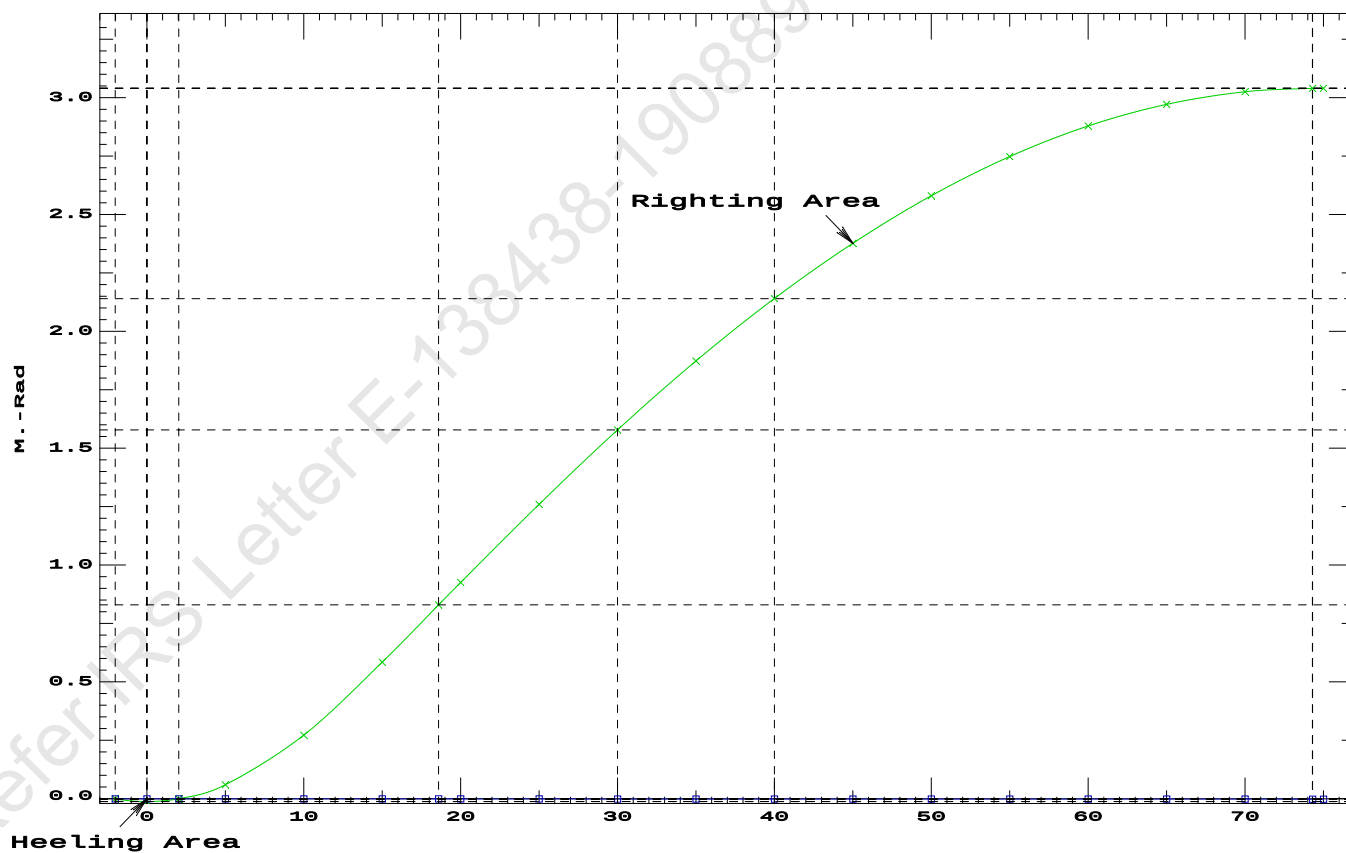
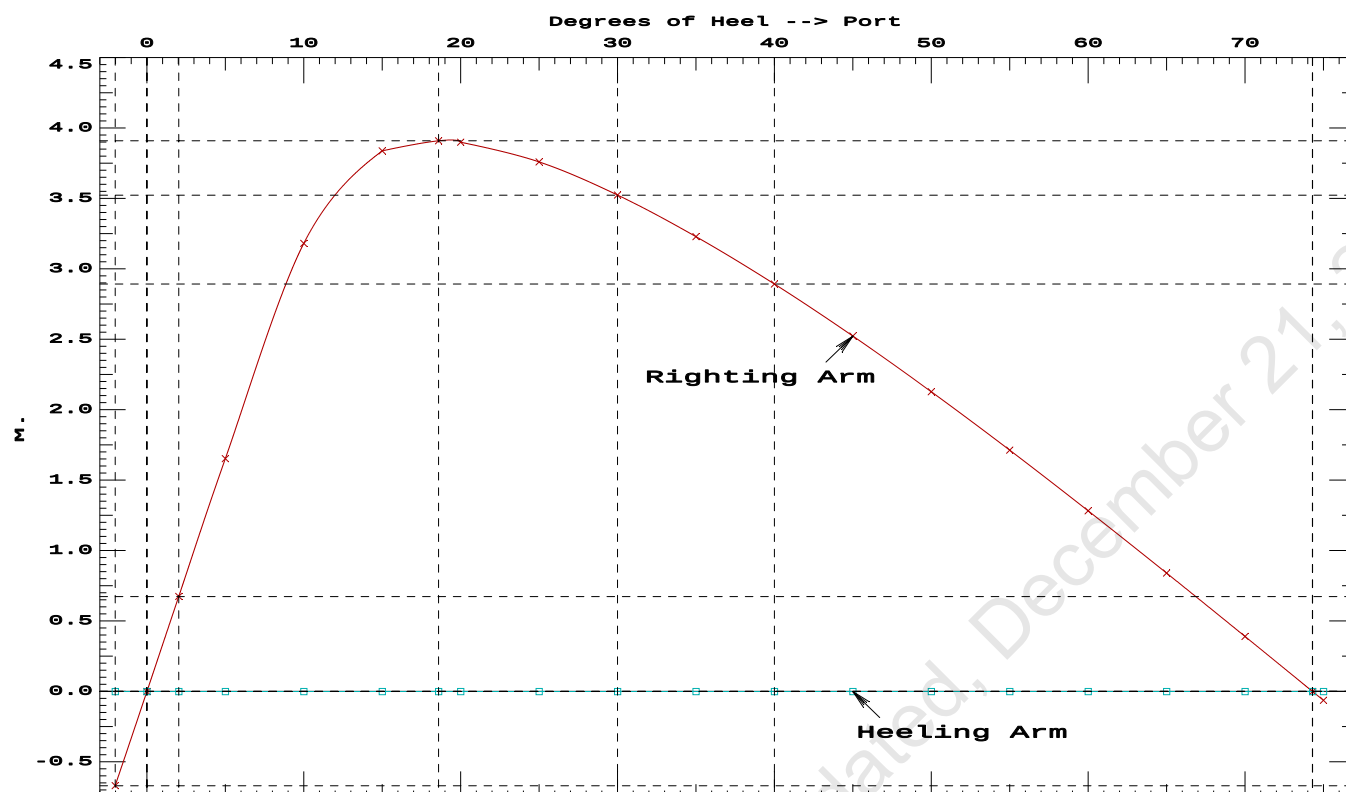
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.238	0.26f	2.02s	1,127.7	0.000	-0.669	0.0000
1.242	0.26f	0.00	1,127.7	0.000	0.000	-0.0118
1.238	0.26f	2.03p	1,127.7	0.000	0.674	0.0002
1.227	0.26f	5.00p	1,127.7	0.000	1.654	0.0605
1.164	0.28f	10.00p	1,127.7	0.000	3.181	0.2722
0.972	0.41f	15.00p	1,127.7	0.000	3.838	0.5849
0.822	0.51f	18.59p	1,128.0	0.000	3.910	0.8295
0.762	0.55f	20.00p	1,127.7	0.000	3.899	0.9254
0.543	0.69f	25.00p	1,127.7	0.000	3.760	1.2620
0.318	0.83f	30.00p	1,127.7	0.000	3.524	1.5806
0.087	0.97f	35.00p	1,127.7	0.000	3.229	1.8757
-0.146	1.11f	40.00p	1,127.7	0.000	2.892	2.1431
-0.381	1.25f	45.00p	1,127.7	0.000	2.523	2.3796
-0.615	1.38f	50.00p	1,127.7	0.000	2.129	2.5828
-0.845	1.51f	55.00p	1,127.7	0.000	1.714	2.7506
-1.069	1.62f	60.00p	1,127.7	0.000	1.283	2.8815
-1.285	1.72f	65.00p	1,127.7	0.000	0.841	2.9742
-1.489	1.80f	70.00p	1,127.7	0.000	0.391	3.0280
-1.654	1.86f	74.31p	1,127.7	0.000	0.000	3.0428
-1.680	1.87f	75.00p	1,127.7	0.000	-0.063	3.0424

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 705.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to RAZero > 1.000 259.249 P
(2) Angle from abs 2 deg to RAZero > 20.00 deg 72.28 P

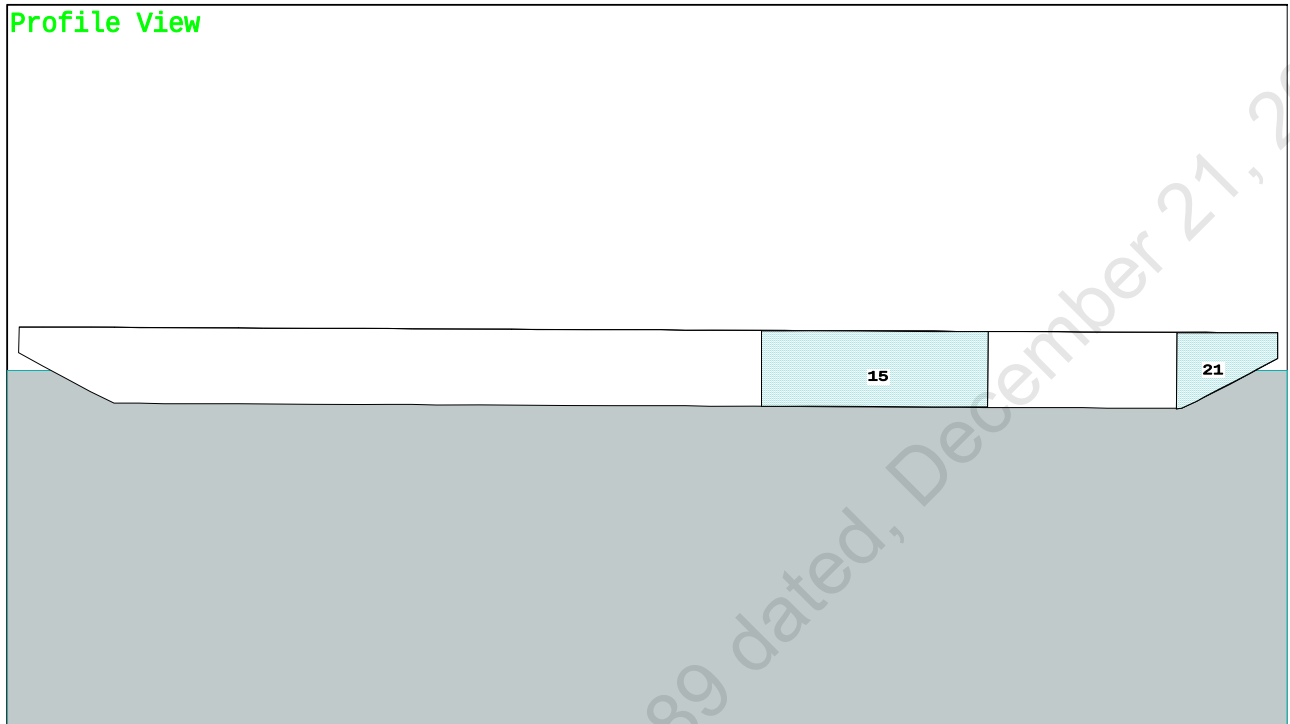
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PONT00N

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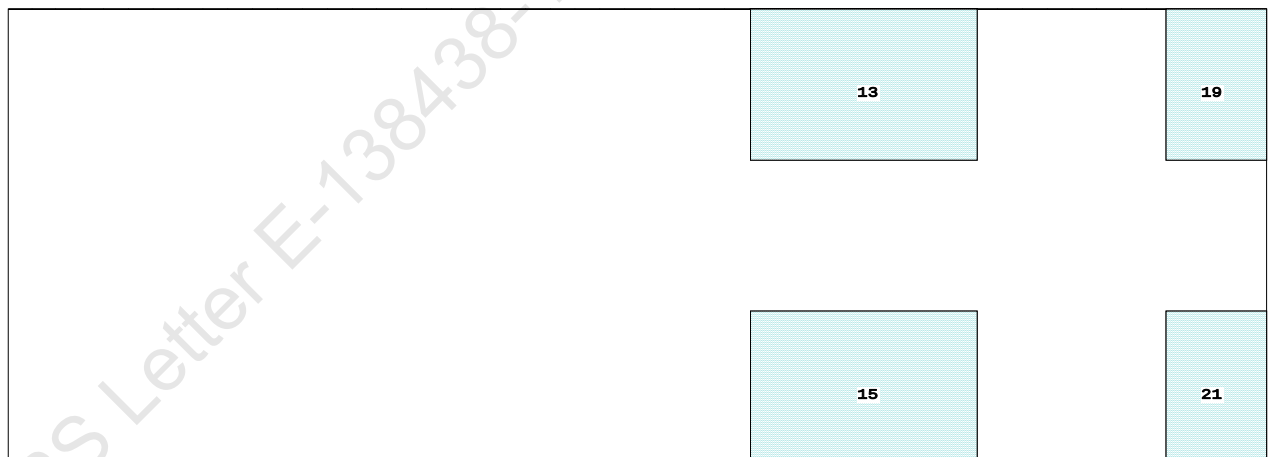
Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

Condition Graphic - Draft: 1.527 @ 50.000f, 1.292 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S 19 07_WBTK.P 21 07_WBTK.S

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.527 @ 50.00f, 1.409 @ 25.00f, 1.292 @ 0.00

Trim: Fwd 0.235/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Total Fixed----->			741.94	19.084f	0.000	7.391	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,164.34	25.659f	0.000	5.289	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,164.30	25.680f	0.000	0.714	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	861.8	25.205f	0.000	144.62	20.083
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.83		32.61	140.04	15.508
Distances in METERS.-----				Moments in m.-MT-----			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.527
DRAFT AT MS (M) = 1.409
DRAFT AT AP (M) = 1.292
DRAFT AT FWD MARK (M) = 1.508
DRAFT AT AFT MARK (M) = 1.310

HYDROSTATIC PROPERTIES

Trim: Fwd 0.235/50.000, No Heel, VCG = 5.289

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT-----
1.410	1,164.30	25.680f	0.714	8.83	25.205f	32.61	140.04	15.508
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.22

PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.527 @ 50.00f, 1.409 @ 25.00f, 1.292 @ 0.00

Trim: Fwd 0.235/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.9	0.814	1.504	0.05500	6.61
CRANE	57.0	4.173	4.863	0.05500	15.25
Total wind heeling moment to starboard-->					21.86
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.659f TCG = 0.000 VCG = 5.289

Free Surface Adjustment: 0.404

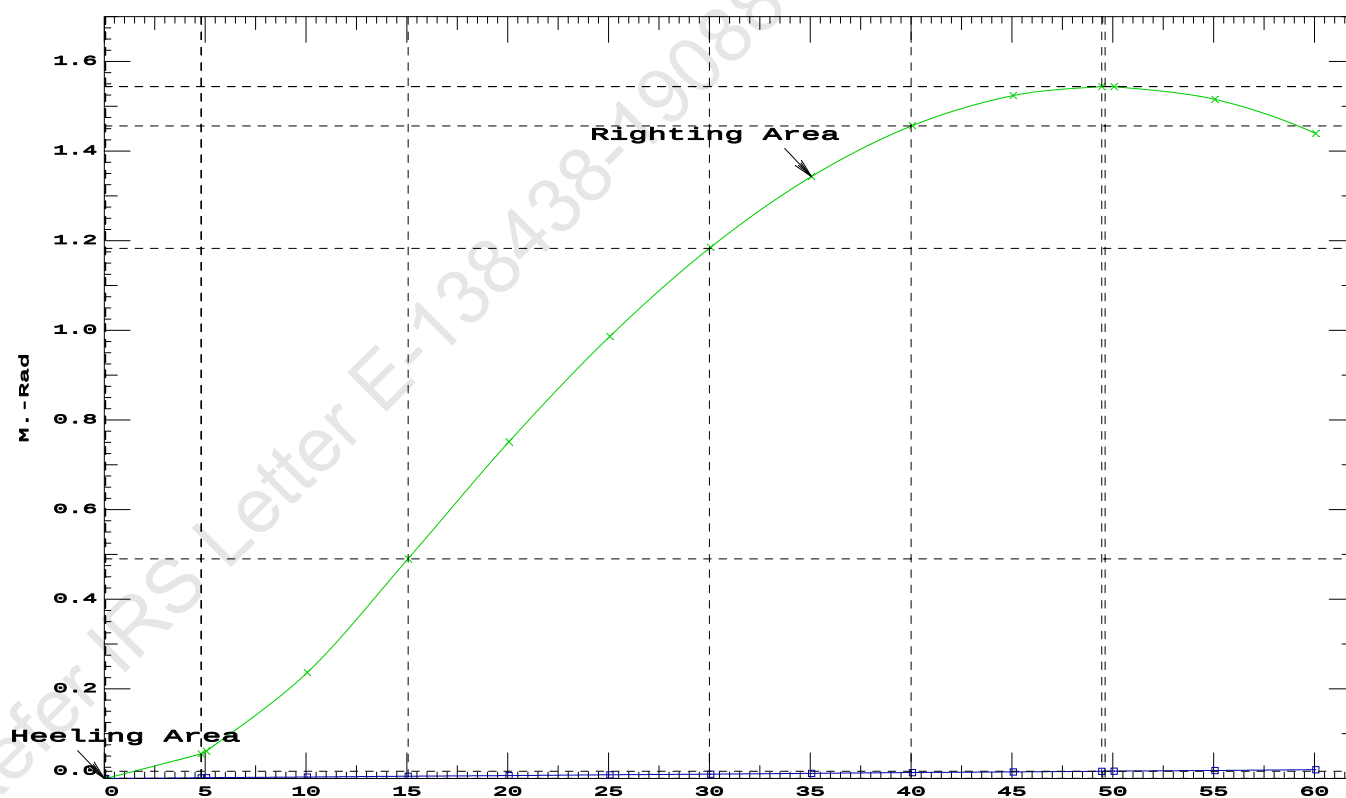
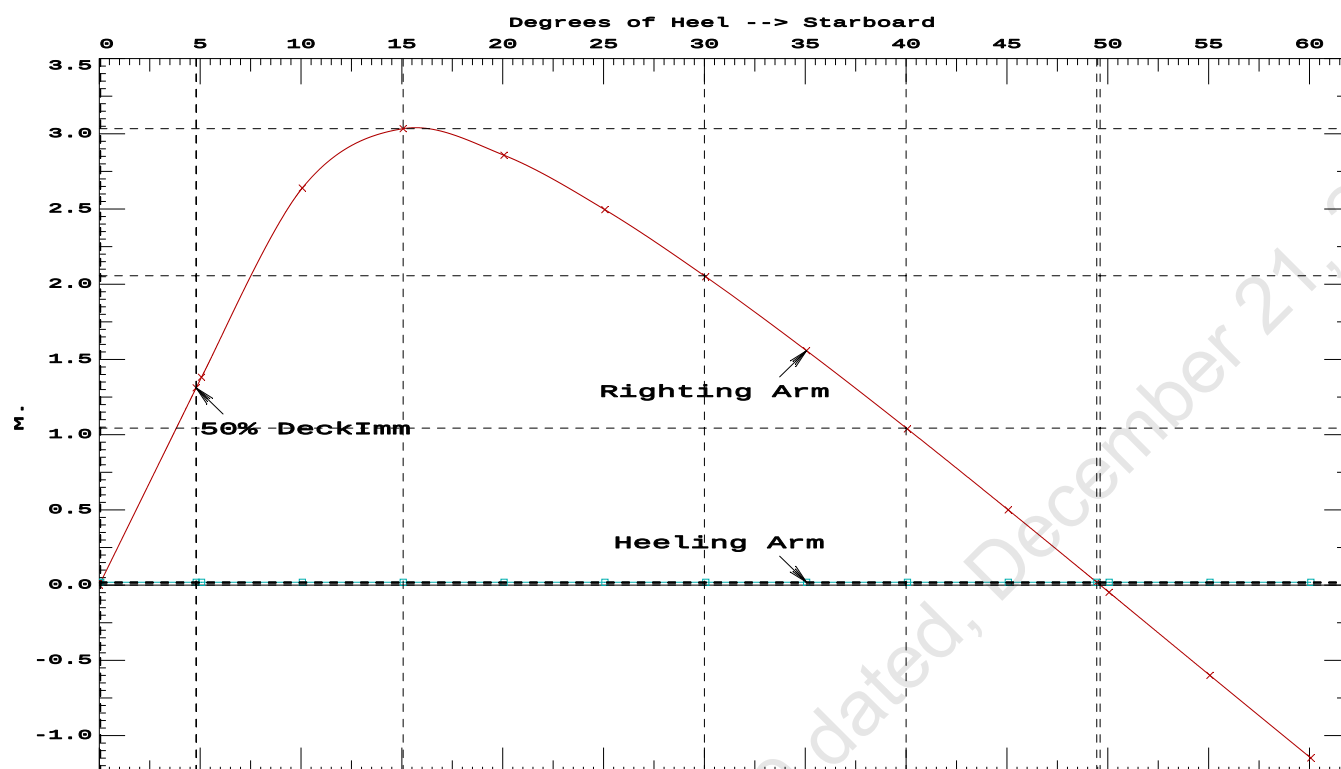
Adjusted CG: LCG = 25.657f TCG = 0.000 VCG = 5.692

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.280	0.27f	0.00	1,164.4	0.000 -0.018	0.0000	4.803
1.280	0.27f	0.06s	1,164.4	0.000 0.000	-0.0000	4.741
1.266	0.27f	4.80s	1,164.3	0.000 1.293	0.0535	0.000
1.264	0.27f	5.06s	1,164.3	0.000 1.363	0.0595	-0.259
1.204	0.29f	10.06s	1,164.3	0.000 2.621	0.2340	-5.259
1.024	0.43f	15.06s	1,164.3	0.000 3.015	0.4872	-10.259
0.829	0.59f	20.06s	1,164.3	0.000 2.839	0.7468	-15.259
0.624	0.74f	25.06s	1,164.3	0.000 2.477	0.9801	-20.259
0.410	0.90f	30.06s	1,164.3	0.000 2.032	1.1774	-25.259
0.190	1.06f	35.06s	1,164.3	0.000 1.540	1.3336	-30.259
-0.034	1.22f	40.06s	1,164.3	0.000 1.019	1.4455	-35.259
-0.260	1.37f	45.06s	1,164.3	0.000 0.481	1.5110	-40.259
-0.458	1.50f	49.45s	1,164.3	0.000 0.000	1.5295	-44.649
-0.485	1.52f	50.06s	1,164.3	0.000 -0.067	1.5291	-45.259
-0.706	1.65f	55.06s	1,164.3	0.000 -0.619	1.4993	-50.259
-0.920	1.77f	60.06s	1,164.3	0.000 -1.168	1.4213	-55.259
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.86 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.4872 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	49.39 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.74 P

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GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTOONCondition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

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GHS 17.22

PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.659f TCG = 0.000 VCG = 5.289
Free Surface Adjustment: 0.404
Adjusted CG: LCG = 25.657f TCG = 0.000 VCG = 5.692

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. in Heel	Res. Area
1.279	0.27f	0.01s	1,164.4	0.000	0.000	0.0000
1.264	0.27f	5.01s	1,164.3	0.000	1.363	0.0595
1.205	0.29f	10.01s	1,164.3	0.000	2.625	0.2342
1.026	0.43f	15.01s	1,164.3	0.000	3.029	0.4883
0.831	0.58f	20.01s	1,164.3	0.000	2.856	0.7493
0.626	0.74f	25.01s	1,164.3	0.000	2.495	0.9841
0.412	0.90f	30.01s	1,164.3	0.000	2.051	1.1831
0.192	1.06f	35.01s	1,164.3	0.000	1.559	1.3409
-0.032	1.22f	40.01s	1,164.3	0.000	1.039	1.4545
-0.258	1.37f	45.01s	1,164.3	0.000	0.501	1.5218
-0.464	1.50f	49.58s	1,164.3	0.000	0.000	1.5418
-0.483	1.52f	50.01s	1,164.3	0.000	-0.047	1.5416
-0.704	1.65f	55.01s	1,164.3	0.000	-0.599	1.5135
-0.918	1.77f	60.01s	1,164.3	0.000	-1.148	1.4373

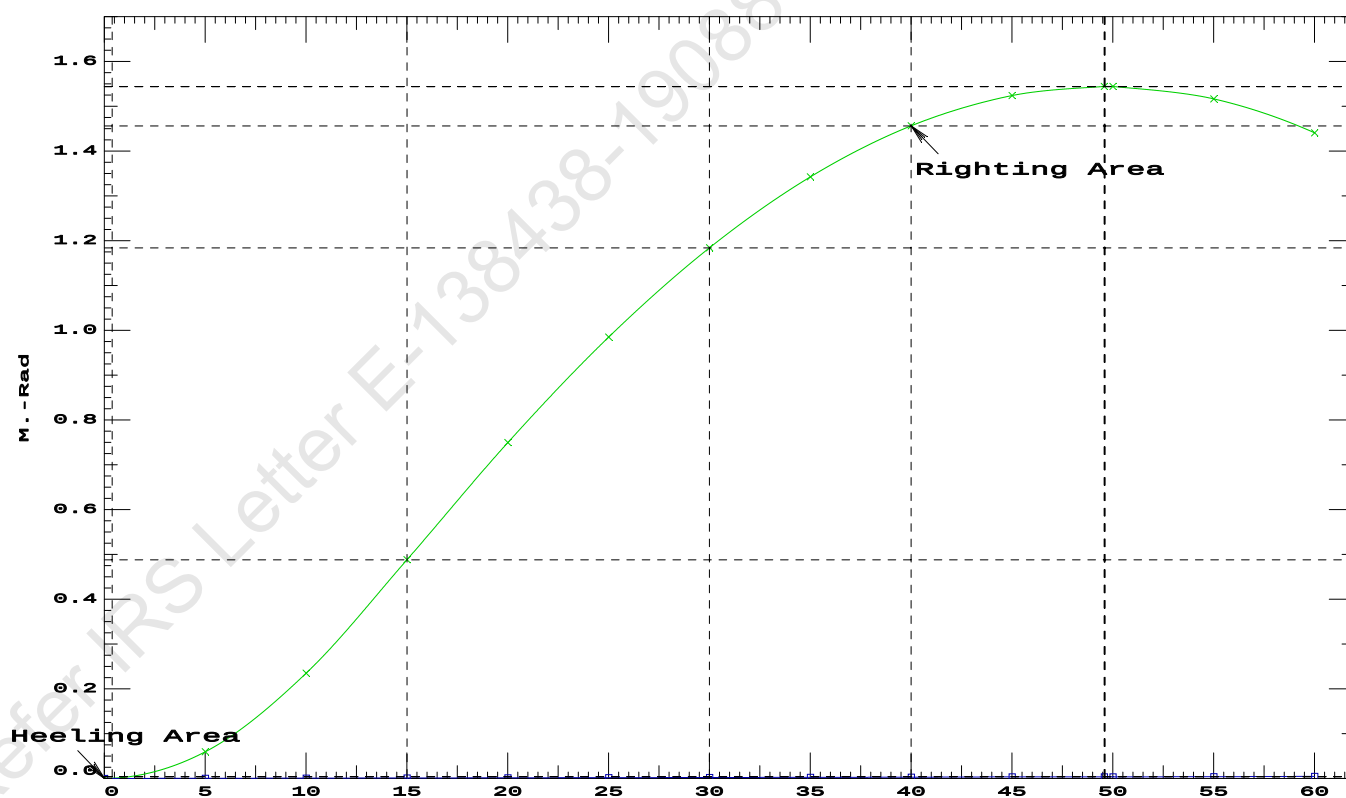
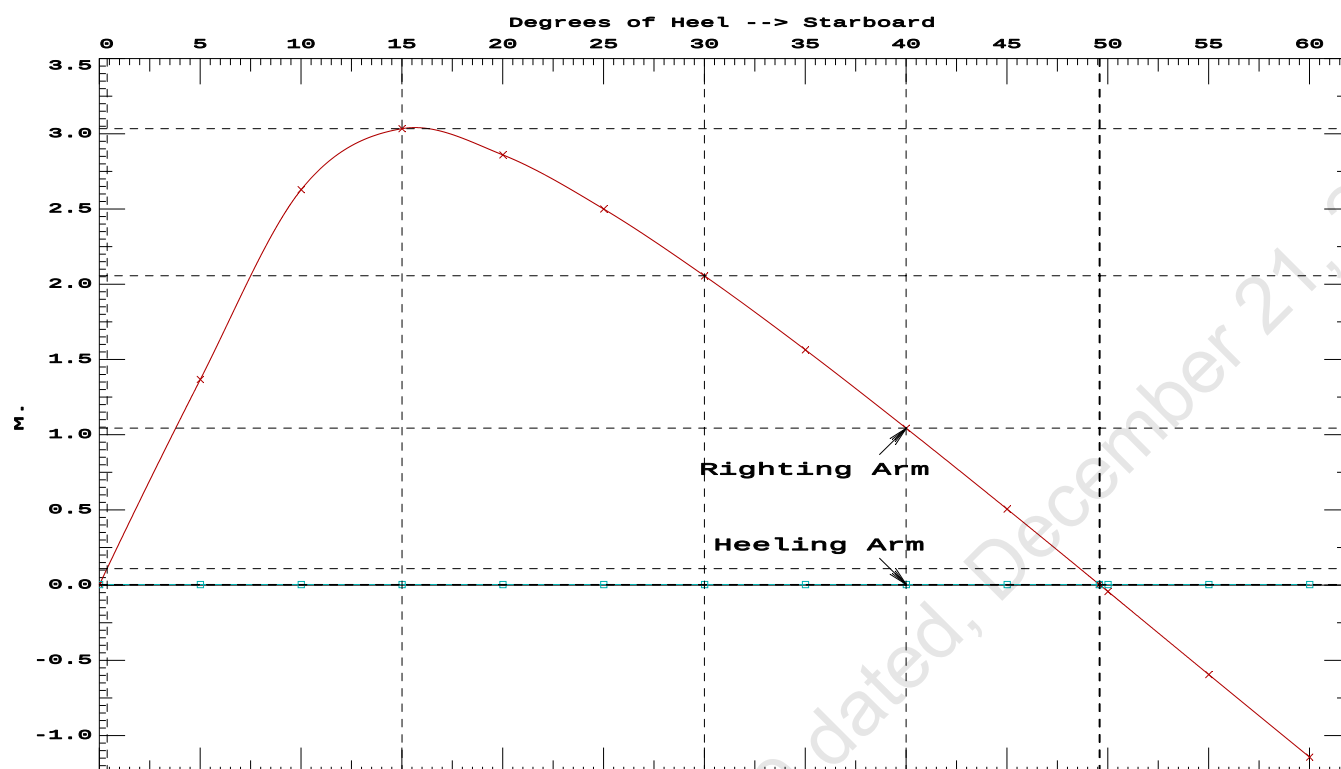
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.01 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	70547.8 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	415.991 P

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GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTOONCondition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

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GHS 17.22 PONT00N

Condition 12 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.527 @ 50.00f, 1.409 @ 25.00f, 1.292 @ 0.00

Trim: Fwd 0.235/50.000, Heel: Stbd 0.01 deg.

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	455.34	24.928f	0.000	3.000	
Crane	250.00	7.500f	0.000	5.200	
Crane Load	36.60	25.500f	0.000	76.977	
Total Fixed----->	741.94	19.084f	0.000	7.391	
Load-----	SpGr-----	Weight(MT)	LCG	TCG	VCG
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		422.40	37.208f	0.000	1.596
Total Weight----->		1,164.34	25.659f	0.000	5.289
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,164.34	25.680f	0.005s	0.714

Righting Arms:		0.000	0.004s		
External Arms:		0.000	0.004s		
Residual Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA	LCF	TCF	BML
Total Waterplane----->	1.025	858.2	25.302f	0.026s	142.83
		MT/cm			
		8.80			

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

PONTOON

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.751f TCG = 0.000 VCG = 3.780

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.242	0.26f	0.01s	1,127.8	0.000	-0.004	0.0000
1.242	0.26f	0.00	1,127.8	0.000	0.000	-0.0000
1.239	0.26f	0.97p	1,127.7	0.000	0.324	0.0027
1.227	0.26f	5.00p	1,127.7	0.000	1.654	0.0723
1.164	0.28f	10.00p	1,127.7	0.000	3.181	0.2842
0.972	0.41f	15.00p	1,127.7	0.000	3.838	0.5968
0.822	0.51f	18.59p	1,128.0	0.000	3.910	0.8415
0.762	0.55f	20.00p	1,127.7	0.000	3.899	0.9374
0.543	0.69f	25.00p	1,127.7	0.000	3.760	1.2740
0.318	0.83f	30.00p	1,127.7	0.000	3.524	1.5925

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 705.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 7490.49 P

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.751f TCG = 0.000 VCG = 3.780

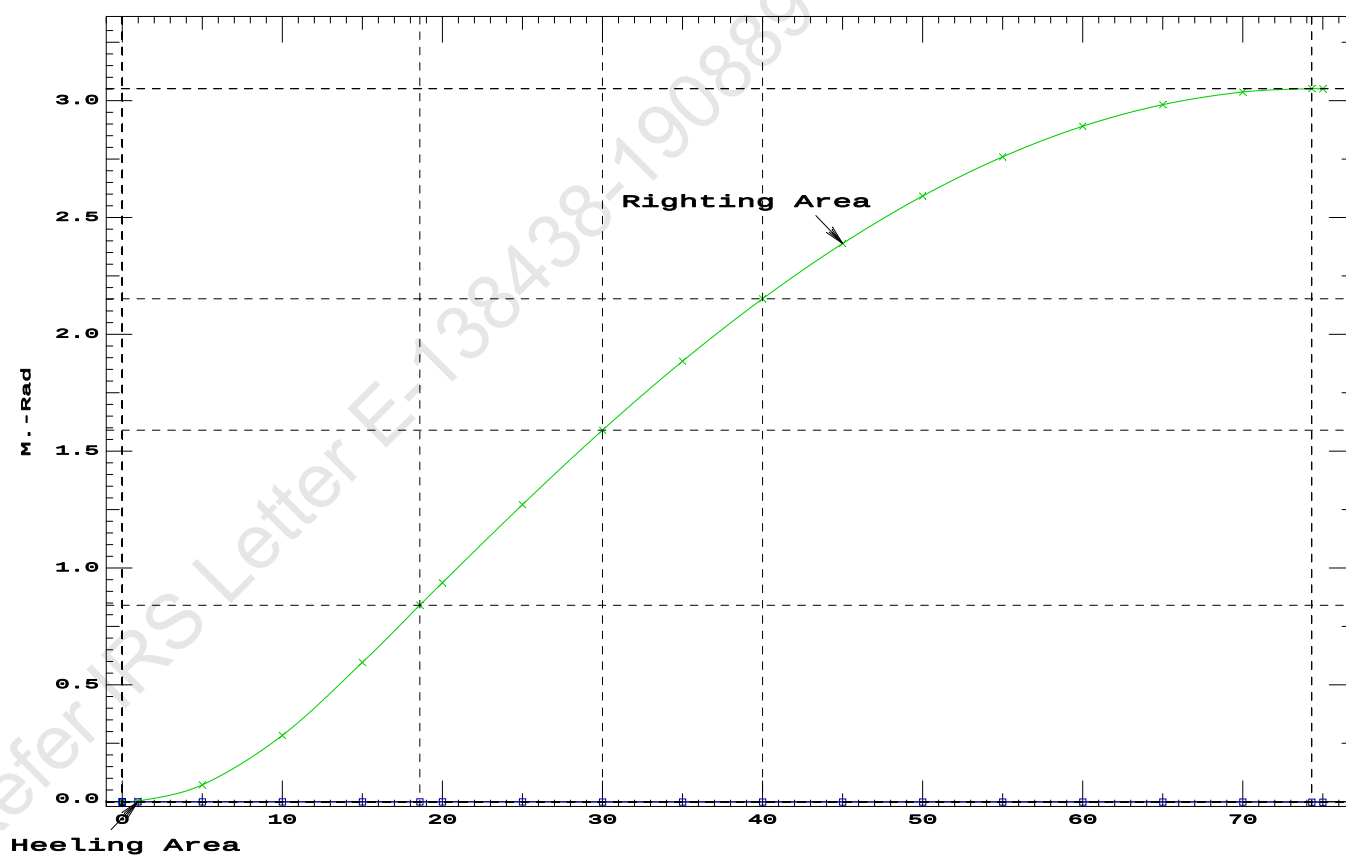
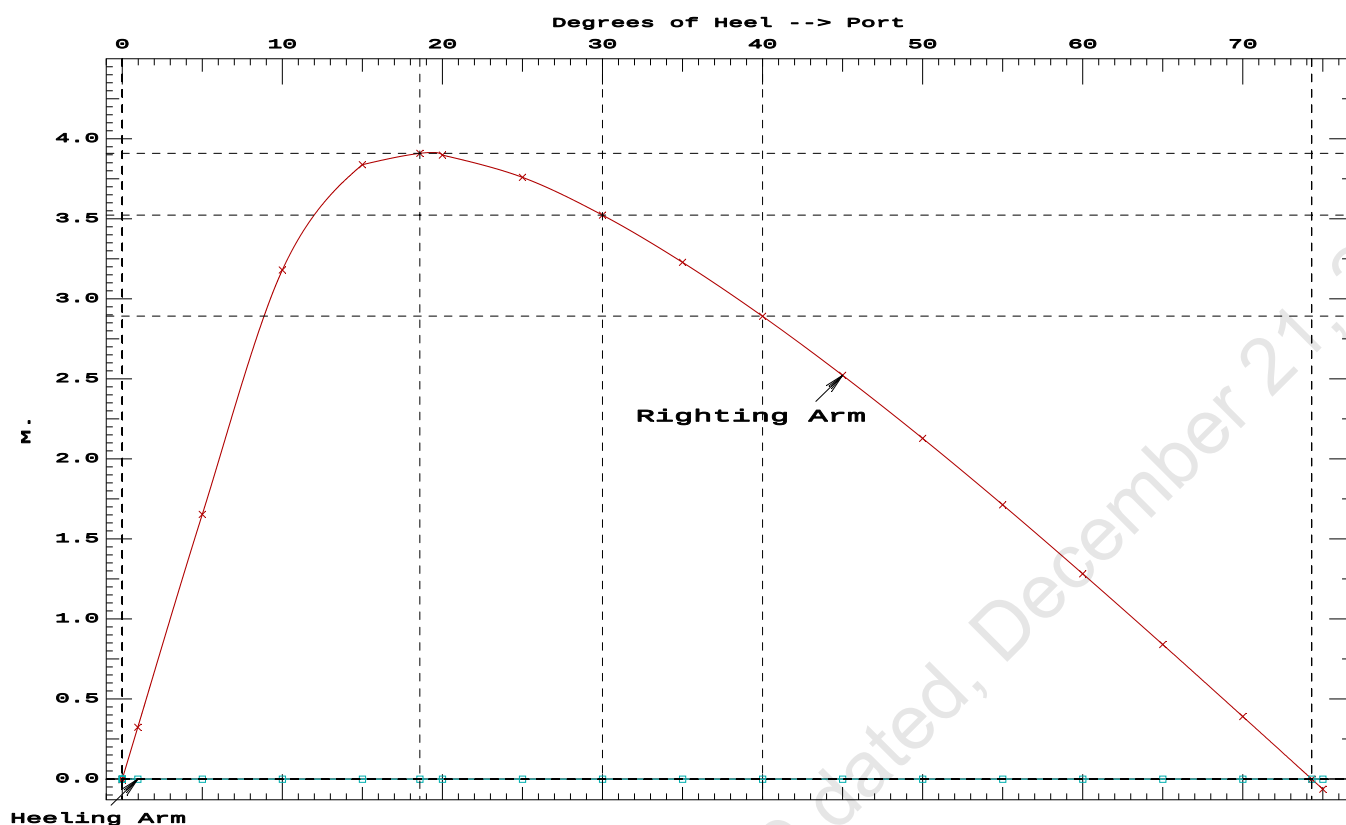
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.242	0.26f	0.01s	1,127.8	0.000	-0.004	0.0000
1.242	0.26f	0.00	1,127.8	0.000	0.000	-0.0000
1.239	0.26f	0.97p	1,127.7	0.000	0.324	0.0027
1.227	0.26f	5.00p	1,127.7	0.000	1.654	0.0723
1.164	0.28f	10.00p	1,127.7	0.000	3.181	0.2842
0.972	0.41f	15.00p	1,127.7	0.000	3.838	0.5968
0.822	0.51f	18.59p	1,128.0	0.000	3.910	0.8415
0.762	0.55f	20.00p	1,127.7	0.000	3.899	0.9374
0.543	0.69f	25.00p	1,127.7	0.000	3.760	1.2740
0.318	0.83f	30.00p	1,127.7	0.000	3.524	1.5925
0.087	0.97f	35.00p	1,127.7	0.000	3.229	1.8876
-0.146	1.11f	40.00p	1,127.7	0.000	2.892	2.1550
-0.381	1.25f	45.00p	1,127.7	0.000	2.523	2.3915
-0.615	1.38f	50.00p	1,127.7	0.000	2.129	2.5947
-0.845	1.51f	55.00p	1,127.7	0.000	1.714	2.7625
-1.069	1.62f	60.00p	1,127.7	0.000	1.283	2.8934
-1.285	1.72f	65.00p	1,127.7	0.000	0.841	2.9862
-1.489	1.80f	70.00p	1,127.7	0.000	0.391	3.0400
-1.655	1.86f	74.31p	1,127.7	0.000	0.000	3.0547
-1.680	1.87f	75.00p	1,127.7	0.000	-0.063	3.0543

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 705.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 8344294 P
(2) Angle from abs 1 deg to RAZero > 20.00 deg 73.34 P

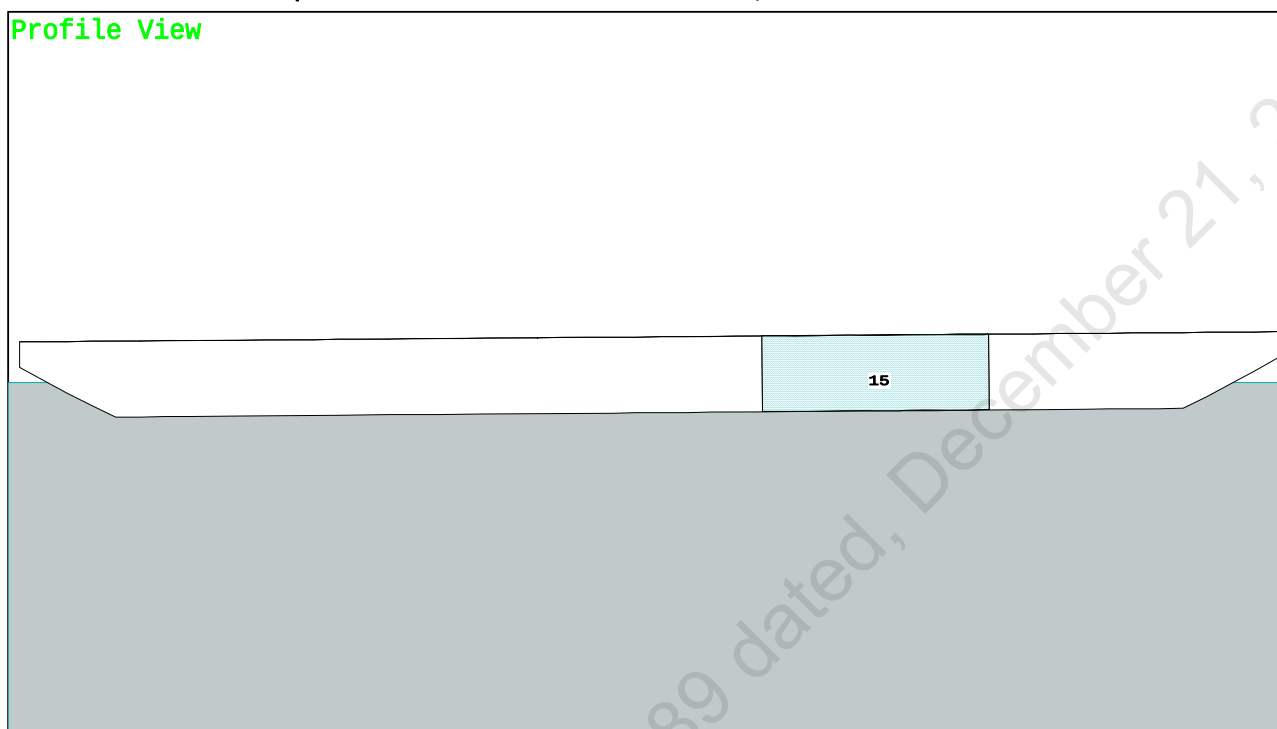
22-10-21 12:43:42
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

22-10-21 12:43:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

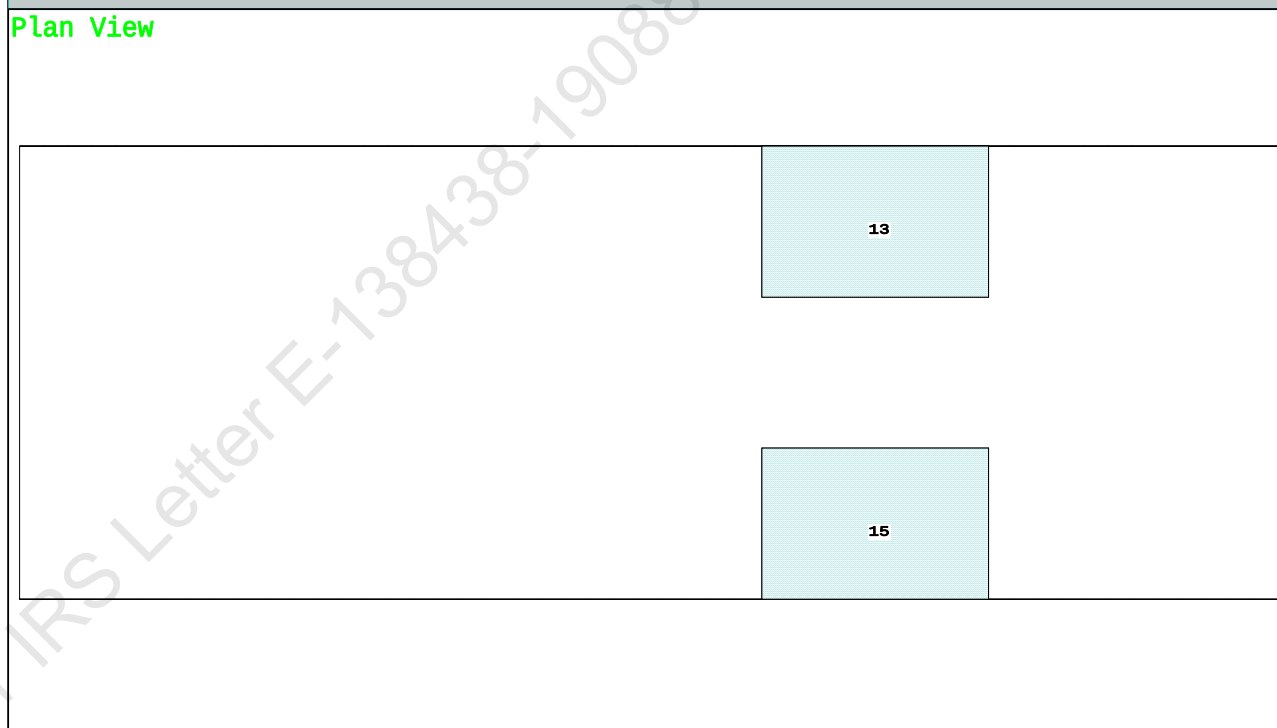
Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

Condition Graphic - Draft: 0.996 @ 50.000f, 1.399 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S

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GHS 17.22

PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.996 @ 50.00f, 1.198 @ 25.00f, 1.399 @ 0.00

Trim: Aft 0.403/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Total Fixed----->			656.64	18.648f	0.000	5.863	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			979.84	23.712f	0.000	4.427	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	979.74	23.681f	0.000	0.608	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	848.0	24.639f	0.000	163.73	23.632
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.69		31.33	159.91	19.812
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 0.996
DRAFT AT MS (M) = 1.198
DRAFT AT AP (M) = 1.399
DRAFT AT FWD MARK (M) = 1.028
DRAFT AT AFT MARK (M) = 1.367

HYDROSTATIC PROPERTIES

Trim: Aft 0.403/50.000, No Heel, VCG = 4.427

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----
1.200	979.74	23.681f	0.608	8.69
				24.639f
				31.33
				159.91
				19.812

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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GHS 17.22

PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.996 @ 50.00f, 1.198 @ 25.00f, 1.399 @ 0.00

Trim: Aft 0.403/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.9	0.925	1.517	0.05500	7.50
CRANE	57.0	4.194	4.786	0.05500	15.00
Total wind heeling moment to starboard-->					22.51
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.712f TCG = 0.000 VCG = 4.427

Free Surface Adjustment: 0.332

Adjusted CG: LCG = 23.714f TCG = 0.000 VCG = 4.759

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.387	0.46a	0.00	979.83	0.000 -0.023	0.0000	5.261
1.387	0.46a	0.06s	979.84	0.000 0.000	-0.0000	5.204
1.372	0.46a	5.06s	979.83	0.000 1.732	0.0756	0.204
1.371	0.46a	5.26s	979.78	0.000 1.801	0.0818	0.000
1.325	0.52a	10.06s	979.73	0.000 3.180	0.2922	-4.796
1.194	0.70a	15.06s	979.84	0.000 3.701	0.5985	-9.796
1.153	0.78a	16.66s	979.84	0.000 3.721	0.7025	-11.402
1.066	0.94a	20.06s	979.84	0.000 3.643	0.9219	-14.796
0.935	1.19a	25.06s	979.84	0.000 3.362	1.2300	-19.796
0.802	1.44a	30.06s	979.84	0.000 2.978	1.5074	-24.796
0.667	1.68a	35.06s	979.84	0.000 2.535	1.7484	-29.796
0.532	1.93a	40.06s	979.84	0.000 2.052	1.9488	-34.796
0.396	2.16a	45.06s	979.84	0.000 1.542	2.1058	-39.796
0.258	2.39a	50.06s	979.84	0.000 1.013	2.2175	-44.796
0.120	2.59a	55.06s	979.84	0.000 0.472	2.2824	-49.796
-0.001	2.76a	59.37s	979.83	0.000 0.000	2.3002	-54.112
-0.020	2.78a	60.06s	979.84	0.000 -0.075	2.2998	-54.796
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

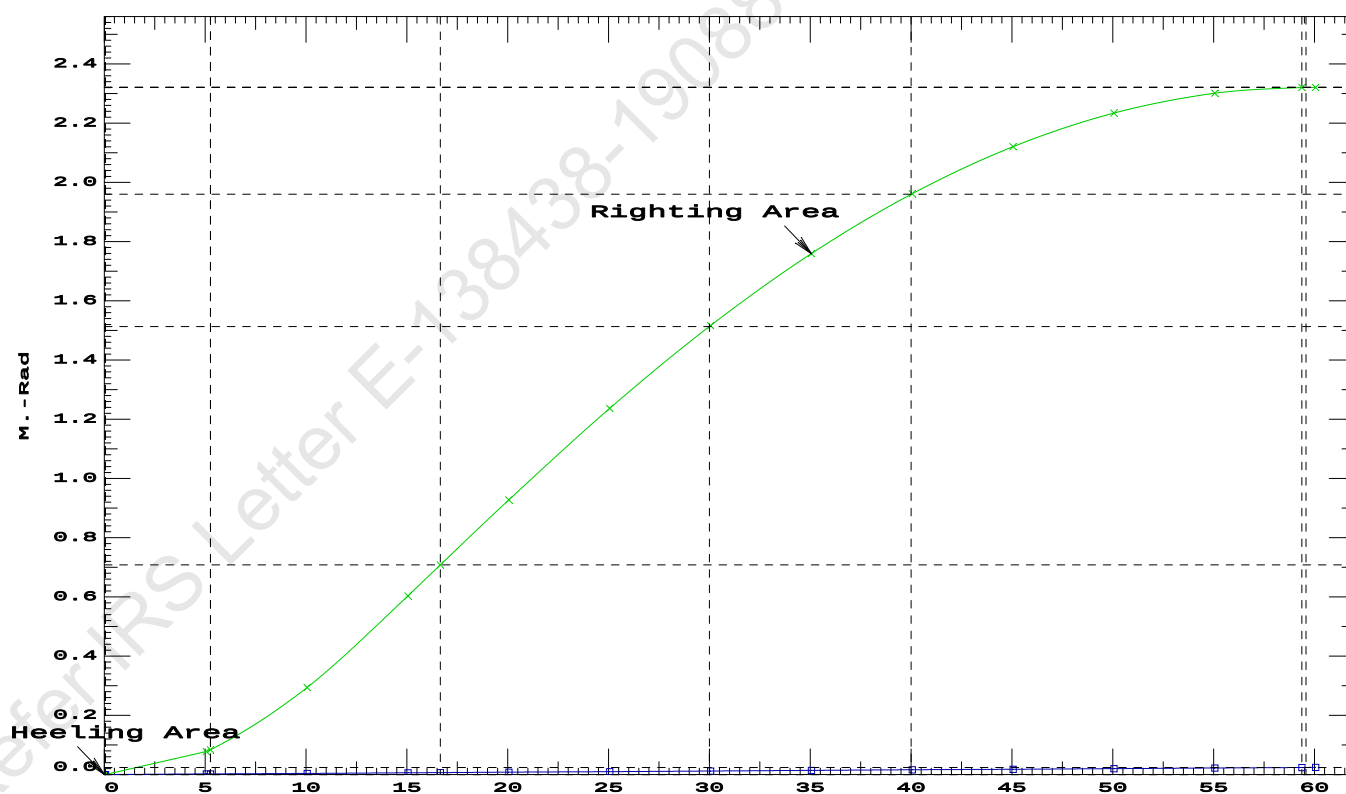
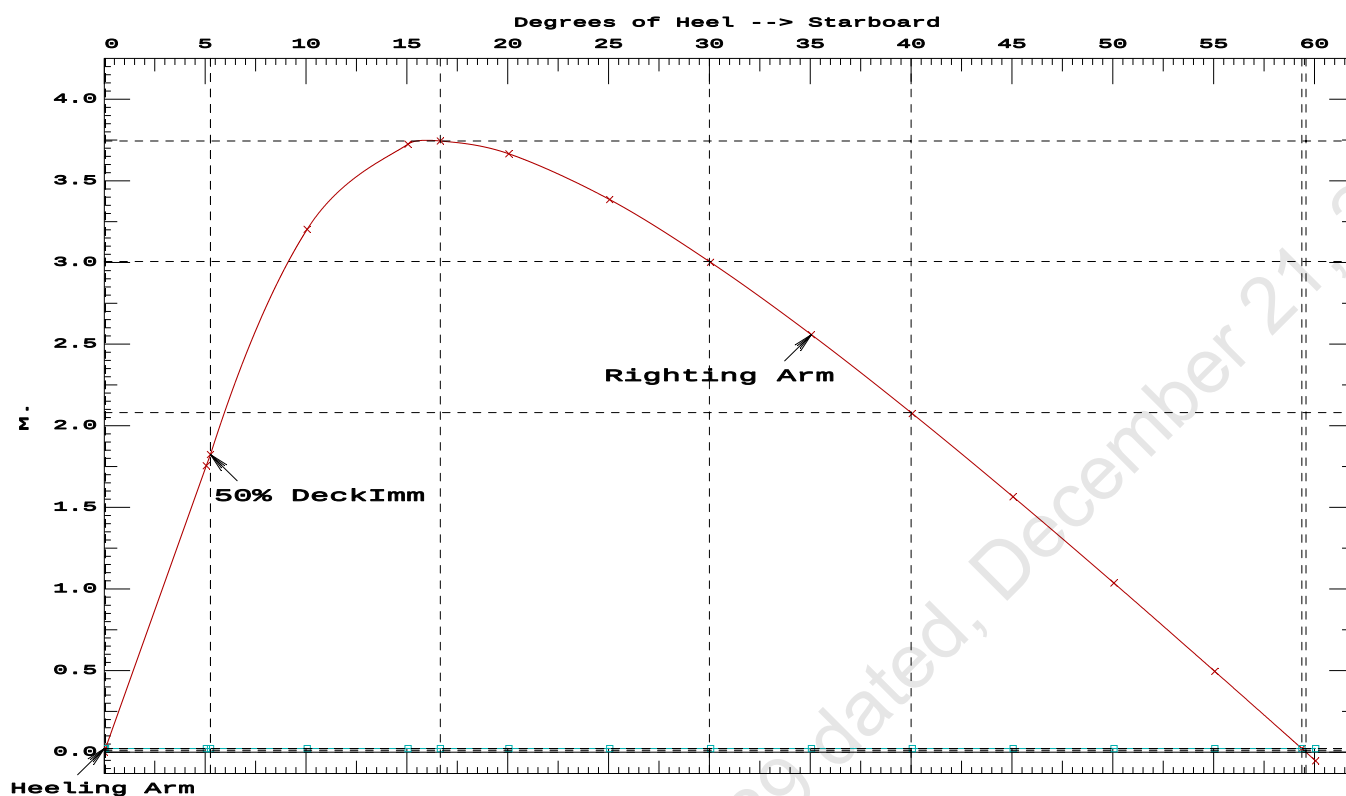
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.51 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.7025 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	59.32 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	5.20 P

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GHS 17.22 PONT00N

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft



22-10-21 12:43:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.712f TCG = 0.000 VCG = 4.427

Free Surface Adjustment: 0.332

Adjusted CG: LCG = 23.714f TCG = 0.000 VCG = 4.759

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area
1.387	0.46a	0.01s	979.84	0.000	0.000	0.0000
1.373	0.46a	5.01s	979.84	0.000	1.732	0.0756
1.326	0.52a	10.01s	979.73	0.000	3.188	0.2923
1.195	0.70a	15.01s	979.84	0.000	3.718	0.5997
1.153	0.78a	16.67s	979.84	0.000	3.739	0.7077
1.068	0.94a	20.01s	979.84	0.000	3.662	0.9246
0.937	1.19a	25.01s	979.84	0.000	3.383	1.2345
0.803	1.43a	30.01s	979.84	0.000	3.000	1.5138
0.669	1.68a	35.01s	979.84	0.000	2.557	1.7567
0.533	1.92a	40.01s	979.84	0.000	2.075	1.9591
0.397	2.16a	45.01s	979.84	0.000	1.565	2.1181
0.260	2.38a	50.01s	979.84	0.000	1.037	2.2318
0.121	2.59a	55.01s	979.84	0.000	0.496	2.2987
-0.006	2.76a	59.53s	979.84	0.000	0.000	2.3183
-0.019	2.78a	60.01s	979.84	0.000	-0.052	2.3181

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

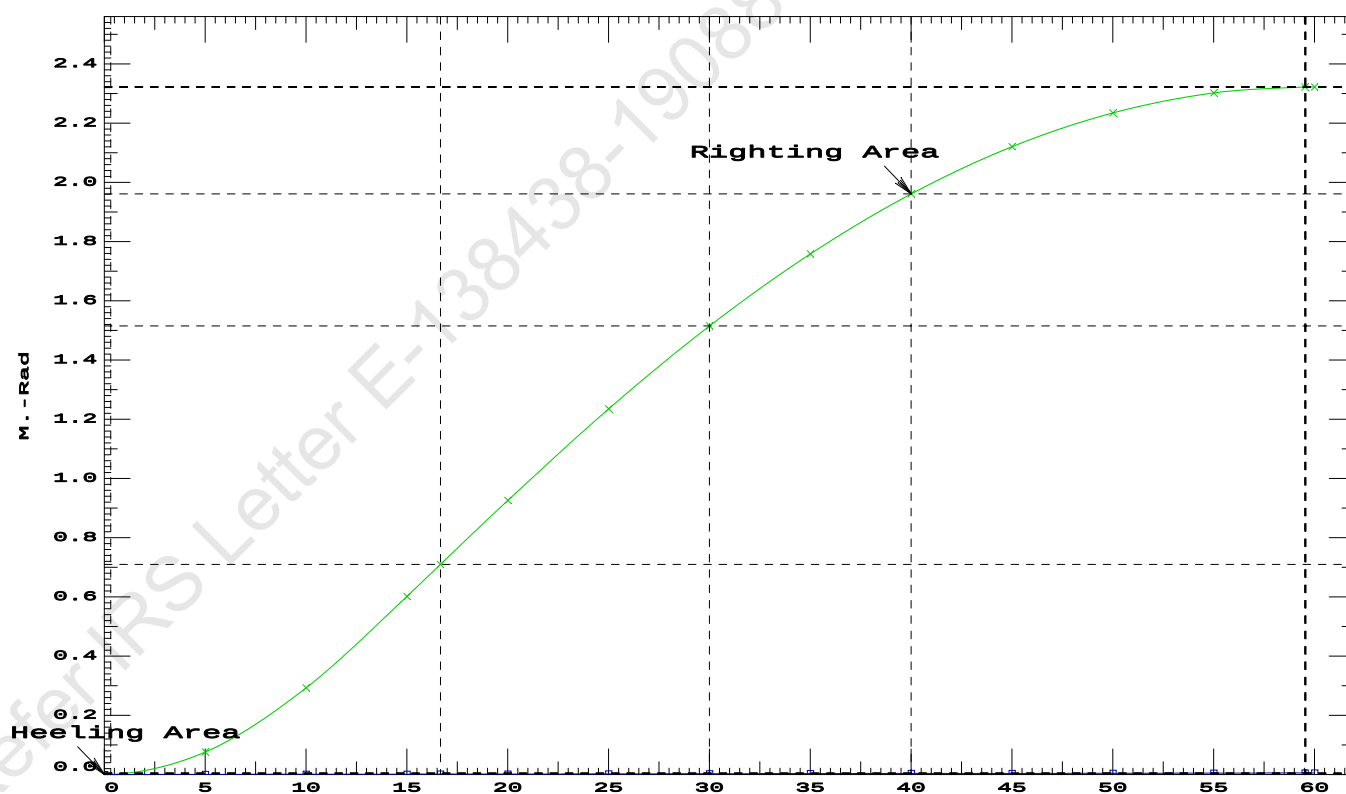
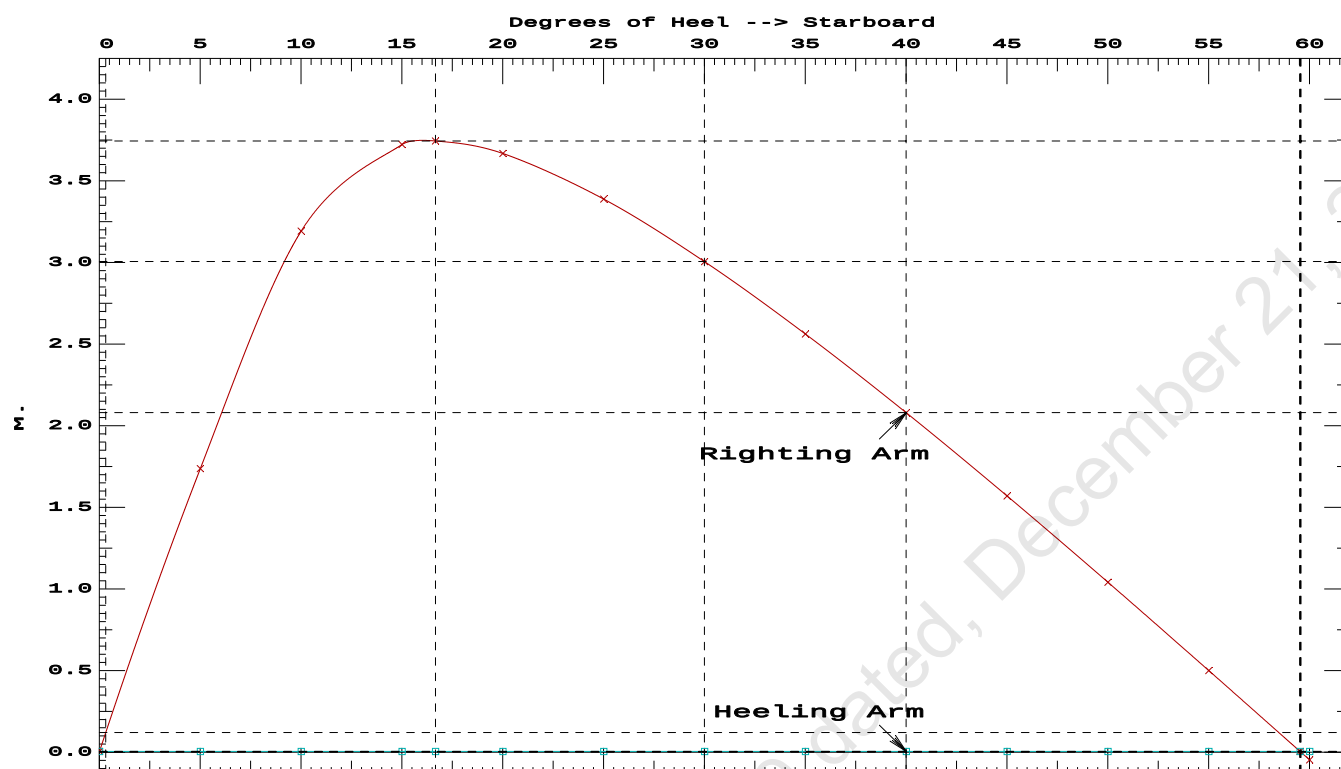
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	73273.0 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	438.282 P

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GHS 17.22 PONT00N

**Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft**



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Condition 14 : Operating Condition of 150T Crane

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.996 @ 50.00f, 1.198 @ 25.00f, 1.399 @ 0.00

Trim: Aft 0.403/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	455.34	24.928f	0.000	3.000	
Crane	150.00	7.500f	0.000	5.200	
Crane Load	51.30	4.500a	0.000	33.218	
Total Fixed----->	656.64	18.648f	0.000	5.863	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
Total Tanks----->			323.20	34.000f	0.000
Total Weight----->			979.84	23.712f	0.000
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	979.83	23.681f	0.003s	0.608

Righting Arms:		0.000	0.003s		
External Arms:		0.000	0.001s		
Residual Righting Arms:		0.000	0.002s		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025	842.1	24.797f	0.037s	160.37
		MT/cm			23.469
		8.63			
Distances in METERS.-----					
Moments in m.-MT.					

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.180 @ 50.00f, 1.140 @ 25.00f, 1.100 @ 0.00
Trim: Fwd 0.081/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Total Fixed----->	605.34	20.609f	0.000	3.545			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			928.54	25.270f	0.000	2.837	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		928.54	25.274f	0.001s	0.572	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001s			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	841.4	25.047f	0.018s	168.80	24.676	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.62	30.93	166.54	22.412		
Distances in METERS.-----					Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.609f TCG = 0.000 VCG = 3.545

Origin Depth	Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.087	0.09f	0.00	928.54	0.000	0.000	0.0000
1.087	0.09f	0.07s	928.59	0.000	0.028	0.0000
1.073	0.09f	5.00s	928.54	0.000	2.004	0.0874
0.986	0.11f	10.00s	928.53	0.000	3.659	0.3371
0.741	0.14f	15.00s	928.54	0.000	4.369	0.6943
0.498	0.18f	19.32s	928.54	0.000	4.493	1.0309
0.459	0.18f	20.00s	928.54	0.000	4.490	1.0845
0.172	0.23f	25.00s	928.54	0.000	4.370	1.4737
-0.116	0.28f	30.00s	928.54	0.000	4.134	1.8456

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 605.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.609f TCG = 0.000 VCG = 3.545

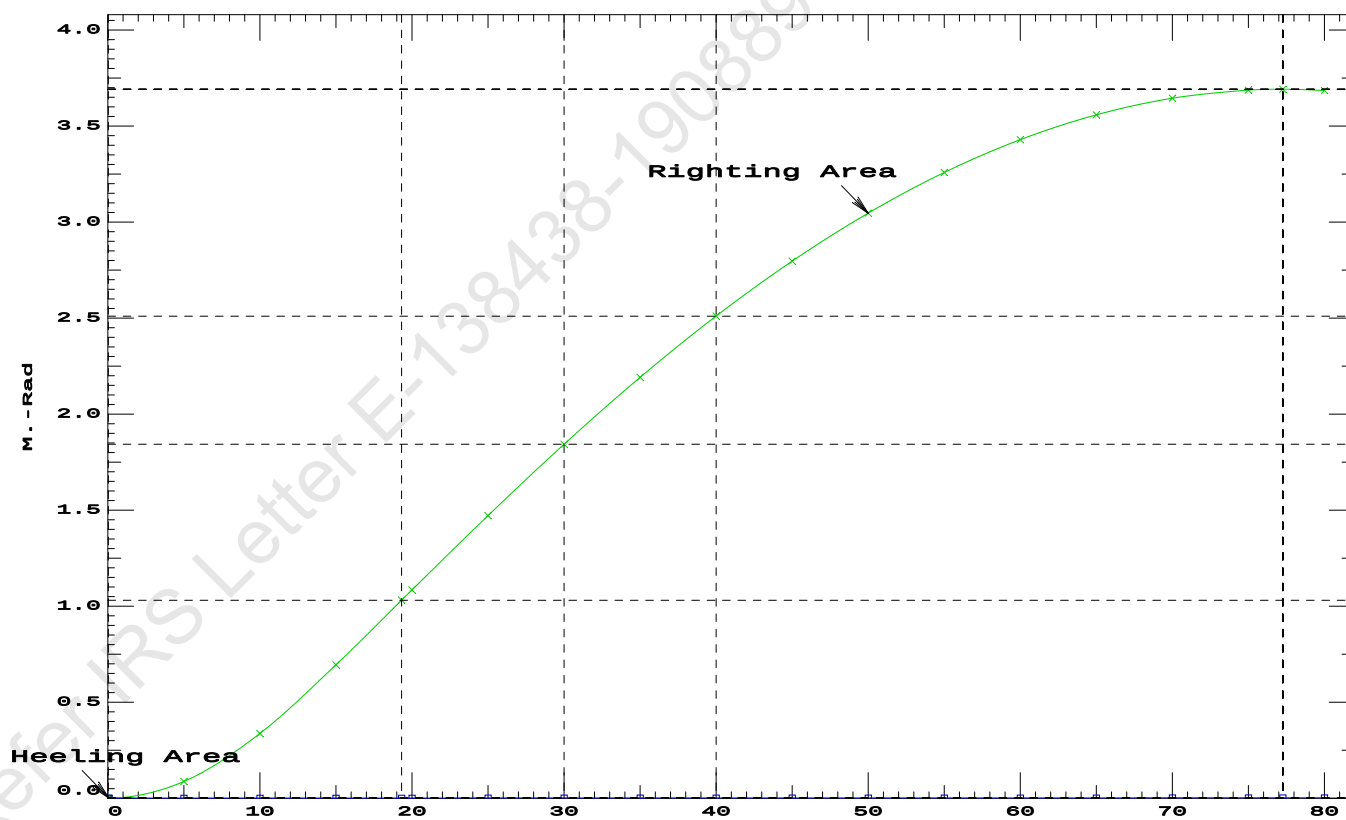
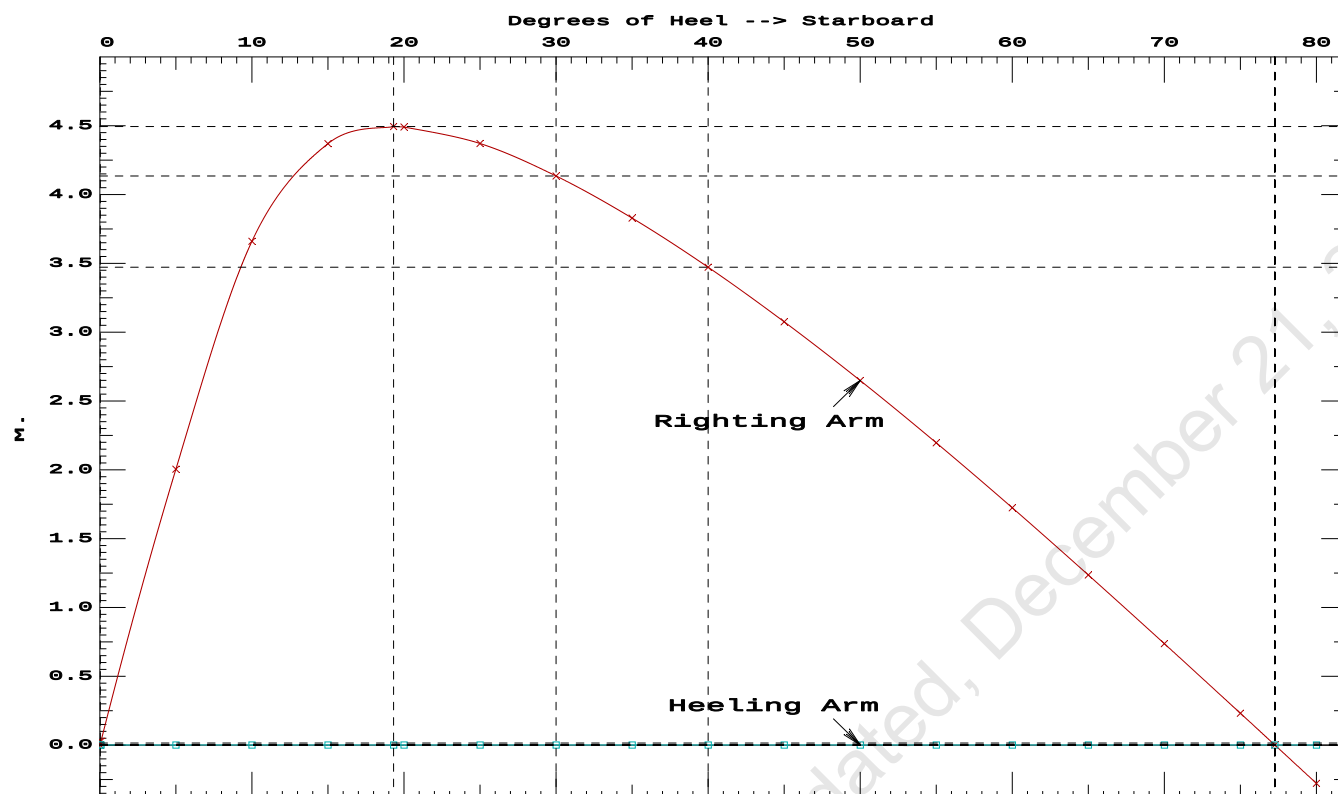
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.087	0.09f	0.00	928.54	0.000	0.000	0.0000
1.087	0.09f	0.07s	928.59	0.000	0.028	0.0000
1.073	0.09f	5.00s	928.54	0.000	2.004	0.0874
0.986	0.11f	10.00s	928.53	0.000	3.659	0.3371
0.741	0.14f	15.00s	928.54	0.000	4.369	0.6943
0.498	0.18f	19.32s	928.54	0.000	4.493	1.0309
0.459	0.18f	20.00s	928.54	0.000	4.490	1.0845
0.172	0.23f	25.00s	928.54	0.000	4.370	1.4737
-0.116	0.28f	30.00s	928.54	0.000	4.134	1.8456
-0.404	0.32f	35.00s	928.54	0.000	3.828	2.1935
-0.690	0.37f	40.00s	928.54	0.000	3.471	2.5123
-0.972	0.41f	45.00s	928.54	0.000	3.075	2.7982
-1.246	0.46f	50.00s	928.54	0.000	2.648	3.0481
-1.512	0.50f	55.00s	928.54	0.000	2.196	3.2596
-1.766	0.53f	60.00s	928.54	0.000	1.723	3.4308
-2.007	0.57f	65.00s	928.54	0.000	1.235	3.5600
-2.232	0.59f	70.00s	928.54	0.000	0.736	3.6461
-2.439	0.61f	75.00s	928.54	0.000	0.230	3.6883
-2.527	0.62f	77.27s	928.54	0.000	0.000	3.6928
-2.627	0.63f	80.00s	928.53	0.000	-0.278	3.6862

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 605.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 77.20 P

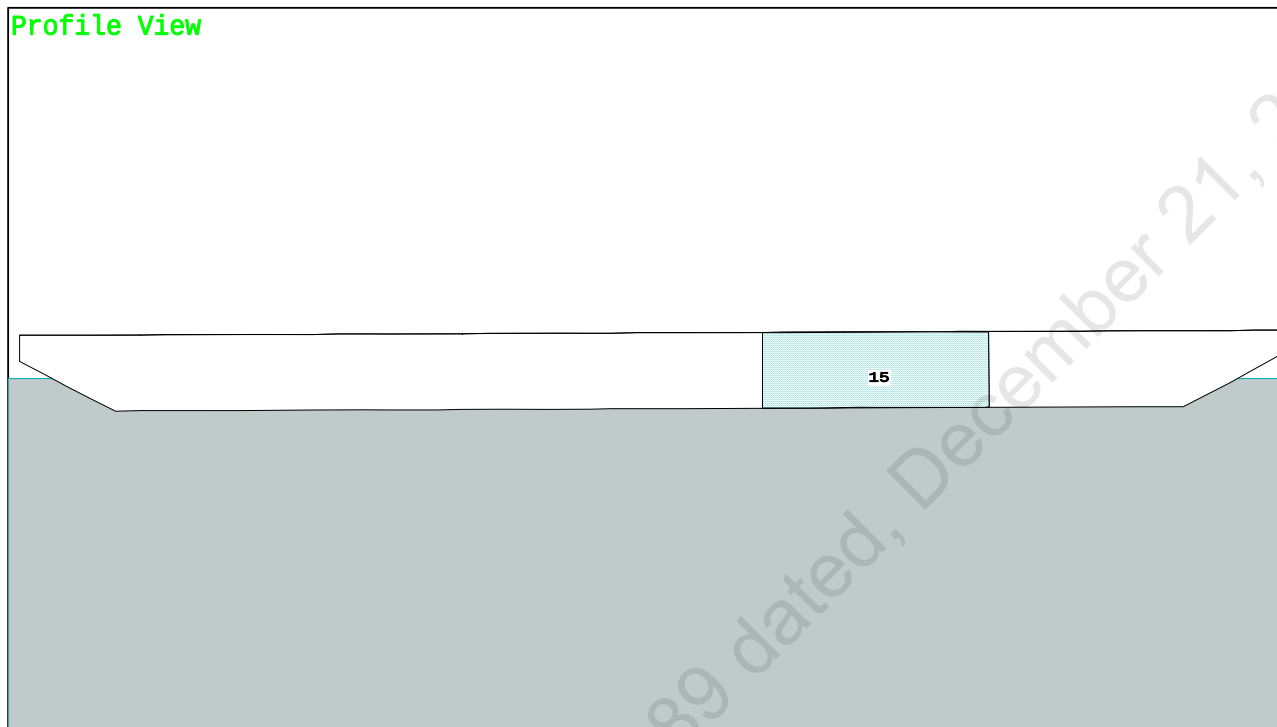
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PONT00N

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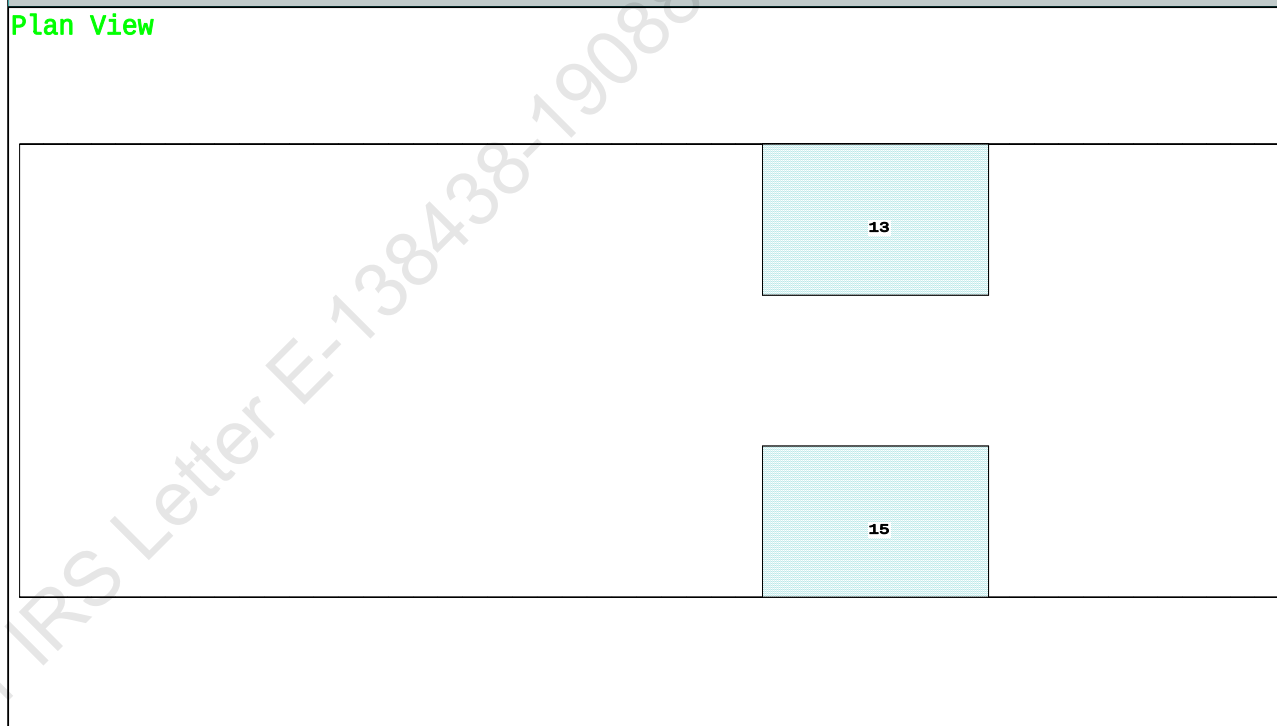
Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

Condition Graphic - Draft: 1.095 @ 50.000f, 1.302 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S

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GHS 17.22

PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.095 @ 50.00f, 1.199 @ 25.00f, 1.302 @ 0.00

Trim: Aft 0.206/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Total Fixed----->			656.64	19.585f	0.000	5.863	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			979.84	24.340f	0.000	4.427	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	979.80	24.325f	0.000	0.604	

	Righting Arms:			0.001	0.000		
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		847.7	24.816f	0.000	163.54	23.609
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.69	31.30	159.72	19.786	
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 1.095
DRAFT AT MS (M) = 1.199
DRAFT AT AP (M) = 1.302
DRAFT AT FWD MARK (M) = 1.112
DRAFT AT AFT MARK (M) = 1.285

HYDROSTATIC PROPERTIES

Trim: Aft 0.206/50.000, No Heel, VCG = 4.427

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----
1.199	979.80	24.325f	0.604	8.69
				24.816f
				31.30
				159.72
				19.786

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.095 @ 50.00f, 1.199 @ 25.00f, 1.302 @ 0.00

Trim: Aft 0.206/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.9	0.922	1.511	0.05500	7.47
CRANE	57.0	4.252	4.840	0.05500	15.17
Total wind heeling moment to starboard-->					22.64
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.340f TCG = 0.000 VCG = 4.427

Free Surface Adjustment: 0.332

Adjusted CG: LCG = 24.341f TCG = 0.000 VCG = 4.759

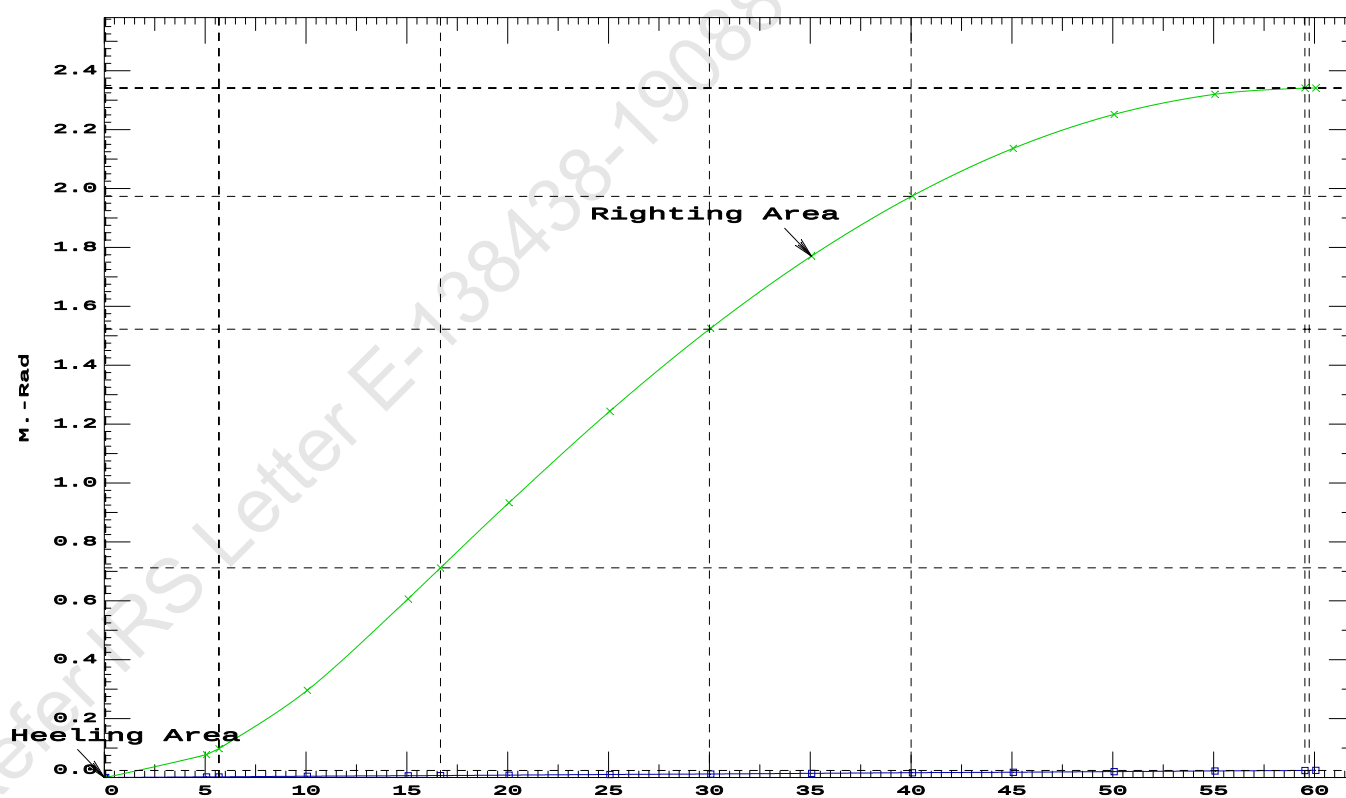
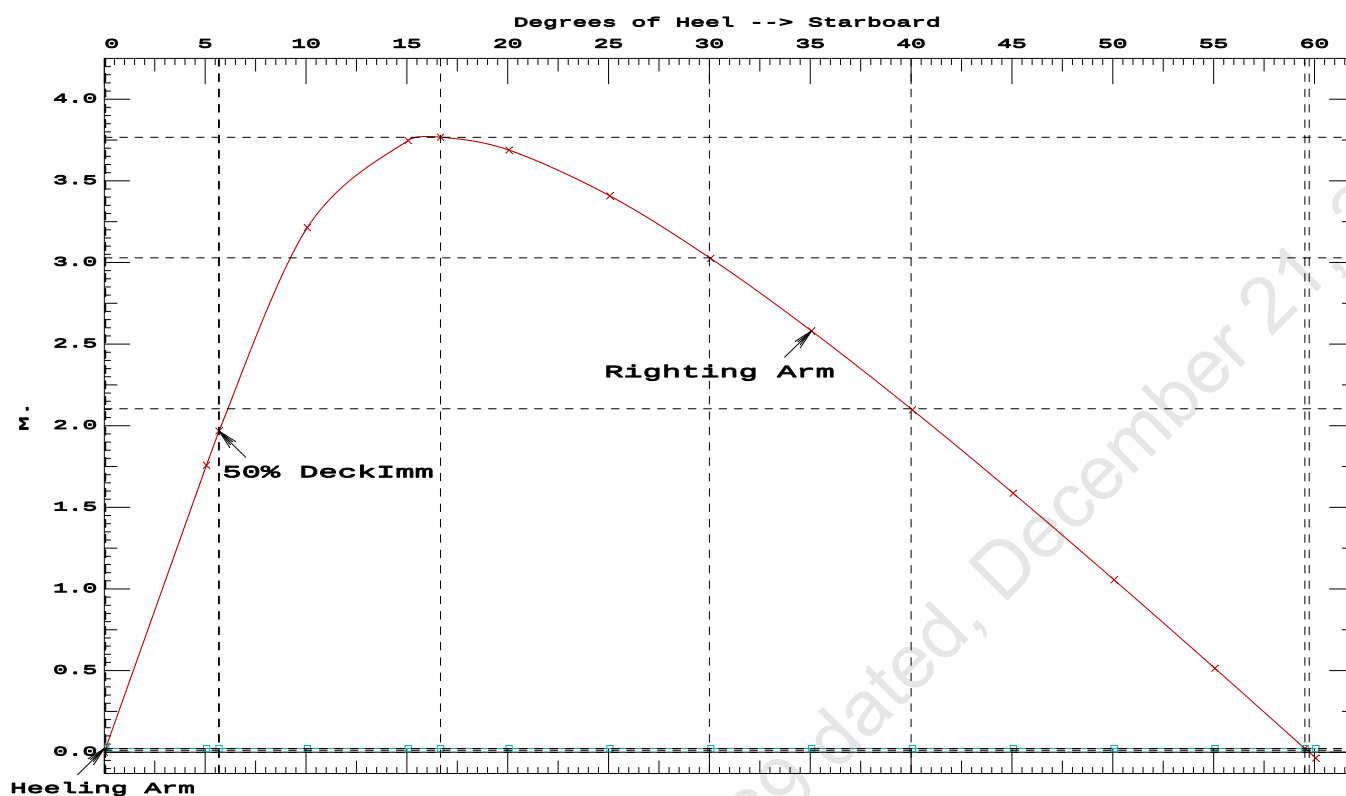
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.290	0.24a	0.00	979.83	0.000	-0.023	0.0000	5.682
1.290	0.24a	0.06s	979.84	0.000	0.000	-0.0000	5.621
1.275	0.24a	5.06s	979.84	0.000	1.735	0.0757	0.621
1.271	0.24a	5.68s	979.81	0.000	1.945	0.0956	0.000
1.215	0.27a	10.06s	979.81	0.000	3.191	0.2934	-4.379
1.045	0.36a	15.06s	979.84	0.000	3.724	0.6003	-9.379
0.988	0.40a	16.67s	979.84	0.000	3.744	0.7049	-10.985
0.866	0.48a	20.06s	979.84	0.000	3.666	0.9256	-14.379
0.683	0.61a	25.06s	979.84	0.000	3.385	1.2358	-19.379
0.497	0.74a	30.06s	979.84	0.000	3.001	1.5152	-24.379
0.310	0.86a	35.06s	979.84	0.000	2.558	1.7582	-29.379
0.122	0.99a	40.06s	979.84	0.000	2.074	1.9605	-34.379
-0.064	1.11a	45.06s	979.84	0.000	1.564	2.1195	-39.379
-0.249	1.22a	50.06s	979.84	0.000	1.033	2.2329	-44.379
-0.432	1.33a	55.06s	979.84	0.000	0.491	2.2995	-49.379
-0.593	1.42a	59.52s	979.84	0.000	0.000	2.3187	-53.841
-0.612	1.43a	60.06s	979.84	0.000	-0.059	2.3184	-54.379
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.64 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.7049 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	59.46 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.62 P

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PONTOONCondition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

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GHS 17.22

PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.340f TCG = 0.000 VCG = 4.427

Free Surface Adjustment: 0.332

Adjusted CG: LCG = 24.341f TCG = 0.010p VCG = 4.759

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area -->
1.286	0.23a	1.77s	979.84	0.000	0.000	0.0000
1.263	0.24a	6.77s	979.84	0.000	1.706	0.0745
1.161	0.29a	11.77s	979.78	0.000	2.872	0.2782
0.992	0.40a	16.53s	979.84	0.000	3.149	0.5343
0.984	0.40a	16.77s	979.84	0.000	3.148	0.5471
0.804	0.53a	21.77s	979.84	0.000	2.990	0.8181
0.619	0.65a	26.77s	979.84	0.000	2.666	1.0661
0.433	0.78a	31.77s	979.84	0.000	2.259	1.2816
0.246	0.91a	36.77s	979.84	0.000	1.799	1.4590
0.059	1.03a	41.77s	979.84	0.000	1.305	1.5947
-0.127	1.15a	46.77s	979.84	0.000	0.786	1.6862
-0.312	1.26a	51.77s	979.84	0.000	0.250	1.7315
-0.396	1.31a	54.07s	979.84	0.000	0.000	1.7366
-0.493	1.37a	56.77s	979.84	0.000	-0.296	1.7296
-0.672	1.46a	61.77s	979.84	0.000	-0.848	1.6797

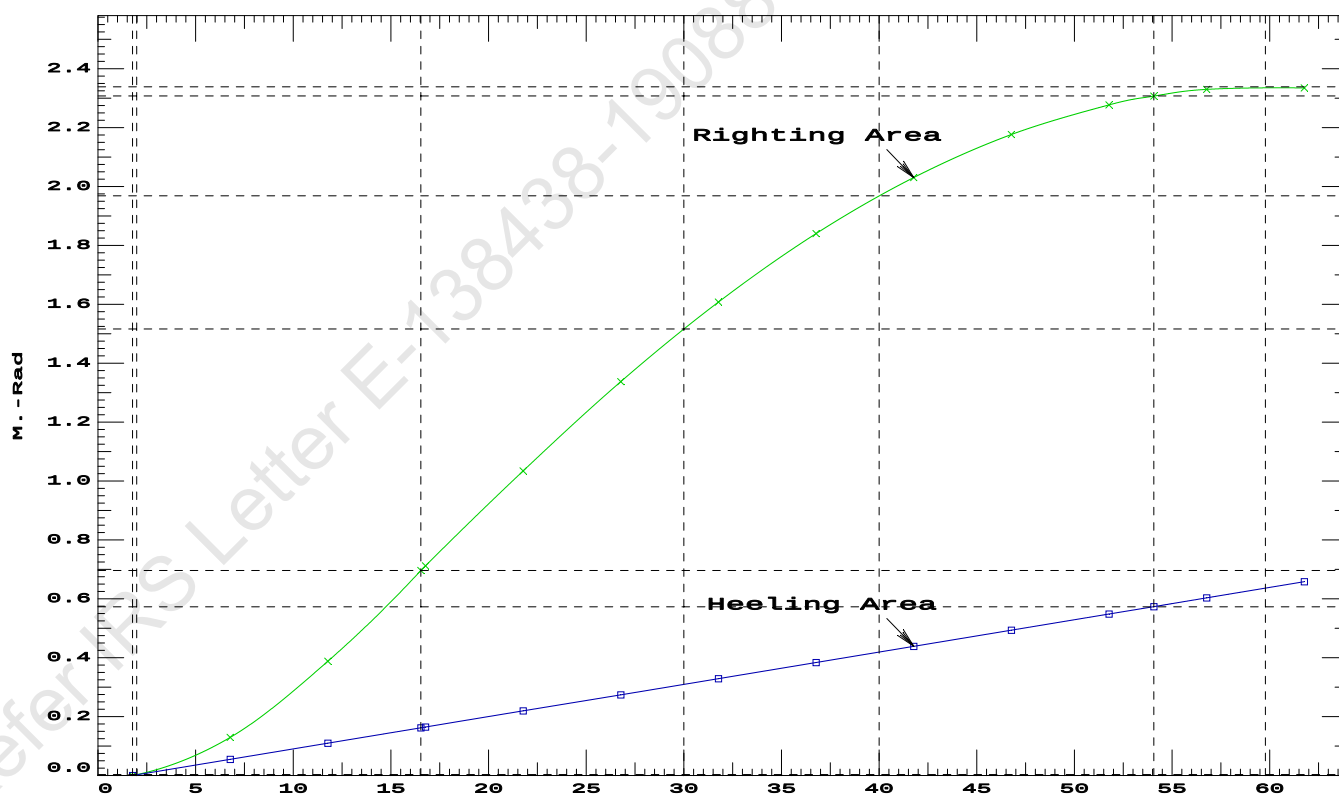
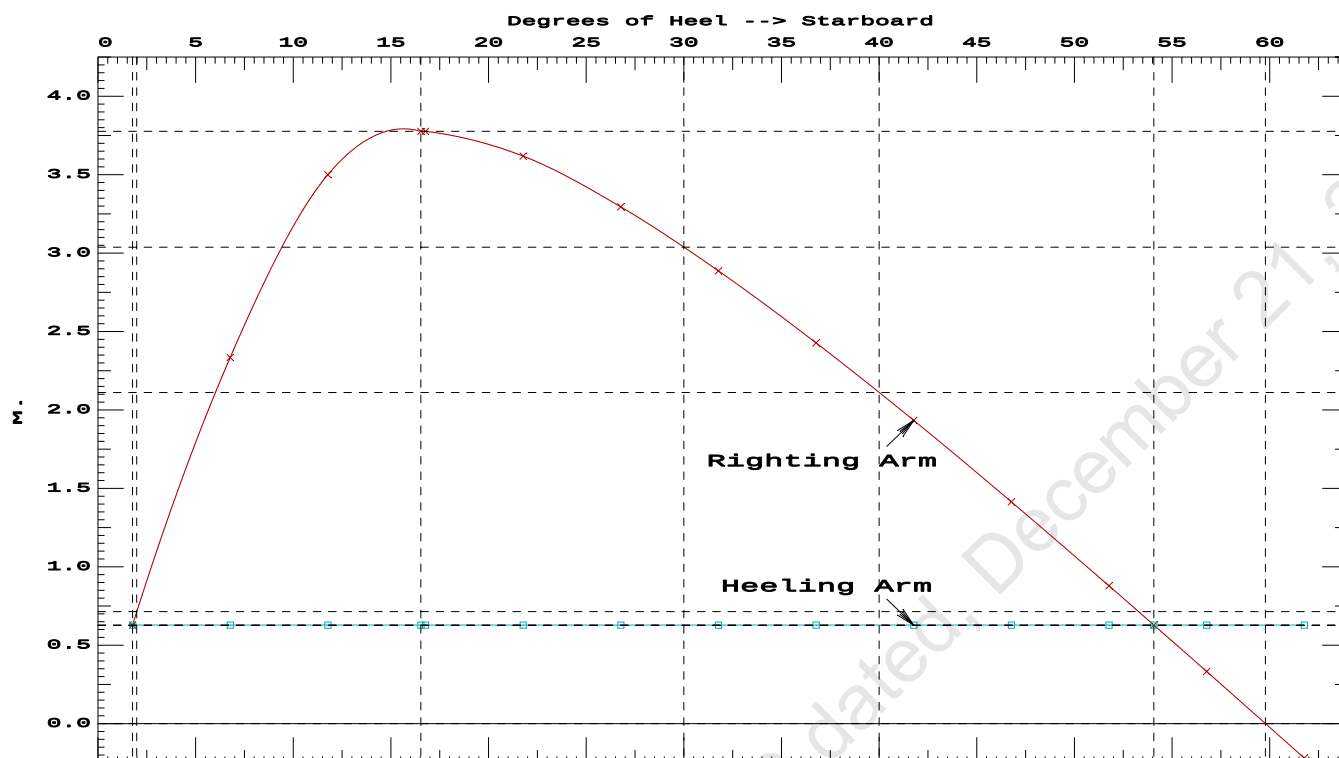
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	1.77 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	501.2 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	4.028 P

22-10-21 12:43:20
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00NCondition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

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GHS 17.22 PONTON

Condition 14 : Operating Condition of 150T Crane**Crane Lifting 51.3T @ 12m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.094 @ 50.00f, 1.196 @ 25.00f, 1.299 @ 0.00

Trim: Aft 0.205/50.000, Heel: Stbd 1.77 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Total Fixed----->			656.64	19.585f	0.000	5.863	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			979.84	24.340f	0.000	4.427	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	979.86	24.324f	0.746s	0.616	

	Righting Arms:			0.000	0.628s		
	External Arms:			0.000	0.628s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		846.8	24.843f	0.102s	163.02	23.583
			MT/cm				
			8.68				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.77

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.180 @ 50.00f, 1.140 @ 25.00f, 1.100 @ 0.00
Trim: Fwd 0.081/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
LIGHT SHIP	455.34	24.928f	0.000	3.000		
Crane	150.00	7.500f	0.000	5.200		
Total Fixed----->	605.34	20.609f	0.000	3.545		
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td>	LCG-----	TCG-----	VCG-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
Total Tanks----->			323.20	34.000f	0.000	1.509
Total Weight----->			928.54	25.270f	0.000	2.837
		Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025	928.54	25.274f	0.001s	0.572	
Righting Arms:		0.000	0.001s			
External Arms:		0.000	0.001s			
Residual Righting Arms:		0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	841.4	25.047f	0.018s	168.80	24.676
		MT/cm-----	m.-MT/cm-----	GML-----	GML-----	GML-----
		8.62	30.93	166.54	22.412	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 16670.26; t3: 30.00; t: 15.00
limmarg: 6209.43; t3: 15.00; t: 7.50
limmarg: 1680.10; t3: 7.50; t: 3.75
limmarg: 349.20; t3: 3.75; t: 1.88
limmarg: 12.45; t3: 1.87; t: 0.94
limmarg: -72.11; t3: 0.93; t: 0.47
limmarg: -36.87; t3: 1.40; t: 0.23
limmarg: -14.48; t3: 1.63; t: 0.12
limmarg: -1.47; t3: 1.75; t: 0.06
limmarg: 5.37; t3: 1.81; t: 0.03
limmarg: 1.92; t3: 1.78; t: 0.01

Theta3 is 1.78 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.609f TCG = 0.000 VCG = 3.545

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.086	0.09f	1.77s	928.52	0.000	-0.715	0.0000
1.088	0.09f	0.00	928.53	0.000	0.000	-0.0110
1.086	0.09f	1.78p	928.54	0.000	0.722	0.0002
1.073	0.09f	5.00p	928.56	0.000	2.005	0.0769
0.986	0.11f	10.00p	928.53	0.000	3.661	0.3263
0.741	0.14f	15.00p	928.54	0.000	4.371	0.6836
0.498	0.18f	19.32p	928.54	0.000	4.495	1.0206
0.459	0.18f	20.00p	928.54	0.000	4.492	1.0741
0.173	0.23f	25.00p	928.54	0.000	4.372	1.4634
-0.116	0.28f	30.00p	928.54	0.000	4.136	1.8355

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 605.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to abs 1.8 > 1.000 1.019 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.609f TCG = 0.000 VCG = 3.545

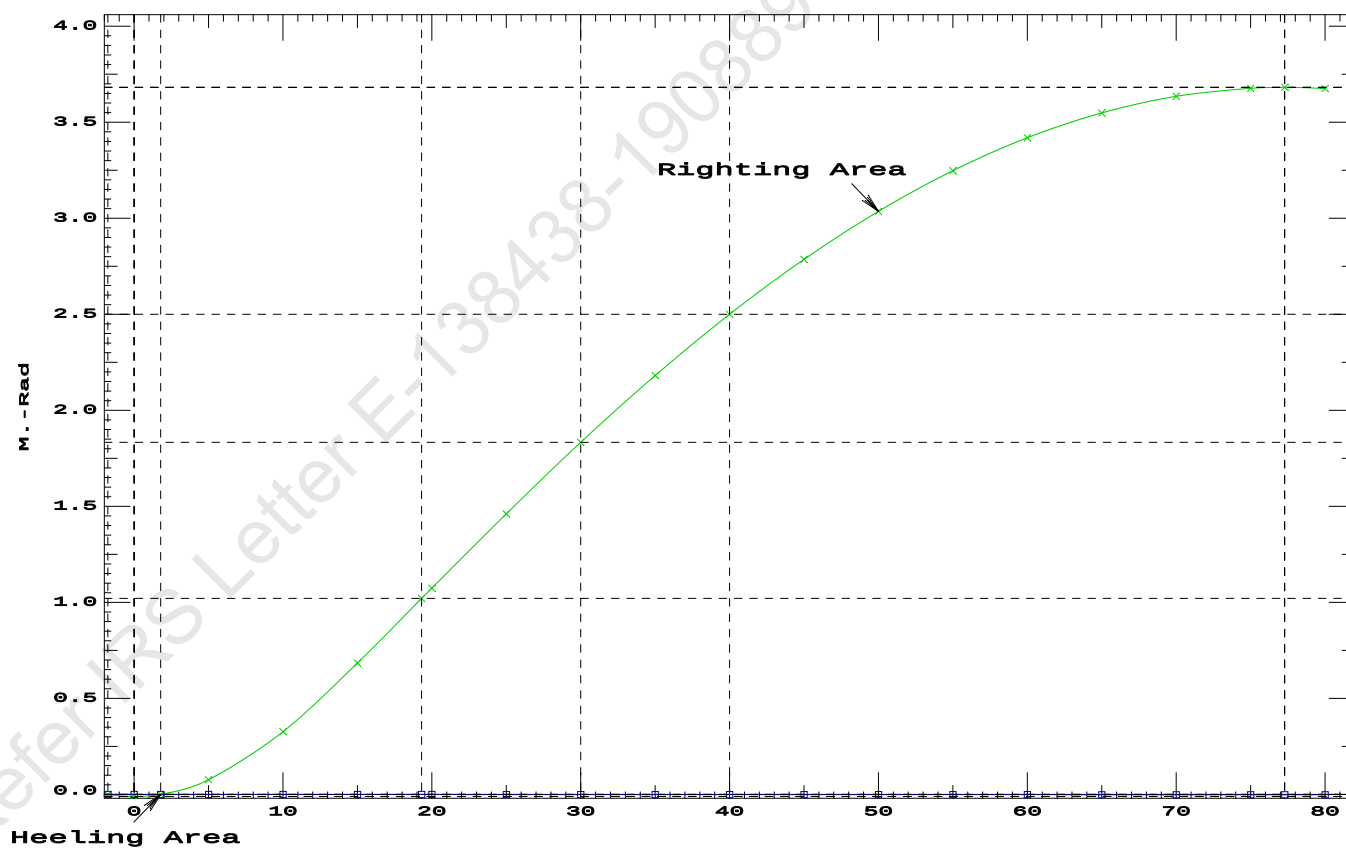
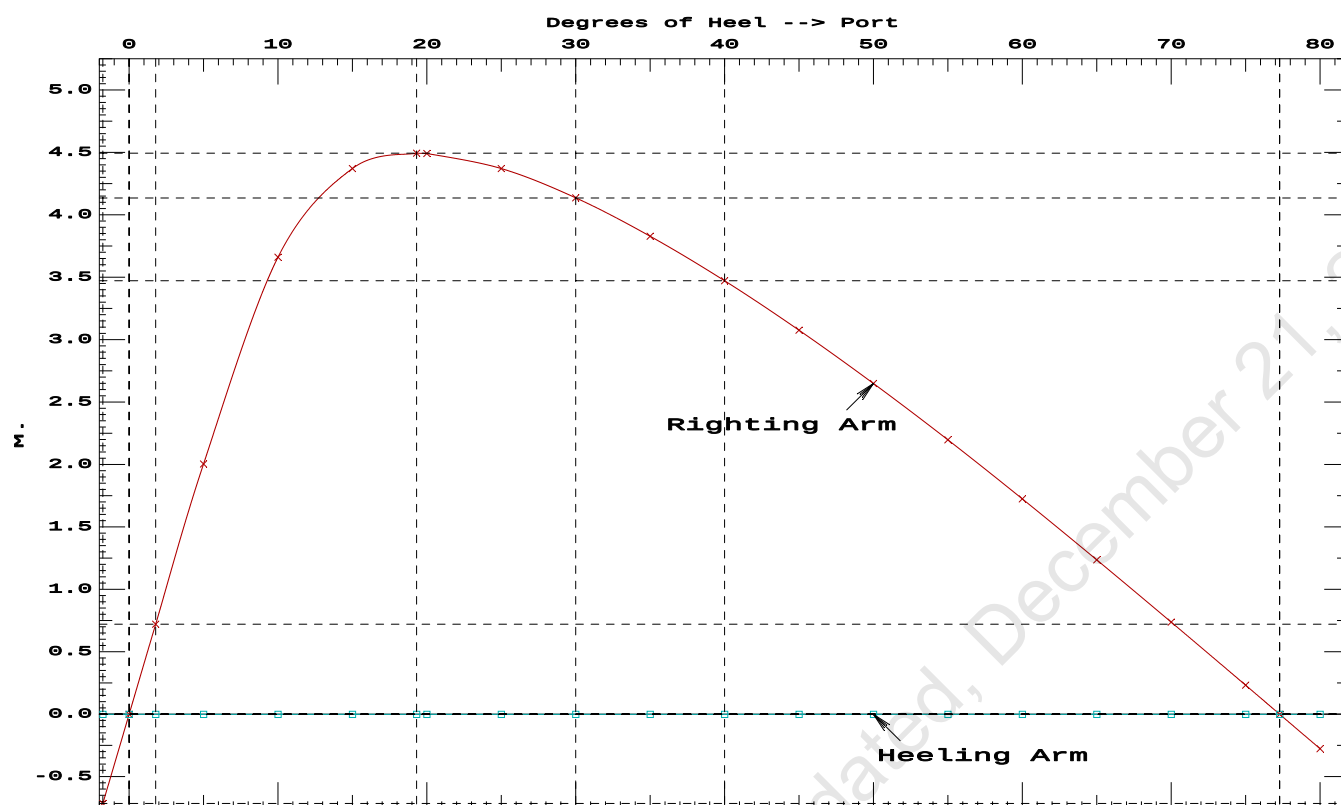
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.086	0.09f	1.77s	928.52	0.000	-0.715	0.0000
1.088	0.09f	0.00	928.53	0.000	0.000	-0.0110
1.086	0.09f	1.78p	928.54	0.000	0.722	0.0002
1.073	0.09f	5.00p	928.56	0.000	2.005	0.0769
0.986	0.11f	10.00p	928.53	0.000	3.661	0.3263
0.741	0.14f	15.00p	928.54	0.000	4.371	0.6836
0.498	0.18f	19.32p	928.54	0.000	4.495	1.0206
0.459	0.18f	20.00p	928.54	0.000	4.492	1.0741
0.173	0.23f	25.00p	928.54	0.000	4.372	1.4634
-0.116	0.28f	30.00p	928.54	0.000	4.136	1.8355
-0.404	0.32f	35.00p	928.54	0.000	3.830	2.1837
-0.690	0.37f	40.00p	928.54	0.000	3.473	2.5027
-0.972	0.41f	45.00p	928.54	0.000	3.077	2.7888
-1.246	0.46f	50.00p	928.54	0.000	2.650	3.0389
-1.512	0.50f	55.00p	928.54	0.000	2.198	3.2506
-1.766	0.53f	60.00p	928.54	0.000	1.726	3.4220
-2.007	0.57f	65.00p	928.54	0.000	1.238	3.5514
-2.232	0.59f	70.00p	928.54	0.000	0.739	3.6377
-2.439	0.61f	75.00p	928.54	0.000	0.233	3.6801
-2.528	0.62f	77.29p	928.54	0.000	0.000	3.6848
-2.627	0.63f	80.00p	928.53	0.000	-0.276	3.6782

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 605.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to RAzero > 1.000 335.623 P
(2) Angle from abs 1.8 deg to RAzero > 20.00 deg 75.51 P

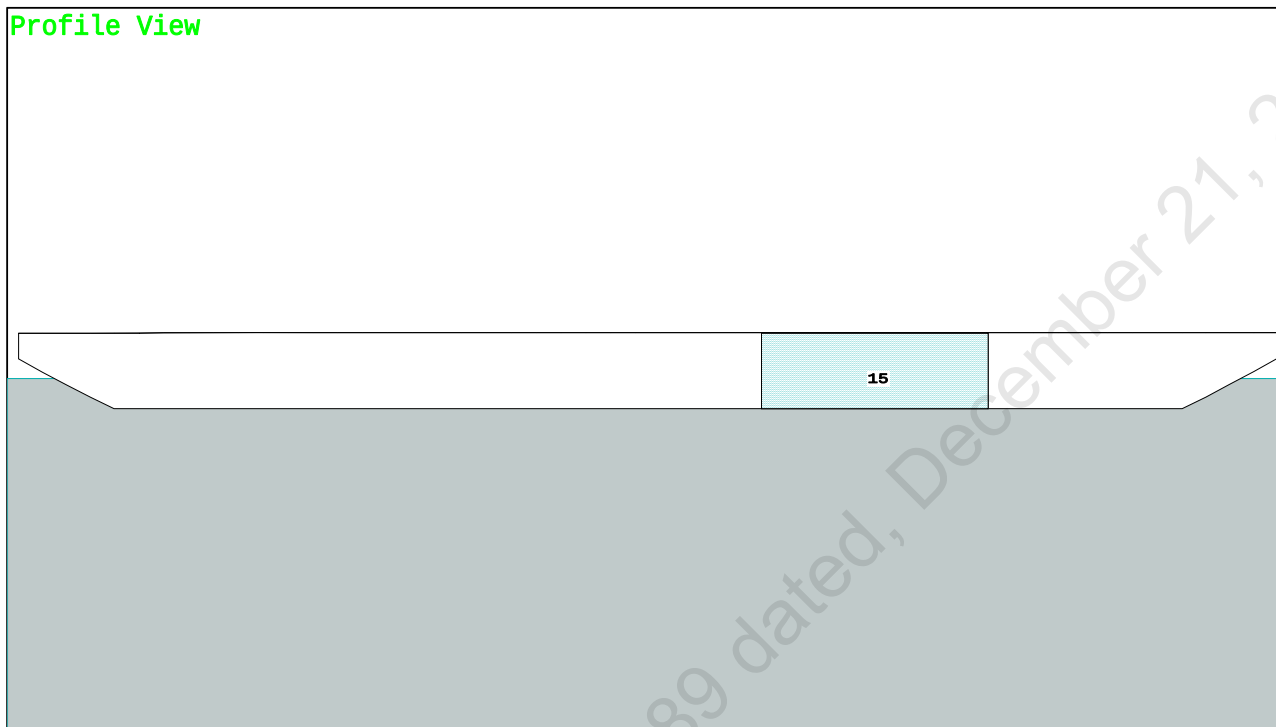
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GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00N

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GHS 17.22 PONT00N

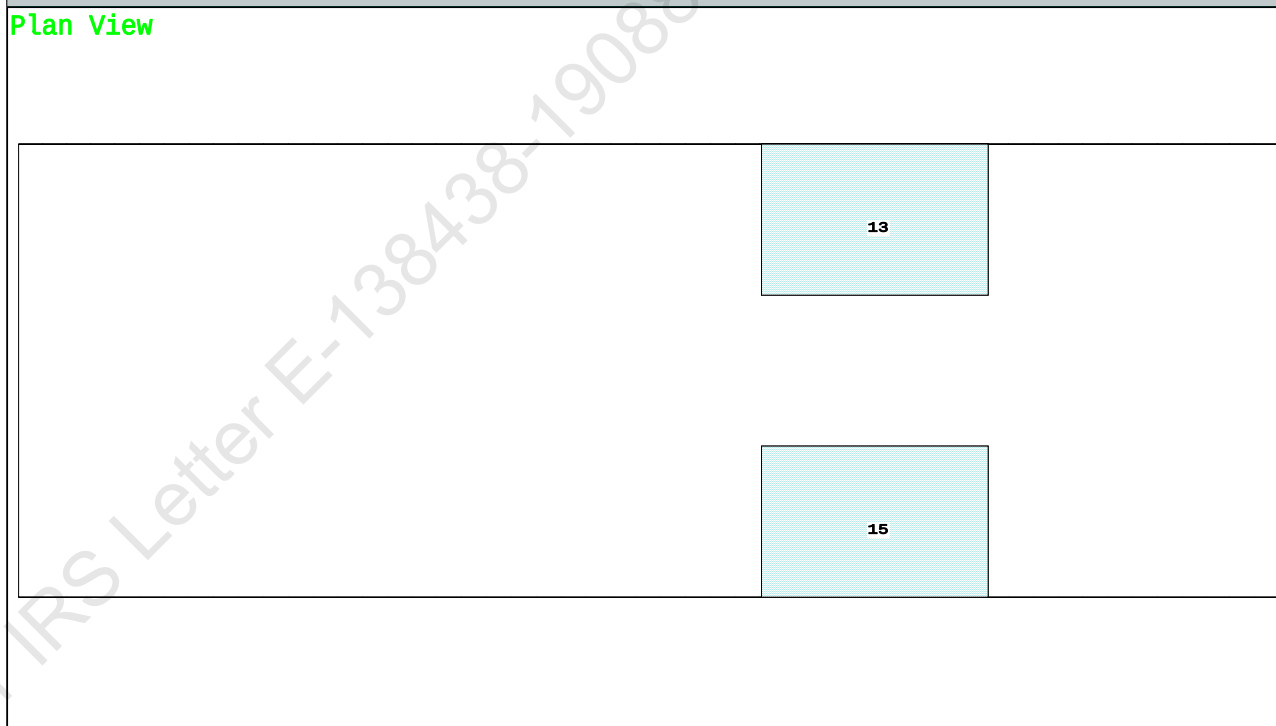
Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

Condition Graphic - Draft: 1.194 @ 50.000f, 1.204 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P 15 05_WBTK.S

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.194 @ 50.00f, 1.199 @ 25.00f, 1.204 @ 0.00

Trim: Aft 0.010/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Total Fixed----->			656.64	20.523f	0.000	5.863	
	Load----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			979.84	24.968f	0.000	4.427	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	979.84	24.967f	0.000	0.603	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	847.6	24.991f	0.000	163.47	23.601
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.69		31.29	159.65	19.777
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 1.194
DRAFT AT MS (M) = 1.199
DRAFT AT AP (M) = 1.204
DRAFT AT FWD MARK (M) = 1.195
DRAFT AT AFT MARK (M) = 1.203

HYDROSTATIC PROPERTIES

Trim: Aft 0.010/50.000, No Heel, VCG = 4.427

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----
1.199	979.84	24.967f	0.603	8.69
				24.991f
				31.29
				159.65
				19.777

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.194 @ 50.00f, 1.199 @ 25.00f, 1.204 @ 0.00

Trim: Aft 0.010/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	89.9	0.921	1.509	0.05500	7.46
CRANE	57.0	4.310	4.897	0.05500	15.35
Total wind heeling moment to starboard-->					22.81
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.968f TCG = 0.000 VCG = 4.427

Free Surface Adjustment: 0.332

Adjusted CG: LCG = 24.968f TCG = 0.000 VCG = 4.759

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.192	0.01a	0.00	979.84	0.000	-0.023	0.0000
1.192	0.01a	0.06s	979.84	0.000	0.000	-0.0000
1.177	0.01a	5.06s	979.84	0.000	1.737	0.0758
1.169	0.01a	6.16s	979.84	0.000	2.103	0.1124
1.105	0.01a	10.06s	979.83	0.000	3.195	0.2941
0.895	0.02a	15.06s	979.84	0.000	3.732	0.6015
0.822	0.02a	16.67s	979.84	0.000	3.752	0.7063
0.665	0.02a	20.06s	979.84	0.000	3.673	0.9276
0.429	0.03a	25.06s	979.61	0.000	3.394	1.2385
0.191	0.04a	30.06s	979.84	0.000	3.009	1.5186
-0.049	0.04a	35.06s	979.84	0.000	2.565	1.7623
-0.288	0.05a	40.06s	979.84	0.000	2.082	1.9653
-0.525	0.05a	45.06s	979.84	0.000	1.571	2.1249
-0.758	0.06a	50.06s	979.84	0.000	1.040	2.2390
-0.985	0.06a	55.06s	979.84	0.000	0.497	2.3062
-1.184	0.07a	59.58s	979.83	0.000	0.000	2.3258
-1.205	0.07a	60.06s	979.84	0.000	-0.054	2.3256
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

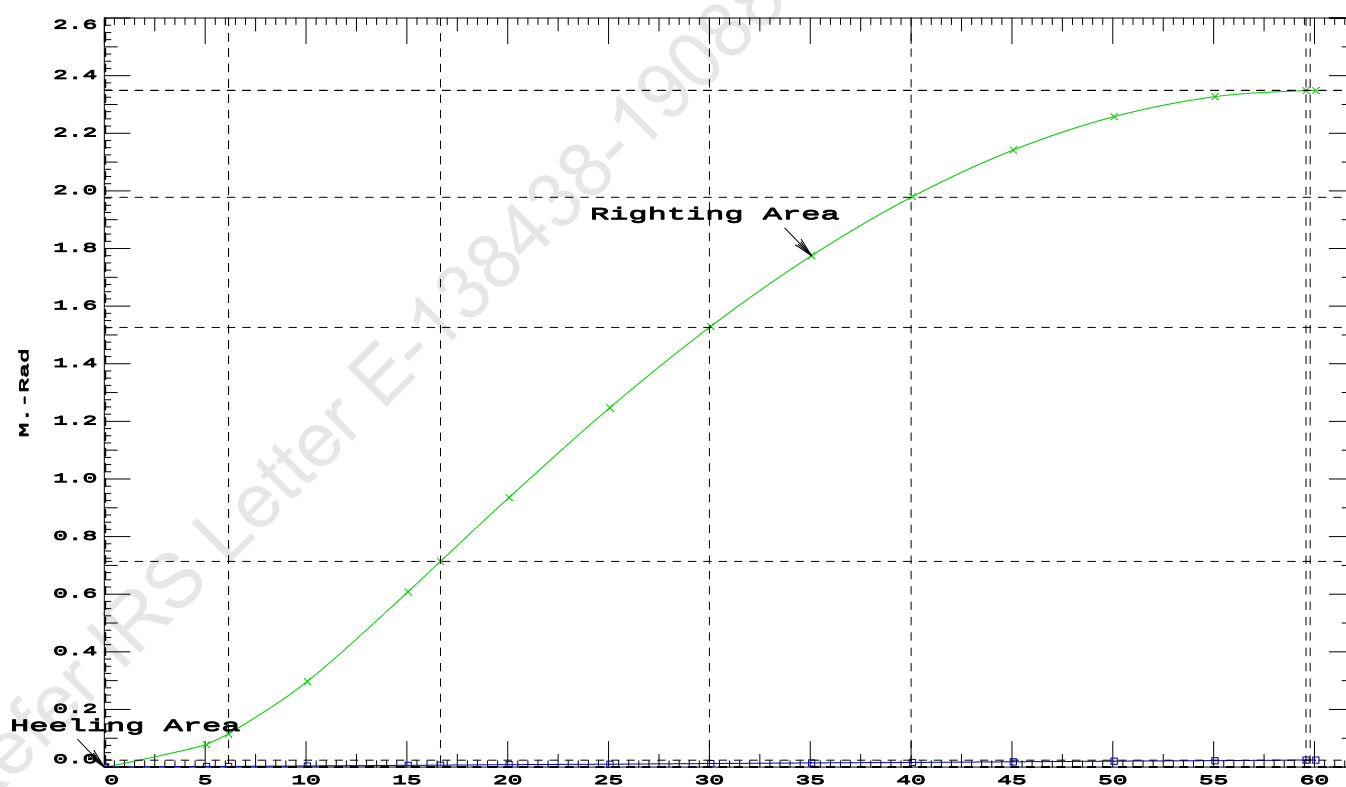
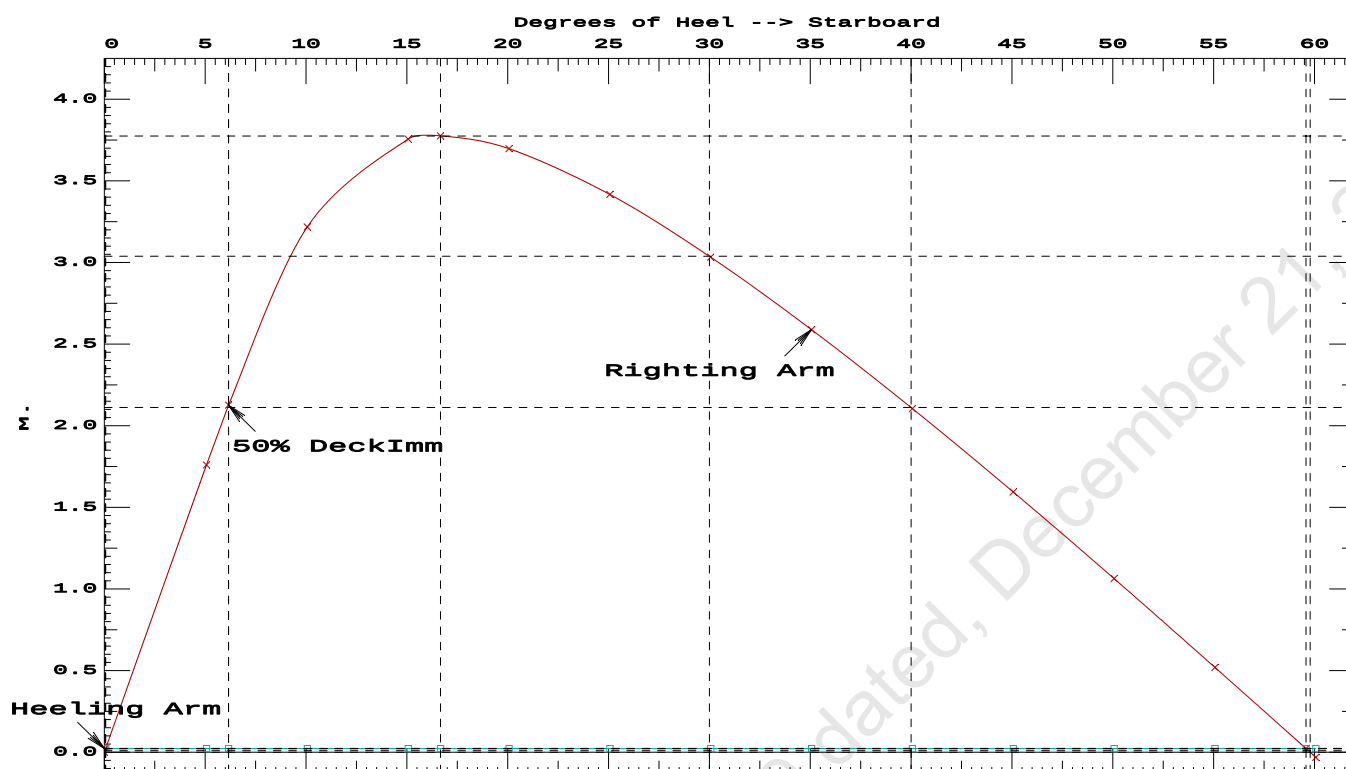
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.81 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.7063 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	59.51 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	6.09 P

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Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd



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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.968f TCG = 0.000 VCG = 4.427
Free Surface Adjustment: 0.332
Adjusted CG: LCG = 24.968f TCG = 0.000 VCG = 4.759

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.192	0.01a	0.00	979.83	0.000	0.000	0.0000
1.178	0.01a	5.00s	979.84	0.000	1.737	0.0758
1.107	0.01a	10.00s	979.83	0.000	3.205	0.2934
0.898	0.02a	15.00s	979.84	0.000	3.752	0.6036
0.822	0.02a	16.67s	979.84	0.000	3.774	0.7136
0.668	0.02a	20.00s	979.84	0.000	3.698	0.9318
0.432	0.03a	25.00s	979.61	0.000	3.420	1.2449
0.194	0.04a	30.00s	979.84	0.000	3.036	1.5274
-0.046	0.04a	35.00s	979.84	0.000	2.593	1.7735
-0.285	0.05a	40.00s	979.84	0.000	2.110	1.9790
-0.522	0.05a	45.00s	979.84	0.000	1.600	2.1411
-0.755	0.06a	50.00s	979.84	0.000	1.069	2.2576
-0.982	0.06a	55.00s	979.84	0.000	0.526	2.3273
-1.193	0.07a	59.78s	979.84	0.000	0.000	2.3493
-1.202	0.07a	60.00s	979.84	0.000	-0.025	2.3492

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

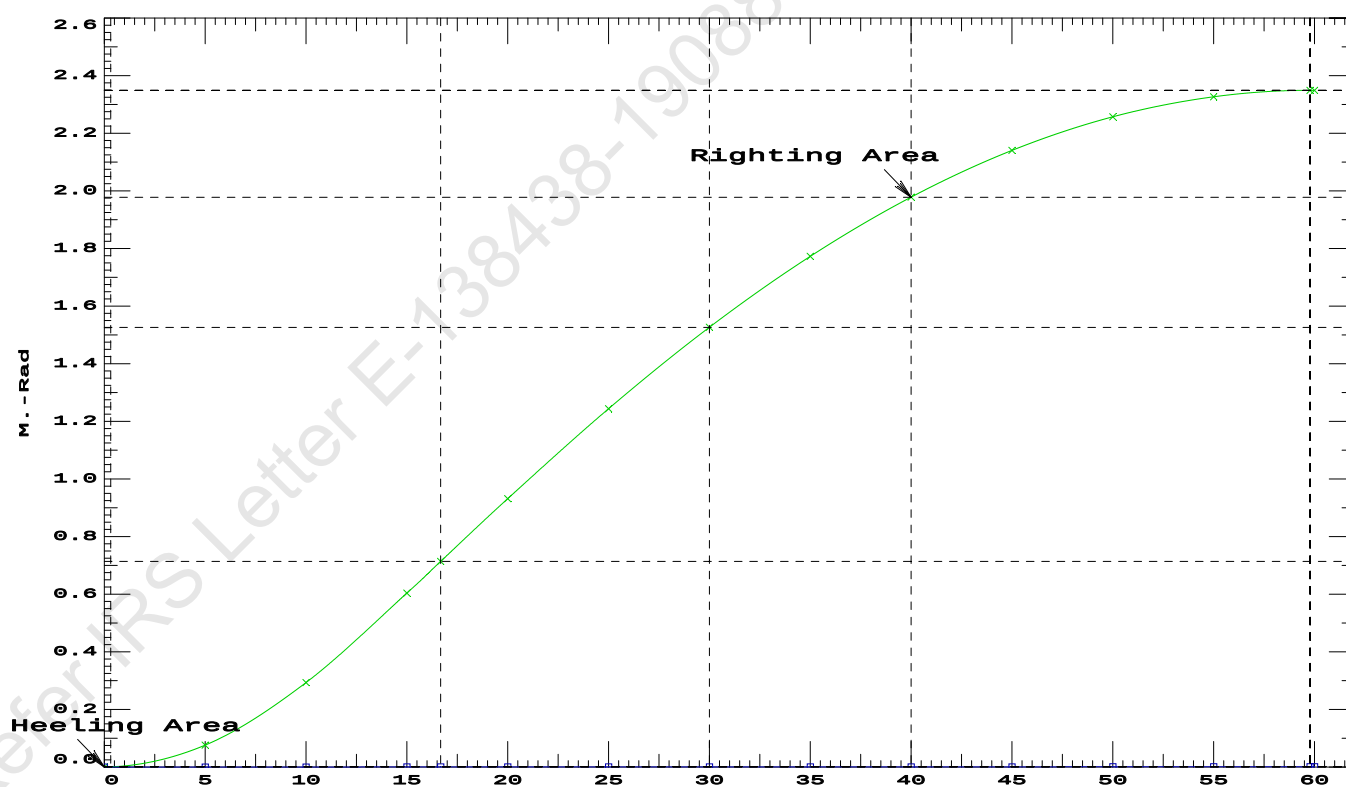
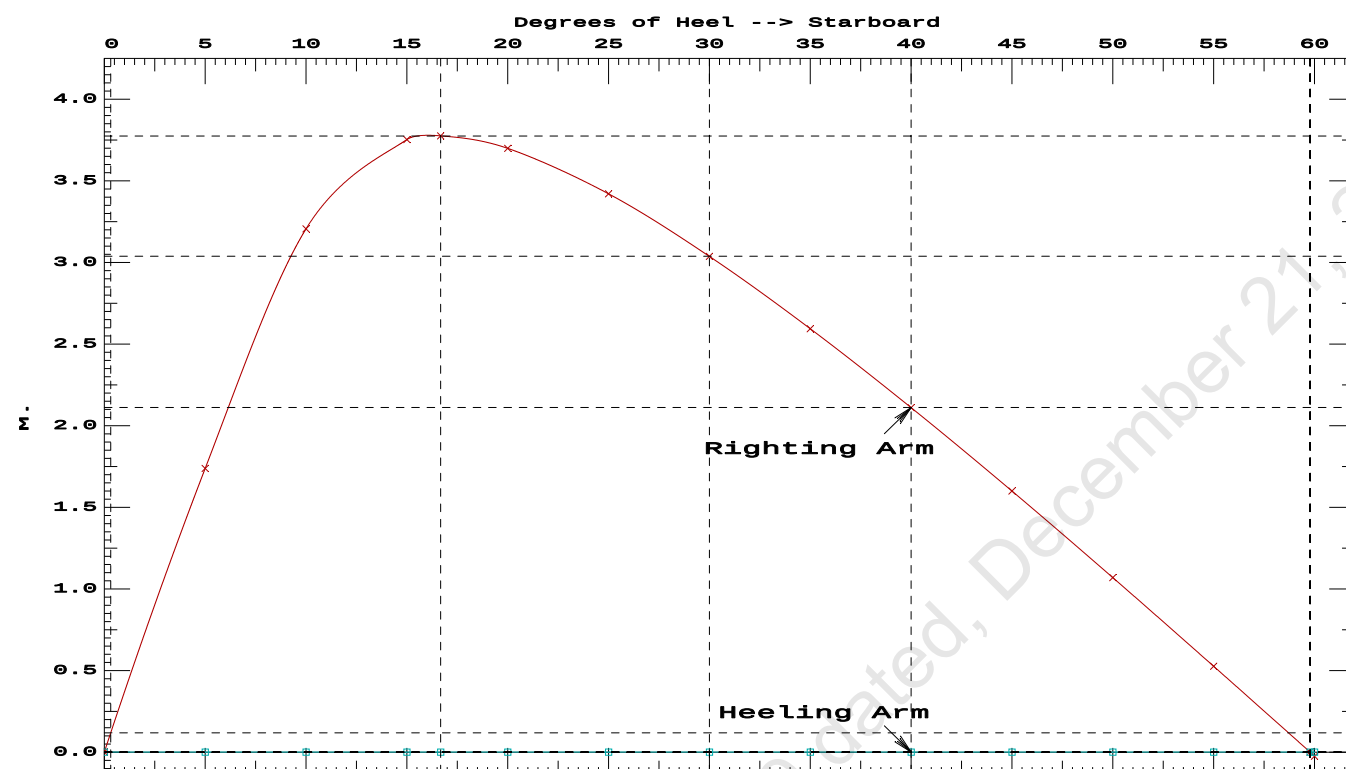
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.00 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	369798 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	2207.40 P

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Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd



22-10-21 12:43:28 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

Condition 15 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.194 @ 50.00f, 1.199 @ 25.00f, 1.204 @ 0.00

Trim: Aft 0.010/50.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			455.34	24.928f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Total Fixed----->			656.64	20.523f	0.000	5.863	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			979.84	24.968f	0.000	4.427	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	979.84	24.968f	0.001s	0.603	

Righting Arms:	0.000	0.001s			
External Arms:	0.000	0.001s			
Residual Righting Arms:	0.000	0.000s			
Part-----SpGr-----WPA-----LCF-----TCF-----BML-----BMT					
Total Waterplane----> 1.025	844.0	24.995f	0.028s	161.42	23.446
	MT/cm-----m.-MT/cm-----GML-----GMT				
	8.65		30.88	157.60	19.622
Distances in METERS.-----				Moments in m.-MT.	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 17.22 PONT00N

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.180 @ 50.00f, 1.140 @ 25.00f, 1.100 @ 0.00
Trim: Fwd 0.080/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	455.34	24.928f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Total Fixed----->	605.34	20.609f	0.000	3.545			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			928.54	25.270f	0.000	2.837	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		928.54	25.274f	0.001s	0.572	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001s			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	841.4	25.047f	0.018s	168.80	24.676	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.62	30.93	166.54	22.412		
Distances in METERS.-----					Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

22-10-21 12:43:28 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.609f TCG = 0.000 VCG = 3.545

Origin Depth	Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.087	0.09f	0.00	928.54	0.000	0.000	0.0000
1.087	0.09f	0.97s	928.54	0.000	0.392	0.0033
1.073	0.09f	5.00s	928.54	0.000	2.004	0.0876
0.986	0.11f	10.00s	928.53	0.000	3.659	0.3372
0.741	0.14f	15.00s	928.54	0.000	4.369	0.6944
0.498	0.18f	19.32s	928.54	0.000	4.493	1.0311
0.459	0.18f	20.00s	928.54	0.000	4.490	1.0847
0.172	0.23f	25.00s	928.54	0.000	4.370	1.4739
-0.116	0.28f	30.00s	928.54	0.000	4.134	1.8458

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 605.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

22-10-21 12:43:28 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.609f TCG = 0.000 VCG = 3.545

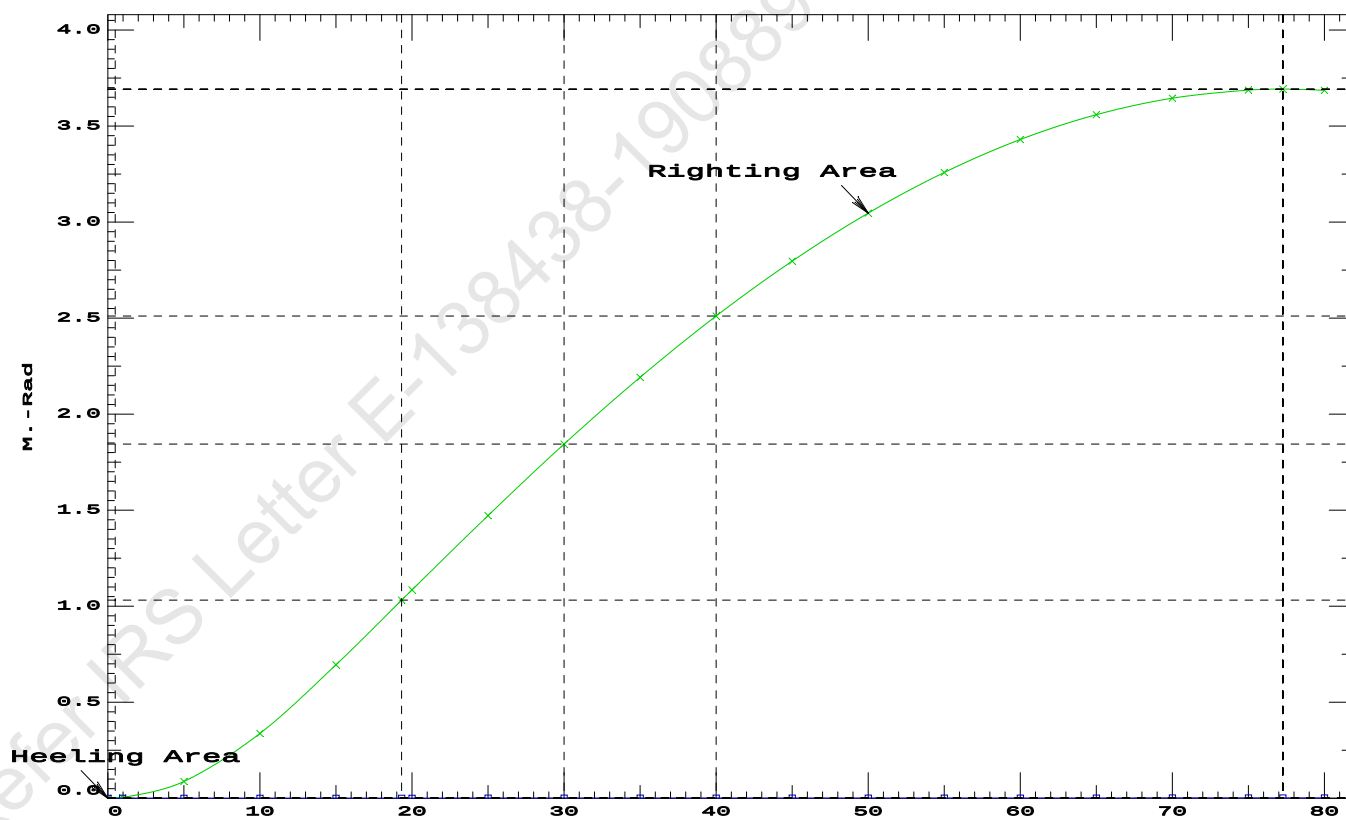
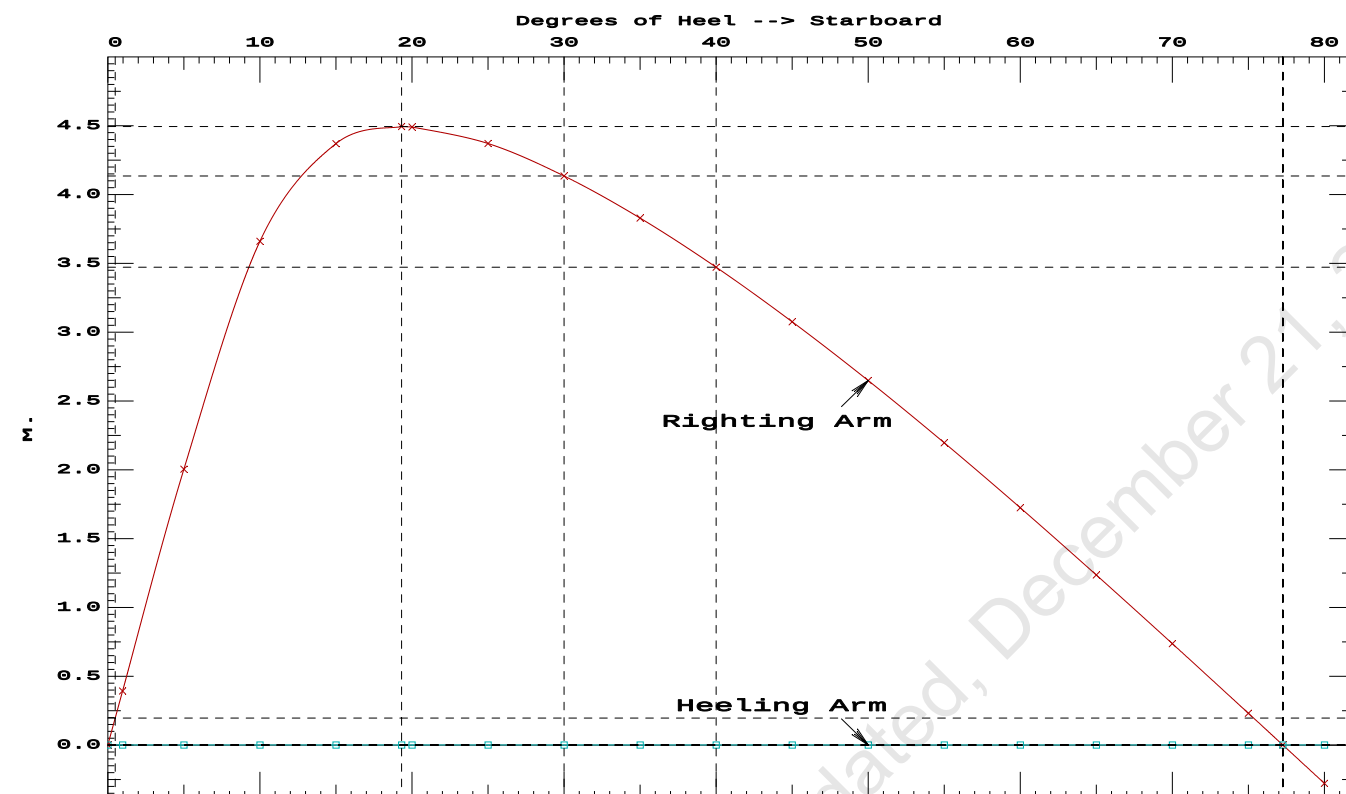
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.087	0.09f	0.00	928.54	0.000	0.000	0.0000
1.087	0.09f	0.97s	928.54	0.000	0.392	0.0033
1.073	0.09f	5.00s	928.54	0.000	2.004	0.0876
0.986	0.11f	10.00s	928.53	0.000	3.659	0.3372
0.741	0.14f	15.00s	928.54	0.000	4.369	0.6944
0.498	0.18f	19.32s	928.54	0.000	4.493	1.0311
0.459	0.18f	20.00s	928.54	0.000	4.490	1.0847
0.172	0.23f	25.00s	928.54	0.000	4.370	1.4739
-0.116	0.28f	30.00s	928.54	0.000	4.134	1.8458
-0.404	0.32f	35.00s	928.54	0.000	3.828	2.1937
-0.690	0.37f	40.00s	928.54	0.000	3.471	2.5125
-0.972	0.41f	45.00s	928.54	0.000	3.075	2.7984
-1.246	0.46f	50.00s	928.54	0.000	2.648	3.0483
-1.512	0.50f	55.00s	928.54	0.000	2.196	3.2598
-1.766	0.53f	60.00s	928.54	0.000	1.723	3.4309
-2.007	0.57f	65.00s	928.54	0.000	1.235	3.5601
-2.232	0.59f	70.00s	928.54	0.000	0.736	3.6462
-2.439	0.61f	75.00s	928.54	0.000	0.230	3.6884
-2.527	0.62f	77.27s	928.54	0.000	0.000	3.6930
-2.627	0.63f	80.00s	928.53	0.000	-0.278	3.6863

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 605.34 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 76.30 P

22-10-21 12:43:28
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONT00N

LIGHTSHIP PARTICULARS

Lightship Weight	=	455.34	T
VCG	=	3.000	M above Baseline
LCG	=	24.928	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)

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GHS 17.22 PONT00N

USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

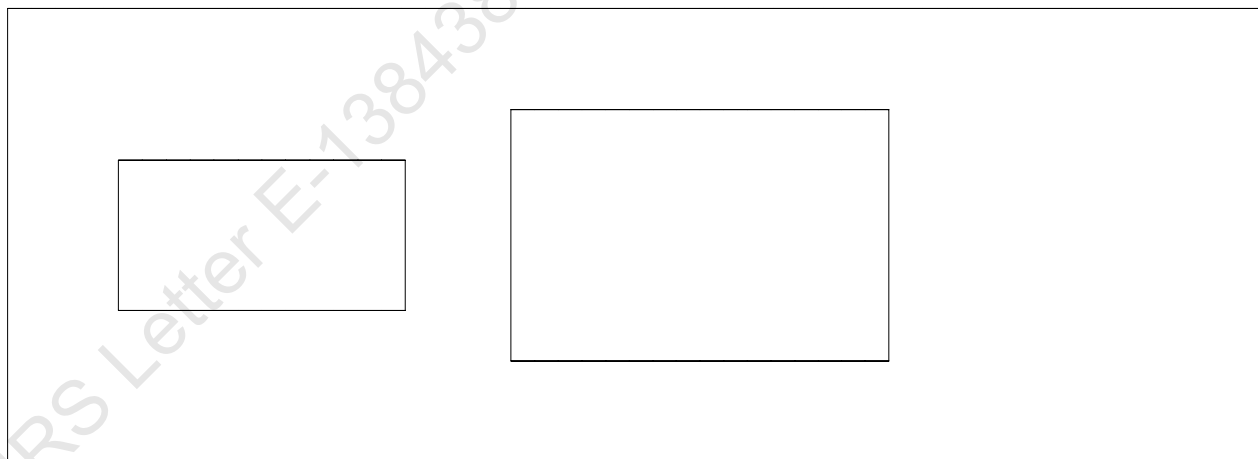
WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic

Profile View



Plan View



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GHS 17.22 PONT00N

LATERAL PLANE STATUS

USK draft: 0.512 @ 50.00f, 0.512 @ 25.00f, 0.512 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	22.3	25.000f	-0.253	121.4	25.000f	1.281
CRANE				57.0	10.100f	5.000
DECKCARGO.C				29.4	27.500f	3.500
Total Lateral Plane->	22.3	25.000f	-0.253	207.8	21.267f	2.615
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.612 @ 50.00f, 0.612 @ 25.00f, 0.612 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.8	25.000f	-0.302	116.9	25.000f	1.229
CRANE				57.0	10.100f	4.900
DECKCARGO.C				29.4	27.500f	3.400
Total Lateral Plane->	26.8	25.000f	-0.302	203.3	21.184f	2.572
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.712 @ 50.00f, 0.712 @ 25.00f, 0.712 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	31.3	25.000f	-0.351	112.4	25.000f	1.176
CRANE				57.0	10.100f	4.800
DECKCARGO.C				29.4	27.500f	3.300
Total Lateral Plane->	31.3	25.000f	-0.351	198.8	21.097f	2.529
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.812 @ 50.00f, 0.812 @ 25.00f, 0.812 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.9	25.000f	-0.400	107.8	25.000f	1.123
CRANE				57.0	10.100f	4.700
DECKCARGO.C				29.4	27.500f	3.200
Total Lateral Plane->	35.9	25.000f	-0.400	194.2	21.006f	2.487
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.912 @ 50.00f, 0.912 @ 25.00f, 0.912 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	40.4	25.000f	-0.450	103.3	25.000f	1.070
CRANE				57.0	10.100f	4.600
DECKCARGO.C				29.4	27.500f	3.100
Total Lateral Plane->	40.4	25.000f	-0.450	189.7	20.911f	2.445
Distances in METERS.						

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GHS 17.22 PONTON

LATERAL PLANE STATUS

USK draft: 1.012 @ 50.00f, 1.012 @ 25.00f, 1.012 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	45.0	25.000f	-0.497	98.7	25.000f	1.018
CRANE				57.0	10.100f	4.500
DECKCARGO.C				29.4	27.500f	3.000
Total Lateral Plane->	45.0	25.000f	-0.497	185.1	20.809f	2.405
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.112 @ 50.00f, 1.112 @ 25.00f, 1.112 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	49.7	25.000f	-0.545	94.0	25.000f	0.966
CRANE				57.0	10.100f	4.400
DECKCARGO.C				29.4	27.500f	2.900
Total Lateral Plane->	49.7	25.000f	-0.545	180.4	20.699f	2.367
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.212 @ 50.00f, 1.212 @ 25.00f, 1.212 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	54.4	25.000f	-0.594	89.3	25.000f	0.914
CRANE				57.0	10.100f	4.300
DECKCARGO.C				29.4	27.500f	2.800
Total Lateral Plane->	54.4	25.000f	-0.594	175.7	20.584f	2.329
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.312 @ 50.00f, 1.312 @ 25.00f, 1.312 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	59.1	25.000f	-0.642	84.5	25.000f	0.863
CRANE				57.0	10.100f	4.200
DECKCARGO.C				29.4	27.500f	2.700
Total Lateral Plane->	59.1	25.000f	-0.642	170.9	20.462f	2.291
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.412 @ 50.00f, 1.412 @ 25.00f, 1.412 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	63.9	25.000f	-0.691	79.8	25.000f	0.810
CRANE				57.0	10.100f	4.100
DECKCARGO.C				29.4	27.500f	2.600
Total Lateral Plane->	63.9	25.000f	-0.691	166.2	20.333f	2.255
Distances in METERS.						

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GHS 17.22 PONT00N

LATERAL PLANE STATUS

USK draft: 1.512 @ 50.00f, 1.512 @ 25.00f, 1.512 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	68.7	25.000f	-0.739	75.0	25.000f	0.759
CRANE				57.0	10.100f	4.000
DECKCARGO.C				29.4	27.500f	2.500
Total Lateral Plane->	68.7	25.000f	-0.739	161.4	20.194f	2.220
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.612 @ 50.00f, 1.612 @ 25.00f, 1.612 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	73.6	25.000f	-0.786	70.1	25.000f	0.708
CRANE				57.0	10.100f	3.900
DECKCARGO.C				29.4	27.500f	2.400
Total Lateral Plane->	73.6	25.000f	-0.786	156.5	20.044f	2.188
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.712 @ 50.00f, 1.712 @ 25.00f, 1.712 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	78.5	25.000f	-0.834	65.2	25.000f	0.657
CRANE				57.0	10.100f	3.800
DECKCARGO.C				29.4	27.500f	2.300
Total Lateral Plane->	78.5	25.000f	-0.834	151.6	19.883f	2.157
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.812 @ 50.00f, 1.812 @ 25.00f, 1.812 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	83.4	25.000f	-0.882	60.3	25.000f	0.607
CRANE				57.0	10.100f	3.700
DECKCARGO.C				29.4	27.500f	2.200
Total Lateral Plane->	83.4	25.000f	-0.882	146.7	19.712f	2.128
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.912 @ 50.00f, 1.912 @ 25.00f, 1.912 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	88.3	25.000f	-0.930	55.4	25.000f	0.556
CRANE				57.0	10.100f	3.600
DECKCARGO.C				29.4	27.500f	2.100
Total Lateral Plane->	88.3	25.000f	-0.930	141.8	19.529f	2.100
Distances in METERS.-----						

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

LATERAL PLANE STATUS

USK draft: 2.012 @ 50.00f, 2.012 @ 25.00f, 2.012 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	93.2	25.000f	-0.978	50.5	25.000f	0.505
CRANE				57.0	10.100f	3.500
DECKCARGO.C				29.4	27.500f	2.000
Total Lateral Plane->	93.2	25.000f	-0.978	136.9	19.333f	2.073
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.112 @ 50.00f, 2.112 @ 25.00f, 2.112 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	98.2	25.000f	-1.025	45.5	25.000f	0.455
CRANE				57.0	10.100f	3.400
DECKCARGO.C				29.4	27.500f	1.900
Total Lateral Plane->	98.2	25.000f	-1.025	131.9	19.118f	2.050
Distances in METERS.-----						

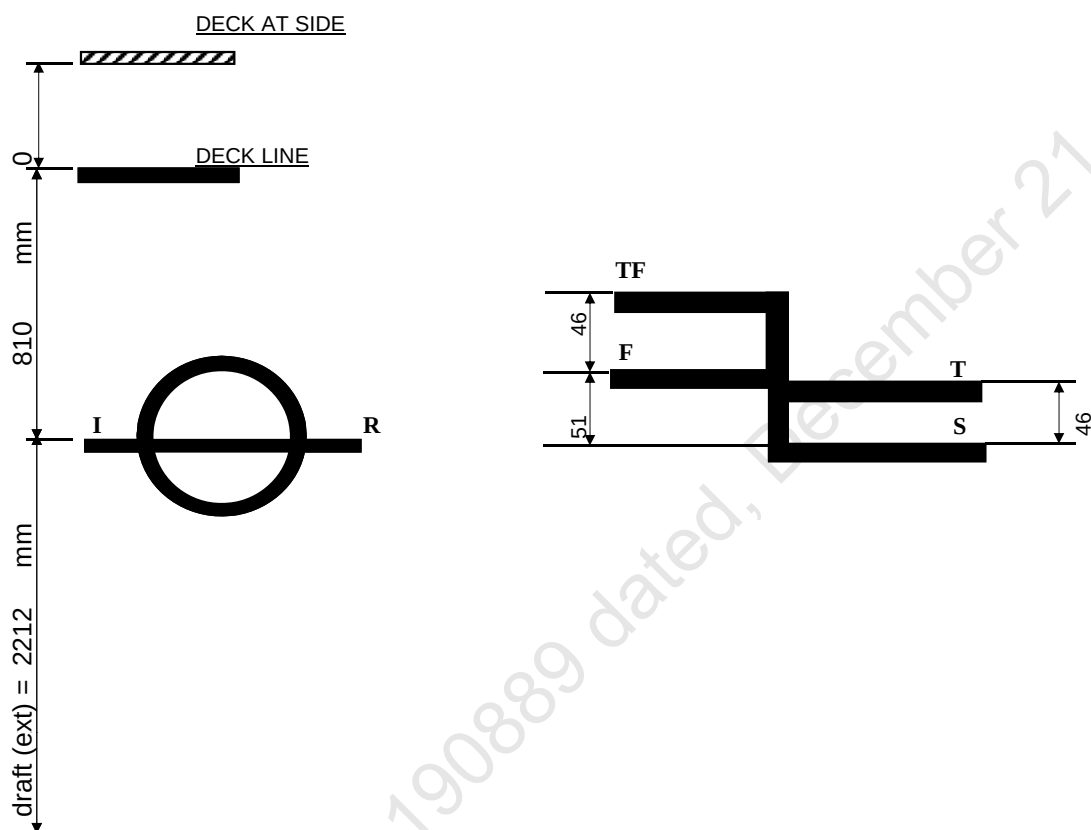
LATERAL PLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	103.2	25.000f	-1.073	40.5	25.000f	0.405
CRANE				57.0	10.100f	3.300
DECKCARGO.C				29.4	27.500f	1.800
Total Lateral Plane->	103.2	25.000f	-1.073	126.9	18.887f	2.029
Distances in METERS.-----						

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1893.06	1437.72
FRESH WATER	759	2.263	1892.81	1437.47
TROPICAL	764	2.258	1935.52	1480.18
TROPICAL FRESH WATER	713	2.309	1934.23	1478.89

Deck Plate Thickness = 10 mm

Keel Plate Thickness = 12 mm



MEMORANDUM OF FREEBOARDS

4 October 2021

Chisti Marine Engineering Works.

Yard No. 018

Type B freeboard

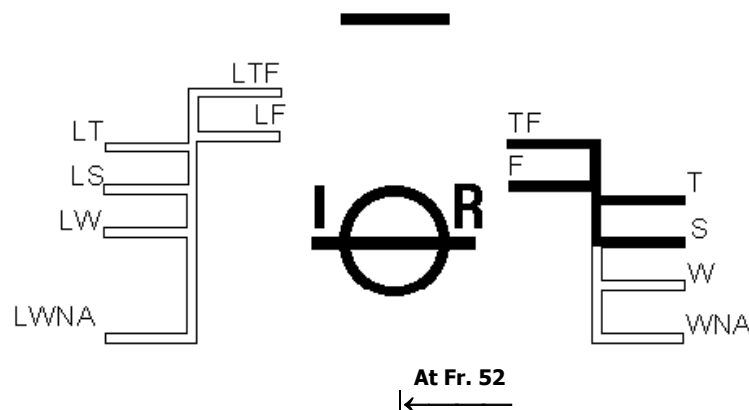
Freeboard Length : 48.000 m

Summer WL Extreme Draught : 2.212 m
- load line amidships

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of
* ~~Wood~~ / * steel Deck * at side / * ~~continued to side~~. (*Delete words not applicable).

Freeboards from Deck Line		Seasonal, Fresh Water, or Timber allowances		
Tropical Fresh Water	713	mm.	97	mm. above summer
Fresh Water	759	mm.	51	mm. above summer
Tropical	764	mm.	46	mm. above summer
Summer	810	mm.	Upper edge of line through centre	
Winter	Not	mm.	Assigned	mm. below summer
Winter North Atlantic	Not	mm.	Assigned	mm. below summer
Timber Tropical fresh water	—	mm.	—	mm. above timber summer
Timber fresh water	—	mm.	—	mm. above timber summer
Timber Tropical	—	mm.	—	mm. above timber summer
Timber Summer	—	mm.	—	mm. above centre of ring
Timber Winter	—	mm.	—	mm. below timber summer
Timber Winter North Atlantic	—	mm.	—	mm. below timber summer

Valid only for unmanned operation while under tow along the Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.





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Indian Register of Shipping
CIN: U61100MH1975NPL018244

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Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-138165-185843

22-October-2021

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002662
Project name	Chisti Marine Engineering Works - CHISTI/018 and CHISTI/019
Specific Yard No(s):	CHISTI/018
Project Manager	Rishi Doshi
Project In-Charge	Veerendra P

Document Details	
Plan No.	CM21-V-102-0207
Plan Title	Draft Survey Report
Rev.No.	0
Upload.No.	3
Submitted Date	21-October-2021
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

The following Lightship particulars are approved based on the Draft Survey conducted on 14 Oct 2021:

Lightship Weight : 455.340 T

Lightship LCG : 24.928 m Forward of A.P (Frame 0)

Lightship VCG : 3.000 m Above Baseline. (VCG taken at the level of Main Deck)

Lightship TCG : 0.000 m from Centerline.

Final Stability Booklet prepared based on the approved Lightship particulars should be submitted for approval.

Copy of the approved Draft Survey report should form part of the Final Stability Booklet.

Thanking you,
Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Pradeepkumar Karayil

Senior Surveyor

This is an electronically published document and does not require signature.